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CHAPTER 1: MTEP OVERVIEW

1.1 MISO Overview



Midcontinent Independent System Operator (MISO) is an independent, not-for-profit organization that delivers safe, cost-effective electric power across 15 U.S. states and the Canadian province of Manitoba. MISO is committed to reliable, nondiscriminatory operation of the bulk power transmission system and collaborating with all stakeholders to create cost-effective and innovative solutions for our changing industry.

Scope of Operations

Generation Capacity

- 183,693 MW (market)
- 198,600 MW (reliability)

Historic Summer Peak Load (set July 20, 2011)

- 127,125 MW (market)
- 130,917 MW (reliability)

Historic Winter Peak Load (set January 6, 2014)

- 109,336 MW (market)
- 117,903 MW (reliability)

Transmission

- Approximately 72,000 miles
(Reliability Footprint)

Balancing Authorities

- 38 Local Balancing Authorities in MISO

Network Model

- 293,671 SCADA data points
- 6,692 generating units

Registered Wind	25,974 MW
Registered In-Service Wind Generation Capacity	22,082 MW
Registered Solar Generation Capacity	464 MW
Registered In-Service Solar Generation Capacity	464 MW

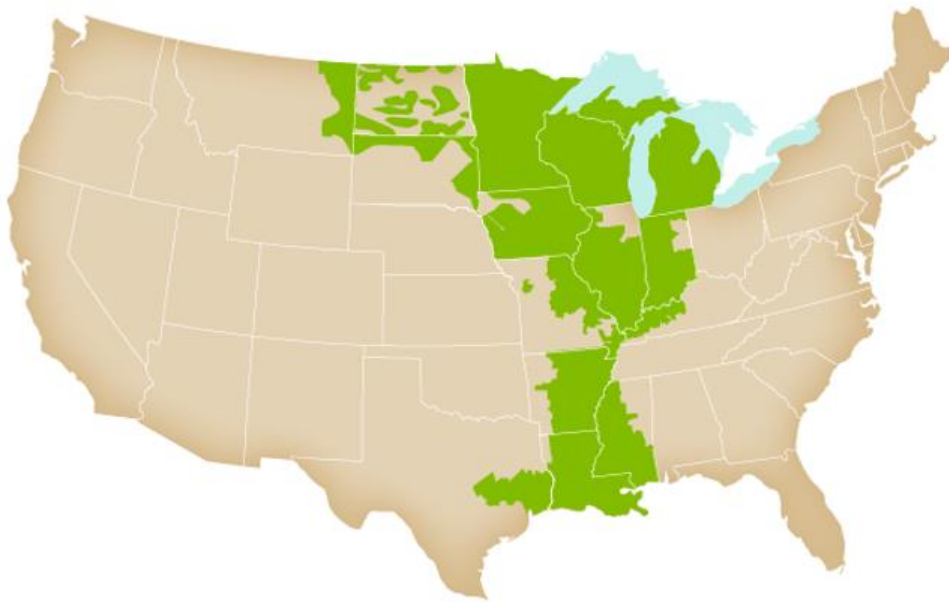
Markets Overview

MISO manages one of the world's largest energy and operating reserve markets using security-constrained economic commitment and dispatch of generation.

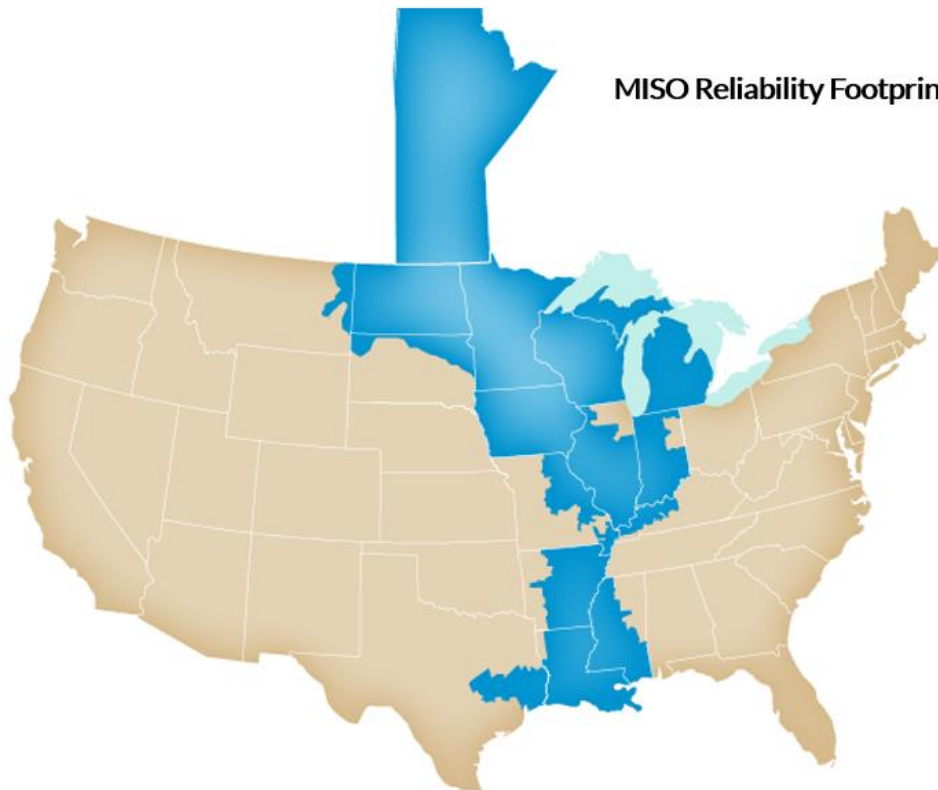
The Energy and Operating Reserves Market includes a Day-Ahead Market, a Real-Time Market, and a Financial Transmission Rights market. These markets are operated and settled separately.

- \$24.7 billion annual gross market charges (2019)
- 471 Market Participants that serve approximately 42 million people

MISO Market Footprint



MISO Reliability Footprint



MISO Transmission Infrastructure Investment

MISO is a not-for-profit organization; it does not own any generation or transmission facilities. MISO strictly manages the generation and flow of electricity across the high-voltage lines within its territory. Through the collaborative efforts of a diverse group of industry participants, MISO manages approximately 72,000 miles of transmission lines across 15 states and the Canadian Province of Manitoba.

This iteration of the MTEP report, MTEP20, builds and expands on the 16 prior years of projects since 2003 for a total of over \$42 billion of investment. The last ten cycles are illustrated below (Figure 1.1-1). MISO's proposed new projects for this MTEP cycle are detailed in Section 1.3, Chapter 4 and Appendix A of this report.

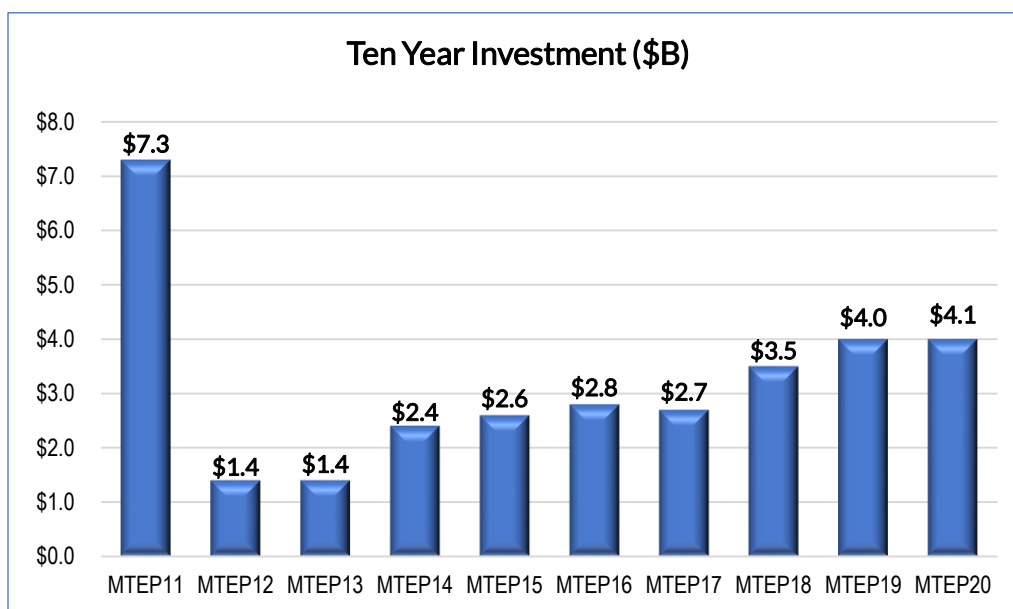


Figure 1.1-1: MTEP investment over the last 10 cycles (\$billion), data as of 10-5-2020

MTEP11 reflects the approval of 17 Multi-Value Projects discussed later in this section and MTEP14 saw an initial rise in investment with the first full year of the South region joining as a MISO member. Consistent growth in infrastructure investment in the MISO footprint over the last several years reflects the predominance in Transmission Owner-driven upgrades to improve efficiency, reliability, and safety in outdated system designs and replace aging assets to make the system more resilient.

MISO's transmission planning responsibilities include the monitoring of previously approved Appendix A projects. MISO surveys all Transmission Owners and Selected Developers every quarter to determine the progress of each project. These [status updates](#) are reported to the MISO Board of Directors and posted quarterly to the MISO Transmission Expansion Plan page at [misoenergy.org](https://www.misoenergy.org)¹.

Full archived files of all [previous MTEP Reports](#) can be accessed via the MISO Transmission Expansion Plan page at [misoenergy.org](https://www.misoenergy.org).

¹ MISO Transmission Expansion Plan website address: <https://www.misoenergy.org/planning/planning/>

Transmission Facility Investment

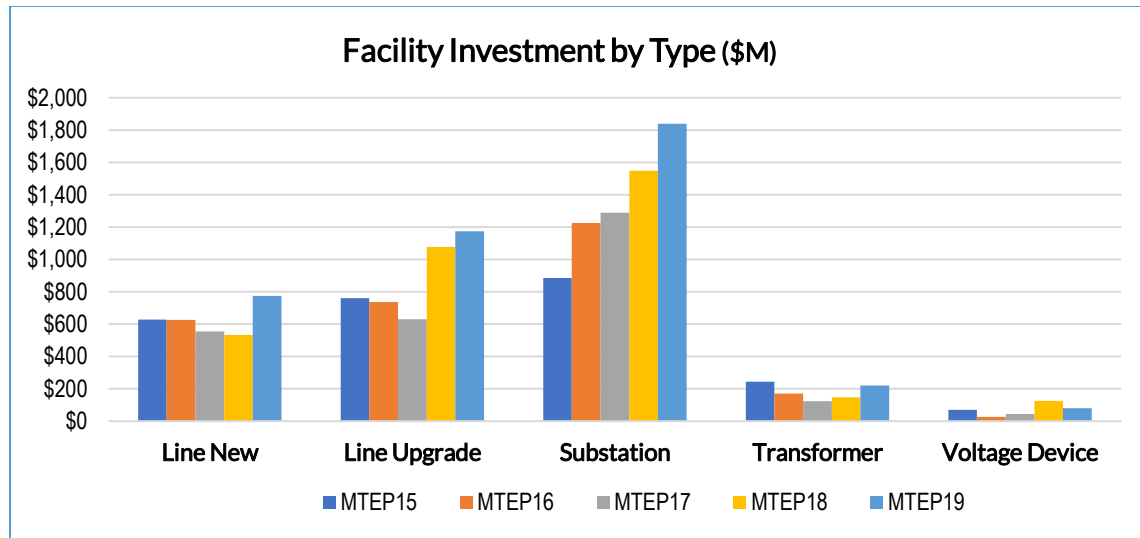


Figure 1.1-2: Appendix A Project data for previously approved five MTEP cycles

MTEP20 Nameplate Fuel Capacity

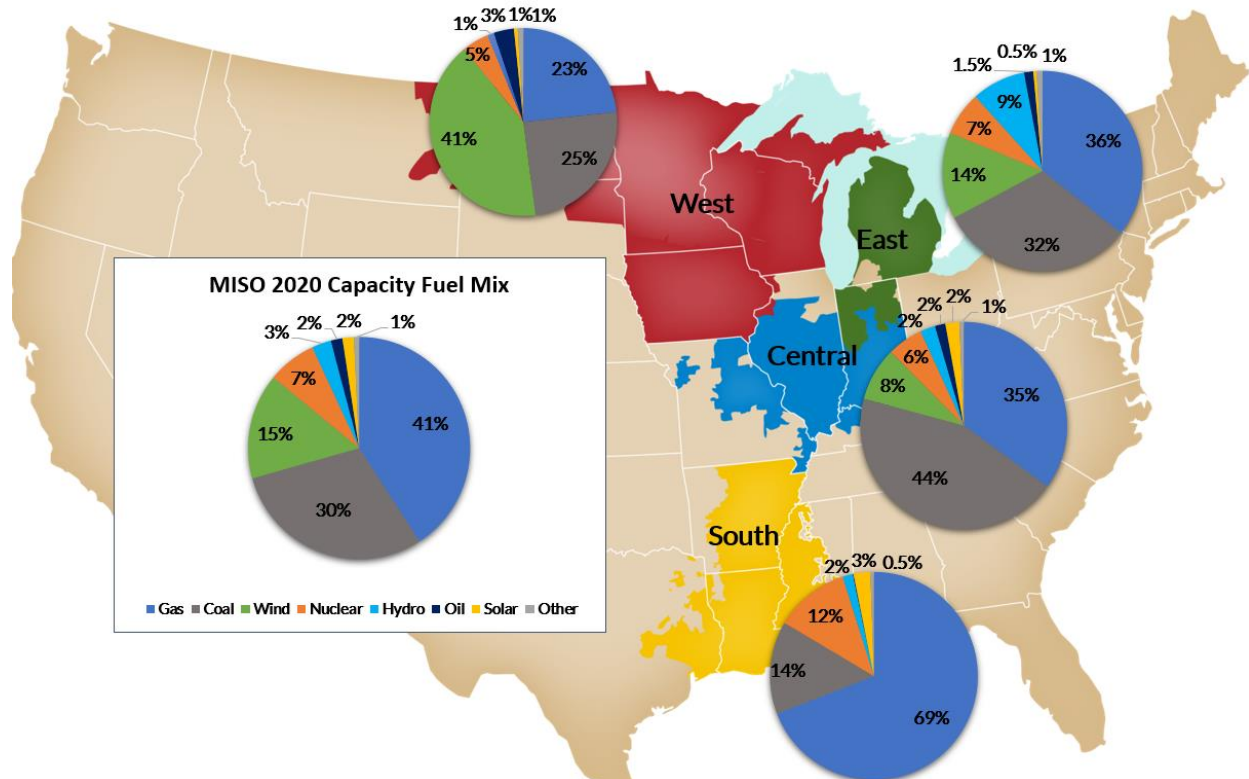


Figure 1.1-3: MTEP20 Nameplate dispatchable capacity (MW) by fuel type in year 2020

Line Miles Summary

MISO has approximately 68,000 circuit-miles of transferred functional control transmission lines serving as the backbone of the footprint (Figure 1.1-4) in the United States. Currently, the West region holds 40% of total footprint line miles, the South region holds 24%, the Central region holds 20%, and the East at 16%.

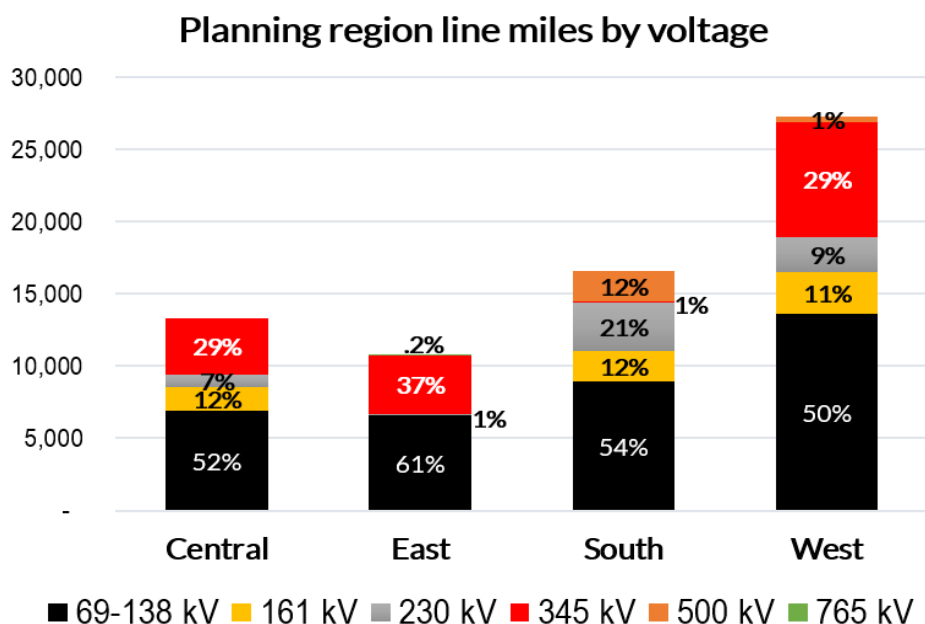


Figure 1.1-4: 2020 in-service transferred circuit miles by voltage class (kV)

Multi-Value Project Portfolio Status Update

The 17 Multi-Value Projects (MVP) are a regional portfolio of high voltage transmission projects, developed and approved in MTEP11 to support public policy, reliability and economic needs throughout the MISO footprint². The cost and schedule of the MVP portfolio is monitored and reported quarterly (see: [MVP Dashboard](#) at [misoenergy.org](#)). As of the second quarter of 2020, 15 MVPs have gone into service.

In accordance with Attachment FF of the Tariff, MISO provides an updated annual view into the projected benefits of the MVP portfolio by recreating the analysis performed for the original business case. MTEP20 will feature a full 'triennial' review, where all the benefit metrics calculated in the original business case are reassessed using the latest MTEP20 system planning information and models. The MVP Review report and detailed business case spreadsheet will be available on the MISO public website (see: [Multi-Value Projects](#) at [misoenergy.org](#)), with an expected completion and posting in Q4 2020.

² When the MVP portfolio was approved, the MISO footprint did not include MISO South

Planning Guiding Principles

- Make the benefits of an economically efficient electricity market available to customers by identifying transmission projects which provide access to electricity at the lowest total electric system cost.
- Develop a transmission plan that meets all applicable NERC and Transmission Owner planning criteria and safeguards local and regional reliability through identification of transmission projects to meet those needs.
- Support state and federal energy policy requirements by planning for access to a changing resource mix.
- Provide an appropriate cost allocation mechanism that ensures that costs of transmission projects are allocated in a manner roughly commensurate with the projected benefits of those projects.
- Analyze system scenarios and make the results available to state and federal energy policy makers and other stakeholders to provide context to inform choices.
- Coordinate planning processes with neighbors and work to eliminate barriers to reliable and efficient operations.

1.2 MISO Transmission Planning Process

A goal of the MTEP report is to satisfy the regulatory requirements as specified in Federal Energy Regulatory Commission Orders and the ISO Agreement. The report provides an annual snapshot with results and recommendations from the continuous planning processes MISO undertakes. MISO's planning process follows established guiding principles to ensure reliability, support policy requirements, and enable a competitive market to benefit all customers (see figure 1.2-1 planning regions).

MISO will continue to follow federal and state policy as well as monitor fuel prices, plant retirements, and announced member plans for any changing industry trends. The ability not only to meet peak demand, but to move bulk power from resource areas to load centers across the footprint in all hours of the day will be needed to maintain system reliability and improve efficiency with the evolving resource fleet. Regional planning solutions will play an essential role in optimizing the natural and geographic diversity of these resources.

Periodically, the System Planning Committee of the Board of Directors provides input into MISO's Planning Guiding Principles. The most recent review and approval occurred in March 2019.

Planning Functions

The planning process includes these functions, which are described in detail in the [Transmission Planning Business Practices Manual](#)³.

- Model development
- Generator interconnection planning
- Transmission service planning
- Cyclical regional expansion planning activities
- Interregional coordination with neighboring transmission planning regions
- System Support Resource studies for unit suspension or retirement
- Transmission-to-Transmission interconnection
- Load interconnections
- Focus studies

MISO addresses current dramatic changes in the projected resource mix in its current strategic vision, which focuses on the key trends of de-marginalization, decentralization and digitalization.

³ <https://www.misoenergy.org/legal/business-practice-manuals/>

Furthermore, MISO identified the critical themes to address the associated challenges and opportunities: availability, flexibility and visibility. Understanding these resource characteristics will be key to understanding the characteristics that will directly influence the composition and volume of new interconnection requests.

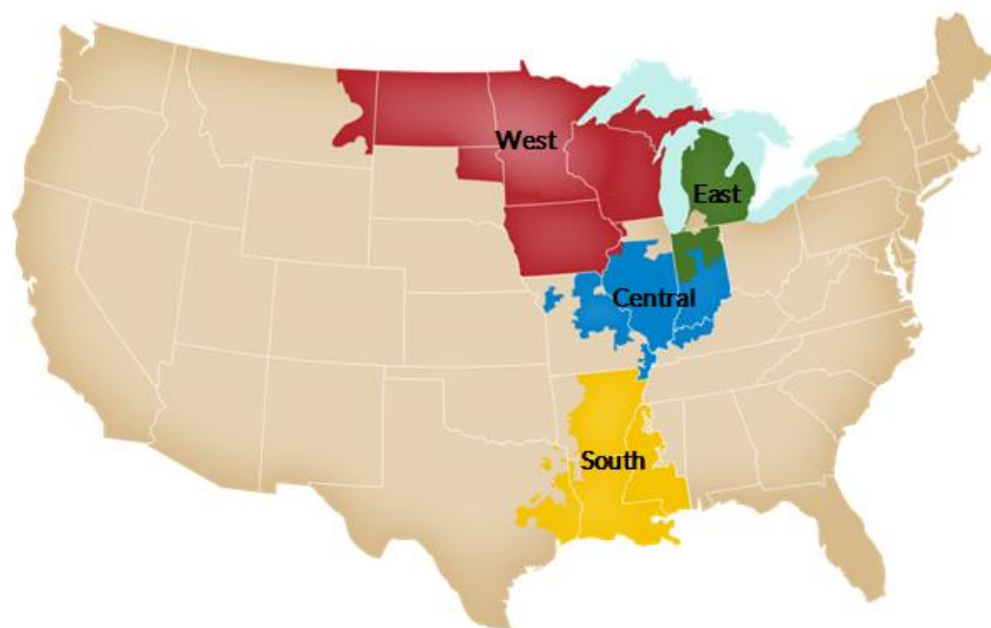


Figure 1.2-1: MISO footprint planning regions

Project Input and Stakeholder Coordination

Each planning cycle commences with regional model development; identification of potential expansions from the local planning processes of the Transmission Owners; identification and selection of transmission needs driven by public policy requirements to be included as transmission issues; and identification by stakeholders or MISO staff of potential expansions that address the transmission issues. Each cycle concludes with recommendations to the MISO Board of recommended solutions to the transmission issues evaluated.

Transmission Owner plans developed through local planning processes are included in the beginning of each regional planning cycle as potential solutions to local transmission issues identified by the Transmission Owners.

MISO's regional planning process makes evaluations — with stakeholder input from the Sub-regional Planning Meetings, the Planning Subcommittee, and the Planning Advisory Committee — throughout the cycle to develop expansion plans to meet the needs of the system. This multi-party collaborative process allows analysis of all projects with regional and inter-regional impact for their combined effects on the Transmission System. Moreover, the design of this collaborative process ensures that the MTEP addresses transmission issues within the applicable planning horizon in an efficient and cost-effective manner, while considering the input of stakeholders.

MISO's planning process ensures local needs are integrated with regional requirements

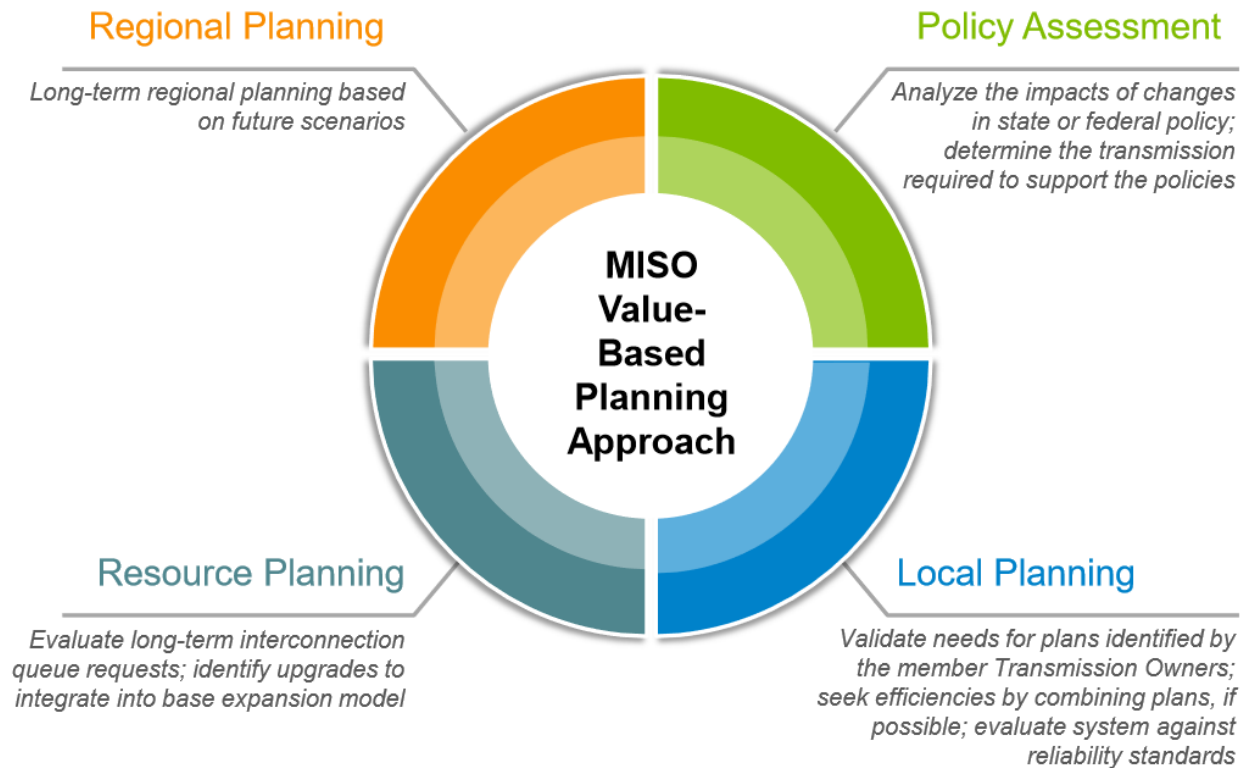


Figure 1.2-2: The MISO Value-Based Planning approach

Key Planning Cycle Milestones

While following the MISO value-based planning approach (Figure 1.2-2), key milestones in the typical MTEP development process include requirements and timelines for data submittal, review, and comment at each of these milestone points as described in the [Transmission Planning Business Practices Manual](#) posted on the MISO public website.

- Model development
- Identification and selection of transmission needs driven by public policy requirements to be included as transmission issues
- Testing models against applicable planning criteria
- Development of possible solutions to identified transmission issues
- Selection of preferred solution
- Determination of funding and cost responsibility
- Monitoring progress on solution implementation

Planning Analysis Methods

Planning analyses performed by MISO test the transmission system under a wide variety of conditions using standard industry applications to model steady state powerflow, angular and voltage stability, short-circuit, and economic parameters, as determined appropriate by MISO to be compliant with applicable criteria and the Tariff. MISO collaborates with Transmission Owners, other transmission providers, transmission customers, and other stakeholders to develop appropriate planning models that reflect expected system conditions for the planning horizon.

[Models](#) are available to stakeholders with security measures as provided for in the Transmission Planning Business Practices Manual. MISO provides opportunity for stakeholders to review and comment on the posted models before commencing planning studies.

Project Approval

MISO staff formally recommends a set of projects to the MISO Board of Directors for review and approval after MISO completes an independent review of all proposed projects and addresses any stakeholder feedback received. These projects make up Appendix A of the MTEP report and represent the preferred solutions to the identified transmission needs of the MISO transmission planning process.

Proposed transmission upgrades with sufficient lead times are included in Appendix B for further review in future planning cycles.

Details of the project proposal process and transmission projects reviewed this cycle are summarized in Section 1.3 and Chapter 4 of the MTEP20 report.

Interregional Coordination and Planning Studies

On an annual basis MISO works with neighboring transmission planning regions Southwest Power Pool (SPP) and PJM Interconnection (PJM) to identify issues on the seams, perform studies, and jointly evaluate transmission solutions that may be more efficient or effective than a corresponding regional solution. While MISO has a separate Joint Operating Agreement (JOA) with both SPP and PJM that details specific processes and criteria, the high-level interregional coordination activities are similar on each seam:

- 1) Exchange modeling data and other system information (typically performed in Q4).
- 2) Review identified issues on the seam (typically performed in Q1).
- 3) Evaluate whether to perform an interregional study based on the identified issues.

MISO performs joint coordinated system plan (CSP) studies with SPP and PJM on a regular basis, in accordance the timelines and frequencies dictated in their respective JOAs. A CSP study may have a targeted scope or a more complex scope requiring a longer study period, and can include reliability, economic and/or public policy issues. All interregional issues and CSP study efforts are coordinated through a public Interregional Planning Stakeholder Advisory Committee (ISPAC) consisting of representatives and interested parties from each RTO community.

In addition to the joint study efforts with SPP and PJM, MISO performs studies as needed with neighboring entities of the Southeastern Regional Transmission Planning (SERTP) group and the Independent Electricity System Operator of Ontario (IESO). While the study process is less formal, MISO and these entities still meet regularly to review interregional issues and possible areas of collaboration.

Details on planning procedures, on-going studies and stakeholder meetings can be found on the [Interregional Coordination](#) page of the MISO public website (misoenergy.org).

1.3 MTEP20 Investment Summary

The MTEP20 cycle proposes 515 new Appendix A projects as justified in this MISO Transmission Expansion Plan and represents \$4.159 billion in transmission infrastructure investment for the MISO region.

Overview of Tariff-defined Project Types

- **Baseline Reliability Project (BRP)** - Projects are Network Upgrades identified in the base case as required to ensure that the Transmission System is in compliance with applicable national Electric Reliability Organization reliability standards and reliability standards adopted by Regional Reliability Organizations, and applicable within the Transmission Provider Region. Baseline Reliability Project costs are allocated to the local Transmission Pricing Zone(s) and recovered through Attachment O by the Transmission Owner(s) developing the projects.
- **Generator Interconnection Project (GIP)** - Projects are New Transmission Access Projects that are associated with interconnection of new generation or the capacity modification of existing generation. Costs are primarily paid for by the interconnection customers with certain exceptions as specified in Attachment FF. Costs of network upgrades rated at 345 kV and above are eligible for 10 percent cost recovery on a system-wide basis.
- **Market Efficiency Project (MEP)** - Projects meet Attachment FF requirements for reduction in market congestion and are eligible for regional cost allocation. Projects qualify as Market Efficiency Projects based on cost and voltage thresholds and are developed to produce a benefit-to-cost ratio of 1.25 or greater. Costs are distributed to benefiting pricing zones, in accordance with Attachment FF of the Tariff.
- **Targeted Market Efficiency Project (TMEP)** - Projects are designed to alleviate historical market-to-market congestion between MISO and PJM Interconnection, while meeting certain cost and construction requirements. The costs of Targeted Market Efficiency Projects are allocated first between MISO and PJM Interconnection by the ratio of each RTO's Day-Ahead and Excess Congestion Fund congestion, offset by historical market-to-market payments. The MISO share of costs for the project is then allocated to beneficiaries using historical nodal load congestion contribution data.
- **Multi-Value Project (MVP)** - Projects meet Attachment FF requirements to provide regional public policy, economic and/or reliability benefits. Costs are shared with loads and export transactions in proportion to metered MWh consumption or export schedules.
- **Other** - Projects included in MTEP20 which do not qualify as Baseline Reliability Projects, New Transmission Access Projects, Targeted Market Efficiency Projects, Market Efficiency Projects, or Multi-Value Projects.
- **Transmission Delivery Service Project (TDSP)** - Projects are required to satisfy a transmission service request. The costs are generally assigned to the requestor.
- **Market Participant Funded Project (MPFP)** - Projects are defined as Network Upgrades fully funded by one or more market participants but owned and operated by an incumbent Transmission Owner.

Overview of MTEP20 Projects

MTEP20 Appendix A Projects

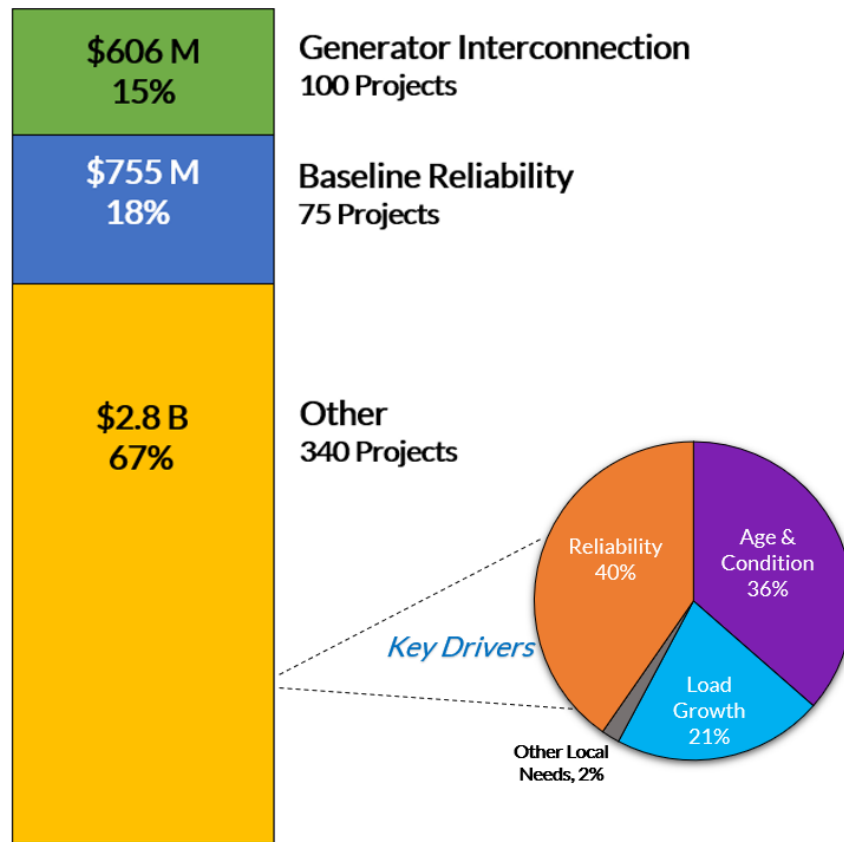
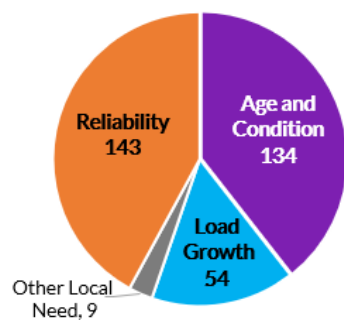


Figure 1.3-1: Appendix A New Project Investment (data as of 10-5-2020)

Of the 515 new Appendix A projects proposed in MTEP20 (Figure 1.3-1), 75 of them are Baseline Reliability; 100 are Generator Interconnection, and 340 are in the Other project category. Investment dollars in the Generator Interconnection category are over double that in the MTEP19 cycle. The remaining categories are similar to MTEP19 investment dollars.

Other project drivers - Count of projects



Other project drivers - Investment (\$M)

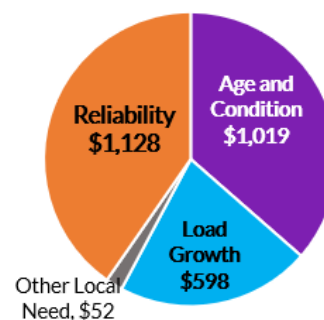


Figure 1.3-2: MTEP20 Appendix A Other category by project count and investment

The majority of Other projects address localized reliability issues, either due to aging transmission infrastructure, which increased from 29% last cycle to 37%, or local non-baseline reliability needs that are not dictated by NERC and regional reliability standards (Figure 1.3-2). Of the 56 projects in MTEP20 that cost \$20 million and above in all categories, 50% of those are categorized as Other to specifically address localized reliability issues.

The remaining projects mostly address distribution concerns, with a small percentage of projects targeting localized economic benefits or line relocations to accommodate other infrastructure.

Breakdown of each project category by planning region (Figure 1.3-3).

MTEP20 Regional Investment by Project Category

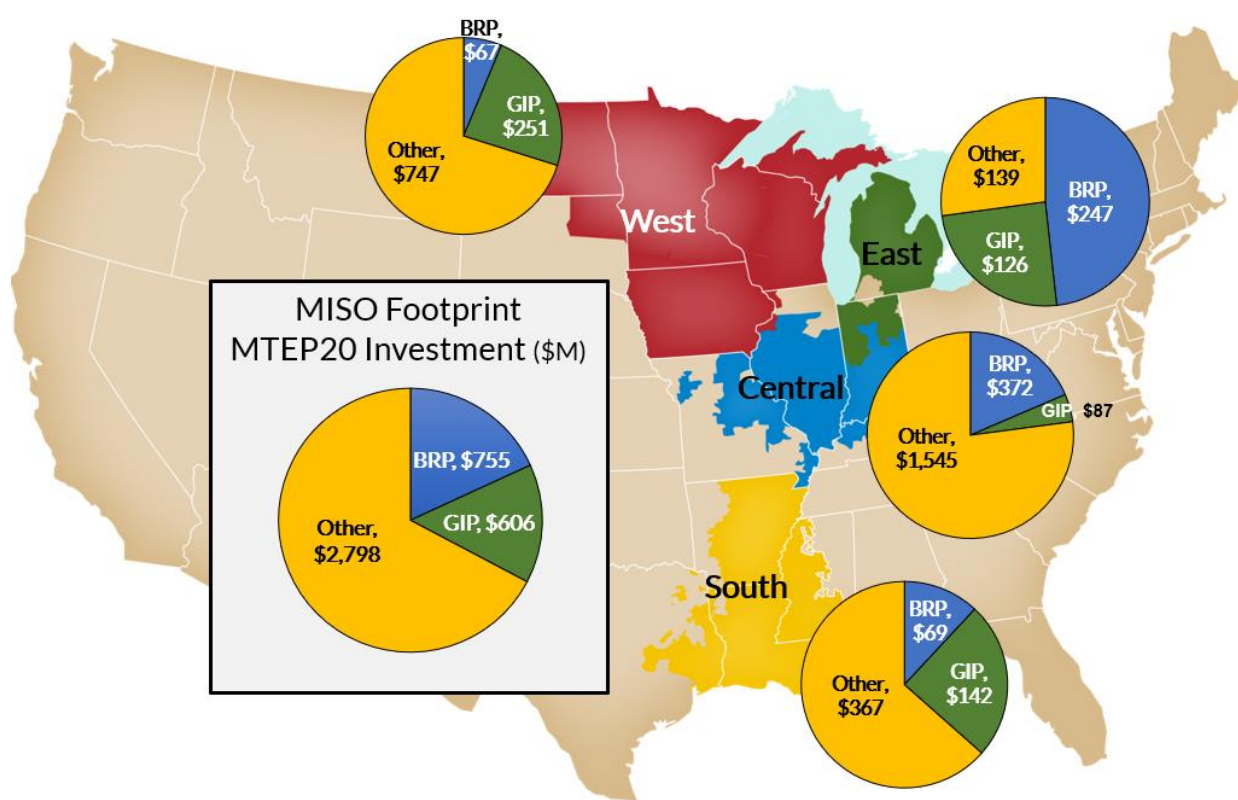


Figure 1.3-3: Regional MTEP20 investment by project category (data as of 10-5-2020)

The new projects recommended for approval in MTEP20 Appendix A are broken down by region and project type (Table 1.3-4).

MISO Region	BRP	GIP	Other	Total
Central	\$371,790,400	\$86,772,496	\$1,545,274,295	\$2,003,837,191
East	\$246,713,000	\$125,980,263	\$138,600,389	\$511,293,652
South	\$69,027,329	\$142,104,417	\$366,891,001	\$578,022,747
West	\$67,333,852	\$251,220,351	\$747,268,406	\$1,065,822,609
Grand Total	\$754,864,581	\$606,077,527	\$2,798,034,091	\$4,158,976,200

Table 1.3-4: MTEP20 Appendix A new project investment by category and planning region (data as of 10-5-2020)

New Appendix A projects are spread over 15 states, with nine states scheduled for more than \$150 million in new investment (Figure 1.3-5; Figure 1.3-6). These geographic trends vary greatly year to year as existing transmission capacity in other parts of the system is consumed and new build becomes necessary.

Consistent with MTEP19, MTEP20 Other projects reflect significant asset replacement in the Central region that implement updated system designs in order to operate more efficiently and reliably. Updating systems from straight buses to ring buses and breaker and a half are a priority for safety and reliability. See Chapter 4 and Appendix A for a detailed list of these projects.

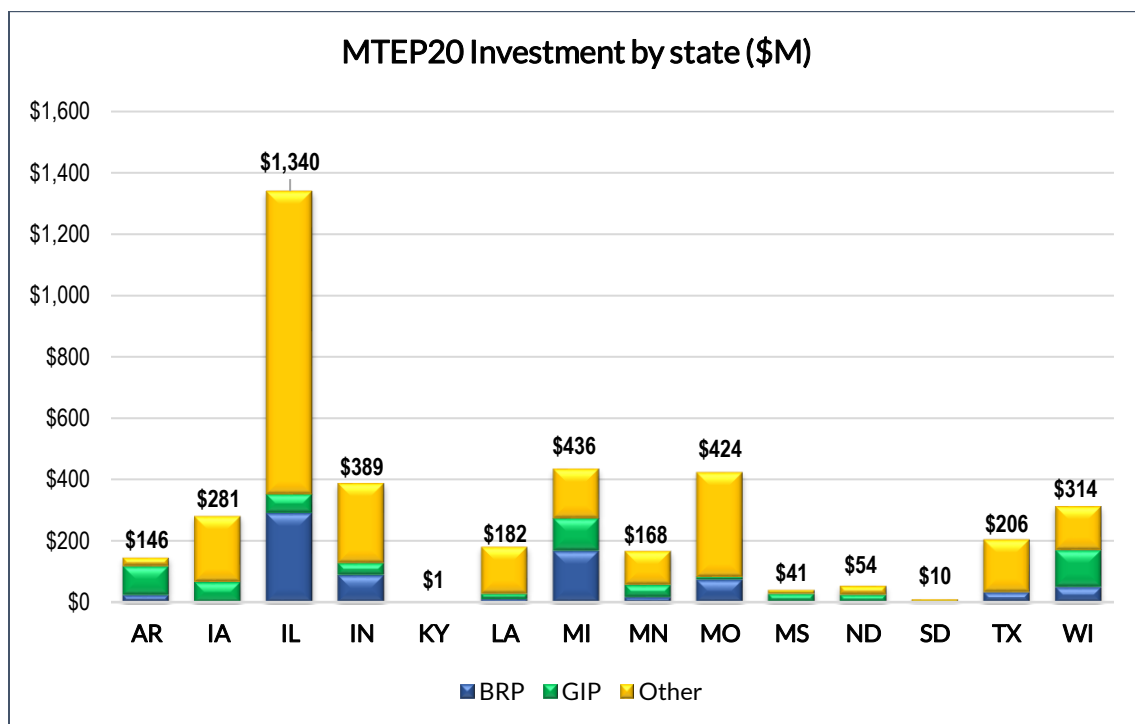


Figure 1.3-5: MTEP20 Appendix A investment categorized by state - (excludes multi-state blanket projects)

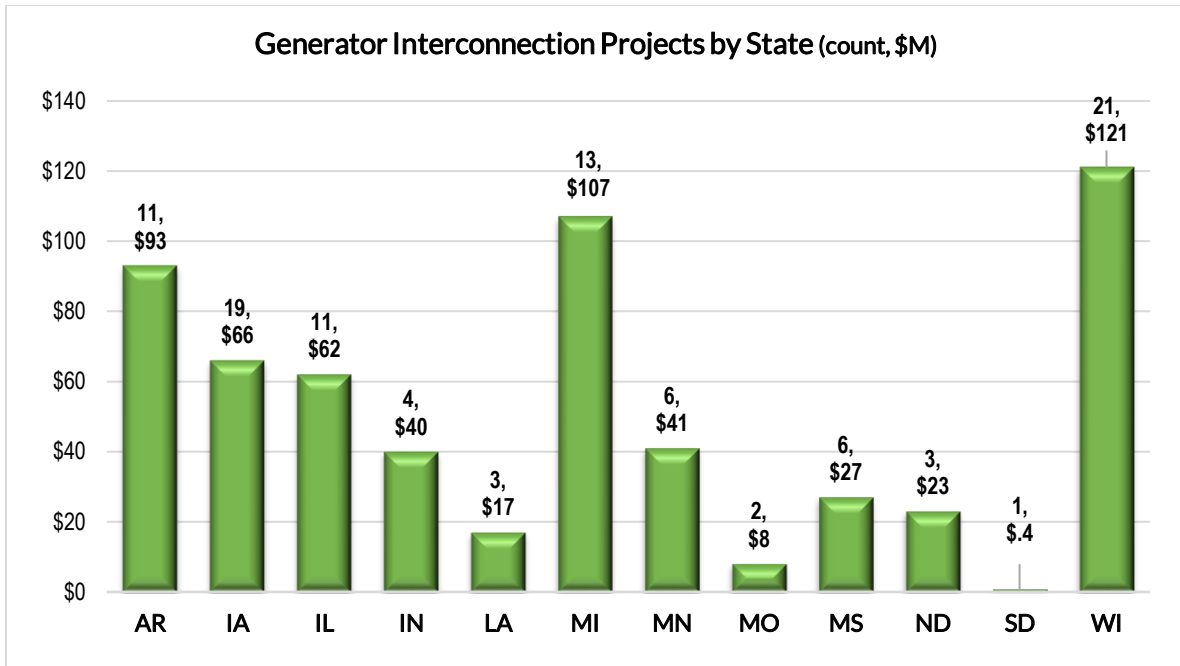


Figure 1.3-6: MTEP20 Appendix A Generator Interconnection projects by state

Facility Type

Each MTEP project is composed of one or more facilities, where each facility represents an individual element of the project. Examples of facilities include substations, transformers, circuit breakers or various types of transmission lines (Figure 1.3-7).

The majority of facility investment in the MTEP20 cycle, 45%, is dedicated to substation or switching station related construction and maintenance. This includes completely new substations as well as terminal equipment work, circuit breaker additions and replacements, or new transformers. Thirty-seven percent of MTEP facility costs go toward line upgrades, which include rebuilds, conversions and relocations. Eleven percent of facility costs are dedicated to new lines on new right-of-way across the MISO footprint.

Eighty-six percent of MTEP20 projects are declared to go into service by the end of 2023, ninety-five percent by the end of 2024.

MTEP20 Transmission Investment by Facility Type

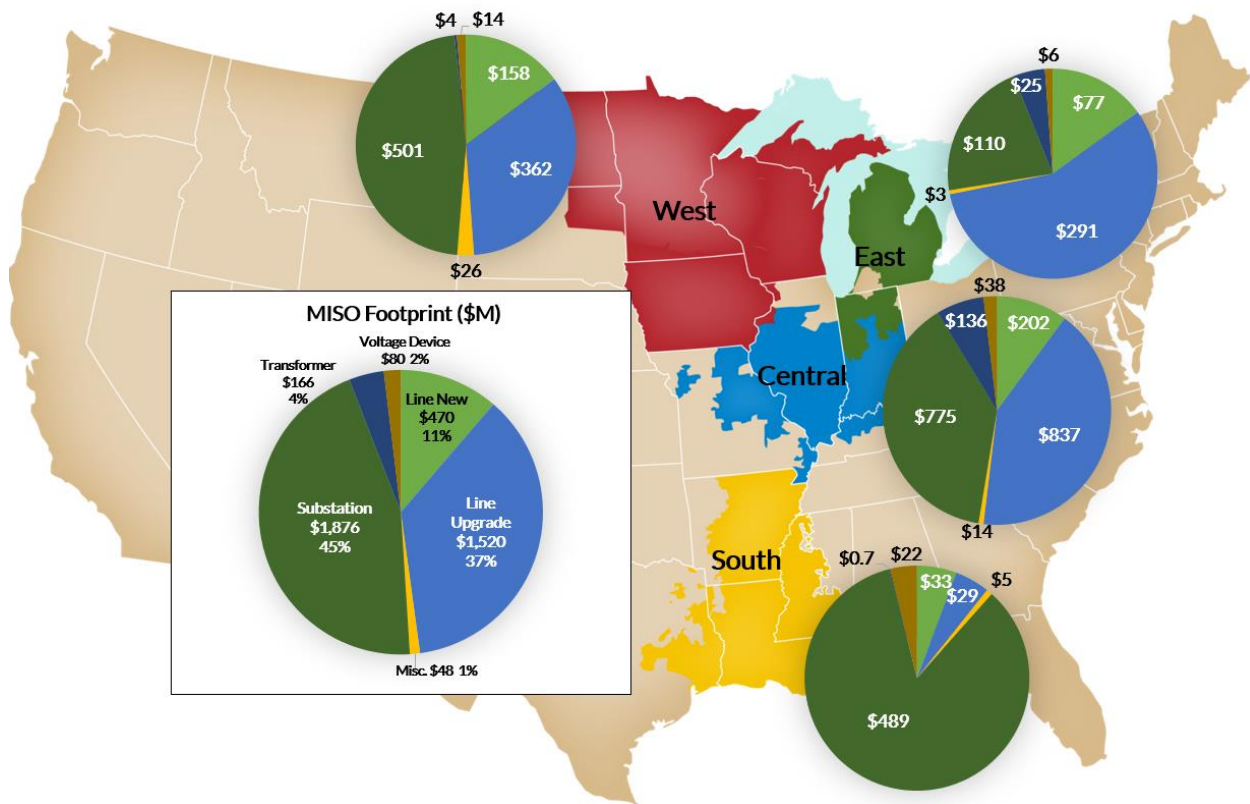


Figure 1.3-7: Facility type for new MTEP20 Appendix A projects by planning region

Allocation of Costs

MTEP20 includes a total of 39 new cost-share eligible generator interconnection projects (GIPs) for Appendix A. GIP costs are primarily paid for by the interconnecting customer (generator), however, a portion of the costs for certain network upgrades are eligible for regional cost allocation under Attachment FF of the MISO Tariff. Detailed allocations by pricing zone are provided in Appendix A1.

Indicative rates related to past MTEP cost-shared projects are calculated on an annual basis. Please refer to the reports posted on the MISO public website⁴ [here](https://www.misoenergy.org/planning/planning/schedule-26-and-26a-indicative-reports/).

⁴ Cost Allocation updates web address: <https://www.misoenergy.org/planning/planning/schedule-26-and-26a-indicative-reports/>

MTEP Appendix B

MTEP Appendix B contains all projects that have been validated by MISO as the preferred solution to address an identified system need based on current information and forecasts, but where it is prudent to defer the final recommendation of a solution to a subsequent MTEP cycle.

This generally occurs when the preferred project does not yet need a commitment based on anticipated lead-time and there is still some uncertainty as to the prudence of selecting this project over an alternative project given potential changes in projected future conditions. MTEP Appendix B is limited to Baseline Reliability Projects and Other Projects and will be reviewed by MISO in subsequent cycles.

CHAPTER 2: PORTFOLIO EVOLUTION

MISO manages the life cycle of an extensive and diverse fleet of resources that continues to experience a shift from conventional fossil fuel generation to advanced and carbon-free technologies. With the shift away from dispatchable generation close to load centers to remote variable energy resources, the transmission system no longer serves the same resources for which it was designed, and transmission upgrades are needed to enable integration of new resources.

In order to enable reliable and efficient effectuation of the resource portfolio evolution, MISO has to first understand and reflect on the past and identify future trends. This understanding is used to develop the spectrum of possibilities in the creation of MISO Futures. The trends and future reliability and efficiency impacts of resource evolution is the impetus behind what MISO coined the Reliability Imperative, a term that is used to describe the broad range of activities that are underway to anticipate and reliably adapt to the rapid changes that are and will be occurring. The Long Range Transmission Plan (LRTP) is one of the key activities that is part of the Reliability Imperative, and progress on that effort will continue to be reflected in future MTEP reports.

2.1 Historical Trends and Retirements

One aspect of the resource evolution that MISO assists its membership in managing is the retirement of generation facilities to ensure that the broader MISO footprint and markets remain reliable after resources are removed from service. Through the process articulated in Attachment Y of the MISO Tariff, resource owners submit a request to retire generation resources for MISO approval, which triggers an assessment into the impact that the requested resource would cause once it is retired from service. As a result of these analyses, any reliability issues are addressed through transmission reinforcements or other needed mitigation measures; and, if the reliability issue cannot be addressed prior to the planned retirement date, MISO may require the resource to remain in service as a system support resource (SSR) until the upgrade is complete, or mitigation is available. In recent years the need for system support resources has diminished and no generation resources are currently operating under a system support resource agreement.

Resource retirements of coal and gas-fired generation have seen a steady increase in the past several years as renewables have become economically and environmentally more attractive sources of energy. In the last 10 years MISO has experienced retirement of 24.9 GW of which 17.5 GW was coal based (Figure 2.1-1). The age of generating facilities retired in 2020 to date has declined to 46 years compared to 57 years in 2010 displaced largely by the interconnection of more economic renewable resources. Advancements in technology and interest in renewables are expected to continue the current trend.

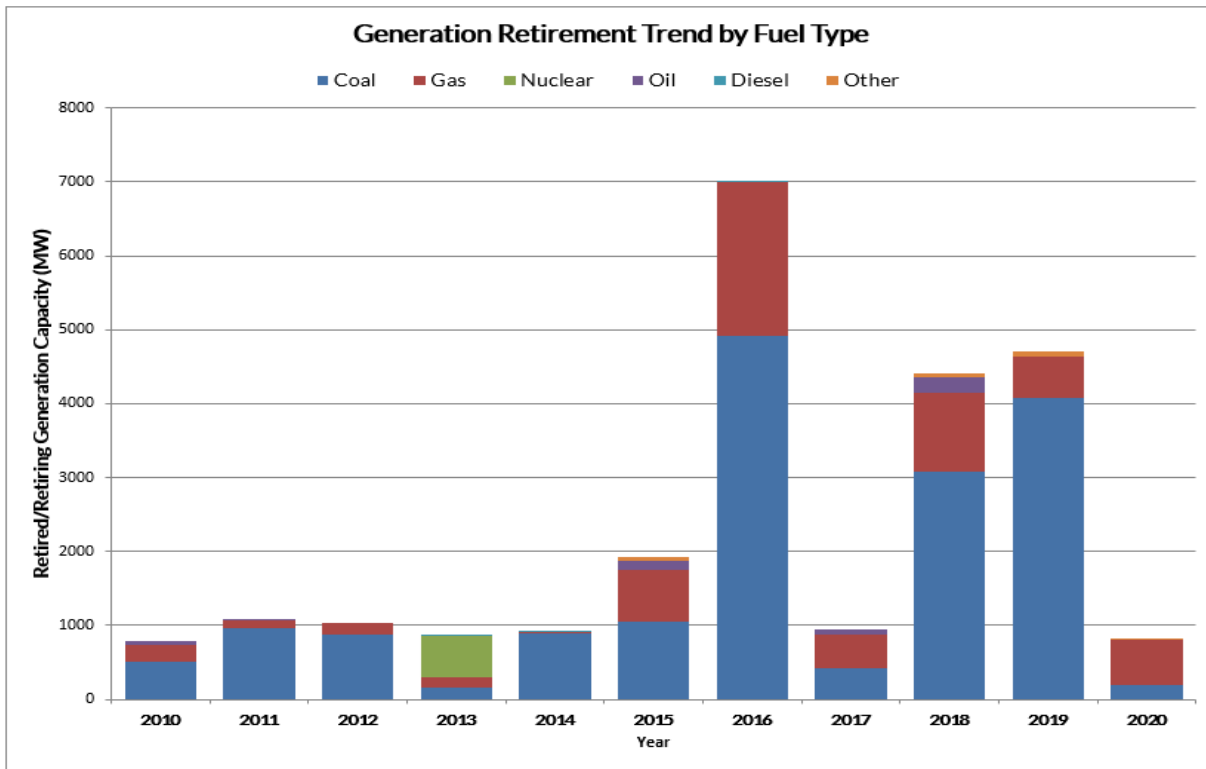


Figure 2.1-1: MW Generation Retirement by Fuel Type

2.2 Current State of Queue

The MISO Generator Interconnection (GI) queue provides an active and competitive mechanism to enable resource interconnections that will serve future energy and capacity needs. Projects submitted in the annual queue cycle are evaluated by MISO through an iterative study process to determine the reliability impacts and to identify transmission upgrades needed to support resource integration. Project viability is often tied to the costs of network upgrades with the most viable candidates successfully executing a Generator Interconnection Agreement (GIA).

The Generator Interconnection queue has experienced high volume over the last several years as a result of growing interest in renewable technology that has benefitted from declining costs of technology, favorable tax incentives and regulatory treatment. Wind has comprised a large portion of the interconnection queue volume in the last decade while solar resources have emerged more recently in part due to advances in solar technology and escalating regional transmission costs associated with integrating new wind development. As battery storage technology advances and interest continues to grow, MISO has seen an increase in number of the projects comprised of standalone storage or hybrid applications.

In 2020, MISO received 353 individual project requests representing a total of 52.5 GW of requested capacity during the application period that ended in June (Figure 2.2-1) marking a continuing trend of aggressive resource development. Solar installations have continued to trend upward, representing 69% of new entries and new wind development has shown a reversal of a downward trend from the prior year. Once the additional requests received in the latest queue cycle were validated, the total volume of active queued projects had reached 756 projects representing 113 GW.

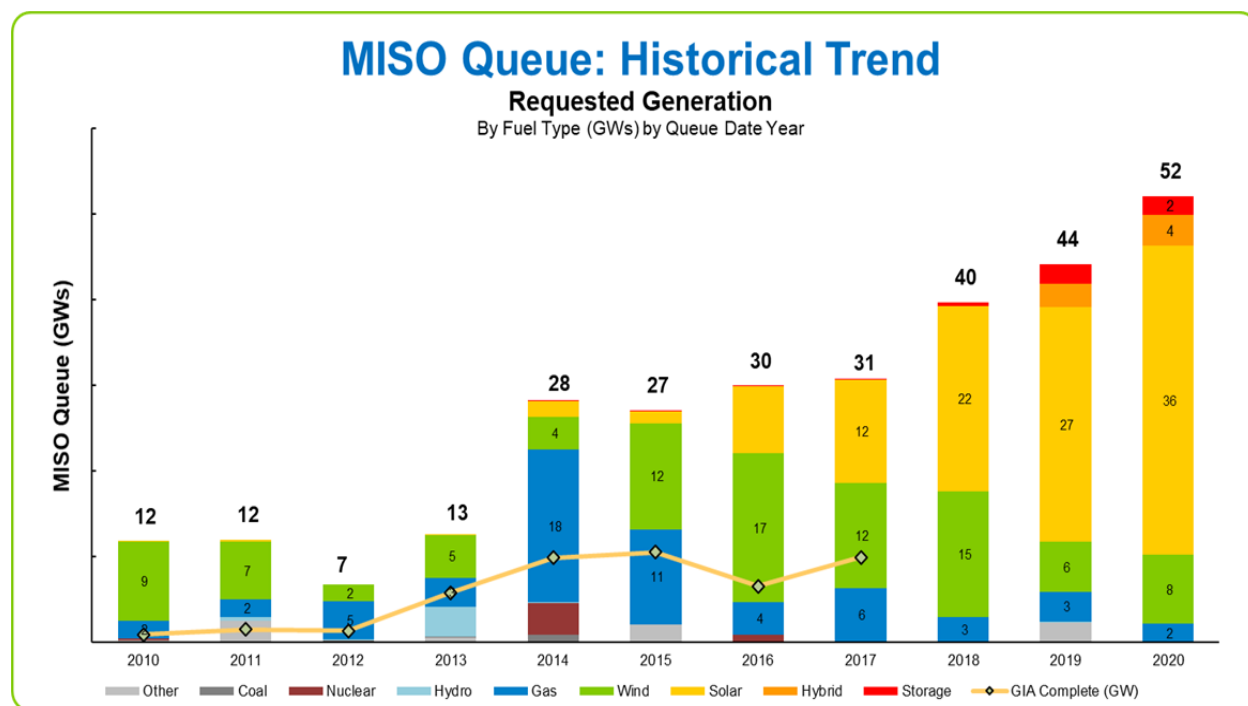


Figure 2.2-1 GI Queue by Fuel Type over the last ten years

As of September 21, the current state of the queue has 719 projects representing 108 GW of total capacity. The drop from our historic peak of 113 GW was mainly driven by the August decision point in the South Region where Interconnection Customers can withdraw from the queue before committing additional milestone payments. Renewables represent over 80% of the remaining capacity. The figure below is updated monthly on the MISO website under the [GIQ Web Overview](#) link on the [Generator Interconnection Queue](#) page. A list of all active projects can also be reviewed on the page. The five study regions in the GI queue currently have 13 active cycles in various stages of the process from the start of the DPP to GIA negotiations.

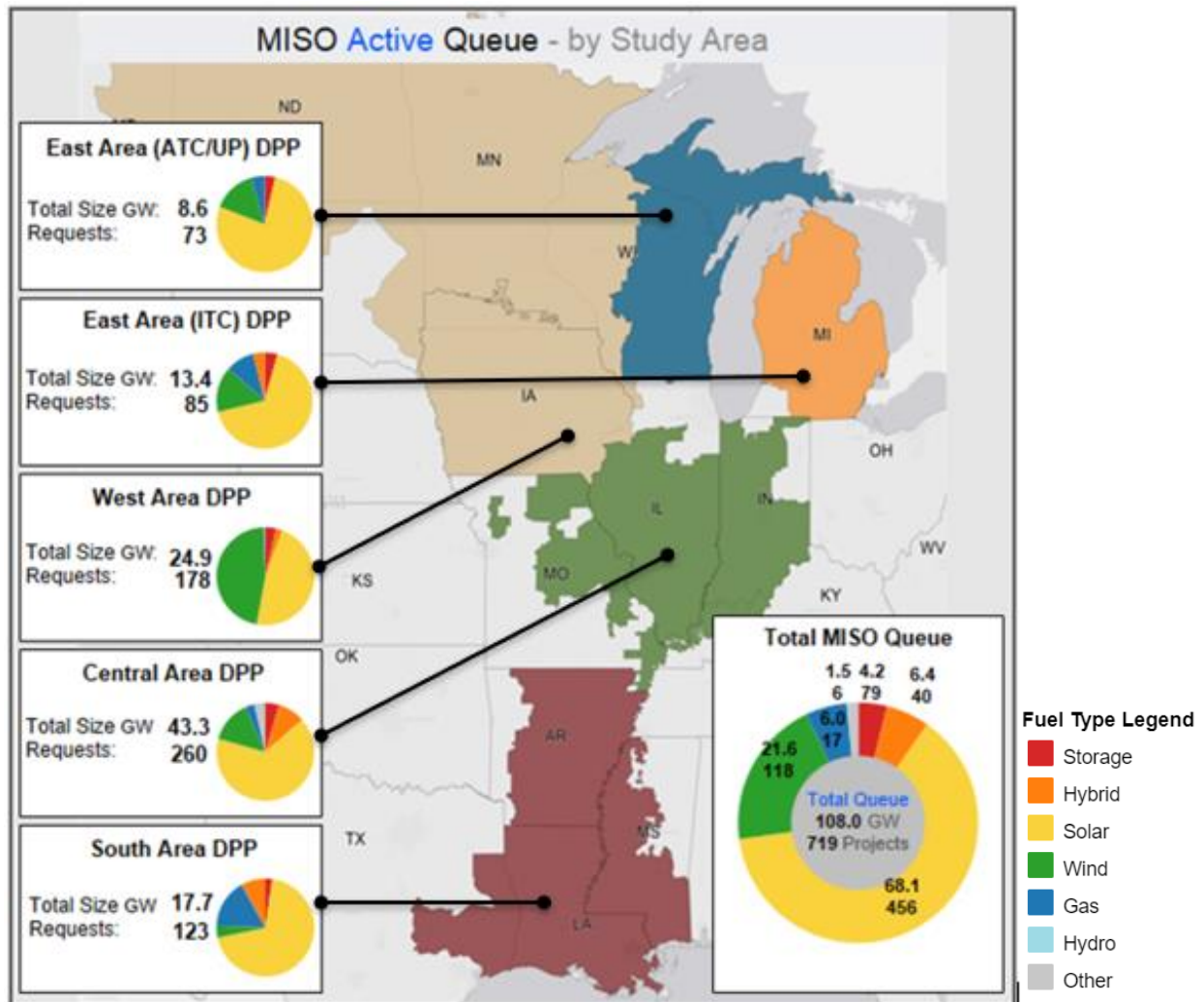


Figure 2.2-2: MISO Active Queue by Fuel Type as of 9-21-2020

2.3 The Cost of New Transmission to Integrate Generation is Changing and Impactful

In the past, the MISO Multi-Value Project (MVP) Portfolio facilitated new renewable resource development that delivered regional benefits by enabling access to lower cost energy. This broad regional approach to transmission planning recognized the benefits of a regional plan that would result in the most cost-effective transmission investment rather than an incremental build-out resulting from the Generator Interconnection process. Over the last decade, the continued interconnection of new resources has fully utilized the additional capacity provided by the MVPs resulting in the need for more network upgrades to support ongoing interconnection requests.

The continued growth of remotely located renewable resources has resulted in the need for major transmission upgrades with a significant increase in transmission costs incurred for resource interconnections. As the industry transitions away from traditional central station generation to more dispersed and variable energy resources, transmission investment will be needed to facilitate the change and support continued reliability. A comprehensive approach to system planning and resource interconnection recognizes broader benefits of transmission investment while facilitating resource evolution in a timely manner.

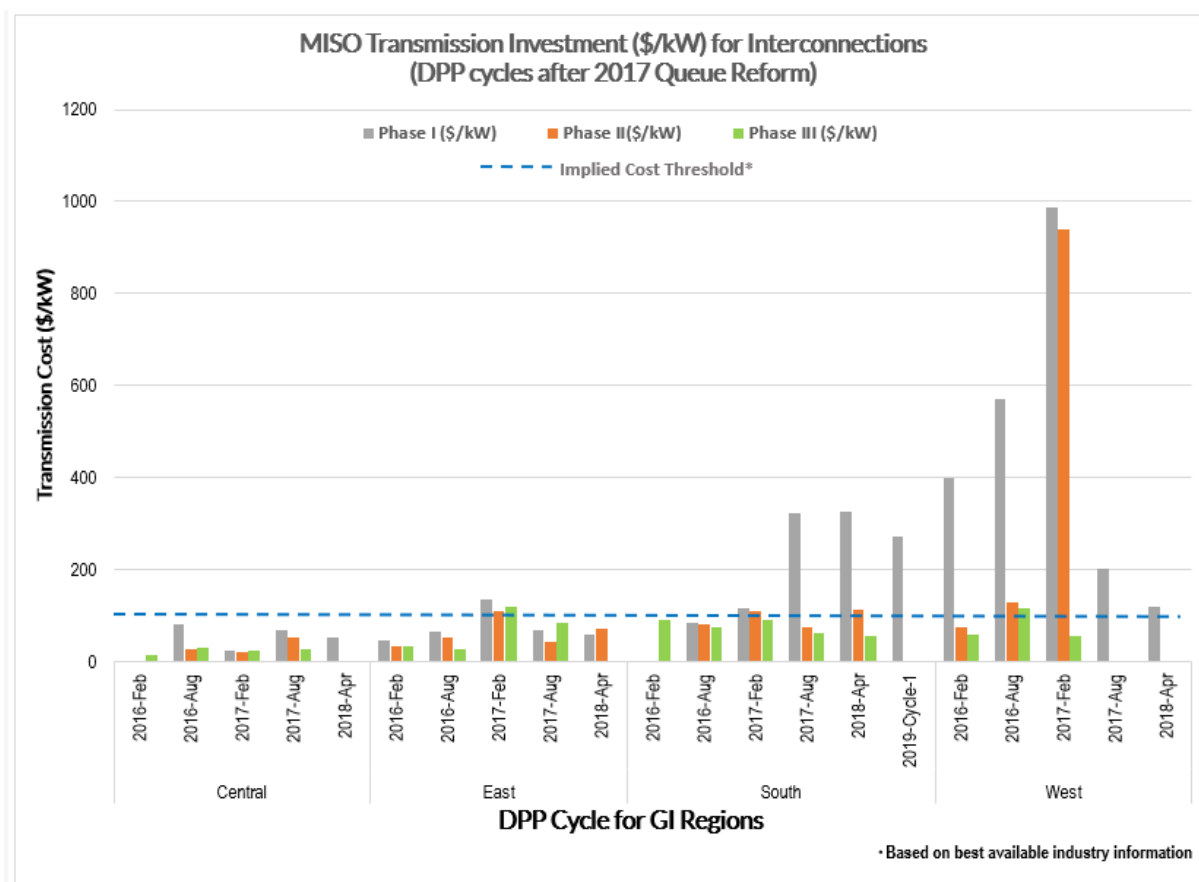


Figure 2.3-1: DPP Cycle Transmission Cost for each Planning Region

2.4 Current Generation Fleet Must Continue to be Deliverable to Load

MISO performs generator deliverability analysis as a part of the annual MTEP process to ensure continued deliverability of existing generating units with firm service, including Network Resource Interconnection Service. The generation deliverability analysis results in the identification of projects which mitigate transmission system constraints that restrict generation output below the established network resource amount.

Results of the assessment are determined on an analysis of near-term (five-year) summer peak scenario.

Observed constraints that restrict generation beyond the established Network Resource amounts require mitigation.

A total of two projects, and a special protection scheme (SPS) were identified to alleviate identified congestion.

MTEP20 Constraints

In MTEP20, constraints are due to major construction efforts in the area. Two MTEP19 projects were identified to mitigate these constraints which restrict generation beyond the established network resource amount (Table 2.4-1). These projects, along with alternatives, were reviewed by stakeholders in the MTEP19 planning process and were approved for construction.

Overloaded Branch	Area	Mitigation MTEP19 ID	Notes
John Junction – Morris 115 kV circuit 1 Ortonville – John Junction 115 kV circuit 1	OTP	17006	Mitigated by an Appendix A project in MTEP19. MISO identified thermal overloads on the John Junction to Morris 115 kV circuit 1, and Ortonville to John Junction 115 kV circuit 1 during the MTEP19 Deliverability Analysis
Big Stone – Browns Valley 230 kV circuit 1	OTP	16965	Mitigated by an Appendix A project in MTEP19. MISO identified thermal overloads on the Big Stone to Browns Valley 230 kV circuit 1 during the MTEP19 Deliverability Analysis
Crossroads – Clarksdale 115 kV circuit 1	EMBA/CLECO	NA	Mitigated by Crossroads SPS which will trip one of the Crossroads units to provide relief

Table 2.4-1: Projects identified to alleviate MTEP20 constraints that limit deliverability of network resources

2.5 Planning for the Future

Due to the long lead time of constructing new transmission infrastructure (identification of project, design, regulatory approval, construction, and energization), effective transmission planning must occur nearly 10 years prior to a significant transmission need, in most cases. To address this, MISO has developed a process to prudently plan transmission over a 10 - to 20-year period that captures a wide array of potential resource fleet changes and conditions due to political, economic, technological, industrial, commercial, and consumer trends. This process utilizes multiple planning scenarios, or “Futures” to bookend the spectrum of potential changes before they happen.

MTEP20 Future Scenarios

The MTEP20 cycle included four Futures. These four Futures include only slight modifications from those developed in MTEP19. The minimal updates to the Futures for MTEP20 include updates reflecting interconnection queue activity, retirements, additions, as well as an updated unit/resource dataset from ASEA Brown Boveri (ABB), the Swiss-Swedish multinational corporation that maintains the PROMOD simulation tool and data.

The four MTEP20 Futures are:

- Limited Fleet Change (LFC)
- Continued Fleet Change (CFC)
- Accelerated Fleet Change (AFC)
- Distributed and Emerging Technologies (DET)

The goal of MTEP Futures is to bookend uncertainty by defining a wide range of potential plausible outcomes.

The Regional Resource Forecasting (RRF) process uses the assumptions defined within each Future to economically identify the least-cost portfolio of new supply-side and demand-side resources. Base data assumptions in the associated ABB PowerBase database are presented in Appendix E along with fuel forecasts, new unit construction costs, emissions constraints, retirement assumptions, renewable energy assumptions and regional demand and energy projections. The resulting resource additions and retirements from the MTEP19 regional resource forecasting process are shown in Figure 2.5-1.



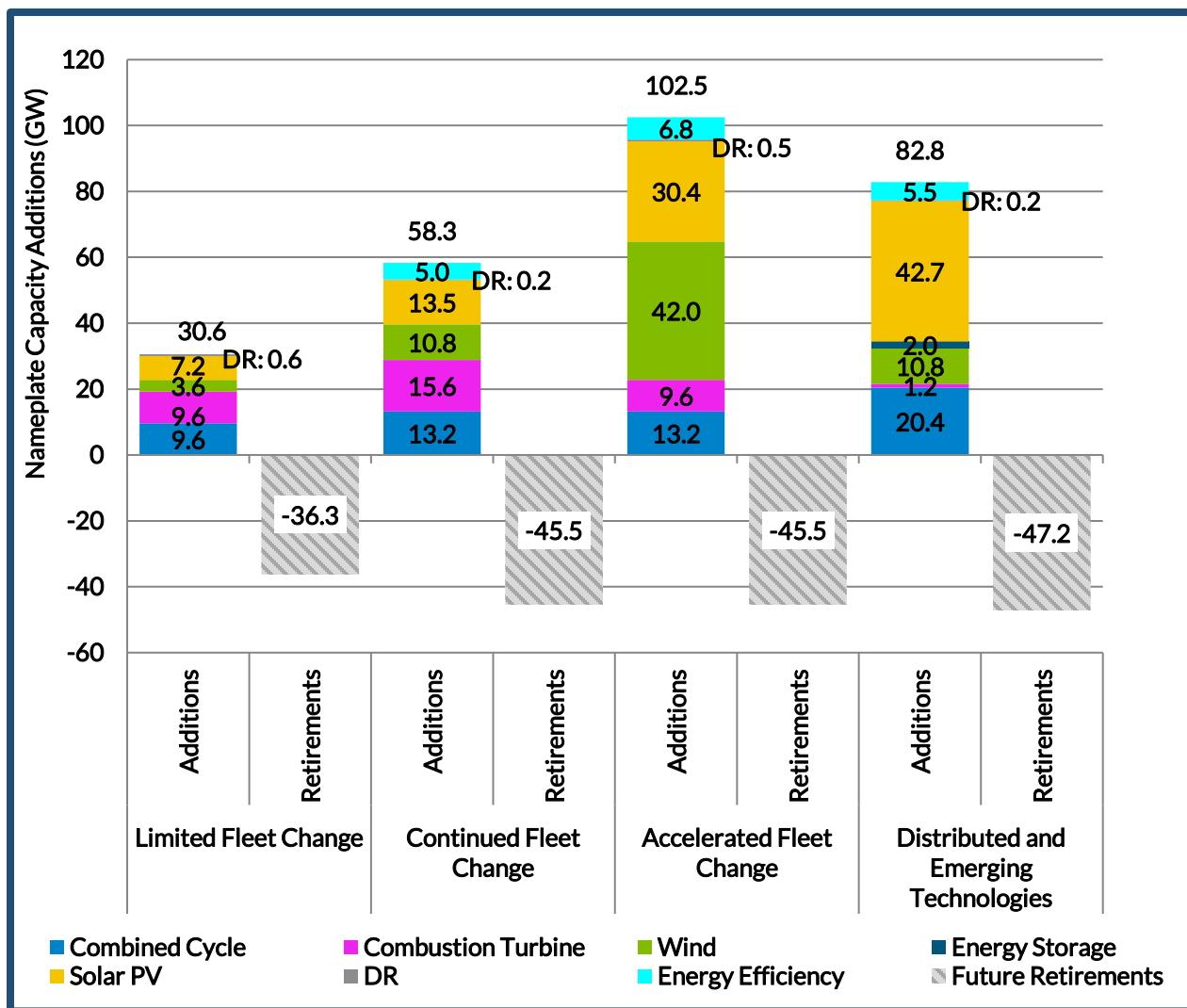


Figure 2.5-1: MTEP20 Futures- Resource additions and retirements by 2033

To produce the capacity mix in 2033 for each Future, the retirements and new resources identified from the regional resource forecasting process must be applied to the existing generation fleet (Figure 2.5-3).

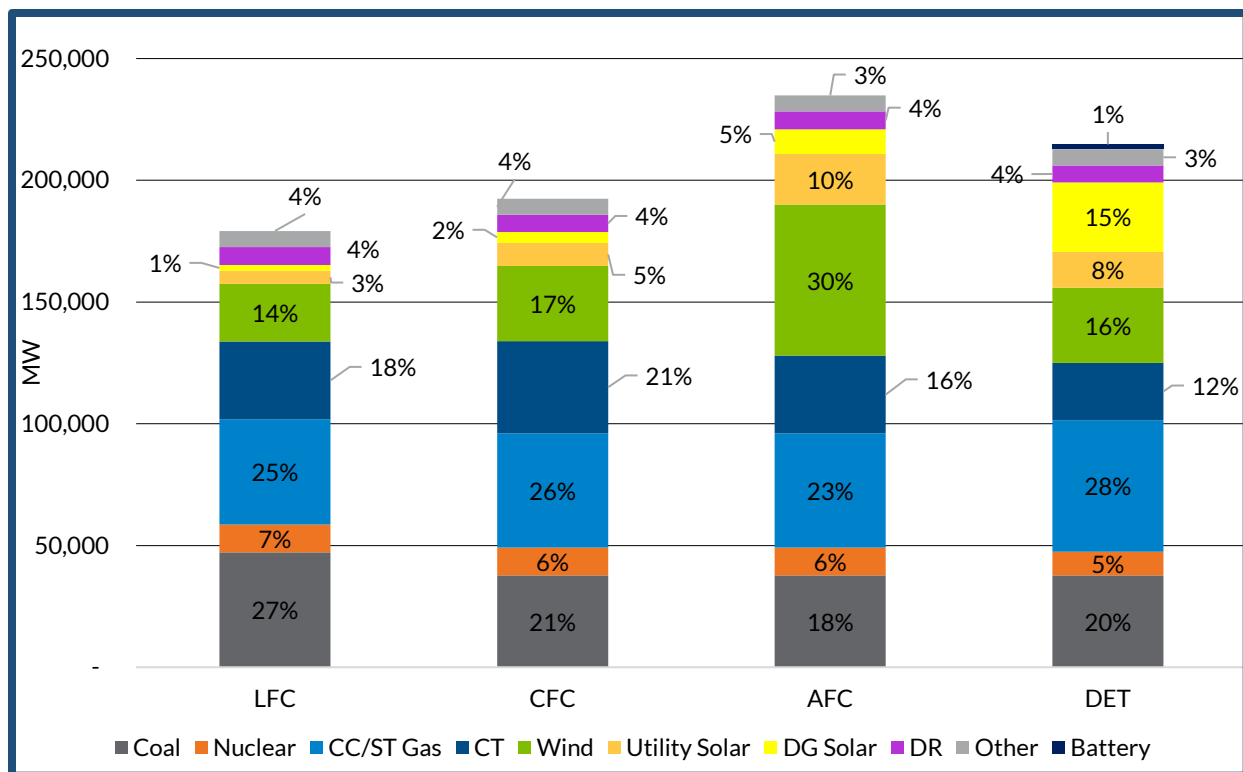


Figure 2.5-2: MTEP20 Futures-MISO 2033 Futures nameplate capacity mix by resource

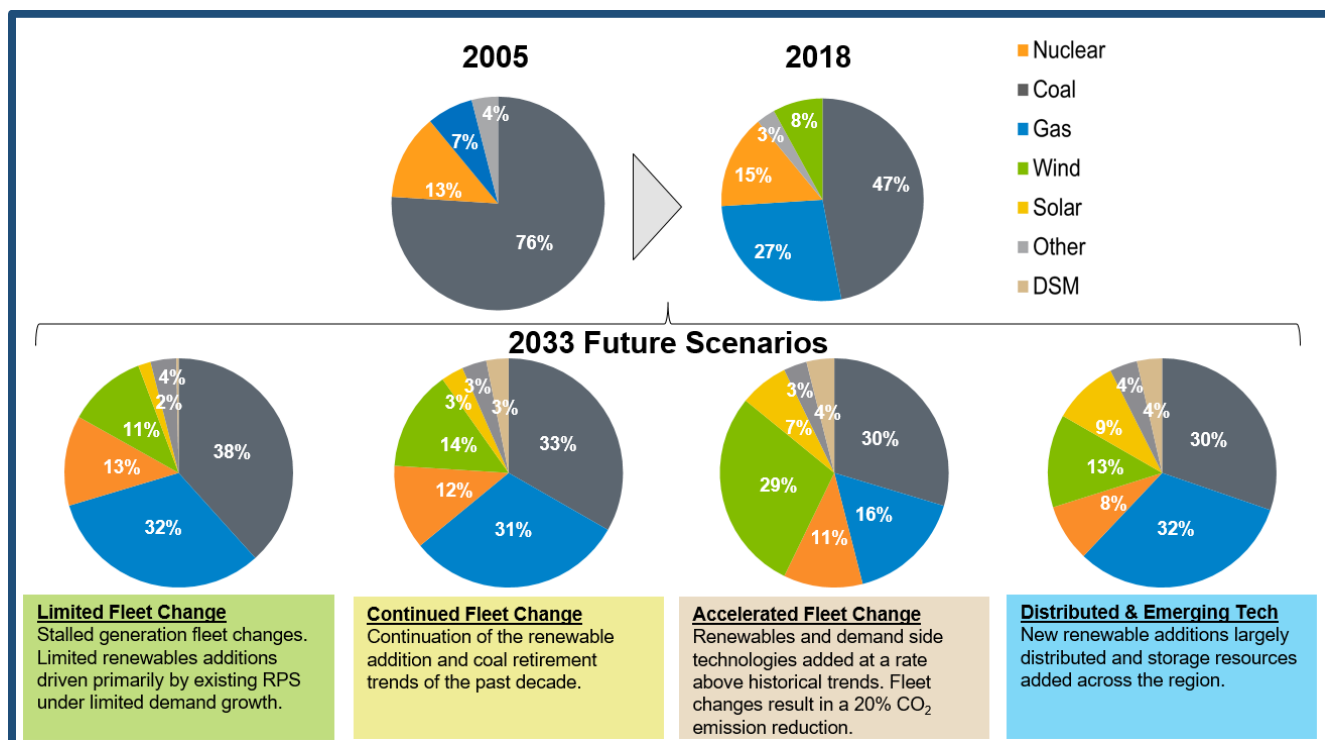


Figure 2.5-3: MTEP20 Futures - MISO 2033 Futures energy utilization mix

The results from the regional resource forecasting (RRF) process identify the type, size and installation date of new resources. However, they do not specify where these units should be located within the MISO footprint. Therefore, new resources identified in the regional resource forecasting process must be sited

within the economic production cost model. The Futures siting process is based on stakeholder-agreed-upon rules and criteria detailed in section 4 of Appendix E (Figure 2.5-4).

Additional details regarding Futures development, resource forecasting, and siting processes are in Appendix E of this report.

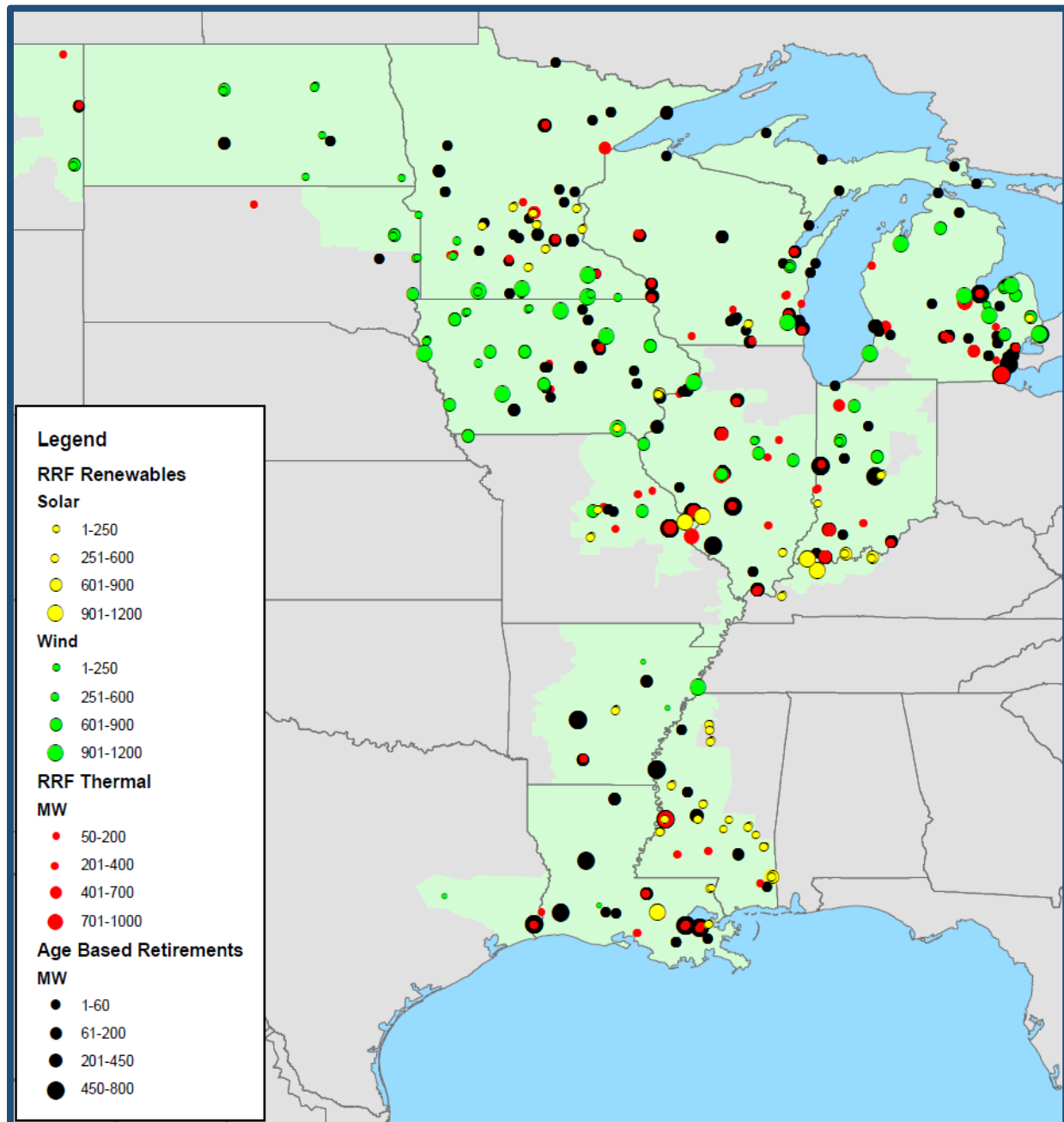


Figure 2.5-4: MTEP20 Futures -MISO Future supply-side resource siting results

2.6 Resource Outlook

The fleet has been steadily changing over the past 10-15 years. Initially this was due to forces such as state Renewable Portfolio Standards, environmental regulations, and fuel competition from historically low natural gas prices. This portfolio evolution continues now to greater levels of renewables and new levels of battery storage based on public interest and support for less reliance on fossil fuels, and historically low costs of renewables. As this change continues, focus on maintaining adequate resources is imperative and becoming increasingly difficult.

The MISO region will have adequate, but tighter, reserve margins for 2021, and continued action will be critical to ensure resource adequacy into the future. For 2021, MISO will have surplus resources to meet the regional resource requirement. In most of the MISO region, load-serving entities with oversight by the applicable state or local jurisdiction are responsible for resource adequacy. Though the 2021 peak demand forecast decreased 300 MWs from last year's survey, the five-year regional demand growth rate is up from 0.2 percent to just under 0.35 percent this year. On the supply side, the [2020 OMS-MISO Survey](#) indicates that increasing resource adequacy risk can be avoided by firming up the commitments of additional potential resources.

The potential for significant generation fleet transformation has prompted MISO to evaluate how system needs will change and how MISO might adapt its planning, markets and operations to maintain reliability with aging and retiring units, higher penetration of intermittent resources, and new load consumption patterns. The MISO membership is rapidly transitioning the resource mix in the footprint.

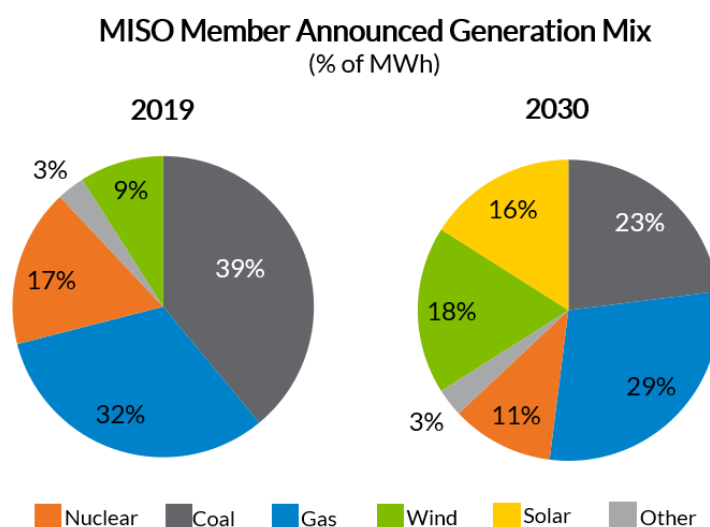


Figure 2.6-1: Member announced energy mix (as of September 2020)

Resource adequacy planning that focuses on summer peak alone will no longer suffice. Resource adequacy analysis will likely need to reflect patterns across the year in order to capture the magnitude of risks.

Effective dialogue amongst stakeholders will be key to this transformation – identifying needs and working with MISO to develop solutions that work across the footprint. MISO will leverage the forums where discussions are already underway on transmission planning, MISO's resource adequacy construct, and pricing enhancements.

The 2020 OMS-MISO survey indicates that the MISO region will have adequate, but tighter, reserve margins for 2021, and that continued action will be needed to ensure resource adequacy in the extended outlook.

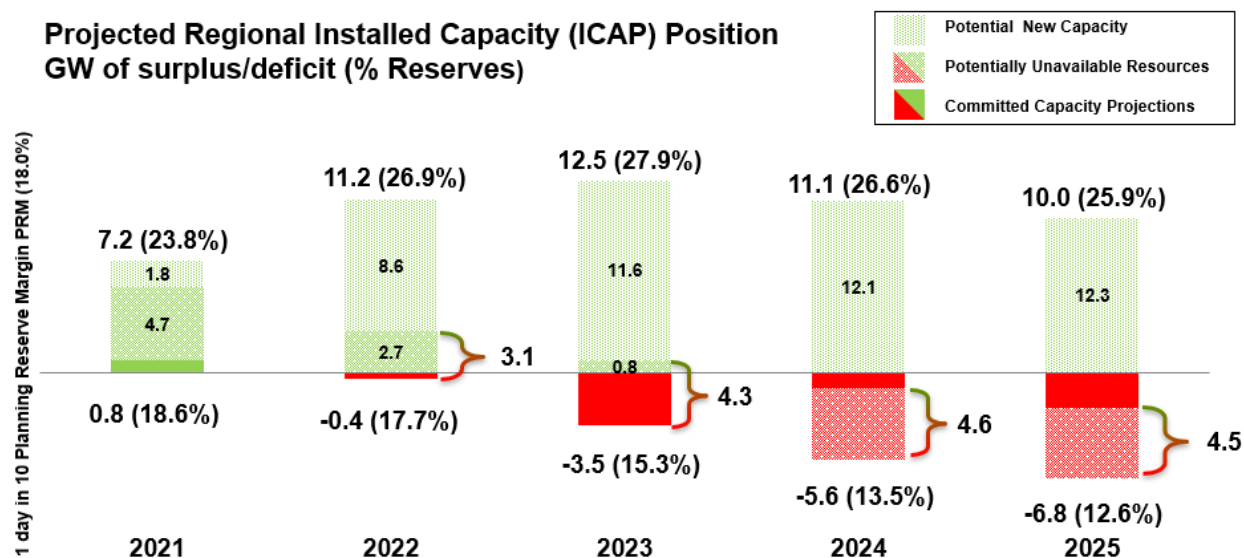


Figure 2.6-2: 2020 OMS-MISO Survey 5-year Results

This year's OMS-MISO survey shows MISO will have 0.8 gigawatts of surplus committed resources beyond the regional requirement, representing a reserve margin of 18.6%. If all potential resources materialize in 2021, there could be as much as 7.2 gigawatts of surplus generation resources, which would result in a reserve margin of 23.8%.

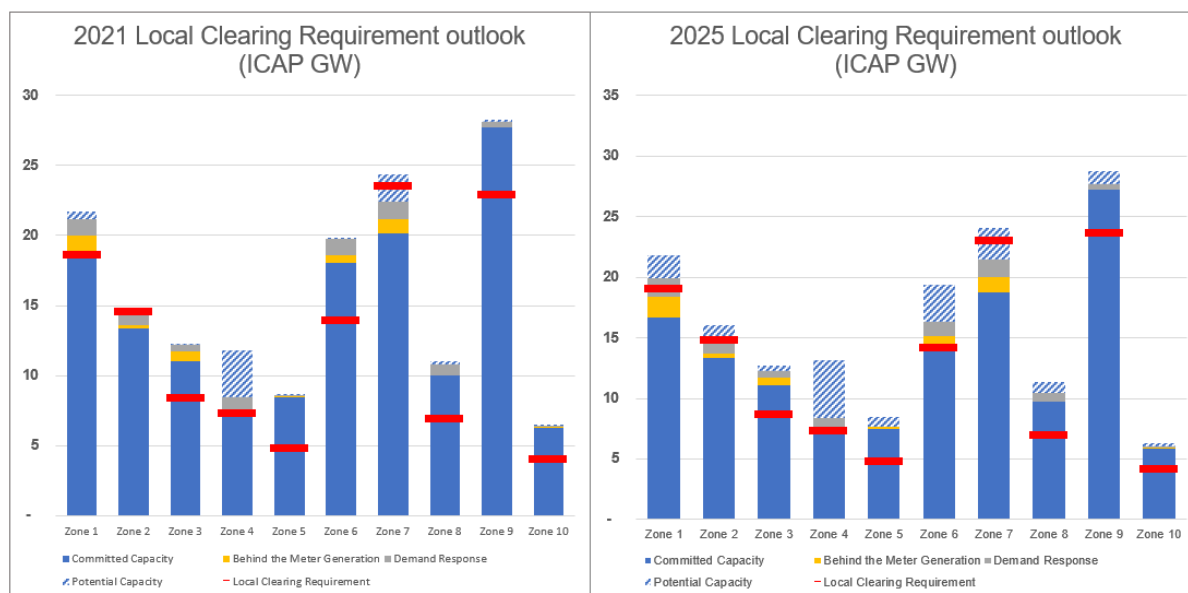


Figure 2.6-3: Potential new generation and retirements, within each zone

Compared to last year, these committed resource reserve margins are lower for both the first year and the full five-year period of the survey, which translates into increased reliance on less certain resources to ensure resource adequacy going forward.

The OMS-MISO Survey shows fleet changes in the next 5-10 years shifting heavily toward renewables. As queue additions are not submitted much beyond 2025, the impacts observed are due more to unit retirement decisions. The increase in battery and hybrid units from zero percent today to two and three percent respectively by 2025 indicates a changing dynamic in the MISO generation fleet. One emphasizing more flexibility and energy-shifting versus traditional fossil-based resources.

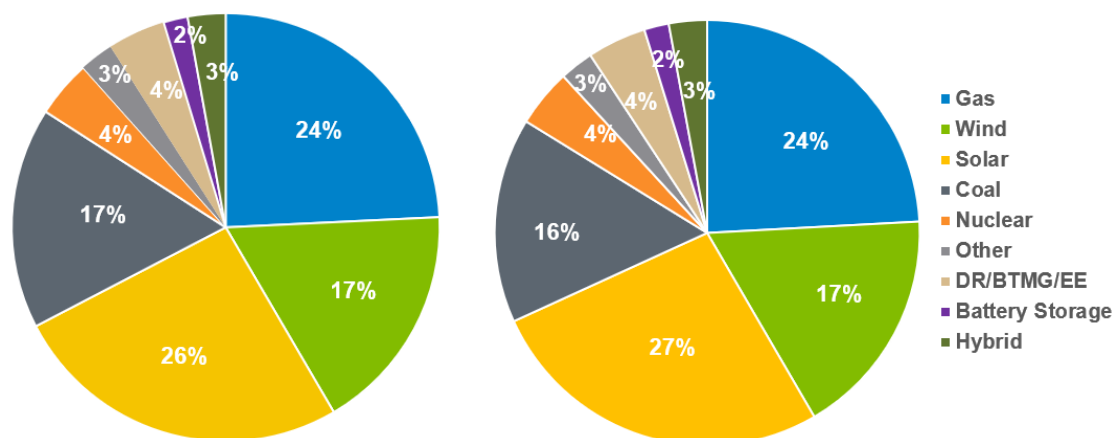


Figure 2.6-4: 2025 and 2030 OMS-MISO Survey Fleet Mix by Nameplate MW

2.7 Continued Future Scenario Development

Looking ahead as it began the MTEP20 cycle, MISO saw increasing momentum in fleet development and many stakeholders noted how new generation could outpace the Futures bookends within the planning horizon. With the accelerated pace of fleet change in mind, MISO initiated conversations to redevelop the Futures with stakeholders in 2019 with the goal of developing a new set of MISO Futures by the end of 2020. These new Futures will bookend a wider range of possibilities and seek to further embed the plans that MISO's members are developing for themselves. The new Futures compounded annual growth rate assumptions and scenario definitions are represented in Figure 2.7-1.

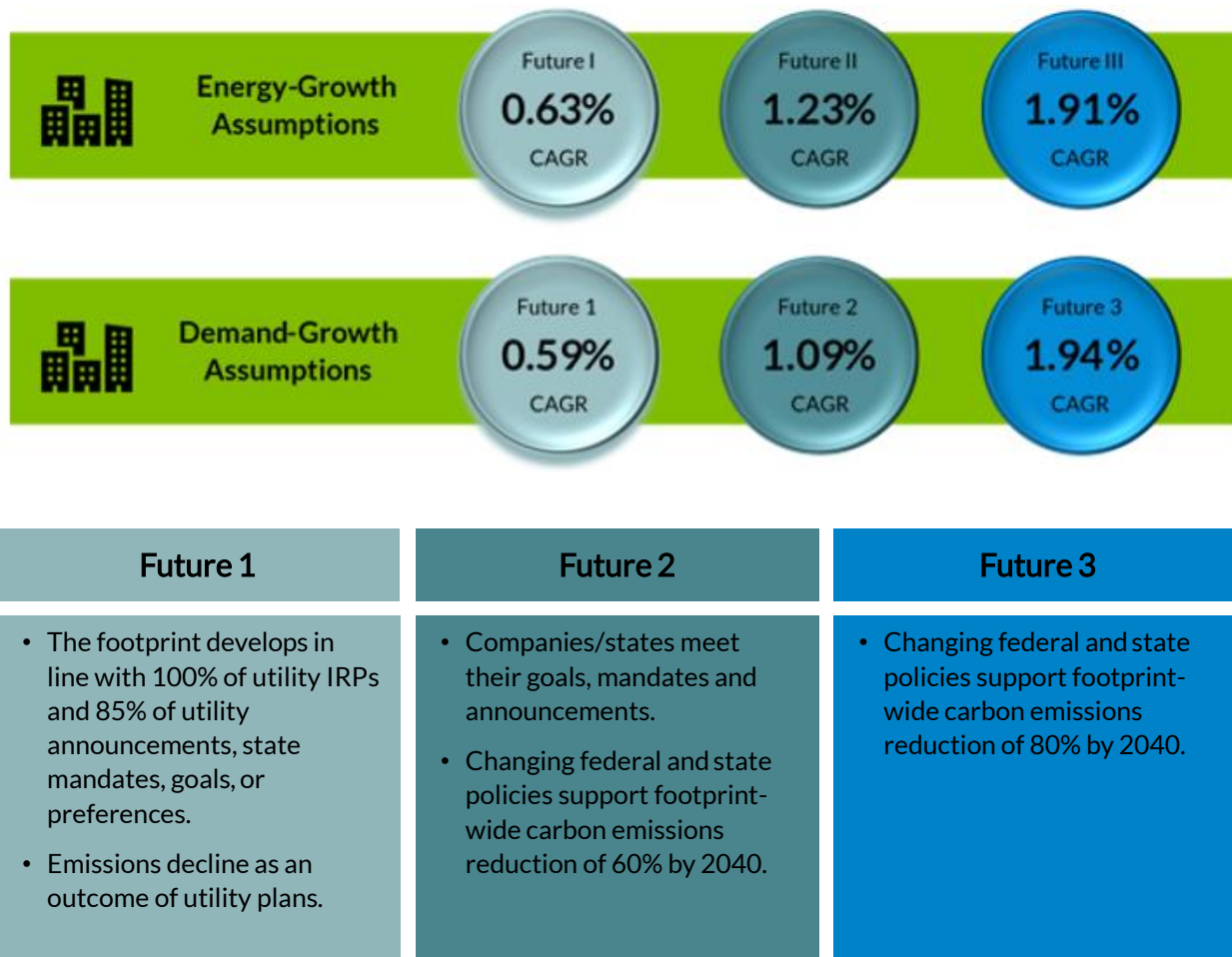


Figure 2.7-1: New Futures CAGR assumptions and descriptions.

In Future 1, the MISO footprint evolves as members’ plans are substantially met, carbon emissions decline 40% from 2005 levels, and current trends of electric vehicle adoption persist.

In Future 2, members’ plans are met or exceeded, carbon emissions decline 60%, and energy increases over the 20-year study period driven by electrification, including increasing adoption of electric vehicles.

Future 3, members’ plans are met or exceeded, carbon emissions decline 80%, energy increases over the 20-year study period driven by electrification, and renewable penetration levels reach a minimum of 50%.

These three Futures allow MISO to further study some of the insights identified in the MISO Forward documents such as the impacts of decentralization and demarginalization; they also support MISO’s vision “to be the most reliable, value-creating RTO.

2.8 Long Range Transmission Planning

MISO has the responsibility and perspective to provide for the enablement of transmission needs determined by the long-range expectations of load and the evolution of the resource portfolio. Transmission system needs and requirements have been and will continue to be driven by Federal and State policy, economics for the MISO region and sub-regions, and driven by the future plans of industry and utilities.

Long Range Transmission Planning (LRTP) takes a long-term view of system development to ensure the values of more near-term solutions are robust in the long-term under a range of Futures, and account for 20- to 40-year benefits.

LRTP is a cornerstone for future operations, but the foundation has many pieces that must work well together:

- Long term resource adequacy that respects the Resource Availability and Need (RAN) program aligning resource availability and need with the changing reliability requirements due to resource evolution;
- Activity in the Interconnection Queue that offers a short- and long-term view of possible future resource siting and identification of sub regional, regional, and interregional transmission limits;
- Interregional reliability and economic planning with MISO neighbors and other entities in the Eastern Interconnect. While MISO's first priority is our members and our footprint it is essential to work together with bordering systems to best develop a nationwide system that supports and works reliably and efficiently together.

LRTP will ensure an optimized regional and interregional system for the changing portfolio across short- and long-term horizons. Regional planning must ensure the system is planned to be reliable, secure, resilient, and efficient over the entire planning horizon; address differences across the MISO region, and holistically incorporate reliability and economic planning with future resource needs/expectations.

The MTEP21 LRTP initiative will build off of the MTEP20 targeted and interregional analyses discussed in Chapter 3 of this report and other identified near term issues developed with stakeholders. LRTP work under MTEP21 will begin identifying the transmission solutions needed to enable the new MISO Futures starting with Future 1 in the MTEP21 transmission planning cycle. Future 1 and MTEP21 will lay the groundwork for continuing efforts with Futures 2 and 3 in MTEP 22 and beyond.

LRTP will seek to establish a transmission road map for the long-term horizon that will be the foundation to drive future investment decisions. Once complete, the LRTP transmission road map will be periodically reviewed and updated over time with select transmission projects pulled into Appendix A as business cases for those projects become robust for impending needs. Some transmission projects may be identified for Appendix A as early as MTEP21.

CHAPTER 3: POLICY AND ECONOMIC STUDIES

3.1 Targeted and Interregional Analyses

MISO is addressing challenges driven by the pace at which the grid is evolving from all directions, including increased renewable integration, retirements, and new technologies. The value of MISO-wide diversity and interconnectedness are key to understanding the best outcome and driving traditional planning processes to change to accommodate this rapidly evolving grid. Many stakeholders share the position that now is the time for MISO to take action to ensure that the grid is positioned to reliably and cost-effectively enable the transitions that are occurring.

For MTEP20, updated MTEP19 Futures were used and instead of performing a Market Congestion Planning Study, four targeted sub-regional issues and areas were evaluated and analyzed, including the completion of the North-South study and an interregional evaluation through a Coordinated System Plan (CSP) with Southwest Power Pool (SPP). These studies help set the stage for further investigation of the long-range transmission needs.

These evaluations highlighted the values of and need for further identification of sub-regional concerns that should be addressed as part of the regional and interregional transmission plan. While there are no proposed Appendix A projects in MTEP20 resulting from these investigations, they provide a significant informational base going into the next planning cycle.



Figure 3.1-1: Focus areas of MTEP20 targeted and interregional analyses

3.2 Analyses Addressing Sub-Regional Needs

North-South Interface

Business Need/Study Scope

In early 2019, MISO submitted updates to the current MEP cost allocation methodology to include the avoided cost of future MISO-SPP Settlement Agreement payments as a benefit metric. Shortly after, MISO received requests from various Stakeholders to examine transmission line alternatives to the MISO-SPP Settlement Agreement. MISO began studying the economic benefit potential of various transmission lines against the cost of future settlement payments. The objective of this analysis was to determine if a transmission project would offer a better value to MISO's membership than the current MISO-SPP Settlement Agreement and if a higher transfer capability between the MISO sub-regions would result in increased economic benefits.

The MISO-SPP Settlement Agreement is an agreement in which MISO financially compensates SPP and The Joint Parties (AECI/TVA/LGE/KU) for real-time energy transfers above 1000 MW (up to 3000 MW North-South and 2500 MW South-North limits) between MISO's North and South sub-regions.

Approach/Methodology

The study began in the middle of the MTEP19 Market Congestion Planning Study (MCPS) with an informational session led by the Economic Studies group which communicated the future value, cost and importance of energy transfers between the North and South sub-regions in MISO. These three concepts were investigated in detail throughout the session to develop a specific framework to identify issues and respective solutions.

MISO collaborated with Stakeholders on various transmission project ideas. These solutions were intended to add additional contract path capacity between MISO's sub-regions, which required that the transmission project possessed termination points in both MISO North and MISO South. In total, 35 transmission solutions passed the screening criteria for adding new contract paths as tested using MISO's MCPS study framework.

The Economic Studies team performed all steps provided in the MCPS process and discovered three projects that maintained a B/C Ratio above 1.0 throughout the analysis. These projects maintained this status due to their low cost and large contract path increase which allowed for sub-regional transfer capability greater than or equal to the limits provided by the MISO-SPP Settlement Agreement.

ID	Transmission Solution	Cost Estimate (2019-\$M)	20-yr B/C Ratio					Assumed Contract Path Increase (MVA)
			AFC	CFC	DET	LFC	Aggregate	
PC-1	New 345kV, 2574MVA line from Jim Hill-Berntie Rebuild 161kV line from Berntie-Stoddard	152	2.33	2.18	1.90	1.75	2.04	2574
PC-2	New 345kV, 2574MVA line from Jim Hill-Lutesville	249	1.43	1.13	1.01	1.08	1.16	2574
PC-3	New 345kV, 1793MVA line from Jim Hill-Kelso. New 161kV, 509MVA lines from Jim Hill-Berntie. Upgrade existing Stoddard-Berntie, Stoddard-Morley, Oran-Morley and Oran-Kelso lines to 509MVA.	262	1.19	1.14	1.04	1.02	1.10	2302

Table 3.2-1: North-South Interface projects meeting MCPS thresholds

Outlook/Next Steps

Although MISO concluded the [North-South Interface Study](#) in the spring of 2020 with the MTEP19 project scope, MISO anticipates further analysis in MTEP21. The three main reasons for concluding the current study are (1) MISO has continuously collaborated with SPP and The Joint Parties and plans to continue that communication into the MTEP21 time period (2) Stakeholders would like MISO to consider benefit metrics beyond APC savings and avoided settlement costs to capture all of the benefits a transmission solution would provide (3) At the time of conclusion, the Tariff updates which included settlement agreement cost avoidance as a benefit metric were still pending FERC approval.

North Region Targeted Study

Business Need/Study Scope

As the MISO fleet continues to shift away from large thermal units to geographically dispersed inverter-based renewables, it is expected that addressing non-thermal system limitations will begin playing a larger role in ensuring reliable operation of the system. MISO's economic planning process has historically focused on thermal limits for its market constraint modeling, with non-thermal limits simply being modeled at their currently defined ratings. To accurately reflect the future MISO energy market, reliability assessments must be performed to determine if any updated or new non-thermal limits should be incorporated into the economic models.

The North Region Targeted Study is an informational study in which MISO evaluates system limitations caused by non-thermal constraints between the high renewable penetration northwestern areas of MISO and load centers in eastern areas of MISO. Findings from this study will be used to inform future transmission planning initiatives and improve existing economic evaluation processes such as the Market Congestion Planning Study (MCPS).

Three scenarios representing varying levels of renewable penetration and different siting methods were developed for this study. These scenarios were then simulated in the production cost modeling software PROMOD with the outputs from these simulations converted to powerflow model inputs for reliability analysis in PSS/E and VSAT. Once non-thermal limits were determined, they were brought back to the PROMOD models where they were included as market constraints. Any non-thermal constraints that resulted in substantial levels of congestion were identified as economic issues, and potential transmission upgrades were tested to resolve them.

MISO targets to use findings from this study effort to scope studies in the upcoming MTEP21 cycle, where it will provide a more holistic view of how the future grid will operate and what transmission upgrades are necessary to continue to reliably and efficiently deliver energy across the MISO footprint.

Approach/Methodology

The current MCPS process emphasizes analysis of congestion related to thermal system limitations on the MISO system. While some non-thermal constraints are included in MCPS economic models (e.g. MWEX), they are modeled using system-intact limits that are currently used in the MISO market. One of the main objectives of the North Region Targeted Study (NRTS) is to develop a process to identify and incorporate non-thermal system limitations as they would exist in the futures developed for the MCPS process.

The following steps were performed to create a standardized process in determining non-thermal system limitations in economic study models:

1. Created three scenarios with varying levels of renewable penetration and different siting methodologies
2. Simulated the three scenarios using production cost models
3. Created powerflow models for each scenario based on the Security Constrained Economic Dispatch (SCED) performed in the production cost models
4. Performed transfer analyses on each powerflow model to identify voltage stability issues and their associated limits
5. Included newly evaluated non-thermal interfaces and limits in production cost models to determine if they resulted in economic congestion issues
6. Evaluated congestion mitigation and economic benefit of project ideas, considering both thermal and non-thermal issues

Evaluating non-thermal issues in economic modeling will allow future economic studies to better capture how the market would operate in any potential future system scenario. As MISO's generation fleet converts from traditional large thermal units to inverter-based intermittent and distributed renewables, this process will become increasingly important in maintaining the most efficient and reliable bulk electric system.

Outlook/Next Steps

The MISO generator fleet is projected to change significantly in the coming years, moving away from large thermal units with high inertia to inverter-based technologies. The [North Region Targeted Study](#) was a first step in re-evaluating MISO's standard economic planning processes to ensure that all issues that will be encountered in the future are considered. An iterative process between reliability and economic analyses is required to address system needs due to changing generation fleet.

The process and findings from this study will be used to further investigate non-thermal issues in the MISO North region as a part of the Long-Range Transmission Planning effort in MTEP21.

In upcoming studies, additional analysis will be conducted to identify solutions that improve voltage stability in the MISO North region. MISO will conduct both voltage and transient stability analysis to reveal weak grid areas that have high penetration of inverter-based resources. Concurrently, options to quantify reliability benefits provided by any projects that mitigate these non-thermal issues will be explored.

Michigan Capacity Import Export Limit Expansion Study

Business Need/Study Scope

When comparing the Local Clearing Requirements (LCR) to regional requirements, Local Resource Zone 7 must source nearly their entire resource adequacy requirement from within Local Resource Zone 7. As Michigan is experiencing a significant number of planned and future power plant retirements, additional import capacity may help Michigan access diverse and economical supplies of power, assist with reliability and resiliency during emergency conditions, and enhance MISO's annual resource adequacy needs, particularly with respect to LRZ 7's ability to meet the MISO Local Clearing Requirement.

In the Michigan State Energy Assessment report issued on September 11, 2019, the Michigan Public Service Commission (MPSC) recommended that the value of resiliency should be incorporated in future electric infrastructure planning and investment decisions. On November 7, 2019, the Governor of Michigan and the MPSC, respectively, requested MISO to conduct analysis on the impacts of increasing Capacity Import Limits (CIL) and Capacity Export Limits (CEL) for MISO's Local Resource Zone 7 (LRZ 7). LRZ 7 geographically covers the majority of Michigan's Lower Peninsula (LP).

Specifically, the MPSC seeks to better understand transmission and non-transmission solution options available to increase the limits into and out of LRZ 7 in the near and long term (e.g., the possibility of considering additional high voltage infrastructure coming into LRZ7), as well as an estimate of the corresponding costs and benefits.

The MPSC requested from MISO to first analyze increasing the CIL in the near term at increments such as 500 MW and 1,500 MW. The goal is to determine the infrastructure needed to accommodate cost-effective increases in the near term, with corresponding costs and benefits to LRZ 7 and other LRZs as applicable.

The MPSC's second request from MISO is to determine what types of projects could facilitate an increase in the CIL in LRZ 7 by larger increments over the next decade to accommodate additional renewable energy and other changes in the generation mix.

Approach/Methodology

The Michigan Capacity Import/Export Limit Expansion Study is an informational study to determine transmission expansion and non-transmission options to increase the capacity import and export limits for MISO's Local Resource Zone 7 (Lower Michigan). As this is an informational study, MTEP Appendix A project recommendations will not be made as a direct result of this study – any further action would be that of the MPSC or MISO stakeholders.

To determine an LRZ's capacity import or export limit, a transfer is modeled by ramping generation up in a source subsystem and ramping generation down in a sink subsystem. The LRZ studied for import limits is the sink subsystem and the adjacent MISO areas are the source subsystem. The LRZ studied for export limits is the source subsystem and the rest of MISO is the sink subsystem. Limited redispatch is applied after performing transfer analyses to mitigate constraints. Redispatch ensures constraints are not caused by the base dispatch and aligns with potential actions that can be implemented for the constraint in MISO operations.

Transfer capability measures the ability of interconnected power systems to reliably transfer power from one area to another under specified system conditions. The incremental amount of power that can be transferred is determined through First Contingency Total Transfer Capability (FCTTC) analysis, which

identifies the total amount of transferrable power before a reliability constraint is identified. Reliability constraints consider steady-state thermal and voltage issues based on the NERC Transmission Planning Reliability Standards.

As initially scoped, the Michigan Capacity Import/Export Limit Expansion study will focus on identifying multiple potential transmission expansion scenarios:

- **Local Smaller-Scope:** Approximately 500 MW incremental increase in capacity import limit
- **Local Larger-Scope:** Approximately 1,500 MW incremental increase in capacity import limit
- **Regional:** 3,000 MW+ incremental increase in capacity import limit

Transmission expansions will be optimized to provide the highest capacity import limit increase to capital cost of the project ratio, while considering the targeted scope ranges. Capacity export limits will be calculated on final transmission expansion options. Additionally, final transmission expansions will be qualitatively evaluated in the context of broader regional needs (i.e. determination of how options fit with additional topology changes being considered through concurrent MTEP20 studies).

As the current Michigan Capacity Import/Export Limit Expansion Study is focused on increasing the capacity import limit (decreasing the Local Clearing Requirement) which is MISO intra-zonal, transmission options will be primarily focused inside the MISO footprint or those which increase the LRZ7 CIL. As MISO continues evaluation of the Long Range Transmission Plan, MISO will consider further opportunities to enhance access and delivery to the broader MISO market for LRZ7.

Outlook/Next Steps

A Capacity Import Limit achieving the targeted near-term megawatt increase was identified for Scenario 1. MISO is asking stakeholders to identify alternative projects to increase the Michigan Capacity Import Limit for Scenario 2. A report will be released by the end of the calendar year.

3.3 Analyses Addressing Interregional Needs

MISO-SPP Coordinated System Planning

In March of 2020, MISO and SPP formally initiated a CSP study based on the recommendation of the MISO-SPP IPSAC. The study scope focused on economic evaluation of top congested seams flowgates identified in each RTO's respective regional planning process (Figure 3.3-1). The complete results of the study will be published in a standalone 2020 CSP study report (anticipated Q4 2020), as well as presented to stakeholders at a conclusionary IPSAC meeting. Any project recommended by the 2020 CSP study will require approval by each RTO's Board of Directors for inclusion in the respective regional transmission plan.

2020 SPP-MISO CSP Study Areas of Congestion:

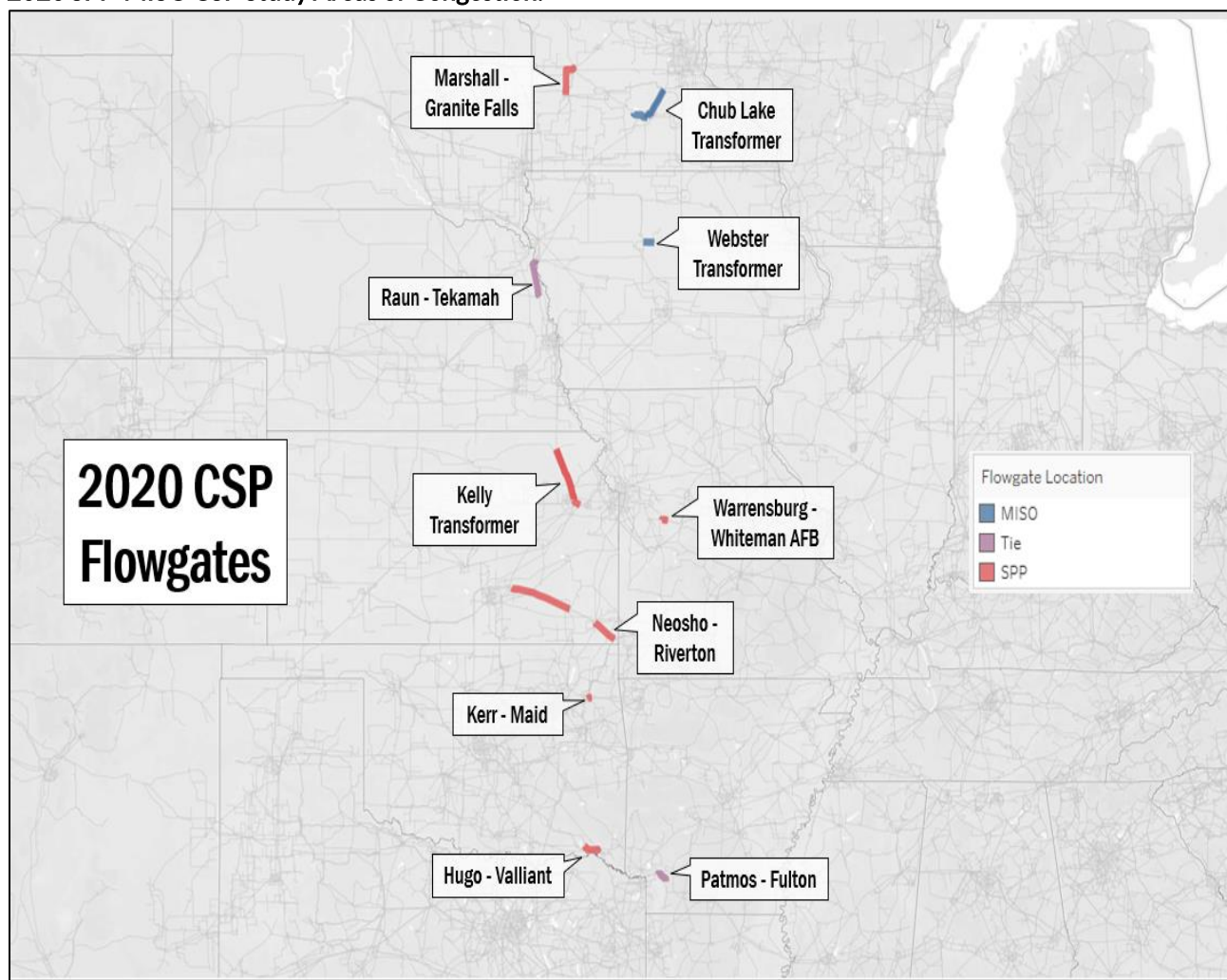


Figure 3.3-1: MISO-SPP 2020 CSP flowgates

On September 14, 2020 it was jointly [announced](#) that MISO and SPP will conduct a Joint Study Targeting Interconnection Challenges. MISO and SPP will collaborate on a year-long transmission study designed to identify transmission projects. This joint study will be a simultaneous evaluation of benefits focused on solutions that the RTOs believe will offer benefits to both their interconnection customers and end use consumers. The study is expected to formally kick off in December 2020 and include several joint stakeholder meetings. The study will be reported on in the MTEP21 Report.

MISO-PJM Coordinated System Planning

In Q1 of 2020, MISO and PJM presented their respective regional issues at an IPSAC meeting to help determine the need for a CSP study. RTOs also reviewed the 2018-2019 historical market-to-market congestion costs for the possibility of developing a Targeted Market Efficiency Project (TMEP) study. After careful consideration, MISO and PJM mutually determined not to initiate a CSP study based on the following rationale:

- Interregional Market Efficiency Project (IMEP) study (2-year study process): Coordinated models are currently under development in each region. A study will be pursued in the near future if the regional models identify constraint(s) that could justify such a project
- Targeted Market Efficiency Project Study (TMEP): Very limited opportunities were identified; study will not be conducted in 2020
- Interregional Reliability Project: No constraints have been identified for study in 2020
- Interregional Public Policy Project Study: No drivers were identified, or studies currently planned
- Ad Hoc: No drivers were identified, and no ad hoc studies are currently planned under the CSP

The next MISO-PJM IPSAC meeting is expected to be scheduled in the fourth quarter of 2020 and will focus on the potential for a two-year cycle IMEP study. The RTOs will evaluate the need for a two-year cycle study as the economic models are completed and results are reviewed later this year.

CHAPTER 4: RELIABILITY STUDIES

4.1 Reliability Assessment and Compliance

MISO, in collaboration with its transmission-owning members and stakeholders, performs annual reliability assessments to identify transmission infrastructure upgrades needed to ensure the continued system reliability in compliance with applicable local and regional reliability standards. The reliability assessment process for MTEP20 (shown in Figure 4.1-1) began with a roll-up of issues and potential solutions from the local planning processes of the Transmission Owners (TOs), followed by an independent reliability assessment conducted by MISO to evaluate and integrate the TO local planning information into the development of the overall MISO Transmission Expansion Plan.

MISO closely coordinates the annual reliability assessment with other planning efforts, such as the Market Congestion Planning Study (MCPS) process to ensure the transmission expansion plan is identified in an efficient and cost-effective fashion. A variety of factors are considered as part of MISO's transmission expansion plan development, including but not limited to, urgency of needs, cost effectiveness of solutions, system performance of solution alternatives to address identified transmission issues, and other considerations such as lead time to develop a project, right-of-way (ROW) or substation impacts, expandability, operational flexibility, etc.

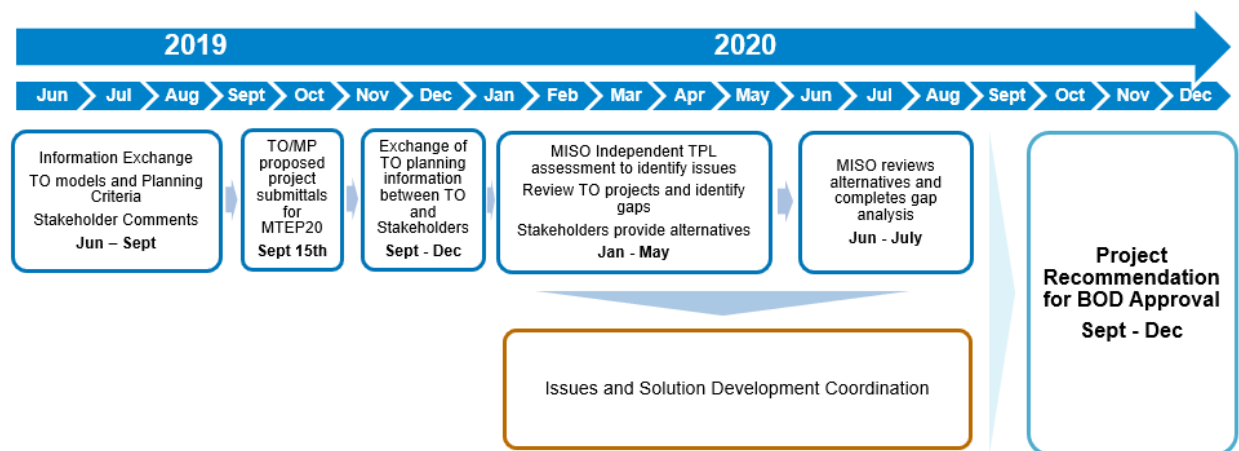


Figure 4.1-1: MTEP20 reliability assessment process

In conjunction with the MTEP planning process, an inclusive, transparent stakeholder process is utilized to facilitate open discussions and allow stakeholders to provide early and meaningful inputs into the development of transmission solutions in each planning cycle. The results of MISO's independent reliability assessments, along with proposed solution alternatives, are presented to stakeholders through a series of public Sub-regional Planning Meetings (SPM), and additional Technical Study Task Force (TSTF) meetings as needed, for each of the four MISO planning sub-regions; Central, East, South, and West.

After MISO completes its independent review of all proposed projects and associated alternatives and addresses stakeholder feedback received through SPM discussions, MISO staff formally recommends a set of projects to the MISO Board of Directors for review and approval. These projects make up Appendix A of the MTEP report and represent the preferred solutions to the identified transmission needs of the MISO

reliability assessments. Proposed transmission upgrades with sufficient lead times are included in Appendix B for further review in future planning cycles.

The complete results of MTEP20 reliability assessments are detailed in Appendices D3-D10 of the MTEP20 report, which are available on the MISO ShareFile site at <https://misoenergy.sharefile.com/>, subject to Critical Energy Infrastructure Information (CEII) and non-disclosure agreements. These results serve as compliance evidence for a variety of NERC planning standards listed on the MISO public website at <https://www.misoenergy.org/planning/transmission-planning/reliability-planning>.

MTEP20 project recommendations

As the result of the MTEP20 reliability assessments, 75 baseline reliability projects totaling \$755 million are included in the MTEP20 Appendix A, accounting for 18% of total transmission infrastructure investment in MTEP20. Seven Expedited Project Review requests were received and evaluated. The vast majority of the recommended projects are driven by reliability (either baseline or local reliability), load growth, and age condition, and are expected to be in service within five years. Project justification details of the recommended Appendix A projects are summarized in the following subsections for each of the four MISO planning sub-regions.

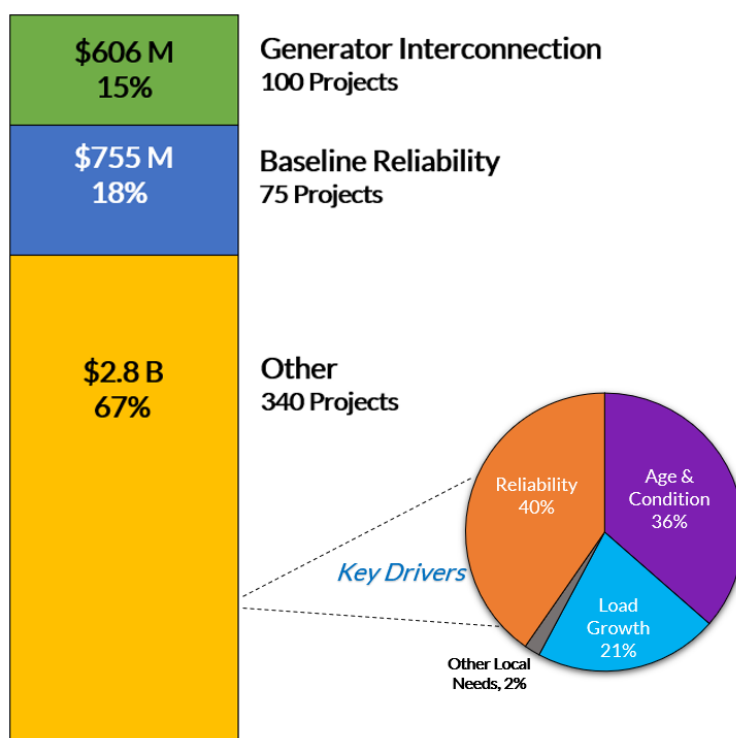


Figure 4.1-2: Project count and cost by project types for MTEP20 cycle, as of 10-5-2020.

4.2 Project Justifications – Central Region

Central Region Overview

The MISO Central planning region resides in four states: Missouri, Illinois, Indiana, and Kentucky and consists of twelve transmission owners: Ameren (AMIL/AMMO); Big Rivers Electric Corp. (BREC); City of Columbia, Mo. (CWLD); City of Springfield, Ill. (CWLP); Duke Energy Corp. (DEI); Henderson Municipal Power & Light (HMPL); Hoosier Energy REC Inc. (HE); Indianapolis Power & Light (IPL); Prairie Power Inc. (PPI); Southern Indiana Gas & Electric (SIGE); Southern Illinois Power Cooperative (SIPC); and Wabash Valley Power Association Inc. (WVPA).

The bulk power system within these states consists of an extensive 345 kV, 230 kV, 161 kV, and 138 kV networked transmission system. The 345 kV network spans Missouri, Illinois and Indiana, both north to south and east to west. The 161 kV and 138 kV networks span both north and south, and east to west. All of Ameren, BREC, CWLD, CWLP and SIPC belong entirely in the SERC Region. All of Duke, Hoosier Energy, IPL, and SIGE belong entirely in the ReliabilityFirst Region.

Major load pockets in the MISO Central planning region are St. Louis, MO; Peoria, IL; Springfield, IL; Evansville, IN; and Indianapolis, IN (shown in Figure 4.2-1).

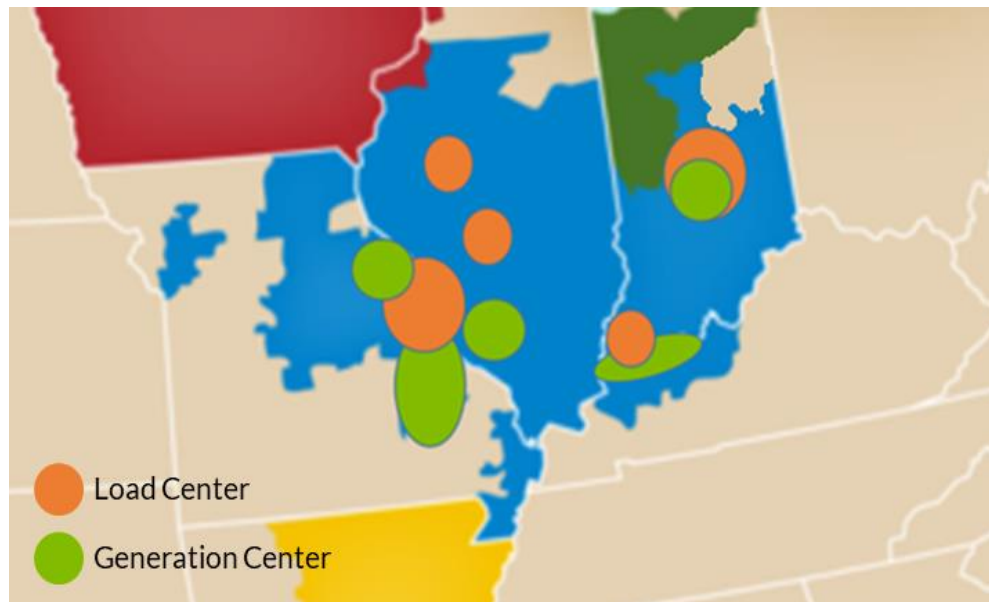


Figure 4.2-1: Major load and generation pockets in the MISO Central Region

For MTEP20, MISO Transmission Planning is recommending 231 projects from the Central region for inclusion in Appendix A at an estimated cost of \$2 billion. Of these, 24 are Baseline Reliability Projects and 14 are Generator Interconnection Projects. The remaining 193 projects are classified as Other Projects because they do not meet the criteria to be considered as Baseline Reliability Projects, New Transmission Access Projects, Market Efficiency Projects, or Multi-Value Projects.

Of the 231 projects, that are being recommended to be included in MTEP20, 51 have an estimated cost of less than \$1 million, 80 have an estimated cost between \$1 million and \$5 million, and the remaining 100 projects are estimated to cost greater than \$5 million (indicated in Figure 4.2-2).

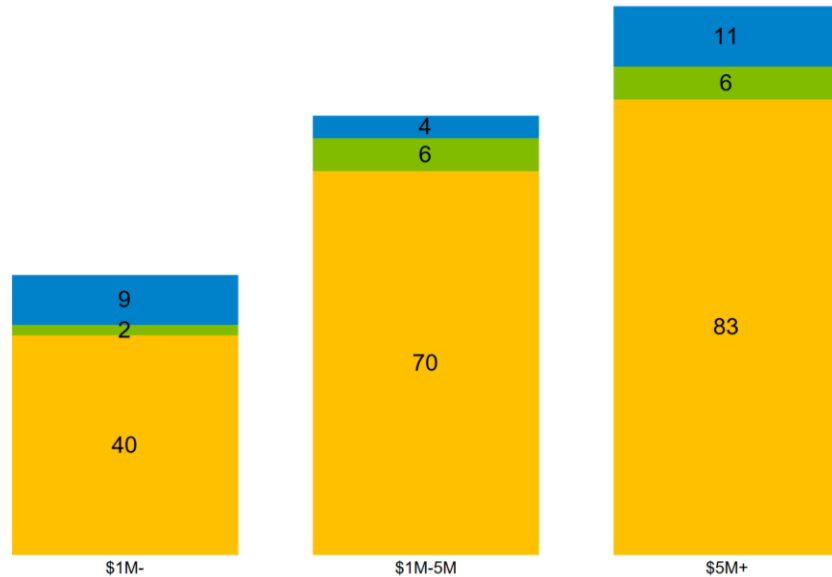


Figure 4.2-2: Project counts by cost category of MISO Central region MTEP20 projects (As of 8-3-2020)

The majority of the projects in the MISO Central planning region are expected to go in service in the next three years (shown in Figure 4.2-4). There were a handful of projects that are being approved in 2020 that either already went into service in 2019 or are expecting to go into service in 2020.

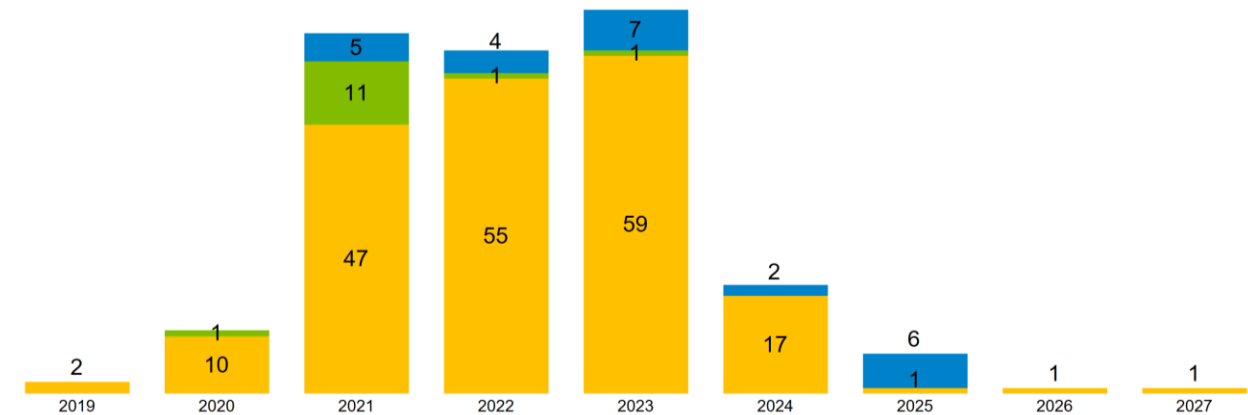


Figure 4.2-3: Central region MTEP20 projects, by in-service date

In accordance with Attachment FF of the tariff, in the event a Transmission Owner determines that system conditions warrant the urgent development of system enhancements an expedited review of the impacts of the project can be requested. MISO shall use a streamlined approval process for reviewing and approving projects proposed by the Transmission Owner(s) and decisions will be provided to the Transmission Owner within 30 Days of the project's submittal to MISO unless a longer review period is mutually agreed upon. During the MTEP20 cycle, MISO received the following projects through the Expedited Project Review (EPR) process.

1. Project ID 18705 Ameren Retire Coffeen 345 kV Switchyard
2. Project ID 18706 Ameren Retire Duck Creek 345 kV Switchyard
3. Project ID 19045 Ameren New Rivian Load Interconnection

In MTEP20, the issues identified in the MISO Central planning region were mainly reliability in nature as well as age and condition issues. There are a number of load-driven projects across Illinois, Indiana and Kentucky.

The ten highest cost projects in the MISO Central Region represent \$690 million (17.25%) of the \$4 billion total recommended projects for the whole MISO region in MTEP20. The locations of these projects are shown in Figure 4.2-4 and the investment is spread across the Central Planning region. Projects that are blanket expenditures (relays, physical security, etc.) are excluded from this list.

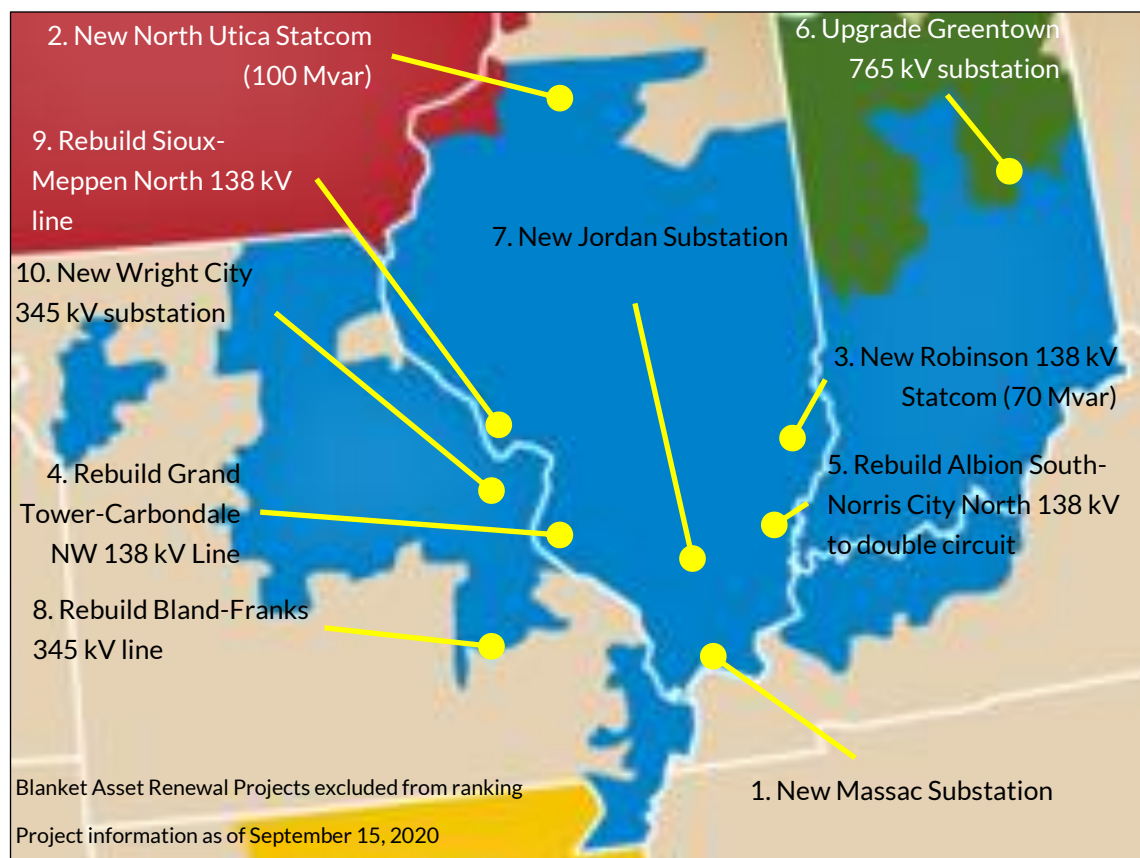


Figure 4.2-4: Central region top ten projects, by cost

4.2.1 American Electric Power Service Corporation (AEP)

American Electric Power Service Corporation did not submit any new projects for MTEP20. MISO has not identified any issues in AEP area.

4.2.2 Ameren Services Company (AMIL & AMMO)

Ameren proposed 151 projects at an estimated cost of \$1.4 billion targeting MTEP20 Appendix A. After MISO's independent reliability analysis of the Ameren area, 151 of these projects were recommended by MISO to be included in MTEP20 Appendix A at an estimated cost of \$1.4 billion. Of these projects, 11 are Baseline Reliability Projects, 129 are Other Projects, and 11 are Generation projects with signed Generator Interconnection Agreements. All the projects' expected in-service dates and estimated costs are provided as of July 30, 2020.

Baseline Reliability Projects

According to Attachment FF of the MISO Tariff, "Baseline Reliability Projects are Network Upgrades identified in the base case as required to ensure that the Transmission System is in compliance with applicable national Electric Reliability Organization (ERO) reliability standards and reliability standards adopted by Regional Reliability Organizations and applicable within the Transmission Provider Region."

Project 9721 - Reconductor [AMMO] Bland—[AMMO] Tegeler 138 kV line

Project Description

The project will Reconductor [AMMO] Bland—[AMMO] Tegeler 138 kV line. This reconductor will mitigate an overload on the same line for a P6-1-1 contingency event. The total estimated cost of this project is approximately twenty-seven (\$27M) million dollars. The expected in-service date for this project is December 2, 2022.

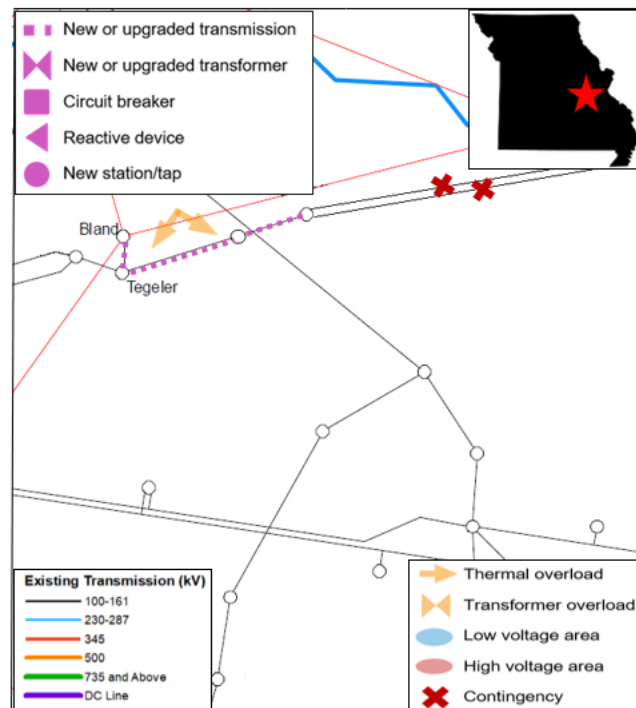


Figure 4.2.2-1: P9721 Geographic transmission map of project area

Project Need

The [AMMO] Bland—[AMMO] Tegeler 138 kV line, the [AMMO] Tegeler—[AMMO] Clover Bottom 138 kV line, and the [AMMO] Clover Bottom—[AMMO] Franklin 138 kV lines become overloaded beyond one-hundred (100%) percent in year 2022 for a NERC defined category P6-1-1 contingency event. Upgrading this line will increase the summer emergency rating from 151 MVA to 287 MVA (Table 4.2.2-1).

Cont. Type	Limiting Element	Rating (MVA)	Pre-Project Loading (%)	Post-Project Loading (%)
n-0	[AMMO] Bland—[AMMO] Tegeler 138 kV line	151	32	32
P6-1-1	[AMMO] Bland—[AMMO] Tegeler 138 kV line	287	172	91
n-0	[AMMO] Tegeler—[AMMO] Clover Bottom 138 kV line	151	23	23
P6-1-1	[AMMO] Tegeler—[AMMO] Clover Bottom 138 kV line	287	108	57
n-0	[AMMO] Clover Bottom—[AMMO] Franklin 138 kV line	151	26	26
P6-1-1	[AMMO] Clover Bottom—[AMMO] Franklin 138 kV line	287	101	53

Table 4.2.2-1: P9721 Thermal loading drivers

Alternatives Considered

Generation redispatch or load shed is not feasible for voltage violations due to P6 contingency events if there is no applicable short-time emergency rating. No other alternatives were considered. These line terminal upgrades are the best and cheapest option to address this reliability issue.

Project 12176 - New [AMIL] West Frankfort East & [AMIL] Norris City 345 kV reactors

Project Description

The addition of these new [AMIL] West Frankfort East 345 kV and [AMIL] Norris City 345 kV reactors will decrease voltages to acceptable levels in and around the West Frankfort East area during light load conditions. This project will mitigate high voltages during light load conditions during a P6-1-1 contingency event. The total estimated cost of this project is twenty (\$20M) million dollars. The expected in-service date for this project is December 1, 2024.

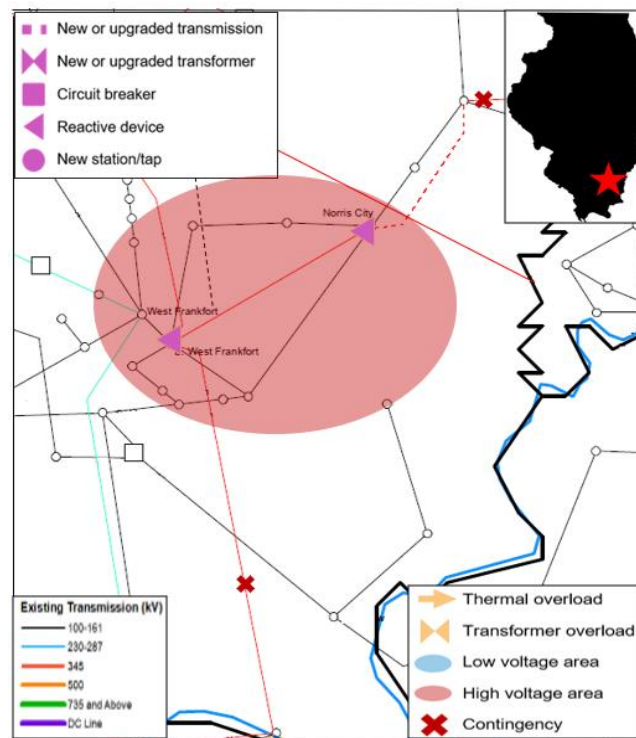


Figure 4.2.2-2: P12176 Geographic transmission map of project area

Project Need

The addition of both the [AMIL] West Frankfort East 345 kV reactor and the [AMIL] Norris City 345 kV reactor helps reduce local high voltages in and around [AMIL] West Frankfort East and [AMIL] Norris 345 kV buses during light load conditions in year 2023 and beyond for a NERC defined category P6-1-1 contingency event (Table 4.2.2-2).

Cont. Type	Limiting Element	Voltage Limit (pu)	Pre-Project Voltage (pu)	Post-Project Voltage (pu)
n-0	[AMIL] W. Frankfort E 345 kV & [AMIL] Norris City 345 kV	1.05	1.047	1.043
P6-1-1	[AMIL] W. Frankfort E 345 kV & [AMIL] Norris City 345 kV	1.05	1.06	1.047

Table 4.2.2-2: P12176 Voltage performance drivers

Alternatives Considered

Generation redispatch or load shed is not feasible for voltage violations due to P6 contingency events if there is no applicable short-time emergency rating. No other alternatives were considered. These line terminal upgrades are the best and cheapest option to address this reliability issue.

Project 17284 - New [AMIL] Woodhall—[AMIL] Hallock 138 kV line No. 2

Project Description

The project will install a second [AMIL] Woodhall—[AMIL] Hallock 138 kV line No. 2 on the vacant position on the existing [AMIL] Woodhall—[AMIL] Hallock 138 kV line No. 1 (line 1570). Also, Ameren will install 3 position initial, 6 position ultimate ring bus at the [AMIL] Hallock 138 kV substation. This project will mitigate low voltages in the area for a P6-1-1 contingency event. The total estimated cost for this project is fourteen (\$14M) million dollars. The expected in-service date for this project is December 1, 2021.

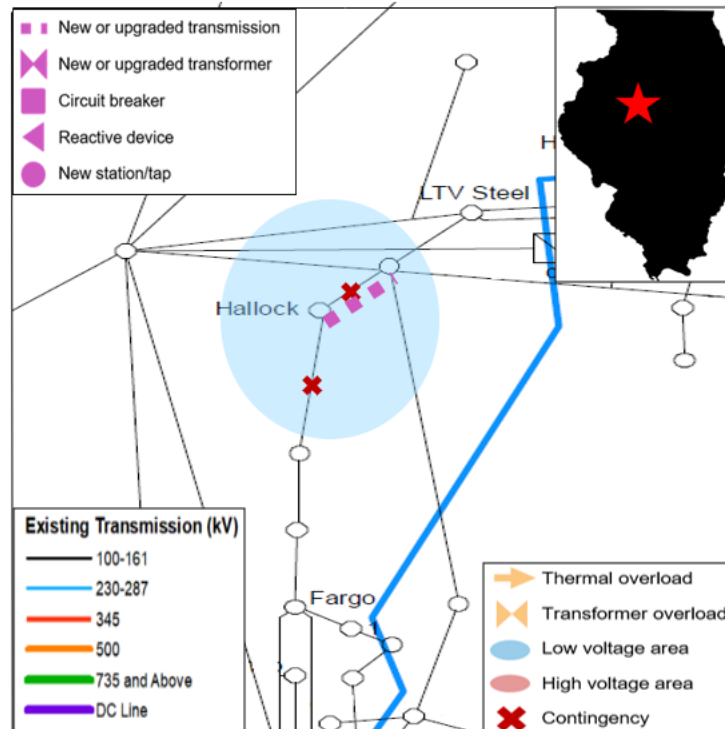


Figure 4.2.2-3: P17284 Geographic transmission map of project area

Project Need

Low voltages at the [AMIL] Hallock 138 kV substation by year 2022 for a NERC-defined category P6-1-1 contingency event. Adding a second [AMIL] Woodhall—[AMIL] Hallock 138 kV line No. 2 will allow for the [AMIL] Hallock 138 kV substation to remain energized during this P6-1-1 event (Table 4.2.2-3).

Cont. Type	Limiting Element	Voltage Limit (pu)	Pre-Project Voltage (pu)	Post-Project Voltage (pu)
Base case	[AMIL] Hallock 138 kV	0.95	1.01	1.02
P6-1-1	[AMIL] Hallock 138 kV	0.90	0.87	1.01

Table 4.2.2-3: P17284-1 Voltage performance drivers

Alternatives Considered

Generation redispatch or load shed is not feasible for voltage violations due to P6 contingency events if there is no applicable short-time emergency rating. No other alternatives were considered. These line terminal upgrades are the best and cheapest option to address this reliability issue.

Project 17565 - New [AMIL] Oreana—[AMIL] Decatur N 27th St. 138 kV line

Project Description

The project will build a new [AMIL] Oreana—[AMIL] Decatur N 27th St. 138 kV line. This new line will mitigate overloads in and around Decatur Illinois for a P6-1-1 contingency event. The total estimated cost of this project is approximately seventeen (\$17M) million dollars. The expected in-service date for this project is June 1, 2025.

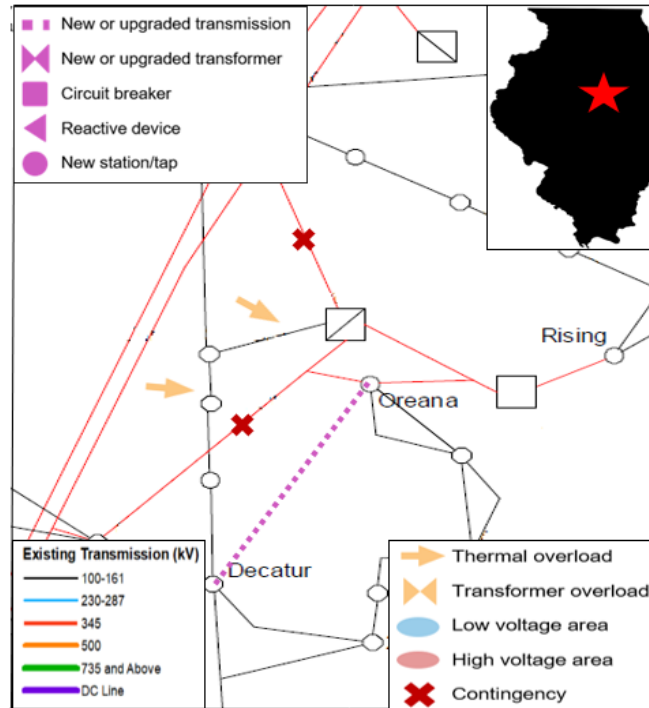


Figure 4.2.2-4: P17565 Geographic transmission map of project area

Project Need

Several BES lines in Decatur Illinois become overloaded beyond one hundred (100%) percent in year 2025 for a NERC defined category P6-1-1 contingency event. Building this new line will relieve some heavily loaded lines in Decatur Illinois for the coincident P6-1-1 events (Table 4.2.2-4).

Cont. Type	Limiting Element	Rating (MVA)	Pre-Project Loading (%)	Post-Project Loading (%)
n-0	[AMIL] ADM North—[AMIL] Caterpillar 138 kV line	477	.06	.06
P6-1-1	[AMIL] ADM North—[AMIL] Caterpillar 138 kV line	477	102	26

Table 4.2.2-4: P17565 Thermal loading drivers

Alternatives Considered

Generation redispatch or load shed is not feasible for voltage violations due to P6 contingency events if there is no applicable short-time emergency rating. No other alternatives were considered. These line terminal upgrades are the best and cheapest option to address this reliability issue.

Project 18139 - Upgrade [AMMO] Labadie 345 kV substation

Project Description

The project will upgrade the [AMMO] Labadie 345 kV substation with new switches and CTs that can handle 3000 A. This upgrade will mitigate overloads on the [AMMO] Labadie 345 kV 2-3 bus tie for a P6-1-1 contingency event along with the specific generation of Labadie No. 1 and No. 2 at Pmax and units No. 3 and No. 4 turned completely off. The total estimated cost of this project is approximately one (\$1M) million dollars. The expected in-service date for this project is December 1, 2022.

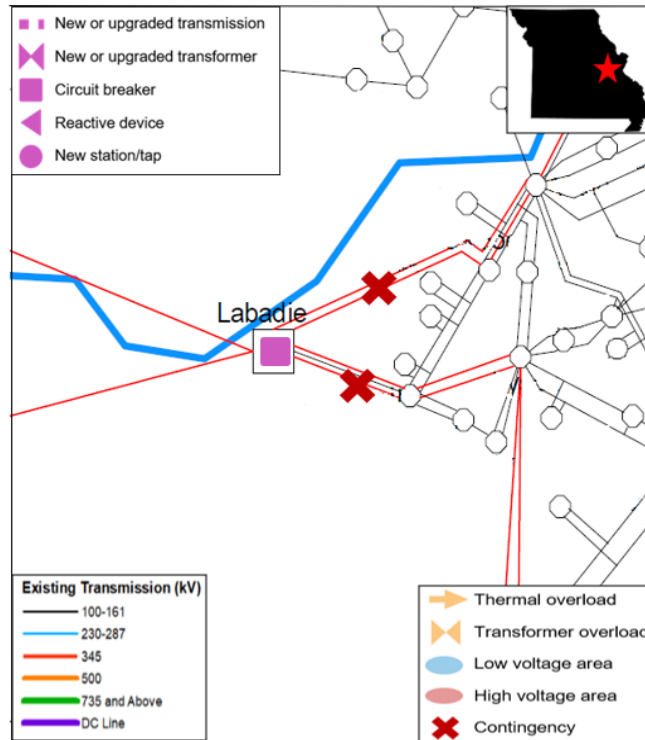


Figure 4.2.2-5: P18139 Geographic transmission map of project area

Project Need

There is an overload on the [AMMO] Labadie 2–[AMMO] Labadie 3 345 kV bus tie in year 2025 summer light load conditions for a NERC defined category P6-1-1 contingency event. Upgrading switches and CTs to 3000A will increase the summer emergency rating of this line from 1195 MVA to 1793 MVA during the summer season (Table 4.2.2-5).

Cont. Type	Limiting Element	Rating (MVA)	Pre-Project Loading (%)	Post-Project Loading (%)
n-0	[AMMO] Labadie2–[AMMO] Labadie3 345 kV bus tie line	1195	113	113
P6-1-1	[AMMO] Labadie2–[AMMO] Labadie3 345 kV bus tie line	1793	113	75

Table 4.2.2-5: P8139 Thermal loading drivers

Alternatives Considered

Generation redispatch or load shed is not feasible for thermal violations due to P6 contingency events if there is no applicable short-time emergency rating. No other alternatives were considered. This line reconductoring project is the best and cheapest option to address this reliability issue.

Project 18249 - New [AMIL] N. Utica 138 kV Statcom (100 Mvar)

Project Description

The project will add a new [AMIL] N. Utica 138 kV STATCOM device. This upgrade will provide voltage support to a soft spot on the system for the loss of a P6-1-1 contingency event. This project,

along with P18269, combine to mitigate low voltages and thermal overloads in Peoria, Illinois for the same P6-1-1 event. The total estimated cost of this project is approximately ninety-one (\$91M) million dollars. The expected in-service date for this project is December 1, 2023.

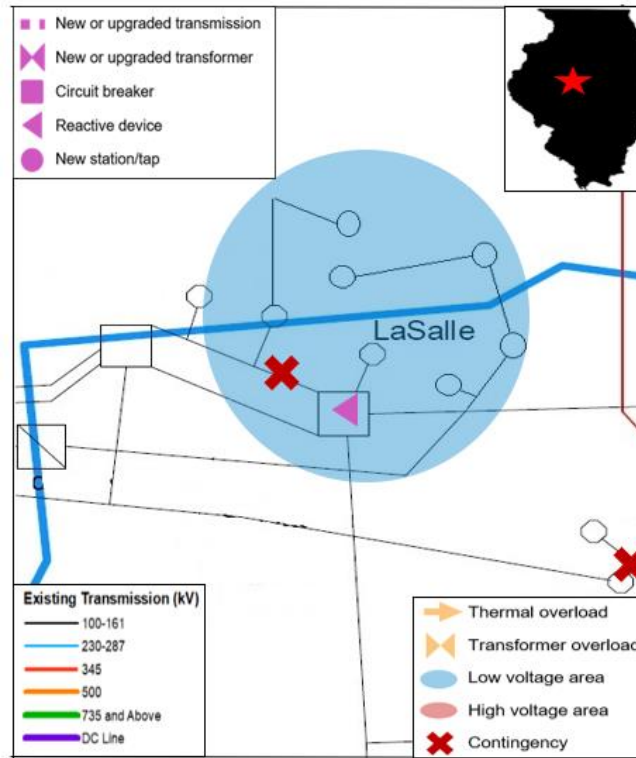


Figure 4.2.2-5: P18249 Geographic transmission map of project area

Project Need

There are a number of 138 kV buses that go below their low voltage limit of 0.90 p.u. in year 2024 summer peak (90/10 load probability) conditions for a NERC defined category P6-1-1 contingency event. Adding a new [AMIL] N. Utica 138 kV STATCOM will help prop the low voltages back up during the 2024 summer peak season (Table 4.2.2-6).

Cont. Type	Limiting Element	Voltage Limit (pu)	Pre-Project Voltage (pu)	Post-Project Voltage (pu)
n-0	[AMIL] Local 138 kV Buses	0.90	0.80	0.97
P6-1-1	[AMIL] Local 138 kV Buses	0.90	0.83	0.98

Table 4.2.2-6: P18249 Voltage performance drivers

Alternatives Considered

Ameren considered a SATOA, but there was not enough justification for such. Ameren also submitted P18270: New [AMIL] Sandburg–[AMIL] LaSalle 345 kV line, but this was not the better option.

Project 18250 - New [AMIL] Robinson 138 kV Statcom (70 Mvar)

Project Description

The project will add a new [AMIL] Robinson 138 kV STATCOM device. This upgrade will provide voltage support to a soft spot on the system for the loss of a P6-1-1 contingency event. The total

estimated cost of this project is approximately seventy-eight (\$78M) million dollars. The expected in-service date for this project is June 1, 2022.

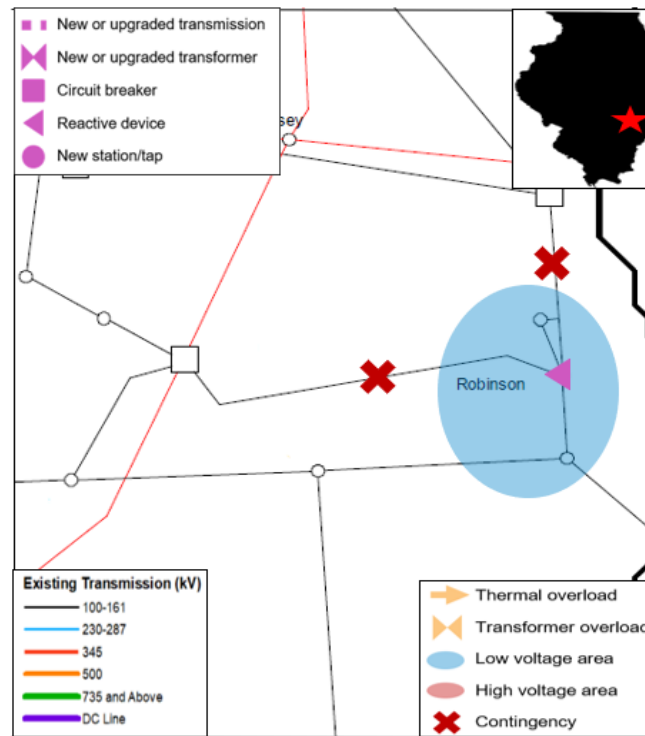


Figure 4.2.2-7: P18250 Geographic transmission map of project area

Project Need

There are a number of 138 kV buses that go below their low voltage limit of 0.90 p.u. in year 2022 summer peak conditions for a NERC defined category P6-1-1 contingency event. Adding a new [AMIL] Robinson 138 kV STATCOM will help prop the low voltages back up during the summer peak season (Table 4.2.2-7).

Cont. Type	Limiting Element	Voltage Limit (pu)	Pre-Project Voltage (pu)	Post-Project Voltage (pu)
n-0	[AMIL] Local 138 kV Buses	0.95	0.00	1.02
P6-1-1	[AMIL] Local 138 kV Buses	0.90	0.05	0.99

Table 4.2.2-7: P18250 Voltage performance drivers

Alternatives Considered

Ameren considered a SATOA, but there was not enough justification for such.

Project 18263 - New [AMIL] Fargo–[AMIL] Hines 138 kV line

Project Description

The project will add a new [AMIL] Fargo–[AMIL] Hines 138 kV line. This new line will mitigate an overload on the local 138 kV lines for a P6-1-1 contingency event. The total estimated cost of this project is approximately eighteen (\$18M) million dollars. The expected in-service date for this project is June 1, 2025.

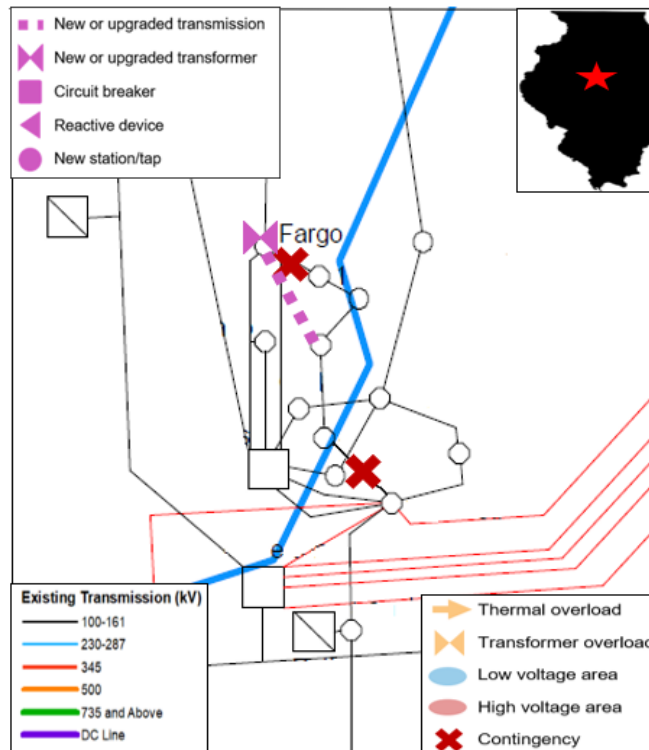


Figure 4.2.2-8: P18263 Geographic transmission map of project area

Project Need

There is an overload on the [AMIL] Fargo 138/69 kV transformer for the loss of this P6-1-1 event. Also, there are a number of 138 kV buses that become isolated and this puts more than 300 MW of load at risk for this same event. Per Ameren's transmission planning criteria, no load loss greater than 300 MW shall be allowed for a P6-1-1 event. Adding this new [AMIL] Fargo—[AMIL] Hines 138 kV line strengthens the local BES for any P6-1-1 events during the summer 90/10 load probability peak season (Table 4.2.2-8).

Cont. Type	Limiting Element	Rating (MVA)	Pre-Project Loading (%)	Post-Project Loading (%)
Base case	[AMIL] Fargo 138/69 kV transformer	130	72	78
P6-1-1	[AMIL] Fargo 138/69 kV transformer	130	138	78

Table 4.2.2-8: P18263 Thermal loading drivers

Alternatives Considered

Generation redispatch or load shed is not feasible for thermal violations due to P6 contingency events if there is no applicable short-time emergency rating. No other alternatives were considered. This line reconductoring project is the best and cheapest option to address this reliability issue.

Project 18267 - New [AMIL] Wright City 345 kV substation

Project Description

The project will add a new [AMIL] Wright City 345 kV substation. This new substation will mitigate overloads, low voltages and load loss on the local system for the loss of a P6-1-1 event. The total

estimated cost of this project is approximately thirty-nine (\$39M) million dollars. The expected in-service date for this project is December 2, 2023.

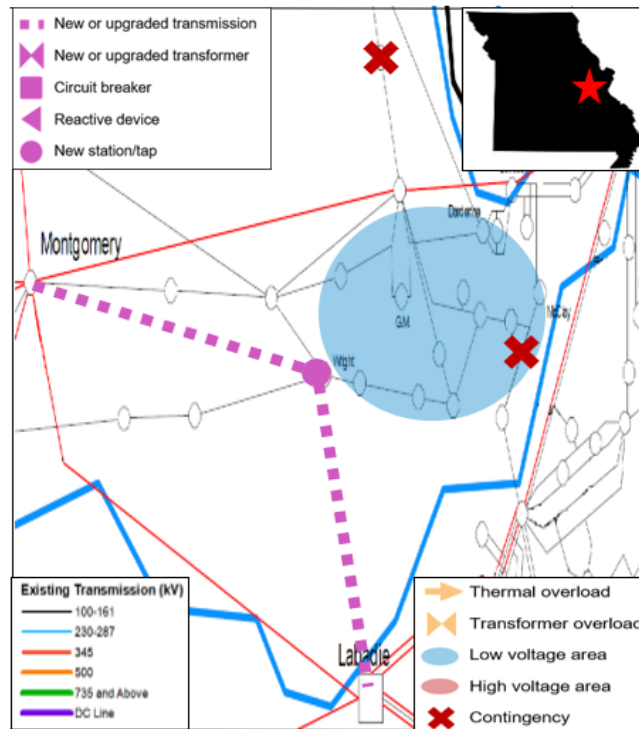


Figure 4.2.2-9: P18267 Geographic transmission map of project area

Project Need

There are a number of BES buses that go below their low voltage limit of 0.90 p.u. in year 2025 as well as a thermal overload during year 2025 summer peak conditions for a P6-1-1 contingency event. Adding a new [AMIL] Wright City 345 kV substation will help prop up the low voltages and reduce the overloaded BES lines during the summer peak season (Table 4.2.2-9 and Table 4.2.2-10).

Cont. Type	Limiting Element	Rating (MVA)	Pre-Project Loading (%)	Post-Project Loading (%)
Base case	[AMMO] Big Creek–[AMMO] Pointe Prairie 161 kV line	319	13	24
P6-1-1	[AMMO] Big Creek–[AMMO] Pointe Prairie 161 kV line	319	110	71

Table 4.2.2-9: P18267 Thermal loading drivers

Cont. Type	Limiting Element	Voltage Limit (pu)	Pre-Project Voltage (pu)	Post-Project Voltage (pu)
Base case	[AMMO] Local 138 kV Buses	0.90	0.89	0.97
P6-1-1	[AMMO] Local 138 kV Buses	0.90	0.88	0.97

Table 4.2.2-10: P18267 Voltage performance drivers

Alternatives Considered

Generation redispatch or load shed is not feasible for thermal violations due to P6 contingency events if there is no applicable short-time emergency rating. No other alternatives were

considered. This line reconductoring project is the best and cheapest option to address this reliability issue.

Project 18269 - New [AMIL] LaSalle–[AMIL] North Utica 138 kV line

Project Description

The project will add a new [AMIL] LaSalle–[AMIL] North Utica 138 kV line. This upgrade will prevent thermal overloads and low voltages that are caused by the loss of a P6-1-1 contingency event. The amount of load that is at risk is around 300 MW. In accordance with Ameren’s planning criteria, no (n-1-1) contingencies shall result in load loss greater than 300 MW. The total estimated cost of this project is approximately twenty-seven (\$27M) million dollars. The expected in-service date for the project is June 1, 2025.

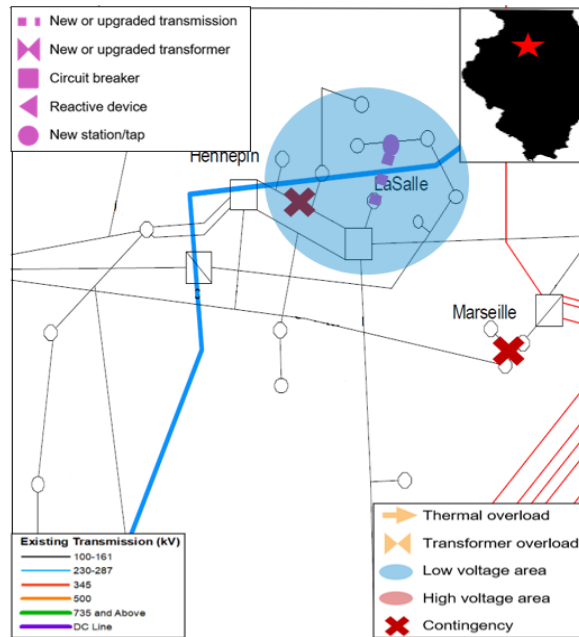


Figure 4.2.2-10: P18269 Geographic transmission map of project area

Project Need

There is a thermal overload on a BES 138 kV line and also a number of 138 kV buses with voltage falling below their low voltage limit of 0.90 p.u. in year 2025 summer peak conditions for NERC defined category P6-1-1 contingency events. Adding this new 138 kV line will mitigate the thermal overload and help prop the low voltages back up during the summer peak season. The low voltages are brought back just above an acceptable operating limit. P18249 – New [AMIL] N. Utica 138 kV STATCOM is also an MTEP20 Target A project that will help bring the voltages back up to a safe operating level (Table 4.2.2-11 and Table 4.2.2-12).

Adding only a STATCOM elevates the voltages but does not address the thermal overload issue. The proposed new 138 kV line addresses the thermal overload and props up the low voltages, but it does not raise the voltages in this area back to a voltage level that is considered acceptable.

Cont. Type	Limiting Element	Rating (MVA)	Pre-Project Loading (%)	Post-Project Loading (%)
Base case	[AMIL] Hickock–[AMIL] N. LaSalle 138 kV line	202	62	40

P6-1-1	[AMIL] Hickock—[AMIL] N. LaSalle 138 kV line	202	141	62
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Table 4.2.2-11: P18269 Thermal loading drivers

Cont. Type	Limiting Element	Voltage Limit (pu)	Pre-Project Voltage (pu)	Post-Project Voltage (pu)
Base case	[AMIL] Local 138 kV Buses	0.90	0.88	0.95
P6-1-1	[AMIL] Local 138 kV Buses	0.90	0.80	0.91

Table 4.2.2-12: P18269 Voltage performance drivers

Alternatives Considered

Generation redispatch or load shed is not feasible for thermal violations due to P6 contingency events if there is no applicable short-time emergency rating. No other alternatives were considered. This line reconductoring project is the best and cheapest option to address this reliability issue.

Project 18275 - New [AMIL] Holland NW—[AMIL] Neoga South 345 kV line (4509)

Project Description

The project will add a new [AMIL] Holland NW—[AMIL] Neoga South 345 kV line (4509). This upgrade will mitigate an overload on the [AMIL] Pana—[AMIL] Ramsey 138 kV line for the outage of a P6-1-1 contingency event. The total estimated cost of this project is approximately thirty-two (\$32M) million dollars. The expected in-service date for this project is December 1, 2023.

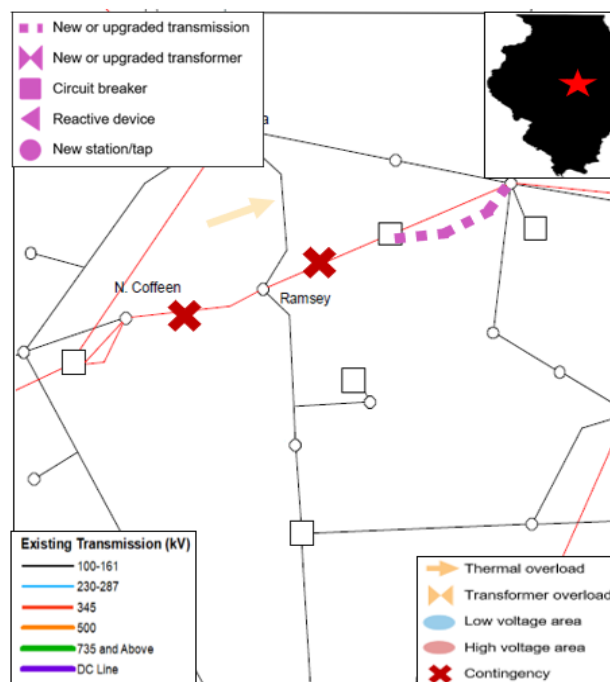


Figure 4.2.2-12: P18275 Geographic transmission map of project area

Project Need

This project will mitigate the heavily loaded [AMIL] Pana—[AMIL] Ramsey 138 kV line in year 2025 summer peak conditions for a NERC defined category P6-1-1 contingency event. Adding this new

345 kV line will reduce the loading on this [AMIL] Pana—[AMIL] Ramsey 138 kV line during the 2025 summer peak season (Table 4.2.2-13).

Cont. Type	Limiting Element	Rating (MVA)	Pre-Project Loading (%)	Post-Project Loading (%)
Base case	[AMIL] Pana—[AMIL] Ramsey 138 kV line	478	99	23
P6-1-1	[AMIL] Pana—[AMIL] Ramsey 138 kV line	478	99	23

Table 4.2.2-13: P18269 Thermal loading drivers

Cont. Type	Limiting Element	Voltage Limit (pu)	Pre-Project Voltage (pu)	Post-Project Voltage (pu)
Base case	[AMIL] Local 138 kV Buses	0.90	0.88	0.95
P6-1-1	[AMIL] Local 138 kV Buses	0.90	0.80	0.91

Table 4.2.2-14: P18269 Voltage performance drivers

Alternatives Considered

Generation redispatch or load shed is not feasible for thermal violations due to P6 contingency events if there is no applicable short-time emergency rating. No other alternatives considered. This line reconductoring project is the best and cheapest option to address this reliability issue.

Other Projects

The “Other” projects category are projects that do not meet the criteria to be considered as Baseline Reliability Projects (BRP), New Transmission Access Projects, Market Efficiency Projects, or Multi-Value Projects. Other projects may include projects to satisfy Transmission Owner and/or state and local planning criteria other than NERC or regional reliability standards, interconnection of new Loads, relocate transmission facilities, address aging transmission infrastructure, replace problematic transmission plant, improve operational performance or address other operational issues, address service reliability issues with end-use consumers, improve aesthetics including but not limited to undergrounding overhead transmission facilities, address localized economic issues, and address other miscellaneous localized needs. The tables of project information are broken down by four general categories of project drivers but note that these four drivers are not defined in the MISO Tariff.

Projects Driven by Local Reliability

The following projects are driven by local planning criteria without Electric Reliability Organization (ERO) standard applicability. These projects could be as a result of operational flexibility purposes.

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
9723	New Fishhook 138 kV substation	Construct 138 kV breaker station in East Quincy-Meredosia-2 138 kV line.	June 1, 2019	\$10.5M
9735	Upgrade Tyson 138 kV substation	Replace overstressed 138 kV breakers,	December 1, 2021	\$2.7M
11925	New Massac Substation - Convert Joppa 230 kV substation to 345 kV substation	Install two (2) 345/161 kV, 560 MVA transformers, remove 230/161 kV equipment. Rebuild Joppa-West Frankfort 230 kV for 345 kV use (Massac-Jordan).	June 1, 2023	\$112.4M
11926	New Jordan Substation - Convert West Frankfort 230	Install 345/138 kV, 560 MVA transformer. Rebuild/reinsulate 20 miles	December 1, 2023	\$56.1M

	kV substation to 345 kV substation	of existing 230 kV line between West Frankfort and St. Johns for use at 345 kV.		
11969	New Miles-Mississippi 138 kV line	Construct new 138 kV line.	December 1, 2020	\$4.0M
12428	Replace Stoddard 161 kV breakers	Replace one (1) 161 kV breaker at Stoddard.	December 1, 2021	\$0.7M
13705	Reconfigure North Nashville 138 kV substation	Install initial 3 position 138 kV ring bus (ultimate 4 position) at North Nashville Substation.	December 1, 2021	\$7.8M
13708	New Athens Dutch Hill 138 kV substation ring bus	Add 138 kV ring Bus. Initial 3-position (ultimate 4-position).	December 1, 2022	\$8.4M
13793	New El Paso 138 kV substation ring bus	Install initial 4-position ring bus at El Paso Substation for in/out supply.	December 1, 2022	\$11.8M
15265	Reconfigure Carbondale NE 138 kV substation to ring bus	Install initial 3-position ring bus at Carbondale NE Substation.	June 1, 2023	\$12.5M
15266	Reconfigure Fayetteville Bee Hollow 138 kV substation to ring bus	Install initial 3-position ring bus at the existing Fayetteville Bee Hollow substation location.	June 1, 2023	\$7.2M
15487	New Vessel 138 kV ring bus	Install 4 position ring bus.	June 1, 2022	TBD
15504	New Dempsey (Blue Mound) 138 kV ring bus addition	Install new ring bus at the tap to Blue Mound on the Pana-Route 51 Decatur-138 kV line (1462).	June 1, 2022	\$11.7M
16504	Reconfigure Viburnum 161/34 kV Substation	Install 2 161 kV line breakers and 1 161 kV circuit switcher on the transformer to split the CLK-CMCO-2 line and create 2 circuits into Viburnum Substation.	December 1, 2021	\$1.3M
16555	Relocate Page 138 kV substation to new Bugle 138 kV Substation	Relocate the existing 138 kV Page substation to the new Bugle 138 kV site to be built as BAAH arrangement.	June 1, 2023	\$27.5M
16704	Rebuild Salem West-Mt Vernon West 138 kV line (1332)	Rebuild line to 2000A. Upgrade terminal equipment at Salem West to 2000A including 1272 ACSR bus conductor and switch 1330.	December 1, 2023	\$13.2M
16705	Rebuild Pana-Shelbyville South 138 kV line (1430)	Rebuild line to 200A capability. Upgrade terminal equipment at Shelbyville South to 2000A including re-tapping CTs and replacing 954 ACSR bus conductor.	December 1, 2021	\$11.9M
16706	Rebuild Goose Creek 345 kV ring bus	Rebuild the Goose Creek 345 kV ring bus to 3000A capability. Replace all limiting terminal equipment and bus conductors with equipment capable of 3000A.	December 1, 2022	\$2.6M
16707	New Kazad 138 kV Substation in St. Joseph area	Install new ring bus substation at the intersection of VERM-NCMP 138 kV line (1396) and GIFF-SDNY 138 kV line (1588). Install 4 positions initial, 6-positions ultimate. Rebuild the existing Vermillion-N. Champaign 138 kV line (1396) 138 kV line to 2000A in both directions from the new substation.	December 1, 2023	\$7.2M
16708	Reconductor Hines-Pioneer 138 kV line (1326)	Replace existing 927 ACAR conductor with ACCR conductor.	December 1, 2023	\$3.5M

16709	Upgrade Ashley-Venice 138 kV line at Venice substation	Replace 2 1200A switches and 1000 Kcmil CU bus conductor with equipment having 2000A capability. Upgrade the position to 2000A.	December 1, 2021	TBD
16944	New Tanner 345/138 kV Transformer	Install new 345 kV ring bus at the Louisville South substation location tapping the Newton-Xenia 345 kV line. Install 560 MVA 345/138 kV transformer and connect to the Louisville South 138 kV bus.	December 1, 2024	\$25.2M
16987	New Benton NW 138 kV Ring bus	Install 6 position ultimate ring bus.	December 1, 2023	\$8.5M
16988	New Fogarty-Limit 138 kV Line	Install new 138 kV line from Fogarty Substation to limit. Install new ring bus at Limit to make the connection.	December 1, 2023	\$24.6M
16989	New Mattoon W 138 kV substation	Install new 6 position ultimate ring bus at Mattoon West Substation.	December 1, 2023	\$8.5M
16990	New Kline 138 kV substation	Install new 6 position ultimate ring bus at Kline Substation (formerly known as Wanda).	December 1, 2022	\$6.3M
17285	New Quiver 138 kV Substation	Install new ring bus installation at the intersection of Mason City West – Fogarty -1346 and San Jose Rail – Interstate – 1318 138 kV line.	December 1, 2023	\$13.0M
17604	Upgrade Stallings-Roxford 345 kV line (4581)	Upgrade 345 kV switches V4A and V4b to 3000A. Remove the 2000A wave trap. Install OPGW.	December 1, 2023	\$1.4M
17624	Rebuild Pike 161 kV Substation	Rebuild the Pike 161 kV substation to a Ring bus configuration.	December 1, 2023	\$13.4M
17625	Upgrade McClay 138 kV Substation	Add breakers to each of the 138 kV lines and upgrade relays.	December 1, 2022	\$1.3M
17626	Rebuild Kickapoo-Limit 138 kV line (1325)	Rebuild line from Kickapoo to Limit to 2000A. Double circuit from Limit to McGrath to supply McGrath radially from Limit.	December 1, 2024	\$3.1M
17644	Upgrade Decatur Route 51 138 kV substation	Install initial 138 kV 4-position (ultimate 6-position) ring bus at the Decatur Route 51 substation location	June 1, 2022	\$9.0M
17805	Rebuild Campbell-Maline 138 kV lines 1/2	Replace existing conductor with composite core conductor on branches 93 and 94.	December 1, 2021	\$1.6M
17829	New Loose Creek 345 kV shunt reactor	Install 345 kV shunt reactor at Loose Creek	December 1, 2021	\$4.8M
17845	New Wittenberg-Trail of Tears 138 kV line	Wabash Valley - Install new Wittenberg ring bus switching station on the Grand Tower-Perryville line. Reconfigure Trail of Tears substation to BAAH. Install new 161 kV line from Trail of Tears to Charmin Bulk Substation. Install 2 breakers at Charmin Bulk Substation. Ameren - Install 12 miles of new 138 kV line from Wittenberg Substation to Trail of Tears Substation. Install new 161/138	December 1, 2023	\$38.5M

		kV transformer at Trail of Tears Substation.		
17864	Upgrade Cahokia 345/138 kV substation	Replace OCB H27 on the C-GRAT-3 position and OCB H35 on the C-RUSL-2 position with symmetrically rated SF6 55 kA circuit breakers.	December 1, 2022	\$0.9M
17865	Upgrade Lakeshire 161 kV substation	Replace 40 kA breakers on positions H and U for cap banks 2A and 2B with 50 kA breakers.	December 1, 2022	\$1.0M
17866	Upgrade Neoga 345/138 kV substation	Replace OCB 1402 on the HUTY-NEGS-1 position with a symmetrically rated 40 kA SF6 breaker.	December 1, 2022	\$0.6M
17867	Upgrade Gray Summit 345/161 kV substation	Replace the Bus-Tie 1-2 breaker at Gray Summit with a symmetrically rated 40 kA circuit breaker.	December 1, 2022	\$0.6M
17904	New Highland-Aviston 138 kV line	Install new Highland area ring bus. Re-configure Aviston Substation to a ring bus. Install new 16.5-mile 138 kV line between Highland and Aviston.	December 1, 2023	\$35.7M
17905	New Hull 161/138 kV transformer	Install new 161 kV Finn ring bus (Hannibal area). Convert the Spalding Substation to a ring bus configuration adding a position for a line to the Hannibal area Substation. Double circuit the PENO-SPLD-2 line from Spalding to the Marion tap and rebuild the Marion Tap 161 kV section of the PENO-SPLD-2 line to the Mississippi River, through the new Hannibal area Subst.	June 1, 2022	\$30.3M
17972	Rebuild Salem West 138 kV substation	Rebuild the Salem West 138 kV substation to a ring bus configuration.	December 1, 2023	\$7.0M
17973	Rebuild Centerville 138 kV substation	Rebuild the Centerville 138 kV substation to a Ring bus configuration.	December 1, 2023	\$8.0M
17974	New Crab Orchard 345/138 kV transformer	Tap the Shawnee-West Frankfort east-4564 line and install 345kV ring bus to make a connection for the installation of a new 345-138 kV transformer.	December 1, 2023	\$28.0M
17975	New Rising 345/138 kV transformer No. 2	Install 2nd 345-138 kV transformer at Rising Substation.	December 1, 2023	\$11.0M
17976	New Redhawk 345/138 kV transformer	Tap the Coffeen-Roxford 345 kV line (4551) line and install 345 kV ring bus to make the connection for a new 345/138 kV transformer at Redhawk Substation.	December 1, 2023	\$22.4M
17977	New Mt. Vernon West 345/138 kV transformer 2	Install 2nd 345-138 kV transformer at Mt. Vernon West Substation.	June 1, 2023	\$8.0M
17978	Rebuild Belleville 17th Street 138 kV substation	Rebuild the Belleville 17th street 138 kV substation to a ring bus configuration.	December 1, 2022	\$10.0M
17981	Rebuild North Champaign 138 kV substation	Rebuild the North Champaign 138 kV substation to a ring bus configuration.	December 1, 2023	\$8.5M
17982	New Maroa 345 kV substation	Install new Maroa 345 kV ring bus substation.	December 1, 2023	\$12.0M
17983	Rebuild South Belleville 138 kV substation	Rebuild the South Belleville 138 kV substation to a ring bus configuration.	June 1, 2023	\$8.0M

18013	New Montgomery 345 kV shunt reactor	Install 50 Mvar shunt reactor at Montgomery 345 kV substation.	December 1, 2025	\$3.8M
18016	Reconductor Berkley South 161 kV lines 1/2	Reconductor Berkley South 1 & 2 on branches (1004) and (1005) with composite core conductor.	December 1, 2023	\$5.6M
18030	New Meredosia 345 kV shunt reactive device	Install a shunt Reactive device to control voltages at IL-Meredosia 345 kV.	April 1, 2021	\$5.7M
18031	New Ipava 345 kV shunt Reactive device	Install a shunt Reactive device to control voltages at Ipava 345 kV.	April 1, 2021	\$4.3M
18032	New Xenia 345 kV shunt Reactive device	Install a shunt Reactive device to control voltages at Xenia 345 kV.	December 1, 2023	\$4.9M
18041	Upgrade Ottawa 138 kV substation	Install new line breaker on the HENN-OGLE-1516 line at Ottawa Substation.	December 1, 2022	\$2.0M
18042	Upgrade Wedron-Fox River 138 kV line (1652)	Install new line breaker on the OTTA-Fox River 138 kV line (1652) circuit at Wedron Fox River Substation.	December 1, 2022	\$2.0M
18044	Reconfigure Caterpillar quad 138 kV circuits	Reconfigure the Tazewell-Caterpillar 2-1375, Edwards-Caterpillar 1-1374, RS Wallace-Keystone-1396 and the Caterpillar 2-Hines-1357 138 kV circuits on a 1.5-mile section where they share a common structure.	December 1, 2024	\$5.0M
18045	Upgrade Murdock 138 kV substation	Install a 138 kV circuit switcher on the high-side of XFMR #1 at Murdock Substation.	June 1, 2023	\$1.0M
18056	New Mason 138 kV shunt reactive device	Install a shunt Reactive device to control voltages at Mason 3 138 kV bus.	December 1, 2023	\$4.5M
18074	Replace Jacksonville Industrial Park 138 kV Breakers	Jacksonville Industrial Park Breaker Replacement: Replace 138 kV breaker JCKI-138KV BT 1-2 BKR 1302 at Jacksonville Industrial Park Substation.	December 1, 2023	\$0.5M
18279	Upgrade South Ottawa 138 kV substation	Install 138 kV bus-tie breaker at the switch 1516 position at South Ottawa.	December 1, 2022	\$0.8M
18309	New Fredericktown 138 kV substation	Install 138 kV ring bus at Fredericktown Substation.	December 1, 2022	\$7.5M
18705	Bypass Coffeen 345 kV Switchyard (EPR)	Bypass the Coffeen 345 kV switchyard by re-terminating the lines at the Coffeen North 345 kV substation.	June 1, 2020	\$1.0M
18706	Bypass Duck Creek 345 kV Switchyard (EPR)	Bypass the Duck Creek 345 kV switchyard.	June 1, 2020	\$0.9M

Projects Driven by Age and Condition

The following projects are proposed to replace aging or degraded equipment.

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
13709	New Putnam BAAH 138 kV substation	Build new 138 kV BAAH substation on new site near Hennepin.	December 1, 2022	\$22.3M
13797	Rebuild Albion South-Norris City North 138 kV to double circuit	Rebuild the existing 138 kV circuit to 2000A. Double circuit the line for new 3000A 345 kV circuit.	December 1, 2022	\$74.0M
15268	Upgrade Selma-Buck Knob 138 kV line (1302)	Upgrade the DPFE-SEL 138 kV line (1302) from Selma Substation to the old Buck	June 1, 2021	\$3.4M

		Knob substation location with conductor capable of 1600A under summer emergency conditions. Upgrade terminal equipment at Selma to 1600A minimum capability.		
16491	Rebuild Sioux-Meppen-4, Page-Sioux-4, Sioux-Huster-1 138 kV lines	Sioux-Huster-1/Sioux-Meppen North-4: Rebuild ~9 miles of triple circuit 138 kV lattice tower line to double-circuit steel poles.	December 1, 2021	\$17.1M
16552	Replace Tyson 345/138 kV Transformer	Replace XFMR 1 with a hardened unit and replace the 138 kV XFMR 1 breaker and Replace XFMR 3 breaker.	December 1, 2022	\$7.6M
16984	Rebuild Sioux-Meppen North 138 kV line (4)	Rebuild existing river crossing (structures 173-179).	December 1, 2021	\$45.7M
16985	Rebuild Sioux-Meppen 138 kV Line (4)	Rebuild 3.7-mile Missouri side section (Str. 146-173) of the old Page-Keokuk line w/ T2 conductor. Page-Sioux-4: Rebuild Missouri River crossing (Str. 112-117) w/ 2000A rated conductor.	December 1, 2021	\$2.8M
16986	Rebuild Page-Sioux 138 kV line (4)	Rebuild existing Missouri River crossing (Str. 112-117)	December 2, 2023	\$26.7M
17046	Upgrade Kelso 345/161 kV substations	Install new 3000 A circuit breaker and motor-operated disconnect switch on 345 kV position V3. Replace the existing Kelso substation 336 MVA auto transformer #1 with a 560 MVA transformer. Replace the existing Kelso substation 161 kV bus tie 1-2 position H7 circuit breaker and bus disconnect switches. Replace the existing Kelso substation 161 kV position H6 circuit breaker and disco.	December 1, 2022	\$5.8M
17065	Upgrade Tegeler 138 kV substation	Install a new 2000 A circuit breaker and associated disconnect switches in the TGLR-BLAN-6756 138 kV line terminal. Replace the existing 1600 A circuit breaker and associated 1200 A disconnect switches and 1590 kcmil AAC bus conductor at position H5 with equipment capable of carrying 2000 A during summer emergency conditions. Increase relay load limit.	December 1, 2021	\$2.1M
17344	Upgrade Berkeley 138 kV Substation	Replace 138 kV bus tie 1-2 and 2-3 breakers. Upgrade position to 2000A.	December 1, 2021	\$1.9M
17345	Rebuild Auburn North-Jacksonville 138 kV line	Scope is broken into 3 sections: - Structures 644-668 will be a complete rebuild to 1600A with 795 T2 conductor. - Structures 19-49 are single pole wood and will require structure replacements. - Jacksonville-Auburn North - 5 structure raises	June 1, 2020	\$2.5M
17346	Upgrade Rivermines 138 kV substation	Replace 138 kV breakers on bus tie 1-2 and bus tie 2-3	December 1, 2022	\$0.6M
17605	Reconductor Page-Berkeley 138 kV lines 1/2	Reconductor the Page-Berkeley-1&2 circuits with ACCR conductor	December 1, 2022	\$11.3M

17627	Rebuild Fogarty-Kickapoo 138 kV line (1672)	Rebuild 138 kV line to 2000A	December 1, 2022	\$14.4M
17804	Rebuild Lutesville-St. Francois 345 kV line	Rebuild 63 miles of 345 kV wood H-frame circuit.	December 1, 2023	\$32.8M
17824	Upgrade Spencer Creek 345 kV substation	Replace switches on 345 kV pos V3, V5, V6, V7 and reactor position.	December 1, 2021	\$0.7M
17825	Reconfigure Lawrenceville South 138 kV substation	Reconfigure the Lawrenceville South substation to a 6-position ultimate ring bus.	December 1, 2023	\$9.2M
17826	Upgrade Greenville McCord-South Centralia 138 kV line (1419)	Replace ~22 miles of existing 5/16" HS Steel Shield Wire with 1 3/8" E.H.S. Steel Shield Wire and 1 OPGW	December 1, 2022	\$1.7M
17827	Upgrade Cottage Hills-Stallings 161 kV line (1622)	Replace ~4 miles of existing 3 #7 Copperweld shield wire from Str. 51-Wanda Sub	December 1, 2020	\$2.0M
17828	Upgrade Kren-Porter Road 138 kV line (1432)	Replace ~6 miles of existing 3 #7 Copperweld shield wire for the entire line.	June 1, 2020	\$2.6M
17830	Upgrade Gasconade-Osage 161 kV line	Replace both shield wires and replace structures as needed.	December 1, 2021	\$9.4M
17908	Rebuild Bland-Franks 345 kV line	Rebuild approximately 44 miles 345 kV line to 3000A.	December 1, 2024	\$48.4M
17909	Upgrade Campbell-Euclid 161 kV line	Replace shield wires on the Campbell-Euclid-4 circuit. Replace structures and crossarms as needed.	December 1, 2021	\$1.6M
17910	Rebuild Grand Tower-Carbondale NW 138 kV line	Rebuild ~36 miles of the Grand Tower-Carbondale NW-W. Frankfort 138 kV line as double circuit 345/138 kV.	December 1, 2024	\$75.6M
17969	Replace Mason 345/138 kV Transformer	Replace 345/138 kV, 560 MVA Transformer #2 with a 700 MVA unit	December 1, 2022	\$6.7M
18034	Upgrade Tazewell 138 kV substation	Replace 14 138 kV breakers at Tazewell Substation including: TAZW-138 KV BKR 1321-73, TAZW-138 KV BKR 1340, TAZW-138 KV BKR 1354, TAZW-138 KV BKR 1323, TAZW-138 KV BKR 1340-63 (TIE), TAZW-138 KV BKR 1350 (XFMR 2), TAZW-138 KV BKR 1363, TAZW-138KV BKR 1375, TAZW-138 KV BKR 1353, TAZW-138 KV BKR 1322, TAZW-138 KV BKR 1367, TA..	December 1, 2022	\$5.6M
18035	Upgrade Eastern 138 kV substation	Replace TAZW-138 KV BKR 1321-73 at Eastern substation	December 1, 2022	\$0.5M
18036	Upgrade Pioneer 138 kV substation	Replace breaker PION-138KV line (1326) at Pioneer substation.	December 1, 2022	\$0.5M
18037	Upgrade Decatur N 27th Street 138 kV substation	Replace breakers N27D-138 KV Bkr 1522 and N27D-138 KV Bkr 1602 at Decatur North 27th street.	December 1, 2022	\$0.8M
18039	Upgrade Havana 138 kV substation	Replace breakers HAV-138 KV BKR 1406, HAV-138 KV BT BKR 1302 and HAV-138KV BKR 1422 at Havana substation.	December 1, 2022	\$1.2M
18075	Upgrade North Decatur 138 kV substation	Replace 138 kV breaker NDEC-138KV BKR 1602 at North Decatur substation.	December 1, 2023	\$0.5M

18076	Upgrade Steeleville 138 kV substation	Replace 138 kV breaker STVL-138 KV BKR 1636 at Steeleville substation.	December 1, 2023	\$0.5M
18077	Upgrade Vermilion 138 kV substation	Replace 138 kV breaker VERM-138 KV BKR 1306 at Vermilion substation.	December 1, 2023	\$0.5M
18078	Upgrade Carbondale NW 138 kV substation	Replace 138 kV breaker CDNW-138 KV LINE 1744 at Carbondale NW substation.	December 1, 2023	\$0.5M
18079	Upgrade Mason City W 138 kV substation	Replace 138 kV breaker MSCW-138 KV LINE 1684 at Mason City West substation.	December 1, 2023	\$0.5M
18080	Upgrade Meredosia E 138 kV substation	Replace breakers MERE-138KV BT 1938, MERE-138 KV BT 1958, MERE-138 KV BKR 1957, MERE-138 KV BKR 1947 and MERE-138 KV BT 1948 at Meredosia East substation.	December 1, 2023	\$2.0M
18081	Upgrade Newton 138 kV substation	Replace 138 kV breakers NWTY-138 KV LT 1869 and NWTY-138 KV LT 1868 at the Newton switchyard.	December 1, 2023	\$0.8M
18085	Replace Tazewell 345/138 kV transformer No. 1	Replace 345/138 kV transformer with new 560 MVA unit.	June 1, 2022	\$3.7M
18086	Replace Tazewell 345/138 kV transformer no. 2	Replace 345/138 kV transformer with new 560 MVA unit.	June 1, 2024	\$3.7M
18088	Replace Neoga South 345/138 kV transformer	Replace 345/138 kV transformer with new 560 MVA unit.	June 1, 2023	\$3.7M
18089	Replace Brokaw 345/138 kV transformer	Replace 345/138 kV transformer with new 560 MVA unit.	December 1, 2023	\$3.7M
18090	Replace Albion South 345/138 kV transformer No. 3	Replace 345/138 kV transformer with new 560 MVA unit.	December 1, 2024	\$3.7M
18091	Upgrade Clinton 345 kV substation	Replace breaker CLNT-345KV BKR 4506 at Clinton switchyard.	December 2, 2023	\$0.5M
18092	Replace East Galesburg 138/69 kV transformer	Replace breaker EGAL-138/69 KV AUTO XFMR 2 at East Galesburg substation.	December 1, 2024	\$0.5M
18093	Upgrade Fargo 138 kV substation	Replace breakers FARG-138 KV BKR & LINE 1394 and FARG-138 KV BKR 1371 at Fargo substation.	December 1, 2024	\$0.8M
18095	Upgrade Gibson City S 138 kV substation	Replace breakers GBCS-138 KV LINE 1524 and GBCS-138 KV LINE 1522 at Gibson City South substation.	December 1, 2024	\$0.8M
18096	Upgrade Goose Creek 345 kV substation	Replace breakers GOSC-345 KV BKR 4545, GOSC-345 KV BKR 4575 and GOSC-345 KV LT BKR 4555 at Goose Creek substation.	December 1, 2023	\$1.8M
18098	Upgrade Grand Tower 138 kV substation	Replace breaker GDTY-138 KV LT 1449 at Grand Tower switchyard.	December 1, 2024	\$0.5M
18102	Upgrade Hines 138 kV substation	Replace breaker HINE-138 KV BKR & LINE 1357 at Hines substation.	December 1, 2023	\$0.5M
18104	Upgrade Latham 345 kV substation	Replace breaker LATM-345 KV POS V19 at Latham substation.	December 1, 2023	\$0.5M
18105	Upgrade Makanda N 138 kV substation	Replace breakers MAKN 138 KV line (1674) and MAKN 138 KV line (1672) at Makanda North substation.	June 1, 2024	\$0.8M
18106	Upgrade Marblehead N 138 kV substation	Replace breakers MRBN-138 KV line 1852 and MRBN-161/138 KV transformer 1 at Marblehead north substation.	June 1, 2024	\$0.8M

18107	Upgrade Tilton Energy Center 138 kV substation	Replace breaker TEC-138 KV tie line BKR (1320) at Tilton Energy Center.	December 1, 2023	\$0.5M
18108	Upgrade Watseka 138 kV substation	Replace breakers WATK-138 KV line (1372) and WATK-138 KV line (1374) at Watseka substation.	December 1, 2024	\$0.8M
18319	Upgrade Frederickstown-Heritage 161 kV line (7011)	Replace poles for OPGW installation. Structure replacements will increase the rating to 120 degrees C.	December 1, 2021	\$7.1M

Projects Driven by Load Growth

The following projects are proposed to replace aging or degraded equipment.

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
9734	New North Utica 138 kV substation	Construct 138 kV breaker station in N. LaSalle-Fox River 138 kV line (1646).	December 1, 2023	\$8.4M
16556	New Bueller 138 kV substation (formerly Grove)	Tap the Macomb West-Niota 138 kV line to install a new ring bus station to supply the PPI Disco Substation.	June 1, 2022	\$6.6M
17224	New Pershall 138/12 kV distribution Substation	Install main-tie-main straight bus substation connecting to Berkeley-Sioux-1 and Berkeley-Sioux-2 lines.	June 1, 2022	TBD
17629	New Ramona 138/12 kV substation	Install a new 138/12 kV substation tapping the Page-Berkeley 1 & 2 lines.	December 1, 2023	TBD
17665	New Barrett Station 138/12 kV Transformer No. 2	Install 3 breakers in a main-tie-main configuration with an open bus tie to facilitate the installation of a 2 nd transformer. Install circuit switchers on the existing and new transformers.	December 31, 2021	TBD
17685	New Creve Coeur 138/12 kV substation	Tap the Mason-Meramec-1 & 2 lines to provide a main-tie-main configuration for new 138/12 kV substation.	December 1, 2022	TBD
17686	New Dougherty Ferry 138/12 kV substation	Tap the Mason-Meramec-1 & 2 lines to provide a main-tie-main configuration for the installation of a new 138/12 kV substation.	December 1, 2023	TBD
17945	New Highway M 138/12 kV substation	Install 2-unit 138/12 kV substation on a site adjacent to the Dardenne Substation.	December 1, 2024	TBD
17947	New Fountain Lakes 138/12 kV transformer No. 2	Add 2nd 138/12 kV unit at Fountain Lakes Substation.	December 1, 2023	TBD
19045	New Rivian Load Interconnection (EPR)	Provide 2 separate 138 kV lines from West Washington St 138 kV BAAH Expansion to serve all load at Diamond Star.	February 1, 2021	\$10.0M

Projects Driven by Other Local Needs

The following projects are proposed due to other local needs.

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
18705	Retire Coffeen 345 kV Switchyard	Retire the Coffeen 345 kV switchyard by re-terminating the lines at the Coffeen North 345 kV substation. This project was submitted by Ameren as EPR.	June 1, 2020	\$0.9M
18706	Retire Duck Creek 345 kV Switchyard	Retire the Duck Creek 345 kV switchyard. This project was submitted by Ameren as EPR.	June 1, 2020	\$0.9M
19045	New Rivian Load Interconnection	Provide 2 separate 138kV lines from West Washington St 138kV BAAH Expansion to serve all load at Diamond Star. This project was submitted by Ameren as EPR.	February 1, 2021	\$10M

Generator Interconnection Projects

According to Attachment FF of the MISO Tariff, "Generator Interconnection Projects are New Transmission Access Projects that are associated with interconnection of new, or increase in generating capacity of existing, generation." The following Generator Interconnection Projects (GIPs) have been evaluated through the Generator Interconnection queue and the associated Generator Interconnection Agreements have been signed.

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
17324	New Mason 138 kV Ring bus (J1055) Glacier Sands interconnection	Interconnection for new 144 MW wind farm at Mason (IL) substation. Tap existing bus at Mason to provide connection.	October 15, 2020	\$10.0M
17907	New Prairie Wolf Solar Farm (J949) interconnection	Add terminal facilities including one 138 kV breaker at Kansas Substation.	October 1, 2021	\$1.2M
18310	New Dressor Plains Solar (J811) interconnection	Add terminal facilities to Ramsey East Substation.	May 28, 2021	\$0.5M
18311	New Maple Flats Solar (J813) interconnection	Re-configure Louisville South 138 kV substation to BAAH bus. Add terminal facilities for J813 connection.	October 29, 2021	\$24.0M
18312	New Blue Bird Solar (J817) interconnection	Add terminal facilities at Warrenton 161 kV substation to interconnect J817,	October 29, 2021	\$3.6M
18314	New McLean County Wind (J826) interconnection	Add breaker position to existing ring bus at Weedman 138 kV Substation for the J826 interconnection.	May 28, 2021	\$1.2M
18315	New Galesburg Wind (J844) interconnection	Add 1 BAAH 138 kV leg to the Sandburg Substation and add 1 breaker position to interconnect J844.	May 28, 2021	\$2.7M
18316	New Big River Solar (J853) interconnection	Add terminal facilities to the Norris City North 138 kV bus to interconnect J853.	May 28, 2021	\$1.2M
18320	New Bennington Wind (J715) interconnection	Install new 138 kV ring bus (Rutland) on the McLean-Oglesby line to make the interconnection for J715.	May 1, 2021	\$6.6M

18322	New City of Roses Wind (J848) interconnection	Add terminal facilities at the Pana 138 kV Substation to make the interconnection for J848.	May 28, 2021	\$0.9M
18323	New Cass County Solar (J859) interconnection	Install a new ring bus (Flanigan) on the Meredosia East-Ipava South 138 kV line to make the interconnection for J859.	May 31, 2022	\$6.8M

4.2.3 Big Rivers Electric Corporation (BREC)

Big Rivers Electric Corporation (BREC) is a member-owned, not-for-profit, generation and transmission cooperative (G&T) with headquarters in Henderson, KY. Big Rivers provides wholesale electric power and services to three distribution cooperative members across twenty-two counties in western Kentucky.

Big Rivers Electric Corporation (BREC) proposed three projects at an estimated cost of \$1.4 million targeting MTEP20 Appendix A. After MISO's independent reliability analysis of the BREC area, all of these projects were recommended by MISO to be included in MTEP20 Appendix A at an estimated cost of \$1.4 million. Of these projects, zero are Baseline Reliability Projects, three are Other Projects, and zero are Generation projects with signed Generator Interconnection Agreements. All the projects' expected in-service dates and estimated costs are provided as of September 15, 2020.

Other Projects

The "Other" projects category do not meet the criteria to be considered as Baseline Reliability Projects (BRP), New Transmission Access Projects, Market Efficiency Projects, or Multi-Value Projects. Other projects may include projects to satisfy Transmission Owner and/or state and local planning criteria other than NERC or regional reliability standards, interconnect new Loads, relocate transmission facilities, address aging transmission infrastructure, replace problematic transmission plant, improve operational performance or address other operational issues, address service reliability issues with end-use consumers, improve aesthetics including but not limited to undergrounding overhead transmission facilities, address localized economic issues, and address other miscellaneous localized needs. The tables of project information are broken down by four general categories of project drivers but note that these four drivers are not defined in the MISO Tariff.

Projects Driven by Local Reliability

The following projects are driven by local planning criteria without Electric Reliability Organization (ERO) standard applicability. These projects could be as a result of operational flexibility purposes.

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
18012	Hardinsburg Upgrade	Add a 161 kV breaker and upgrade protection equipment.	June 1, 2023	\$0.98M
18071	Reid and Reid EHV Capacitor Re-Size	Re-size two existing capacitors by replacing the cans with larger cans.	October 1, 2019	\$0.30M

18765	Relocate Reid EHV 345/161 kV Transformer	Reallocate existing transformers - move one existing Reid EHV 161/345 kV transformer to Otter Creek (steel mill project).	March 1, 2021	\$0.15
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4.2.4 City of Columbia, MO (CWLD)

City of Columbia, MO, (CWLD) is a customer-owned utility company located in Columbia, MO. CWLD provides power to serve its nearly 50,000 residents of the City of Columbia with a peak electric load exceeding 250 MW. Its service area covers the city limits of Columbia. Its transmission system consists of 100 miles of both overhead and underground 161 kV and 69 kV network transmission system. The majority of Columbia's electricity comes from electric producers outside of Columbia.

City of Columbia did not submit any new projects for MTEP20. MISO has not identified any issues in CWLD area.

4.2.5 City of Springfield, IL (CWLP)

City of Springfield, IL, (CWLP) is the municipal electric and water utility for Springfield, IL. CWLP's generation capacity is provided by various fuel mix of generators with a total nameplate capacity of 723 MW. The CWLP electric system's transmission network consists of lines and associated substations operating at voltages of 138 kV and 69 kV. Its 138 kV portion of the transmission network currently includes approximately 63 circuit miles of overhead lines forming a complete loop around the system's service area. The 138 kV transmission lines presently serve nine of the system's substations or switching stations, plus the village of Chatham, IL.

CWLP proposed six projects at an estimated cost of \$1.0 million targeting MTEP20 Appendix A. After MISO's independent reliability analysis of the CWLP area, these projects were recommended by MISO to be included in MTEP20 Appendix A at an estimated cost of \$1.0 million. Of these projects, four are Baseline Reliability Projects, two are Other Projects, and none are Generation projects. All the projects' expected in-service dates and estimated costs are provided as of July 30, 2020.

Baseline Reliability Projects

According to Attachment FF of the MISO Tariff, "Baseline Reliability Projects are Network Upgrades identified in the base case as required to ensure that the Transmission System is in compliance with applicable national Electric Reliability Organization (ERO) reliability standards and reliability standards adopted by Regional Reliability Organizations and applicable within the Transmission Provider Region."

Project 18134 - New [CWLP] Palomino 69 kV capacitor banks

Project Description

The project will add a new [CWLP] Palomino 69 kV capacitor device. This upgrade will mitigate low voltages on the local 138 kV lines for both P3-1 and P6-1-1 contingency events. The total estimated cost of this project is approximately two-hundred fifty (\$250k) thousand dollars. The expected in-service date for this project is March 1, 2021.

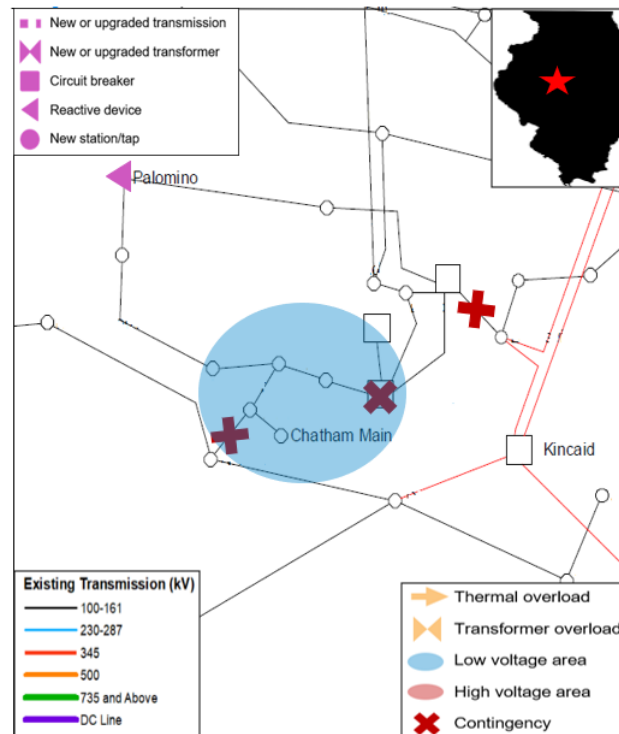


Figure 4.2.5-1: P18134 Geographic transmission map of project area

Project Need

There are a number of 138 kV buses with voltages that go below their low voltage limit of 0.95 p.u. in year 2022 summer peak conditions for two (2) defined NERC contingency events; category P3-1 and category P6-1-1. Adding 40 Mvars of new capacitors at the [CWLP] Palomino 69 kV substation will help prop the low voltages back up during the summer peak season (Table 4.2.5-1).

Cont. Type	Limiting Element	Voltage Limit (pu)	Pre-Project Voltage (pu)	Post-Project Voltage (pu)
n-0	[CWLP] Local 138 kV Buses	0.95	0.94	0.97
P6-1-1	[CWLP] Local 138 kV Buses	0.95	0.94	0.96

Table 4.2.5-1: P18134 Voltage performance drivers

Alternatives Considered

Due to unit retirements, low system voltages across the entire CWLP system were shown in planning studies under certain contingencies. CWLP initially discussed the possibility of adding additional generation or tie line connections to support the decrease in net generation. However, these were not studied in detailed planning models as the costs and complexities were far too great compared to the simple, reliable solution of additional Mvar support from capacitor banks.

Project 18140 - Reconductor [CWLP] Chatham Main—[CWLP] Chatham Ind. 138 kV line

Project Description

The project will reconductor the [CWLP] Chatham Main—[CWLP] Chatham Independence 138 kV line. This reconductor will mitigate an overload of the same line for a P3-1 contingency event during high import conditions. The total estimated cost of this project is approximately three-hundred fifty (\$350k) thousand dollars. The expected in-service date for this project is March 1, 2021.

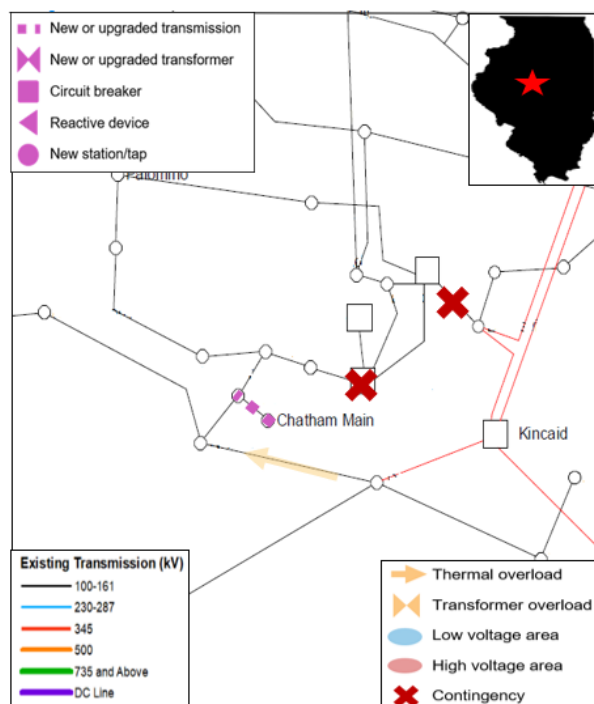


Figure 4.2.5-2: P18140 Geographic transmission map of project area

Project Need

During a high import scenario transferring 500 MW into CWLP across the [CWLP] Chatham Main—[CWLP] Chatham Ind. 138 kV tie line, this tie line will exceed its summer emergency rating of 313 MVA in year 2022 summer peak conditions for a NERC defined category P3-1 contingency event. Reconductoring this line will increase the summer emergency rating to 365 MVA during the summer peak season (Table 4.2.5-2).

Cont. Type	Limiting Element	Rating (MVA)	Pre-Project Loading (%)	Post-Project Loading (%)
Base case	[CWLP] Chatham Main—[AMIL] Chatham Ind. 138 kV line	313	66	101
P3-1	[CWLP] Chatham Main—[AMIL] Chatham Ind. 138 kV line	365	101	86

Table 4.2.5-2: P18140 Thermal loading drivers

Alternatives Considered

Due to unit retirements, higher import levels were shown to overload this tie line in planning studies under certain contingencies. CWLP initially discussed the possibility of adding additional generation or tie line connections to support the decrease in net generation. However, these were not studied in detailed planning models as the costs and complexities were far too great compared

to the cost effective and expedient solution of reconductoring the tie line to cover the additional import capacity.

Project 18141 - Upgrade Interstate 138 kV substation

Project Description

The project will make upgrades at the [CWLP] Interstate 138 kV substation thus increasing the ratings on the [CWLP] Interstate—[AMIL] E. Lanesville 138 kV line. This upgrade will mitigate an overload on the [CWLP] Interstate—[AMIL] E. Lanesville 138 kV tie line for a P3-1 contingency event during high import conditions. The total estimated cost of this project is approximately twenty (\$20k) thousand dollars. The expected in-service date for this project is March 1, 2021.

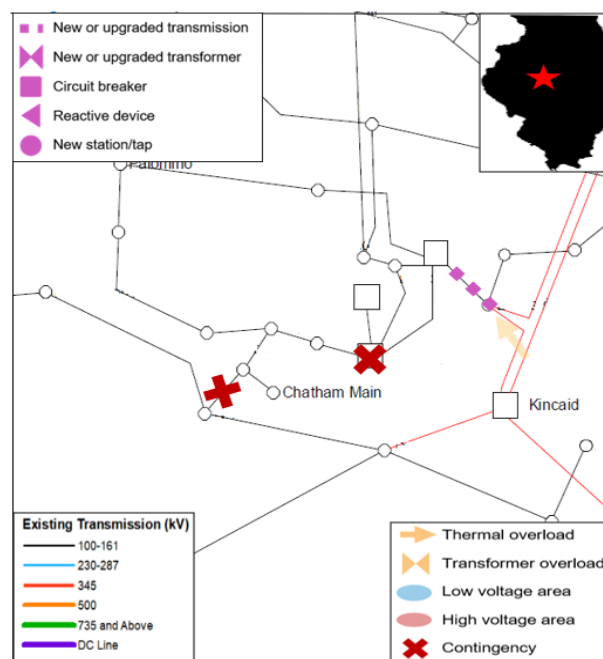


Figure 4.2.5-3: P18141 Geographic transmission map of project area

Project Need

During a high import scenario transferring 500 MW into CWLP across the [CWLP] Interstate—[AMIL] E. Lanesville 138 kV tie line, this tie line will exceed its summer emergency rating of 287 MVA in year 2022 summer peak conditions for a NERC defined category P3-1 contingency event. Replacing the wave trap on the [CWLP] Interstate—[AMIL] E. Lanesville 138 kV tie line with a 2000A wave trap will resolve the overload issue. This will increase the summer emergency rating to 366 MVA during the summer peak season (Table 4.2.5-3).

Cont. Type	Limiting Element	Rating (MVA)	Pre-Project Loading (%)	Post-Project Loading (%)
Base case	[CWLP] Interstate—[AMIL] E. Lanesville 138 kV tie line	313	66	101
P3-1	[CWLP] Interstate—[AMIL] E. Lanesville 138 kV tie line	365	101	86

Table 4.2.5-3: -P18141 Thermal loading drivers

Alternatives Considered

Due to unit retirements, higher import levels were shown to overload this tie line in planning studies under certain contingencies. CWLP initially discussed the possibility of adding additional generation or tie line connections to support the decrease in net generation. However, these were not studied in detailed planning models as the costs and complexities were far too great compared to the cost effective and expedient solution of reconductoring the tie line to cover the additional import capacity.

Project 18171 - Reconductor [CWLP] Chatham Independence—[CWLP] Spaulding 138 kV lines

Project Description

The project will reconductor the [CWLP] Chatham Independence—[CWLP] Spaulding 138 kV line. This upgrade will mitigate an overload on the same line for a P3-1 contingency event. The total estimated cost of this project is approximately one-hundred fifty (\$150k) thousand dollars. The expected in-service date for this project is March 1, 2021.

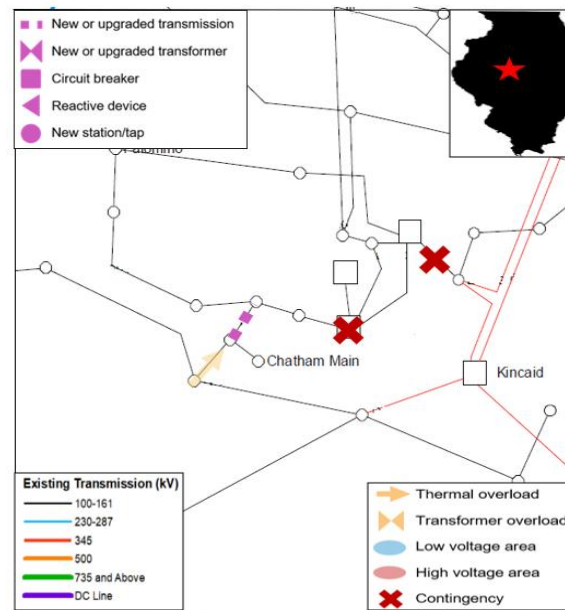


Figure 4.2.5-4: P18171 Geographic transmission map of project area

Project Need

There is an overload of the [CWLP] Chatham Ind.—[CWLP] Spaulding 138 kV line with its summer emergency rating of 313 MVA being exceeded in year 2022 summer peak conditions for a NERC defined category P3-1 contingency event during high transfer scenarios. Reconductoring this line will increase the summer emergency rating to 365 MVA during the summer peak season (Table 4.2.5-4).

Cont. Type	Limiting Element	Rating (MVA)	Pre-Project Loading (%)	Post-Project Loading (%)
Base case	[CWLP] Chatham Ind.—[CWLP] Spaulding 138 kV line	313	61	104
P3-1	[CWLP] Chatham Ind.—[CWLP] Spaulding 138 kV line	365	104	92

Table 4.2.5-4: P18171 Thermal loading drivers

Alternatives Considered

Due to unit retirements, higher import levels were shown to overload this tie line in planning studies under certain contingencies. CWLP initially discussed the possibility of adding additional generation or tie line connections to support the decrease in net generation. However, these were not studied in detailed planning models as the costs and complexities were far too great compared to the cost effective and expedient solution of reconductoring the tie line to cover the additional import capacity.

Other Projects

The “Other” projects category are projects that do not meet the criteria to be considered as Baseline Reliability Projects (BRP), New Transmission Access Projects, Market Efficiency Projects, or Multi-Value Projects. Other projects may include projects to satisfy Transmission Owner and/or state and local planning criteria other than NERC or regional reliability standards, interconnect new Loads, relocate transmission facilities, address aging transmission infrastructure, replace problematic transmission plant, improve operational performance or address other operational issues, address service reliability issues with end-use consumers, improve aesthetics including but not limited to undergrounding overhead transmission facilities, address localized economic issues, and address other miscellaneous localized needs. The tables of project information are broken down by four general categories of project drivers but note that these four drivers are not defined in the MISO Tariff.

Projects Driven by Local Reliability

The following projects are driven by local planning criteria without Electric Reliability Organization (ERO) standard applicability. These projects could be as a result of operational flexibility purposes.

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
18143	Reconductor Amos-Amos PPI 69 kV line	Reconductor the Amos - Amos PPI 69 kV line with 397 ACSR	March 1, 2021	\$0.01M
18172	New Taintor 69 kV capacitor banks (x2 20 Mvar)	Add two (2) 20 Mvar capacitor banks at Taintor Substation	March 1, 2021	\$0.25M

4.2.6 Duke Energy Corporation (DEI)

Duke Energy Corporation (DEI) is a regulated electric and gas delivery company with operations based in Cincinnati, OH. DEI provides power to serve 840,000 electric customers in Indiana across its 23,000 square miles service area. The DEI generation comes from its own coal and gas generation units totaling more than 6,600 MW. DEI also owns a good percentage of renewables and hydroelectric generation plants across Indiana.

Duke Energy Corporation (DEI) proposed 27 projects at an estimated cost of \$183 million targeting MTEP20 Appendix A. After MISO’s independent reliability analysis of the DEI area, all 27 of these projects were recommended by MISO to be included in MTEP20 Appendix A at an estimated cost of \$183 million. Of these projects, one project is a Baseline Reliability Project, 25 are Other Projects, and one project is a Generator Interconnection project with a signed Generator Interconnection Agreement. Most of the projects’ expected in-service dates and estimated costs are provided as of July 30, 2020.

Baseline Reliability Projects

According to Attachment FF of the MISO Tariff, “Baseline Reliability Projects are Network Upgrades identified in the base case as required to ensure that the Transmission System is in compliance with applicable national Electric Reliability Organization (ERO) reliability standards and reliability standards adopted by Regional Reliability Organizations and applicable within the Transmission Provider Region.”

Project 17484 – Rebuild [DEI] Wabash River—[DEI] Water St. 138 kV line (13845)

Project Description

The project will rebuild the [DEI] Wabash River Station—[DEI] Water St. 138 kV line. This line rebuild is due to a thermal overload caused by a P2-4 contingency event. The total estimated cost of this project is \$16,300,000. The expected in-service date for this project is 6/1/2024. The project is shown in Figure 4.2.6-1.

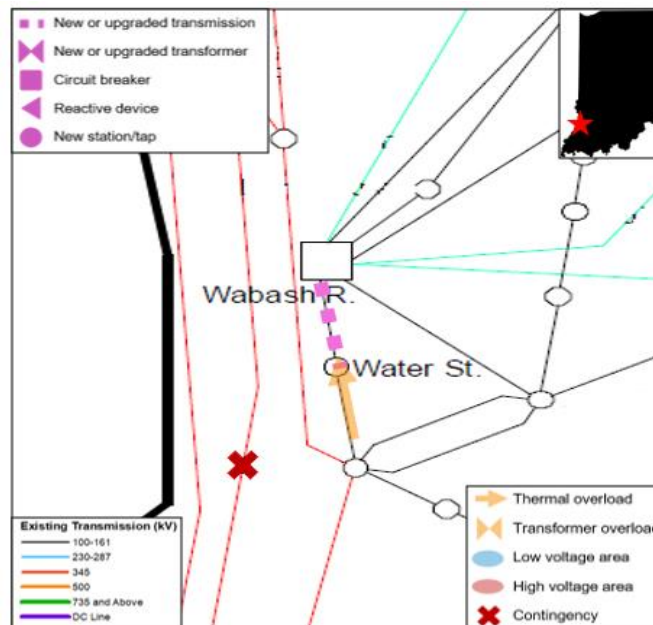


Figure 4.2.6-1: P17484 Geographic transmission map of project area

Project Need

The [DEI] Wabash River Station—[DEI] Water St. 138 kV line becomes overloaded (120%) in year 2024 for a NERC-defined category P2-4 contingency event. The rebuild of the [DEI] Wabash River Station—[DEI] Water St. 138 kV line will increase the summer emergency ratings from 198 MVA to 301 MVA.

Cont. Type	Limiting Element	Rating (MVA)	Pre-Project Loading (%)	Post-Project Loading (%)
Base case	[DEI] Wabash River Station—[DEI] Water St. 138 kV line	198	43	43
P2-4	[DEI] Wabash River Station—[DEI] Water St. 138 kV line	301	127	83

Table 4.2.6-1: P17484 Thermal loading drivers

Alternatives Considered

This line rebuild project is the best and cheapest option to address this reliability issue.

Other Projects

The “Other” projects category are projects that do not meet the criteria to be considered as Baseline Reliability Projects (BRP), New Transmission Access Projects, Market Efficiency Projects, or Multi-Value Projects. Other projects may include projects to satisfy Transmission Owner and/or state and local planning criteria other than NERC or regional reliability standards, interconnect new Loads, relocate transmission facilities, address aging transmission infrastructure, replace problematic transmission plant, improve operational performance or address other operational issues, address service reliability issues with end-use consumers, improve aesthetics including but not limited to undergrounding overhead transmission facilities, address localized economic issues, and address other miscellaneous localized needs. The tables of project information are broken down by four general categories of project drivers but note that these four drivers are not defined in the MISO Tariff.

Projects Driven by Local Reliability

The following projects are driven by local planning criteria without Electric Reliability Organization (ERO) standard applicability. These projects could be as a result of operational flexibility purposes.

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
2864	Reconductor Centerton to Martinsville 69 kV - 69163 ckt.	Reconductor the 69 kV-69163 ckt - 3.47 miles of 366ACSR 80C and 0.23 miles of 477ACSR 80C to 954ACSR 100C.	June 1, 2023	\$7.40M
18285	WVPA McGrawsville to Bunker Hill 69 kV alternate source line	WVPA to build McGrawsville to Bunker Hill 69 kV alternate source line - 7.1 miles to be operated N.O.	June 1, 2021	\$5.51M
18286	Rebuild Brownstown Sw Sta to Pumpkin Center 69 kV	6952 Rbld Pt 2 Brownstown Sw Sta to Pumpkin Center; 10.7 mi. – 477 acsr; Replace the 6952-69123 tie switch on pole #813-1535-02 located just outside Pumpkin Center substation.	December 31, 2021	\$8.36M
18287	Terre Haute S. Vigo to HE Middletown Jct 69 kV Rebuild	Rbld 6901 circuit from TH Haute S. Vigo sub, pole 802-1158, to HE Middletown Jct, pole 802-1315, using light-duty steel poles, 477ACSR conductor & OPGW static. Rpl switch #64LB with two 2000A switches.	June 1, 2022	\$8.77M
18288	Trafalgar Swi Sta to HE Samaria Jct 69 kV Rebuild	Rbld transmission line 6949 from Duke Trafalgar Swi Sta & HE Trafalgar sub to HE Samaria Jct using light duty steel poles, 477 ACSR and OPGW. Approximately 4.9 miles.	December 31, 2021	\$5.28M
18289	HE Needmore Jct to HE Needmore 69 kV Rebuild	6949 Part 2B - Rebuild transmission line using light duty steel poles, 477 ACSR and OPGW from “HE Needmore Jct” (pole #825-3379) to the HE Needmore substation (pole #825-3418); approximately 2 miles.	December 31, 2021	\$3.88M
18290	Purdue University CHP Generation	Purdue University CHP Generation - 15.5MW gas turbines with HRSG unit - non-MISO queue; non-BES.	July 16, 2021	\$13.31M
18304	Batesville 345 kV and 138 kV breaker replacements	Batesville 345: Replace (1) 345 kV circuit switcher with 345 kV SF6 gas circuit breaker; replace 345 kV WT; Replace (6)	June 1, 2022	\$7.33M

		138 kV oil circuit breakers with SF6 gas circuit breakers - 13831, 13833, 13839, 13864, 13874, 138345-3; replace 13833 and 13874 WTs.		
18305	Westwood 345 kV / 138 kV relays and breakers replacement	Westwood misc. station equipment replacement including upgrade relays on 13806, 13808, 138345-1, Bk2, 138 bus 1 and 2, 138 bus tie; Replace OCB 13820, 13883 and relays.	September 2, 2021	\$4.90M
18324	WVPA Delphi South-6988 Tap Line	WVPA to build new 69 kV, 6.3 mile, radial, N.O., alternate source to Delphi South (IPC, Indiana Packers Corp.) from the existing 6988 line tapped between Flora and Rockfield.	June 1, 2023	\$5.51M
18326	Poseyville to HE Winfield 69 kV Rebuild	69139 Rebuild Part 2: Poseyville to HE Winfield TDSIC; 4.4 miles - N.O. tie line	December 31, 2021	\$6.82M
18327	Rebuild Dresser to S. Vigo 69 kV	6901 rebuild from Dresser to S. Vigo with 477ACSR and light-duty steel poles.	July 16, 2021	\$7.16M
18328	Rebuild Greencastle N. to Morton Jct. 69 kV	6933 rebuild from Greencastle N. to Morton Jct. with 477ACSR and light-duty steel poles.	July 22, 2021	\$8.08M
18329	Dover to Thorntown 69 kV Rebuild	69134 rebuild from Dover to Thorntown with 477ACSR and light-duty steel poles.	December 31, 2021	\$5.78M
18330	Westland Jct. to Westland 69 kV Rebuild	69166 rebuild from Westland Jct. to Westland with 477ACSR and light-duty steel poles.	December 31, 2021	\$5.77M
18331	Whitesville to Advance 69 kV Rebuild	69154 rebuild from Whitesville to Advance with 477ACSR and light-duty steel poles.	December 31, 2021	\$14.41M
18332	Knightstown to Carthage 69 kV Rebuild	69140 rebuild from Knightstown to Carthage with 477ACSR and light-duty steel poles.	December 31, 2021	\$7.20M
18333	Greencastle to Greencastle N. 69 kV Rebuild	6933 rebuild from Greencastle to Greencastle N. with 477ACSR and light-duty steel poles.	December 31, 2021	\$5.23M
18334	Rebuild Connersville to Glenwood 69 kV	6920 rebuild from Connersville to Glenwood with 477ACSR and light-duty steel poles.	December 31, 2021	\$9.67M
18335	New Tate and Lyle Generation	Tate and Lyle (old Staley) Generation - existing sub on the 13808 ckt; 2 units - 32.8MW each - non-MISO queue.	April 1, 2021	\$0.25M
18845	Greenwood Clark Twnshp 230 kV Ring bus	Greenwood Clark Twnshp: add (2) CBs to complete a 3-position 230 kV Ring bus; replace misc. 230 kV and 69 kV equipment in-kind.	December 31, 2021	\$8.05M

Projects Driven by Other Local Needs

The following projects are proposed due to other local needs.

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
18306	New H.E. Batesville 138/12 kV new substation	H.E. Batesville 138/12 kV substation: new sub to be located in the 13833 ckt between Batesville 138 kV and Laurel 138 kV.	December 31, 2021	\$0.50M

18307	New Westfield East 69/12 kV substation	Westfield East: new sub - 69/12 kV - 22.4 MVA xfmr bank, single phase regulators SCADA/IVVC, 2 breakers, and space for second bank. Normal feed from 6917 ckt, alternate from 69142.	June 1, 2022	\$3.77M
18308	New Hunts Creek 138/12 kV substation	Hunts Creek 138/12 kV substation: new sub to be located in the 13855 ckt between LGE Ghent 138 kV and DEI Fairview 138 kV.	June 1, 2021	\$3.81M
18325	New NSA Crane ULTF 69/12 kV substation	New sub - "Crane Northwest" to feed NSA Crane ULTF - 17 MW new load to be fed from 69193 ckt - located North of existing NSA owned East and West Substations.	June 1, 2021	\$2.80M

Generator Interconnection Projects

According to Attachment FF of the MISO Tariff, "Generator Interconnection Projects are New Transmission Access Projects that are associated with interconnection of new, or increase in generating capacity of existing, generation." The following Generator Interconnection Projects (GIPs) have been evaluated through the Generator Interconnection queue and the associated Generator Interconnection Agreements have been signed.

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
18284	J754 Gen. Interconnect	J754 Wind Farm - Gen. Interconnect: 34501 ckt. - 300MW Wind farm; 3-breaker ring bus switching station	October 31, 2021	\$19.66M

4.2.7 GridLiance Holdco, LP (EEI)

GridLiance Holdco, LP did not submit any new projects for MTEP20. MISO has not identified any issues in EEI area.

4.2.8 Henderson Municipal Power & Light (HMPL)

Henderson Municipal Power & Light did not submit any new projects for MTEP20. MISO has not identified any issues in HMPL area.

4.2.9 Hoosier Energy REC, Inc. (HE)

Hoosier Energy REC, Inc. did not submit any new projects for MTEP20. MISO has not identified any issues in HE area.

4.2.10 Indianapolis Power & Light Company (IPL)

Indianapolis Power & Light Company (IPL) is an investor-owned utility. IPL provides retail electric service to approximately 500,000 customers in Indianapolis, IN, and the surrounding communities. The IPL transmission system consists of approximately 458 circuit miles of lines at 345 kV, 408 circuit miles of 138 kV transmission and associated substations. IPL owns and operates four generating stations with a total generating capacity of 3,555 MW.

Indianapolis Power & Light Company (IPL) proposed ten projects at an estimated cost of \$22.2 million targeting MTEP20 Appendix A. After MISO's independent reliability analysis of the IPL area, these projects were recommended by MISO to be included in MTEP20 Appendix A at an estimated cost of \$22.2 million. Of these projects, five are Baseline Reliability Projects, five are Other Projects, and there were no Generation projects with signed Generator Interconnection Agreements. All the projects' expected in-service dates and estimated costs are provided as of July 30, 2020.

Baseline Reliability Projects

According to Attachment FF of the MISO Tariff, "Baseline Reliability Projects are Network Upgrades identified in the base case as required to ensure that the Transmission System is in compliance with applicable national Electric Reliability Organization (ERO) reliability standards and reliability standards adopted by Regional Reliability Organizations and applicable within the Transmission Provider Region."

Project 17884 – Replace [IPL] Southeast 138 kV breaker No. 4

Project Description

Replace the existing [IPL] Southeast 138 kV Breaker No. 4. This is due to a short circuit capability issue. The total estimated cost of this project is \$900,000. The expected in-service date is 12/31/2025. Refer to Figure 4.2.10-1 for the geographic location of this project.

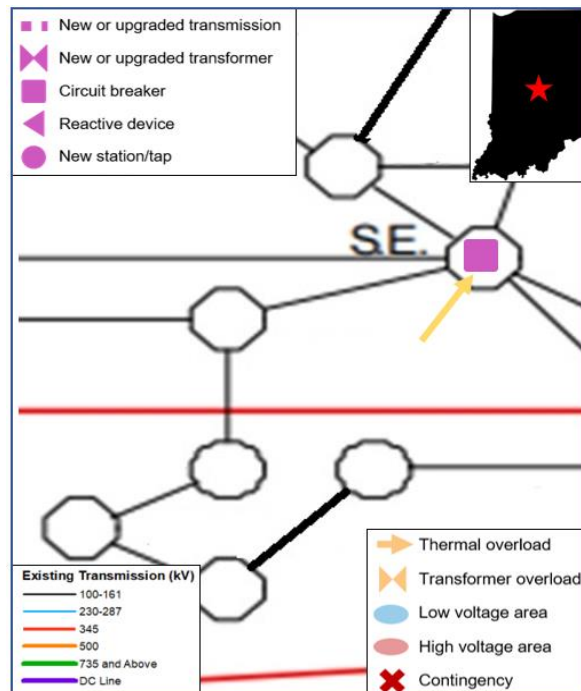


Figure 4.2.10-1: P17884 Geographic transmission map of project area

Project Need

IPL's 2019 short circuit analysis demonstrates that the [IPL] Southeast 138 kV Breaker No. 4 short circuit current interrupting duty exceeds the equipment rating. MISO does not evaluate the need for short circuit issues, therefore, there are no study results to show for this project.

Alternatives Considered

No other alternatives were considered. This terminal upgrade is the standard solution and least cost option to address this reliability issue.

Project 17885 – Replace [IPL] Southeast 138 kV breaker No. 6

Project Description

Replace the existing [IPL] Southeast 138 kV Breaker No. 6. This is due to a short circuit capability issue. The total estimated cost of this project is \$900,000. The expected in-service date is 12/31/2025. Refer to Figure 4.2.10-2 for the geographic location of this project.

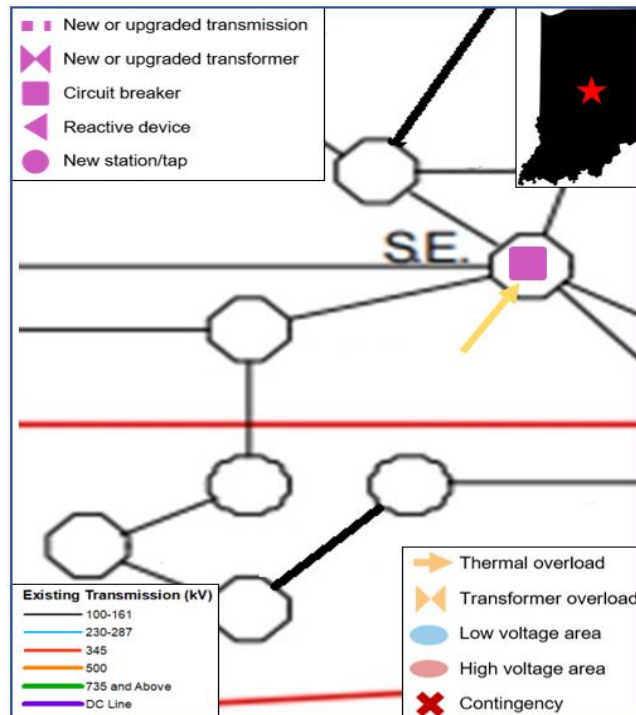


Figure 4.2.10-2: P17885 Geographic transmission map of project area

Project Need

IPL's 2019 short circuit analysis demonstrates that the [IPL] Southeast 138 kV Breaker No. 6 short circuit current interrupting duty exceeds the equipment rating. MISO does not evaluate the need for short circuit issues, therefore, there are no study results to show for this project.

Alternatives Considered

No other alternatives were considered. This terminal upgrade is the standard solution and least cost option to address this reliability issue.

Project 17886 – Replace [IPL] Southeast 138 kV breaker No. 8

Project Description

Replace the existing [IPL] Southeast 138 kV Breaker No. 8. This is due to a short circuit capability issue. The total estimated cost of this project is \$900,000. The expected in-service date is 12/31/2021. Refer to Figure 4.2-13 for the geographic location of this project.

Project Need

IPL's 2019 short circuit analysis demonstrates that the [IPL] Southeast 138 kV Breaker No. 8 short circuit current interrupting duty exceeds the equipment rating. MISO does not evaluate the need for short circuit issues, therefore, there are no study results to show for this project.

Alternatives Considered

No other alternatives were considered. This terminal upgrade is the standard solution and least cost option to address this reliability issue.

Project 17887 – Replace [IPL] Southeast 138 kV breaker No. 1A

Project Description

Replace the existing [IPL] Southeast 138 kV Breaker No. 1A. This is due to a short circuit capability issue. The total estimated cost of this project is \$900,000. The expected in-service date is 12/31/2023. Refer to Figure 4.2-13 for the geographic location of this project.

Project Need

IPL's 2019 short circuit analysis demonstrates that the [IPL] Southeast 138 kV Breaker No. 1A short circuit current interrupting duty exceeds the equipment rating. MISO does not evaluate the need for short circuit issues, therefore, there are no study results to show for this project.

Alternatives Considered

No other alternatives were considered. This terminal upgrade is the standard solution and least cost option to address this reliability issue.

Project 17888 – Replace [IPL] Southeast 138 kV breaker No. 2

Project Description

Replace the existing [IPL] Southeast 138 kV Breaker No. 4. This is due to a short circuit capability issue. The total estimated cost of this project is \$900,000. The expected in-service date is 12/31/2023. Refer to Figure 4.2-13 for the geographic location of this project.

Project Need

IPL's 2019 short circuit analysis demonstrates that the [IPL] Southeast 138 kV Breaker No. 2 short circuit current interrupting duty exceeds the equipment rating. MISO does not evaluate the need for short circuit issues, therefore, there are no study results to show for this project.

Alternatives Considered

No other alternatives were considered; this terminal upgrade is the standard solution and least cost options to address this reliability issue.

Other Projects

The "Other" projects category do not meet the criteria to be considered as Baseline Reliability Projects (BRP), New Transmission Access Projects, Market Efficiency Projects, or Multi-Value Projects. Other projects may include projects to satisfy Transmission Owner and/or state and local planning criteria other than NERC or regional reliability standards, interconnect new Loads, relocate transmission facilities, address aging transmission infrastructure, replace problematic transmission plant, improve operational performance or address other operational issues, address service reliability issues with end-use consumers, improve aesthetics including but not limited to undergrounding overhead transmission facilities, address localized economic issues, and address other miscellaneous localized needs. The tables of project information are broken down by four general categories of project drivers but note that these four drivers are not defined in the MISO Tariff.

Projects Driven by Local Reliability

The following projects are driven by local planning criteria without Electric Reliability Organization (ERO) standard applicability. These projects could be as a result of operational flexibility purposes.

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
17564	Rockville 345 kV Ring Bus	Add one 345 kV breaker to form a ring bus at the Rockville substation.	December 31, 2020	\$3.60M

Projects Driven by Age and Condition

The following projects are proposed to replace aging or degraded equipment.

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
18135	Rockville Auto Replace	Replace 345-138 auto with like-for-like replacement	December 31, 2020	\$3.70M
18138	Center Substation	Add a total of three new 138 kV breakers and replace three existing 138 kV breakers.	December 31, 2021	\$6.00M
18142	Mooresville Substation	Add a total of two new 138 kV breakers and convert the substation to a ring bus.	December 31, 2021	\$2.00M
18145	Hanna Substation	Replace three 345 kV breakers.	December 31, 2021	\$2.40M

4.2.11 Pioneer Transmission, LLC

Pioneer Transmission, LLC did not submit any new projects for MTEP20. MISO has not identified any issues in this area.

4.2.12 Prairie Power, Inc. (PPI)

Prairie Power, Inc. (PPI) is a member-owned, not-for-profit electric generation and transmission cooperative. PPI produces and supplies wholesale electricity to ten electric distribution cooperatives in central Illinois. PPI's distribution cooperatives provide retail electric service to approximately 78,000 members within its local service territories. PPI owns and operates approximately 590 miles of transmission lines at 138 kV, 69 kV and 34.5 kV voltage levels. PPI generates 141 MW of oil and gas-fired peaking units and 79 distribution and transmission substations to serve its members.

PPI proposed 19 projects at an estimated cost of \$92 million targeting MTEP20 Appendix A. After MISO's independent reliability analysis of the PPI area, these 19 projects were recommended by MISO to be included in MTEP20 Appendix A at an estimated cost of \$92 million. Of these projects, one is Baseline Reliability Project, 18 are Other Projects, and there are no Generation projects with signed a Generator Interconnection Agreement. All the projects' expected in-service dates and estimated costs are provided as of July 30, 2020.

Baseline Reliability Projects

According to Attachment FF of the MISO Tariff, "Baseline Reliability Projects are Network Upgrades identified in the base case as required to ensure that the Transmission System is in compliance with

applicable national Electric Reliability Organization (ERO) reliability standards and reliability standards adopted by Regional Reliability Organizations and applicable within the Transmission Provider Region.”

Project 18154 - New [PPI] Alsey 138 kV shunt reactor device

Project Description

The project will add a new [PPI] Alsey 138 kV shunt reactor device. This upgrade will mitigate high voltages on the local 138 kV lines for a variety of P6-1-1 contingency event combinations. The total estimated cost of this project is approximately three (\$3M) million dollars. The expected in-service date for this project is December 31, 2022.

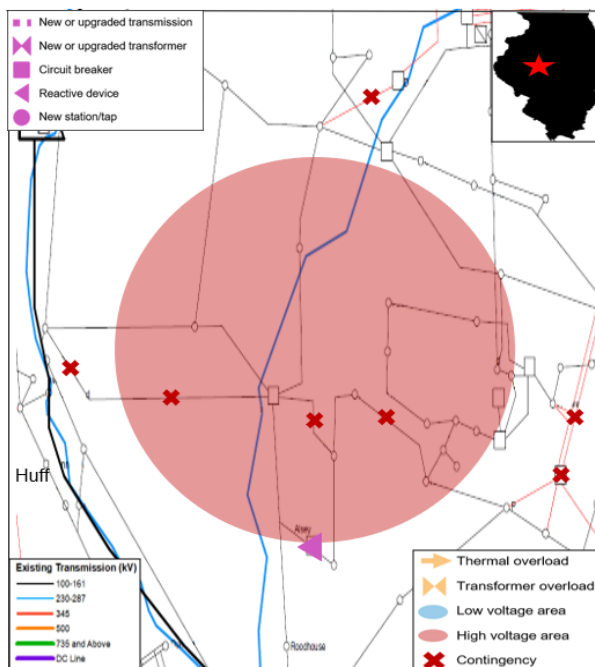


Figure 4.2.12-1: P18154 Geographic transmission map of project area

Project Need

There are a number of 138 kV buses that exceed their 1.05 p.u. high voltage tolerable limit in year 2025 during light load conditions for a NERC variety of defined category P6-1-1 contingency events. Adding a new reactor at Alsey 138 kV substation will help suppress the high voltages during the light load season (Table 4.2.12-1).

Cont. Type	Limiting Element	Voltage Limit (pu)	Pre-Project Voltage (pu)	Post-Project Voltage (pu)
n-0	[PPI] Local 138 kV Buses	0.95	1.01	1.02
P6-1-1	[PPI] Local 138 kV Buses	1.05	> 1.05	0.99

Table 4.2.12-1: P18154-1 Voltage performance drivers

Alternatives Considered

PPI considered proposing a synchronous condenser as an alternative to the Alsey 40 Mvar reactor; however, doing so would have been more expensive, would require more maintenance, and provide less Mvar (27) than the Alsey reactor. Therefore, the synchronous condenser alternative was rejected in favor of installing the Alsey 40 Mvar reactor.

Other Projects

The “Other” projects category do not meet the criteria to be considered as Baseline Reliability Projects (BRP), New Transmission Access Projects, Market Efficiency Projects, or Multi-Value Projects. Other projects may include projects to satisfy Transmission Owner and/or state and local planning criteria other than NERC or regional reliability standards, interconnect new Loads, relocate transmission facilities, address aging transmission infrastructure, replace problematic transmission plant, improve operational performance or address other operational issues, address service reliability issues with end-use consumers, improve aesthetics including but not limited to undergrounding overhead transmission facilities, address localized economic issues, and address other miscellaneous localized needs. The tables of project information are broken down by four general categories of project drivers but note that these four drivers are not defined in the MISO Tariff.

Projects Driven by Local Reliability

The following projects are driven by local planning criteria without Electric Reliability Organization (ERO) standard applicability. These projects could be as a result of operational flexibility purposes.

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
18115	Relocate Kampsville Capacitor Bank to Pearl	Move 2.4 Mvar capacitor bank from Kampsville to Pearl Substation	December 31, 2020	\$0.40M
18116	New Kilbourne-Saidora 69 kV Line	New 9.7-mile 69 kV line in Menard Electric Cooperative	December 31, 2023	\$6.20M
18123	New Payson 69 kV Breaker Station	Establish 69 kV breaker station in Adams Electric Cooperative	December 31, 2022	\$2.94M
18152	Reconductor Andrew-Amos 69 kV Line	Reconductor 8.9-mile 69 kV line from 336.4 MCM ACSR to 795 MCM ACSR	December 31, 2022	\$5.24M
18155	New Burton-Ursa 69 kV Line	New 16.3-mile 69 kV line in Adams Electric Cooperative	June 30, 2021	\$8.95M
18157	New Kuhlman 138/69 kV Substation	1) Establish 138 kV substation connecting to the Ameren Meredosia to Frederick 138 kV line 2) New PPI Kuhlman 138/69 kV substation	December 31, 2022	\$7.43M
18159	Upgrade East Lanesville 138/69 kV transformer	Upgrade existing 37.33 MVA transformer with 84 MVA unit. Existing transformer will be moved to Kuhlman	December 31, 2022	\$1.20M
18160	New Arenzville-Bluff Springs 69 kV Line	New 11-mile 69 kV line tying Illinois Electric Cooperative to Menard Electric Cooperative	December 31, 2023	\$7.46M
18161	New Bluff Springs-Virginia 69 kV Line	New 13-mile 69 kV line in Menard Electric Cooperative	December 31, 2023	\$8.00M
18162	New Kington-Ellisville 69 kV Line	New 18-mile 69 kV line to Spoon River Electric Cooperative	December 31, 2023	\$11.93M
18163	New Ursa-Big Neck 69 kV Line	New 12-mile 69 kV line in Adams Electric Cooperative	December 31, 2023	\$9.02M
18165	New Virginia 69 kV Breaker Station	New 69 kV breaker station in Menard Electric Cooperative	December 31, 2023	\$2.93M
18166	New Virginia-Saidora 69 kV Line	New 14-mile 69 kV line in Menard Electric Cooperative	December 31, 2023	\$8.64M
18167	New Smithfield 69 kV capacitor bank (4.8 Mvar)	Addition of 4.8 Mvar capacitor bank in Spoon River Electric Cooperative	December 31, 2024	\$0.40M
18169	New Macomb 69 kV capacitor bank (4.8 Mvar)	Addition of 4.8 Mvar capacitor bank in McDonough Power Cooperative	December 31, 2026	\$0.40M

Projects Driven by Age and Condition

The following projects are proposed to replace aging or degraded equipment.

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
18120	Upgrade Berlin-Amos 69 kV Line	Upgrade 3-mile portion of Berlin to Amos 69 kV line from 4/0 ACSR to 336 ACSR associated with fiber and distribution installation	December 31, 2020	\$1.20M
18122	Upgrade St. David 69 kV Substation	Replace St. David OCB with SF6 breaker and new relays	December 31, 2020	\$0.28M

Projects Driven by Load Growth

The following projects are proposed to replace aging or degraded equipment.

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
18148	New Riggston 138 kV substation	Establish 138 kV ring bus to serve proposed Data Center in Illinois Electric Cooperative new load between Jacksonville Industrial Park and Winchester.	December 31, 2021	\$6.40M

4.2.13 Southern Indiana Gas & Electric Company (SIGE)

Southern Indiana Gas & Electric Company (SIGE) proposed 12 projects at an estimated cost of \$14.7 million targeting MTEP20 Appendix A. After MISO's independent reliability analysis of the SIGE area, 12 of these projects were recommended by MISO to be included in MTEP20 Appendix A at an estimated cost of \$14.7 million. Of these projects, zero are Baseline Reliability Projects, 11 projects are Other Projects, and one project is a Generation project with a signed Generator Interconnection Agreement. All the projects' expected in-service dates and estimated costs are provided as of July 30, 2020.

Other Projects

The "Other" projects category do not meet the criteria to be considered as Baseline Reliability Projects (BRP), New Transmission Access Projects, Market Efficiency Projects, or Multi-Value Projects. Other projects may include projects to satisfy Transmission Owner and/or state and local planning criteria other than NERC or regional reliability standards, interconnect new Loads, relocate transmission facilities, address aging transmission infrastructure, replace problematic transmission plant, improve operational performance or address other operational issues, address service reliability issues with end-use consumers, improve aesthetics including but not limited to undergrounding overhead transmission facilities, address localized economic issues, and address other miscellaneous localized needs. The tables of project information are broken down by four general categories of project drivers but note that these four drivers are not defined in the MISO Tariff.

Projects Driven by Age and Condition

The following projects are proposed to replace aging or degraded equipment.

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
15860	Broadview 69/12 kV Dist. T2 Repl	Replace 69/12 kV distribution transformer T2 at the Broadview substation.	December 31, 2022	\$1.20M
15940	East Industrial Trans. Repl.	Replace the 69/12 kV distribution transformer at the East Industrial substation.	December 31, 2022	\$1.20M
15948	Fairlawn 69/12 kV Dist. T1 Repl.	Replace the 69/12 kV distribution transformer T1 at the Fairlawn substation.	December 31, 2022	\$1.20M
15949	Fuquay 69/12 kV Dist. T1 Repl.	Replace 69/12 kV distribution transformer T1 at the Fuquay substation.	December 31, 2022	\$1.20M
15950	Hemenway 69/12 kV Dist. T1 Repl.	Replace 69/12 kV distribution transformer T1 at the Hemenway substation.	December 31, 2022	\$1.70M
15952	Culley 138 kV OCB 2200 2277 Repl.	Replace two 138 kV oil circuit breakers (OCBs) with gas circuit breakers (GCBs). Breaker numbers 2200 and 2277.	December 31, 2022	\$0.62M
15953	Replace Heidelberg OCB - 2177 2277	Replace two 138 kV oil circuit breakers (OCBs) with gas circuit breakers (GCBs) at the Heidelberg station - breaker numbers 2177 & 2277.	December 31, 2022	\$0.62M
15958	Rockport 69/12 kV T2 Repl.	Replace 69/12 kV distribution transformer T2 at the Rockport substation.	December 31, 2022	\$1.70M
15960	St. Phillips 69/12 kV T1 Repl.	Replace 69/12 kV distribution transformer T1 at the St. Phillips substation.	December 31, 2022	\$1.20M
15961	Rebuild Y31-1 NW - ORS 69 kV	Rebuild ~2.1 miles of 69 kV overhead transmission between the Northwest and Ohio River Station substations (line Y31-1).	December 31, 2022	\$1.93M
15987	Berry Plastics Dist. XFMR Repl.	Replace one of the distribution transformers serving Berry Plastics. (PSP project)	December 31, 2024	\$1.10M

Generator Interconnection Projects

According to Attachment FF of the MISO Tariff, "Generator Interconnection Projects are New Transmission Access Projects that are associated with interconnection of new, or increase in generating capacity of existing, generation." The following Generator Interconnection Projects (GIPs) have been evaluated through the Generator Interconnection queue and the associated Generator Interconnection Agreements have been signed.

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
18268	Grandview Generation Interconnect	Installation of Interconnection Facilities and Network Upgrades for Generator Interconnection project J783 at the Grandview 69 kV substation.	April 1, 2021	\$1.02M

4.2.14 Southern Illinois Power Cooperative (SIPC)

Southern Illinois Power Cooperative did not submit any new projects for MTEP20. MISO has not identified any issues in the SIPC area.

4.2.15 Wabash Valley Power Association, Inc. (WVPA)

WVPA proposed two projects at an estimated cost of \$5 million targeting MTEP20 Appendix A. After MISO's independent reliability analysis of the WVPA area, these two projects were recommended by MISO to be included in MTEP20 Appendix A at an estimated cost of \$5 million. Of these projects, zero are Baseline Reliability Projects, two are Other Projects, and zero are Generation projects with signed Generator Interconnection Agreements. All the projects' expected in-service dates and estimated costs are provided as of July 30, 2020.

Other Projects

The "Other" projects category do not meet the criteria to be considered as Baseline Reliability Projects (BRP), New Transmission Access Projects, Market Efficiency Projects, or Multi-Value Projects. Other projects may include projects to satisfy Transmission Owner and/or state and local planning criteria other than NERC or regional reliability standards, interconnect new Loads, relocate transmission facilities, address aging transmission infrastructure, replace problematic transmission plant, improve operational performance or address other operational issues, address service reliability issues with end-use consumers, improve aesthetics including but not limited to undergrounding overhead transmission facilities, address localized economic issues, and address other miscellaneous localized needs. The tables of project information are broken down by four general categories of project drivers but note that these four drivers are not defined in the MISO Tariff.

Projects Driven by Local Reliability

The following projects are driven by local planning criteria without Electric Reliability Organization (ERO) standard applicability. These projects could be as a result of operational flexibility.

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
18625	New MJM-Nutwood 69 kV line	New 69 kV loop feed to the MJM-Nutwood 69/12kV substation.	December 31, 2022	\$2.0M

Projects Driven by Load Growth

The following projects are proposed to replace aging or degraded equipment.

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
18626	New Delhi 69 kV substation	69 kV line to new MJM-Delhi substation with 69/12 kV transformer tapping 69kV line near Jerseyville MJM sub.	June 1, 2022	\$3.0M

4.3 Project Justifications – East Region

East Region Overview

The MISO East Planning Region consists of seven Transmission-Ownning members within Michigan and Indiana. These Transmission Owners are ITC Transmission (ITCT); Michigan Electric Transmission Co. (METC); Consumers Energy Transmission Owner (CE); Wolverine Power Supply Cooperative Inc. (WPSC); Michigan Public Power Agency (MPPA); Michigan South Central Power Agency (MSCPA) and Northern Indiana Public Service Co. (NIPSCO).

The region contains 10,770 circuit miles of transmission lines ranging from 120 kV to 345 kV. It also contains 1,524 circuit miles of 69 kV sub-transmission system. The MISO East Region is interconnected with the following non-MISO systems: Hydro One Networks Inc. and American Electric Power to the east and ComEd to the west. The region is also interconnected with the following MISO systems: Duke Energy Indiana in the south and the American Transmission Co. (ATC) to the north. The interconnection with ATC's system is through a High Voltage Direct Current (HVDC) connection.

The 2022 Summer Peak planning model indicates the region contains more than 38.7GW of generation. Installed generation capacity in the region consists mostly of coal, gas, nuclear, and wind. Figure 4.3-1 shows the major load centers and generation pockets within the East Region. The load centers are typically found around larger cities in the region, i.e., Detroit, Lansing, Grand Rapids, and Michigan City. According to the 2022 Summer Peak planning model, the region's load exceeds 25.3 GW.



Figure 4.3-1: Generation and load centers in the East planning region

For MTEP20, MISO Transmission Planning is recommending 85 projects from the East region for inclusion in Appendix A at an estimated cost of \$511 million. Of these, 34 are Baseline Reliability Projects and 15 are Generator Interconnection Projects. The remaining 36 projects are classified as Other Projects as they do not meet the criteria to be considered as Baseline Reliability Projects, New Transmission Access Projects, Market Efficiency Projects, or Multi-Value Projects.

Of the 85 projects, that are being recommended to be included in MTEP20, 27 have an estimated cost of less than \$1 million, 28 have an estimated cost between \$1 million and \$5 million, and the remaining 30 projects are estimated to cost greater than \$5 million (shown in Figure 4.3-2).

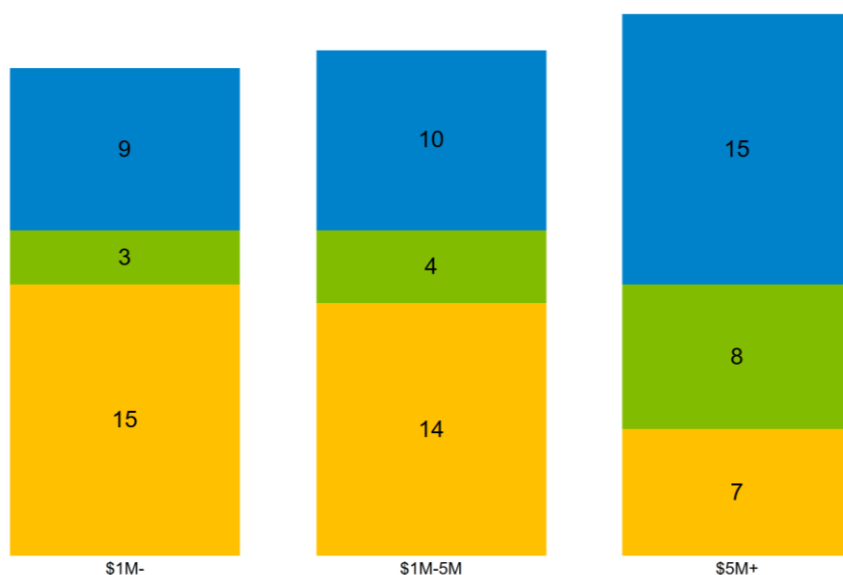


Figure 4.3-2: Project cost category by project type for MTEP20 East projects. (Project data as of September 15, 2020)

A majority of the projects in the MISO East planning region are expected to go into service in the next five years as shown in Figure 4.3-3. There are four GIP projects that are being approved in MTEP20 with in-service dates in 2019 and 2020.

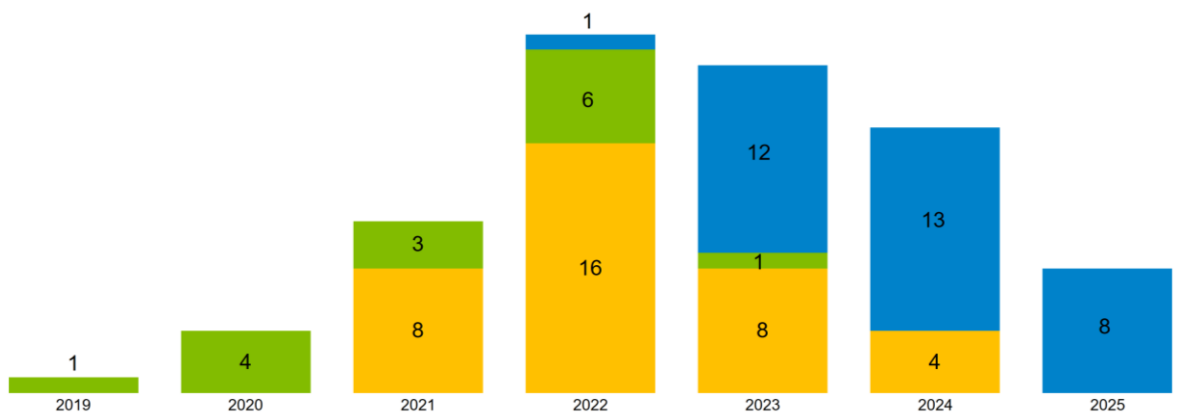


Figure 4.3-3: Project in-service date by project type for MTEP20 East projects. (Project data as of September 15, 2020)

In accordance with Attachment FF of the MISO tariff, if a Transmission Owner determines system conditions warrant the urgent development of system enhancements, MISO will perform an expedited review of the impacts of the project. MISO shall use a streamlined approval process for reviewing and approving such projects proposed by the Transmission Owner(s) so that decisions will be provided to the Transmission Owner within 30 Days of the project's submittal to MISO, unless a longer review period is mutually agreed upon. During the MTEP20 cycle, MISO received the following projects to be expedited through the Expedited Project Review (EPR) process:

1. Project ID 18055, METC Riverview Load Interconnection
2. Project ID 18986, Rebuild NIPSCo BES for Reinforcements

The ten highest cost MISO East projects represent \$260 million of the \$511 million total recommended projects for the region in MTEP20. The locations of these projects are shown in Figure 4.3-4 with the investment spread across the East Planning region. Projects that are blanket expenditures (such as relays, physical security, etc.) are excluded from this list.

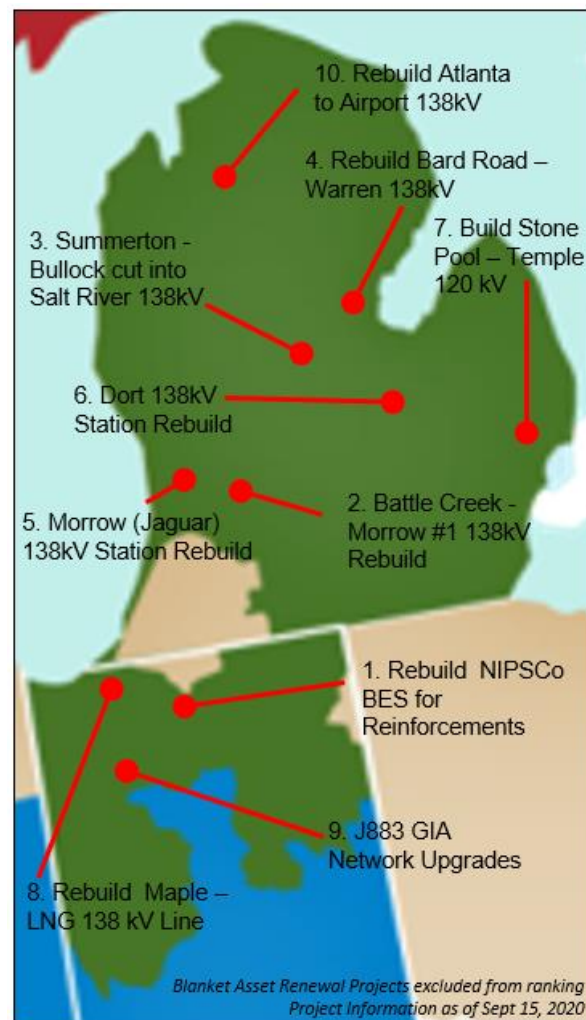


Figure 4.3-4: The Top 10 projects in terms of cost in the East planning region for MTEP20, excluding blanket asset replacement projects. (Project data as of September 15, 2020)

4.3.1 ITC Transmission (ITCT)

ITCT proposed 42 projects at an estimated cost of \$211 million targeting MTEP20 Appendix A. After MISO's independent reliability analysis of ITCT area, 31 of these projects are being recommended by MISO to be included in MTEP20 Appendix A at an estimated cost of \$129 million. Of these projects, 20 are Baseline Reliability Projects, eight are Other Projects, and three are Generation projects with signed Generator Interconnection Agreements. All the projects' expected in-service dates and estimated costs are provided as of September 15, 2020.

Baseline Reliability Projects

According to Attachment FF of the MISO Tariff, "Baseline Reliability Projects are Network Upgrades identified in the base case as required to ensure that the Transmission System is in compliance with applicable national Electric Reliability Organization (ERO) reliability standards and reliability standards adopted by Regional Reliability Organizations and applicable within the Transmission Provider Region."

Project 15890 - Pontiac-Walton 120 kV Rebuild

Project Description

This project will rebuild the Pontiac-Walton 120 kV line. This line rebuild is due to a thermal overload caused by P6 contingency events. The project's estimated cost is \$11.03 million and expected in-service date is December 31, 2025.

Note that Project 15890 was originally proposed in MTEP19.

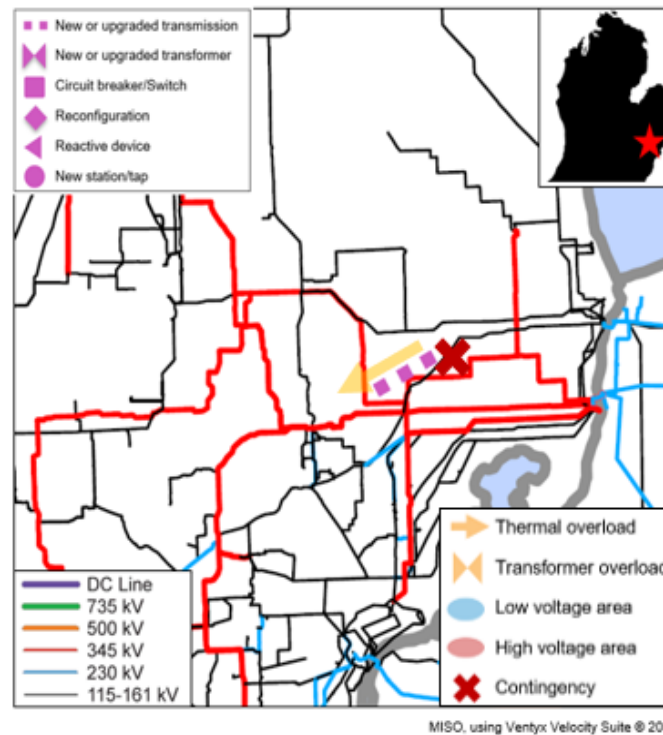


Figure 4.3.1-1: P15890 Geographic transmission map of project area

Project Need

The Pontiac – Giddings 2 – Walton 120 kV circuit is projected to overload for P6 contingencies. The line rebuild will reduce the loading as shown in Table 4.3.1-1.

Cont. Type	Limiting Element	Rating (MVA)	Pre-Project Loading (%)	Post-Project Loading (%)
P6	Giddings - Walton 120 kV - Ckt 1	375	112	67
P6	Pontiac - Giddings 120 kV - Ckt 1	375	115	67

Table 4.3.1-1: P15890 Thermal loading drivers

Alternatives Considered

None.

Project 15880 - Navarre-River Rouge 120 kV Rebuild

Project Description

This project will rebuild the Navarre-River Rouge 120 kV line. This line rebuild is due to a thermal overload caused by P6 contingency events. The project's estimated cost is \$9.73 million and expected in-service date is December 31, 2025.

Note that Project 15880 was originally proposed in MTEP19.

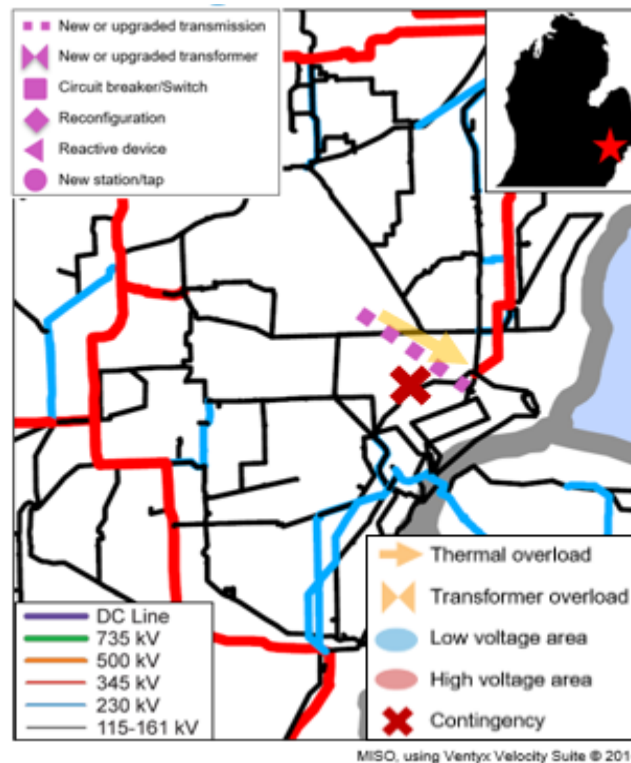


Figure 4.3.1-2: P15880 Geographic transmission map of project area

Project Need

The Navarre – River Rouge, Maron – Deacon Tap and Navarre – Deacon Tap 120 kV lines are projected to overload for P6 contingencies. Rebuilding these lines will reduce the lines loadings as shown in Table 4.3.1-2.

Cont. Type	Limiting Element	Rating (MVA)	Pre-Project Loading (%)	Post-Project Loading (%)
P6	Maron - Deacon Tap 120 kV - Ckt 1	289	99	49
P6	Navarre - Deacon Tap 120 kV - Ckt 1	266	109	68
P6	River Rouge - Maron 120 kV - Ckt 1	266	108	67

Table 4.3.1-2: P15880 Thermal loading drivers

Alternatives Considered

DIG Alternative Project: A reconfiguration of DIG 230 kV facilities, that adds a new 230 kV substation at DIG and a line from DIG to River Rouge 230 kV, was considered as a blanket alternative for projects in MTEP20 and previous cycle's projects. Following reliability and cost evaluation, the DIG Alternative Project was eliminated as a cost-effective solution.

Project 15959 - Rebuild Hager-Sunset 120 kV Rebuild

Project Description

This project will rebuild the Hager-Sunset 120 kV line. This line rebuild is due to a thermal overload caused by P6 contingency events. The project's estimated cost is \$9.66 million and expected in-service date is December 31, 2025.

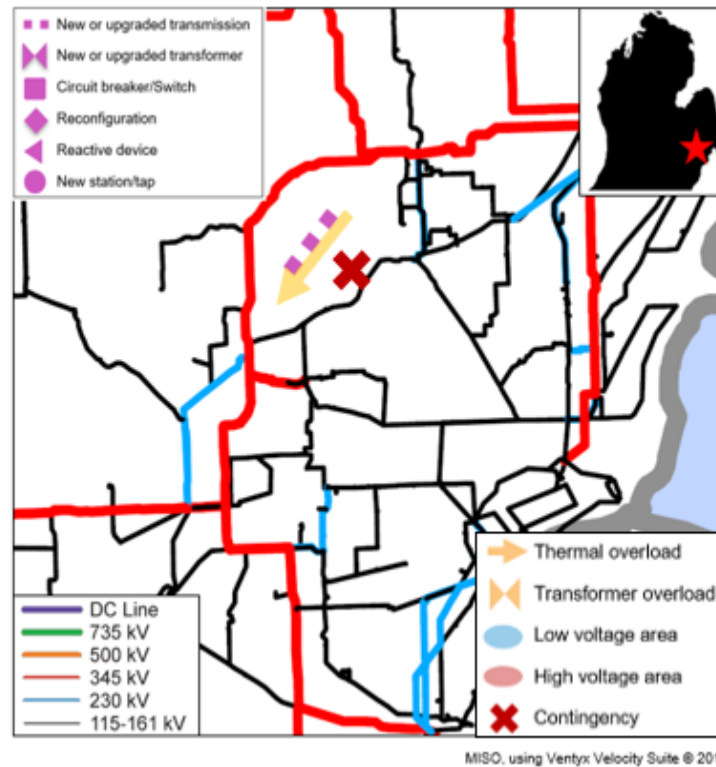


Figure 4.3.1-3: P15959 Geographic transmission map of project area

Project Need

The Sunset – Hager 120 kV line is projected to overload for P6 contingencies. The line rebuild will reduce the loading as shown in Table 4.3.1-3.

Cont. Type	Limiting Element	Rating (MVA)	Pre-Project Loading (%)	Post-Project Loading (%)
P6	Sunset - Hager 120 kV - Ckt 1	291	107	64

Table 4.3.1-3: P15959 Thermal loading drivers

Alternatives Considered

None.

Project 15883 - Bismarck-Stephens #2 120 kV Rebuild

Project Description

This project will rebuild the Bismarck-Stephens #2 120 kV line. This line rebuild is due to a thermal overload caused by a P6 contingency event. The project's estimated cost is \$9.1 million and expected in-service date is December 31, 2024.

Note that Project 15883 was originally proposed in MTEP19.

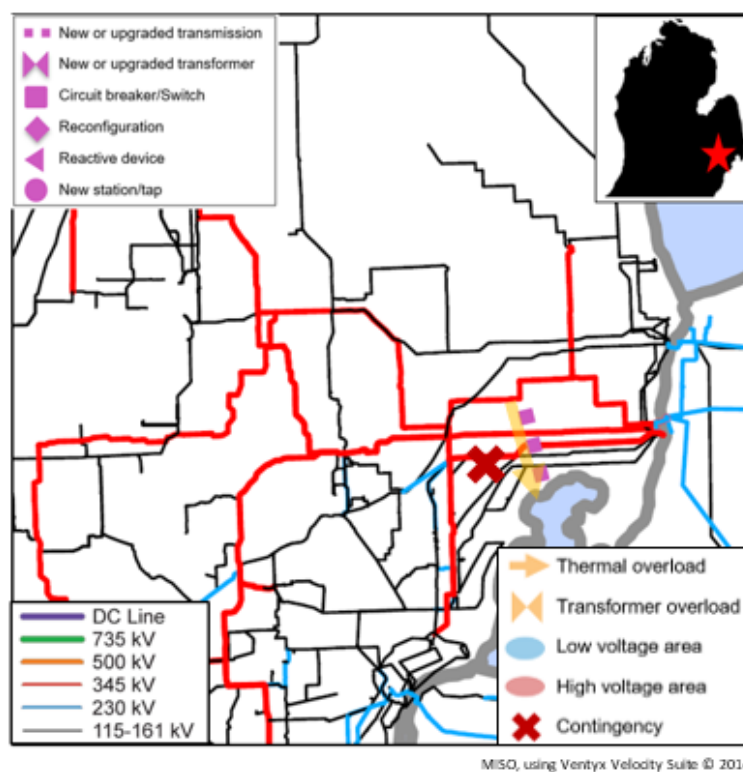


Figure 4.3.1-4: P15883 Geographic transmission map of project area

Project Need

The Bismarck – Stephens #2 120 kV circuit is projected to overload for a P6 contingency. Rebuilding this line will reduce the loading as shown in Table 4.3.1-4.

Cont. Type	Limiting Element	Rating (MVA)	Pre-Project Loading (%)	Post-Project Loading (%)
P6	Stephens - Bismarck 120 kV - Ckt 2	289	119	93

Table 4.3.1-4: P15883 Thermal loading drivers

Alternatives Considered

None.

Project 15900 - Rebuild Wayne-Newburgh #1 120 kV Rebuild

Project Description

This project will rebuild the Wayne-Newburgh #1 120 kV line. This line rebuild is due to a thermal overload caused by P6/P7 contingency events. The project's estimated cost is \$7.72 million and expected in-service date is December 31, 2025.

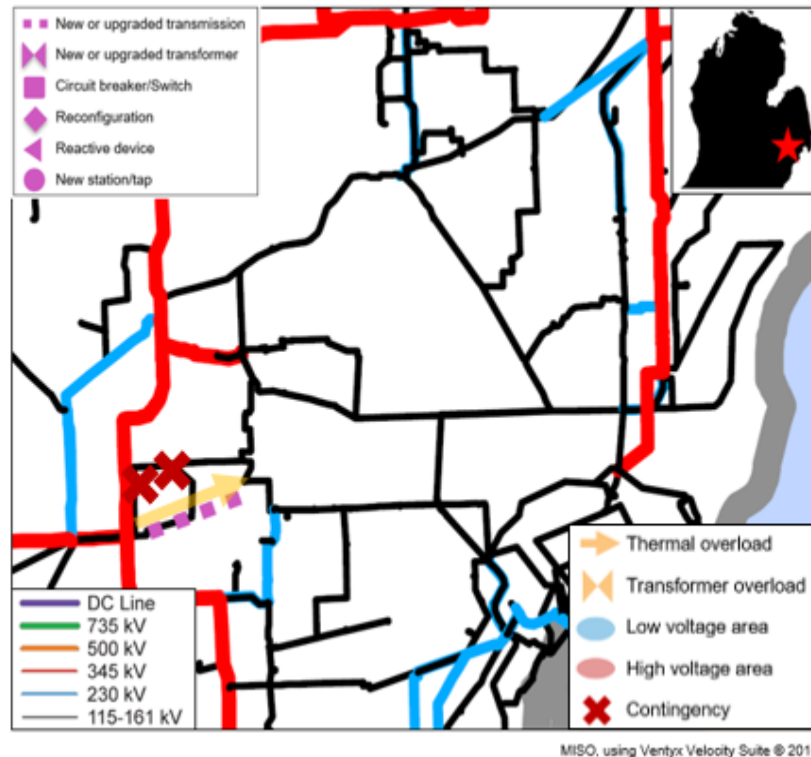


Figure 4.3.1-5: P15900 Geographic transmission map of project area

Project Need

The Wayne - Newburgh #1 120 kV circuit is projected to overload for P6/P7 contingencies. The line rebuild will reduce the loading on the line as shown in Table 4.3.1-5.

Cont. Type	Limiting Element	Rating (MVA)	Pre-Project Loading (%)	Post-Project Loading (%)
P6	Newburgh - Wayne 120 kV - Ckt 1	361	117	85
P7	Newburgh - Wayne 120 kV - Ckt 1	361	102	77

Table 4.3.1-5: P15900 Thermal loading drivers

Alternatives Considered

None.

Project 18184 - Rebuild Red Run-Logan 2-Sloan 120 kV

Project Description

This project will rebuild the Red Run – Logan – Sloan 120 kV line. This line rebuild is due to a thermal overload caused by P6/P7 contingency events. The project's estimated cost is \$7.1 million and expected in-service date is June 1, 2024.

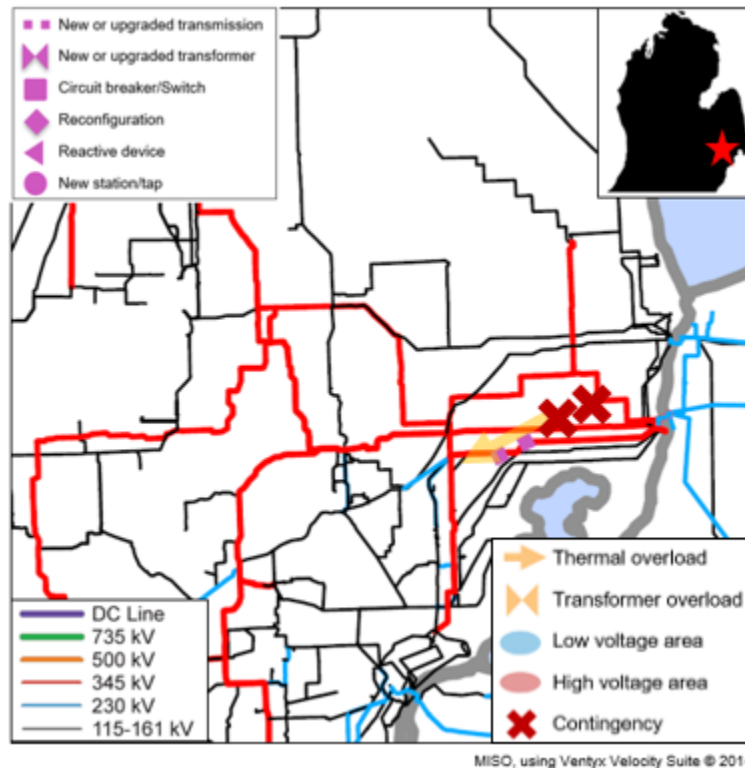


Figure 4.3.1-6: P18184 Geographic transmission map of project area

Project Need

The Red Run – Logan – Sloan 120 kV circuit is projected to become marginally overloaded for various P6/P7 outages. The rebuild will reduce loading on the line sections as shown in Table 4.3.1-6.

Cont. Type	Limiting Element	Rating (MVA)	Pre-Project Loading (%)	Post-Project Loading (%)
P6	Logan - Red Run 120 kV - Ckt 1	266	100	71
P7	Logan - Red Run 120 kV - Ckt 1	266	100	71
P6	Logan – Sloan 120 kV - Ckt 1	266	96	69
P7	Logan – Sloan 120 kV - Ckt 1	266	96	69

Table 4.3.1-6: P18184 Thermal loading drivers

Alternatives Considered

None.

Project 18179 - Rebuild Hood-Lily 120 kV

Project Description

This project will rebuild the Hood – Lily 120 kV line. This line rebuild is due to a thermal overload caused by a P7 contingency event. The project’s estimated cost is \$5.78 million and expected in-service date is June 1, 2024.

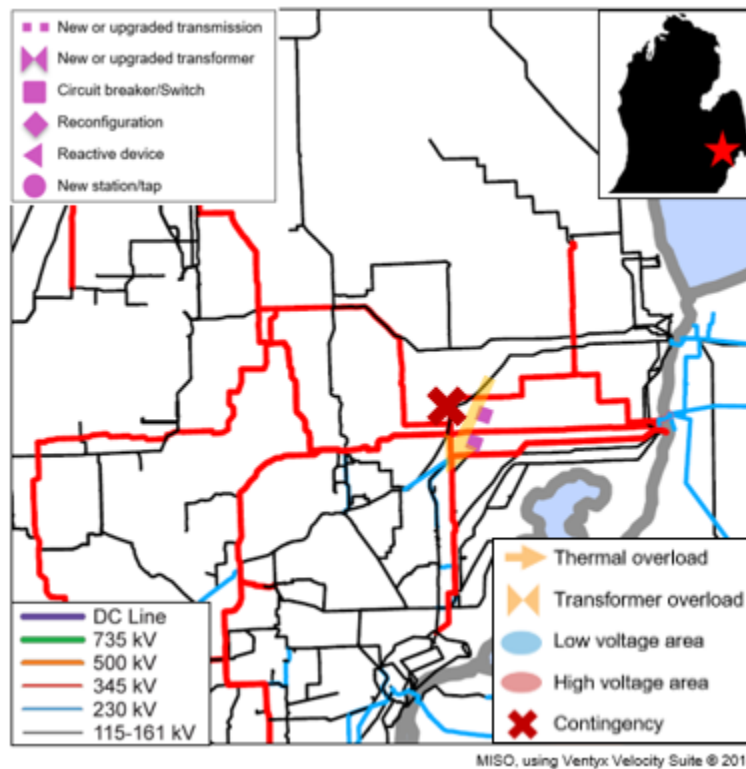


Figure 4.3.1-7: P18197 Geographic transmission map of project area

Project Need

The Hood – Lily 120 kV circuit is projected to become overloaded for a P7 contingency event. The line rebuild will reduce the loading as shown in Table 4.3.1-7.

Cont. Type	Limiting Element	Rating (MVA)	Pre-Project Loading (%)	Post-Project Loading (%)
P7	Lily - Hood 120 kV - Ckt 1	228	102	76

Table 4.3.1-7: P18179 Thermal loading drivers

Alternatives Considered

None.

Project 18178 - Rebuild Hancock-Lily 120 kV

Project Description

This project will rebuild the Hancock – Lily 120 kV line. This line rebuild is due to a thermal overload caused by P6/P7 contingency events. The project’s estimated cost is \$4.68 million and expected in-service date is June 1, 2024.

Project Need

The Hancock – Lily 120 kV circuit is projected to become overloaded for P6/P7 contingency events. The line rebuild will reduce the loading as shown in Table 4.3.1-8.

Cont. Type	Limiting Element	Rating (MVA)	Pre-Project Loading (%)	Post-Project Loading (%)
P6	Hancock - Lily 120 kV - Ckt 1	189	110	62
P7	Hancock - Lily 120 kV - Ckt 1	189	106	60

Table 4.3.1-8: P18178 Thermal loading drivers

Alternatives Considered

None.

Project 18240 - Install Capacitor at Boyne 120 kV

Project Description

This project will install two capacitors at Boyne 120 kV station, 18 Mvar and 33 Mvar. The project's estimated cost is \$2.43 million and expected in-service date is December 31, 2023.

Project Need

There are voltage issues at the Golf 120 kV, Houston 120 kV, and Macomb 120 kV buses driven by P6 contingencies. The capacitor installation at Boyne 120 kV along with the existing distribution capacitors at Boyne and Macomb, will resolve the voltage violations as shown in Table 4.3.1-9.

Cont. Type	Limiting Element	Voltage Limit (pu)	Pre-Project Voltage (pu)	Post-Project Voltage (pu)
P3, P6	Boyne 120 kV Bus 101	>0.92 & <1.07	0.84	0.90
P3, P6	Boyne 120 kV Bus 102	>0.92 & <1.07	0.84	0.90
P6	Golf 120 kV Bus 101	>0.92 & <1.07	0.84	0.90
P6	Golf 120 kV Bus 102	>0.92 & <1.07	0.84	0.90
P6	Golf 120 kV Bus 113	>0.92 & <1.07	0.84	0.90
P6	Houston 120 kV Bus 1	>0.92 & <1.07	0.89	0.95
P6	Houston 120 kV Bus 2	>0.92 & <1.07	0.89	0.95
P6	Houston Tap 1 120 kV	>0.92 & <1.07	0.89	0.95
P6	Houston Tap 2 120 kV	>0.92 & <1.07	0.89	0.95
P6	Macomb 120 kV Bus 101	>0.92 & <1.07	0.84	0.89*
P6	Macomb 120 kV Bus 103	>0.92 & <1.07	0.84	0.89*
P6	Macomb 120 kV Bus 105	>0.92 & <1.07	0.84	0.89*

*Boyne and Macomb distribution capacitors are offline in the shoulder cases. Voltages at Macomb buses ~1.00 pu with the Boyne and Macomb distribution capacitors online.

Table 4.3.1-9: P18240 Voltage performance drivers

Alternatives Considered

None.

Project 18175 - Rebuild Van Dyke 2-Sloan 120 kV

Project Description

This project will rebuild the Van Dyke 2 – Sloan 120 kV line. This line rebuild is due to a thermal overload caused by P6 contingency events. The project's estimated cost is \$2.35 million and expected in-service date is June 1, 2025.

Project Need

The Van Dyke 2 – Sloan 120 kV circuit is projected to become overloaded for various P6 contingencies. The line rebuild will reduce the loading as shown in Table 4.3.1-10.

Cont. Type	Limiting Element	Rating (MVA)	Pre-Project Loading (%)	Post-Project Loading (%)
P6	Vandyke - Sloan 120 kV - Ckt 1	266	97	71

Table 4.3.1-10: P18175 Thermal loading drivers

Alternatives Considered

None.

Project 18168 - Reconductor of Bismarck-Red Run 230 kV

Project Description

This project will reconductor the Bismarck – Red Run 230 kV line. This line reconductor is due to a thermal overload caused by P6 contingency events. The project's estimated cost is \$2.34 million and expected in-service date is June 1, 2024.

Project Need

The Bismarck – Red Run 230 kV circuit is projected to overload for P6 contingency events. The line reconductoring will reduce the loading as shown in Table 4.3.1-11.

Cont. Type	Limiting Element	Rating (MVA)	Pre-Project Loading (%)	Post-Project Loading (%)
P6	Bismarck - Red Run 230 kV - Ckt 1	551	101	78

Table 4.3.1-11: P18168 Thermal loading drivers

Alternatives Considered

None.

Project 16015 - Rebuild Sunset-Quaker 120 kV

Project Description

This project will rebuild the Sunset – Quaker 120 kV line. This line rebuild is due to a thermal overload caused by P6 contingency events. The project's estimated cost is \$2.26 million and expected in-service date is December 31, 2025.

Note that Project 16015 was originally proposed in MTEP19.

Project Need

The Sunset – Quaker 120 kV line is projected to overload for P6 contingencies. The line rebuild will reduce the loading as shown in Table 4.3.1-12.

Cont. Type	Limiting Element	Rating (MVA)	Pre-Project Loading (%)	Post-Project Loading (%)
P6	Quaker - Sunset 120 kV - Ckt 1	375	115	79

Table 4.3.1-12: P16015 Thermal loading drivers

Alternatives Considered

None.

Project 18205 - Upgrade Red Run 120 kV Station Equipment

Project Description

This project will upgrade station equipment at Red Run 120 kV. The equipment upgrade at Red Run 120 kV is due to a thermal overload caused by P6 contingency events. The project's estimated cost is \$1.10 million and expected in-service date is December 31, 2023.

Project Need

The 120 kV bus-tie at Red Run is projected to become overloaded for various P6 contingency events. The equipment upgrade at this station will relieve the loading as shown in Table 4.3.1-13.

Cont. Type	Limiting Element	Rating (MVA)	Pre-Project Loading (%)	Post-Project Loading (%)
P6	Red Run 2 - Red Run 3 120 kV - Ckt Z1	490	100	79

Table 4.3.1-13: P18205 Thermal loading drivers

Alternatives Considered

None.

Project 16024 - River Rouge 120 kV Capacitor

Project Description

This project will install three 18 Mvar capacitors at the River Rouge 120 kV station. The project's estimated cost is \$1.03 million and expected in-service date is December 31, 2023.

Note that this project was originally proposed in MTEP19.

Project Need

There are numerous voltage issues in the region near River Rouge, Riverview, and Trenton Channel 120 kV stations, driven by P6 contingencies. The capacitor install will resolve the voltage violations as shown in Table 4.3.1-14.

Note the post-project voltage levels are due to the combined effect of capacitor banks installed at River Rouge, Trenton Channel (16025), Ironton (16026) and Riverview (16027).

Cont. Type	Limiting Element	Voltage Limit (pu)	Pre-Project Voltage (pu)	Post-Project Voltage (pu)
P6	Adelaide 120 kV Bus 101	>0.92 & <1.07	0.89	0.95
P6	Adelaide 120 kV Bus 102	>0.92 & <1.07	0.89	0.95
P6	Adelaide-Ironton-Navarre 120 kV Bus	>0.92 & <1.07	0.90	0.95
P6	Alloy Tap 120 kV Bus	>0.92 & <1.07	0.90	0.95
P6	Cicot 120 kV Bus	>0.92 & <1.07	0.90	0.95
P6	Deacon Tap 120 kV Bus	>0.92 & <1.07	0.90	0.96
P6	Filmore 120 kV Bus 1	>0.92 & <1.07	0.91	0.96
P6	Filmore Tap 120 kV Bus	>0.92 & <1.07	0.91	0.96
P6	Great Lakes ~M 120kV Bus	>0.92 & <1.07	0.89	0.94
P6	Ironton 120 kV Bus 101	>0.92 & <1.07	0.89	0.96
P6	Ironton 120 kV Bus 102	>0.92 & <1.07	0.89	0.96
P6	Ironton 120 kV Bus 103	>0.92 & <1.07	0.89	0.96
P6	Jupiter 120 kV Bus 101	>0.92 & <1.07	0.91	0.95
P6	Jupiter 120 kV Bus 102	>0.92 & <1.07	0.91	0.95
P6	Maron 120 kV Bus	>0.92 & <1.07	0.90	0.96
P6	Maxwell 120 kV Bus 101	>0.92 & <1.07	0.90	0.95
P6	Navarre 120 kV Bus	>0.92 & <1.07	0.90	0.95
P6	Praxair 120 kV Bus	>0.92 & <1.07	0.89	0.94
P6	Praxair Tap 120 kV	>0.92 & <1.07	0.89	0.96
P6	Praxair 5-6 120 kV Bus	>0.92 & <1.07	0.89	0.95
P6	River Rouge 120 kV Bus 101	>0.92 & <1.07	0.90	0.96
P6	River Rouge 120 kV Bus 1-2-3	>0.92 & <1.07	0.90	0.96
P6	River Rouge 120 kV Bus 121	>0.92 & <1.07	0.89	0.94
P6	Tandem 120 kV Bus 101	>0.92 & <1.07	0.89	0.95
P6	Tandem 120 kV Bus 102	>0.92 & <1.07	0.89	0.95
P6	Tandem 120 kV Bus 103	>0.92 & <1.07	0.89	0.95
P6	Zephyr 120 kV Bus 102	>0.92 & <1.07	0.91	0.96
P6	Zephyr 120 kV Bus 101	>0.92 & <1.07	0.91	0.96

Table 4.3.1-14: P16024 Voltage performance drivers

Alternatives Considered

DIG Alternative, a reconfiguration of DIG 230 kV facilities that adds a new 230 kV substation at DIG and a line from DIG to River Rouge 230 kV, was considered as a blanket alternative for projects in MTEP20 and previous cycle's projects. Following reliability and cost evaluation, the DIG alternative was eliminated as a cost-effective and reliable solution.

Project 18745 - Reconductor Bunce Creek- Scott (B3N) 220 kV

Project Description

This project will reconductor the Bunce Creek – Scott 220 kV circuit. The project's estimated cost is \$0.66 million and expected in-service date is June 1, 2024.

Project Need

ITC has implemented a new methodology for calculating thermal ratings for equipment; and as a result, the current ratings on the Ontario – Michigan interface has seen a reduction in rating on the Bunce Creek- Scott (B3N) 220 kV line.

Both the ITC and Hydro One facilities limit the flow on the Bunce Creek – Scott line. Project 18745 will resolve the issue on ITC's side, while Hydro One will use dynamic rating on their side. This will ensure there are no additional limitations to the capacity of the tie line. The line reconductoring will result in acceptable loading (below line rating) on ITC's side of the interface; however, since it is not possible to implement dynamic ratings in planning studies, the Hydro One's side will cause line loading as shown in Table 4.3.1-15. The loading will be reduced below 103% when the dynamic ratings are implemented.

Cont. Type	Limiting Element	Rating (MVA)	Pre-Project Loading (%)	Post-Project Loading (%)
P6	Scott - Bunce Creek 220 kV Ckt 1	456	103	103

Table 4.3.1-15: P18745 Thermal loading drivers

Alternatives Considered

None.

Project 18201 - Sag Remediation of Hunters Creek-Stratford 120 kV

Project Description

This project will remediate the sag on the Hunters Creek – Stratford 120 kV line. The sag remediation on this line is due to a thermal overload caused by P6 contingency events. The project's estimated cost is \$0.51 million and expected in-service date is December 31, 2023.

Project Need

The Hunters Creek – Stratford 120 kV line is projected to become overloaded for P6 contingency events. The sag remediation on this line will relieve the loading as shown in Table 4.3.1-16.

Cont. Type	Limiting Element	Rating (MVA)	Pre-Project Loading (%)	Post-Project Loading (%)
P6	Hunterscreek - Stratford 120 kV - Ckt 1	140	104	80

Table 4.3.1-16: P18201 Thermal loading drivers

Alternatives Considered

None.

Project 18200 - Sag Remediation of Hemphill-Hunters Creek 120 kV

Project Description

This project will remediate the sag on the Hemphill – Hunters Creek 120 kV line. The sag remediation on this line is due to a thermal overload caused by P6 contingency events. The project's estimated cost is \$0.43 million and expected in-service date is December 31, 2023.

Project Need

The Hemphill – Hunters Creek 120 kV line is projected to become overloaded for various P6 contingency events. The sag remediation on this line will relieve the loading as shown in Table 4.3.1-17.

Cont. Type	Limiting Element	Rating (MVA)	Pre-Project Loading (%)	Post-Project Loading (%)
P6	Hemphill - Hunters Creek 120 kV - Ckt 1	134	104	70

Table 4.3.1-17: P18200 Thermal loading drivers

Alternatives Considered

None.

Project 19607 - Brownstown - Navarre #2 230 kV Sag Remediation

Project Description

This project will remediate the sag on the Brownstown – Navarre #2 230 kV circuit up to the conductor limit and upgrade station equipment at Navarre 230 kV Station. The project’s estimated cost is \$1.07 million and expected in-service date is June 1, 2024.

Project Need

The Brownstown – Navarre # 2 230 kV circuit is projected to become overloaded for P6 contingency events. The sag remediation on this line will relieve the loading as shown in Table 4.3.1-18.

Cont. Type	Limiting Element	Rating (MVA)	Pre-Project Loading (%)	Post-Project Loading (%)
P6	Navarre - Brownstown 230 kV - Ckt 2	637	104	86

Table 4.3.1-18: P19607 Thermal loading drivers

Alternatives Considered

Project 19607 proposed during MTEP20 as a reliable and cost-effective alternative to Project 18191 (Rebuild of Brownstown – Navarre #2 230 kV).

Note DIG Alternative, a reconfiguration of DIG 230 kV facilities that adds a new 230 kV substation at DIG and a line from DIG to River Rouge 230 kV, was considered as a blanket alternative for Project 18191, other projects in MTEP20 and previous cycle’s projects. Following reliability and cost evaluation, the DIG alternative was eliminated as a cost-effective solution.

Project 19565 - Upgrade Trenton Channel 120 kV HH Station Equipment

Project Description

This project will upgrade Trenton Channel 120 kV HH Station Equipment. This equipment upgrade is due to a thermal overload on the Brownstown – Trenton Channel #2 120 kV line, caused by P6 contingency events. The project’s estimated cost is \$0.34 million and expected in-service date is June 1, 2024.

Project Need

The Brownstown – Trenton Channel #2 120 kV line is projected to become overloaded for several P6 contingency events. The sag remediation on this line will relieve the loading as shown in Table 4.3.1-19.

Cont. Type	Limiting Element	Rating (MVA)	Pre-Project Loading (%)	Post-Project Loading (%)
P6	Trenton Channel - Brownstown 120 kV - Ckt 2	349	98	93

Table 4.3.1-19: P19565 Thermal loading drivers

Alternatives Considered

Project 19565 was proposed during MTEP20 as a reliable and cost-effective alternative to Project 18887 (Build Brownstown-Trenton Channel #4 120 kV).

Note that Project 18887 was proposed in MTEP20 as an alternative to Project 18204 (Rebuild Brownstown – Trenton Channel #2 120 kV) and Project 18189 (Reconductor of Rebuild Brownstown – Trenton Channel #3 120 kV). Also, DIG Alternative, a reconfiguration of DIG 230 kV facilities that adds a new 230 kV substation at DIG and a line from DIG to River Rouge 230 kV, was considered as a blanket alternative for Project 18887, other projects in MTEP20 and previous cycle's projects. Following reliability and cost evaluation, the DIG Alternative Project was eliminated as a cost-effective solution.

Project 18198 - Sag Remediation of Airport-Tulsa 120 kV

Project Description

This project will remediate the sag on the Airport – Tulsa 120 kV line. The sag remediation on this line is due to a thermal overload caused by P6 contingency events. The project's estimated cost is \$0.23 million and expected in-service date is December 31, 2023.

Project Need

The Airport – Tulsa 120 kV line section is projected to become overloaded for various P6 contingency events. The sag remediation on this line will relieve the loading as shown in Table 4.3.1-20.

Cont. Type	Limiting Element	Rating (MVA)	Pre-Project Loading (%)	Post-Project Loading (%)
P6	Airport - Tulsa 120 kV - Ckt 1	133	123	80

Table 4.3.1-20: P18198 Thermal loading drivers

Alternatives Considered

DIG Alternative, a reconfiguration of DIG 230 kV facilities that adds a new 230 kV substation at DIG and a line from DIG to River Rouge 230 kV, was considered as a blanket alternative for projects in MTEP20 and previous cycle's projects. Following reliability and cost evaluation, the DIG Alternative Project was eliminated as a cost-effective and reliable solution.

Other Projects

The "Other" projects category do not meet the criteria to be considered as Baseline Reliability Projects (BRP), New Transmission Access Projects, Market Efficiency Projects, or Multi-Value Projects. Other

projects may include projects to satisfy Transmission Owner and/or state and local planning criteria other than NERC or regional reliability standards, interconnect new Loads, relocate transmission facilities, address aging transmission infrastructure, replace problematic transmission plant, improve operational performance or address other operational issues, address service reliability issues with end-use consumers, improve aesthetics including but not limited to undergrounding overhead transmission facilities, address localized economic issues, and address other miscellaneous localized needs. For convenience, tables of project information are broken down by four general categories of project drivers but note that these four drivers are not defined in the MISO Tariff.

Projects Driven by Local Reliability

The following projects are driven by local planning criteria without Electric Reliability Organization (ERO) standard applicability or by operational flexibility.

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
18525	Brownstown 3 345 kV - Upgrade 345 kV Switches	The Fermi 2 Generation Replacement project approved by MISO, is expected to go in-service by April 30, 2020. This Generation Replacement project will increase the maximum gross active power output of Fermi 2 generation. The study results in the 2019 Fermi 2 Nuclear Power Plant NPOA Transmission Grid Reliability Analysis show there are thermal violations due to the Fermi 2 generation replacement. To resolve these issues, this project will upgrade the ratings of five (5) 345 kV switches ("ED", "FD", "FH", "FK", "FO") at Brownstown 3 345 kV station.	Dec. 31, 2021	\$1.56M
19526	Stephens 120 kV Install Breaker GP	Install the GP breaker between Bus 105 and 106 at the Stephens 120 kV station including the associated GQ disconnect, and replace the GO disconnect.	Dec. 31, 2022	\$0.76M

Projects Driven by Age and Condition

The following projects are proposed to replace aging or degraded equipment.

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
18241	Stephens Transformer 303 Replacement	Replace Stephens 345/120 kV Transformer #303 with a new transformer.	Dec. 31, 2023	\$4.80M
18242	Wayne Transformer 303 Replacement	Replace Wayne 345/120 kV Transformer #303 with a new transformer.	Dec. 31, 2023	\$4.40M

Projects Driven by Load Growth

Growing load is the principal driver for the following projects.

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
17504	Abrams Interconnection	Install a new 120 kV substation. Extend the Northeast-Skylark line by adding 0.1 miles of 954 ACSR conductor to loop in Abrams interconnection station. Add three 120 kV breakers and two bus sections.	Oct. 31, 2022	\$4.52M
17744	ITCT Customer Interconnections - Year 2023	Customer interconnection requests and retirements less than \$1 million with in-service date in year 2023.	Dec. 31, 2023	\$2.00M

Projects Driven by Other Local Needs

The following projects are proposed due to other local needs such as city mandate and distribution needs.

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
17124	Frisbie - Midtown 120kV Cable Relocation	Relocate the Frisbie-Midtown underground circuit out of the current location in the Milwaukee Ave bridge over I-75, as mandated by the Michigan Department of Transportation. Utilize 954 ACSR on overhead structures from each side of I-75.	June 1, 2021	\$1.50M
17998	Build Stone Pool - Temple 120 kV	The Frisbie – Resource and Resource – Alfred 120 kV underground cables are projected to overload for NERC Category P2 and P6 conditions. The proposed solution is to build a new circuit (approximately 1.25 miles) from the Stone Pool 120 kV Station to the Temple 120 kV station utilizing 3200 Kcmil underground cable.	June 1, 2024	\$18.70M

Generator Interconnection Projects

According to Attachment FF of the MISO Tariff, “Generator Interconnection Projects are New Transmission Access Projects that are associated with interconnection of new, or increase in generating capacity of existing, generation.” The following Generator Interconnection Projects (GIPs) have been evaluated through the Generator Interconnection queue and the associated Generator Interconnection Agreements have been signed.

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
17364	J793 Generator Interconnection	ITCT will install a new 345 kV station with 3-345 kV breakers and associated disconnects in a ring bus configuration. ITCT will loop the Belle River - Fitz #2 345kV circuit into the new J793 substation. ITCT will upgrade relay panels at the Fitz and Belle River Power Plant Stations to accommodate the new generator J793.	May 14, 2021	\$9.73M
17995	J752 Network Upgrades	Install Relay and communication cutoff switch at Ringle 345 kV station for the J752 expansion of J161.	Oct. 31, 2019	\$0.02M
18313	Sag Remediation of Bauer - Pontiac 345 kV	Raise the sag limit on the 2-954 ACSR 54/7 conductor span of the Bauer - Pontiac 345 kV circuit to achieve a summer emergency rating of at least 1258 MVA.	Dec. 31, 2020	\$0.96M

4.3.2 Michigan Electric Transmission Co. (METC)

METC proposed 38 projects at an estimated cost of \$349 million targeting MTEP20 Appendix A. After MISO's independent reliability analysis of METC area, 29 of these projects were recommended by MISO to be included in MTEP20 Appendix A at an estimated cost of \$252 million. Of these projects, 11 are Baseline Reliability Projects, 9 are Other Projects, and 9 are Generation projects with signed Generator Interconnection Agreements. All the projects' expected in-service dates and estimated costs are provided as of September 15, 2020.

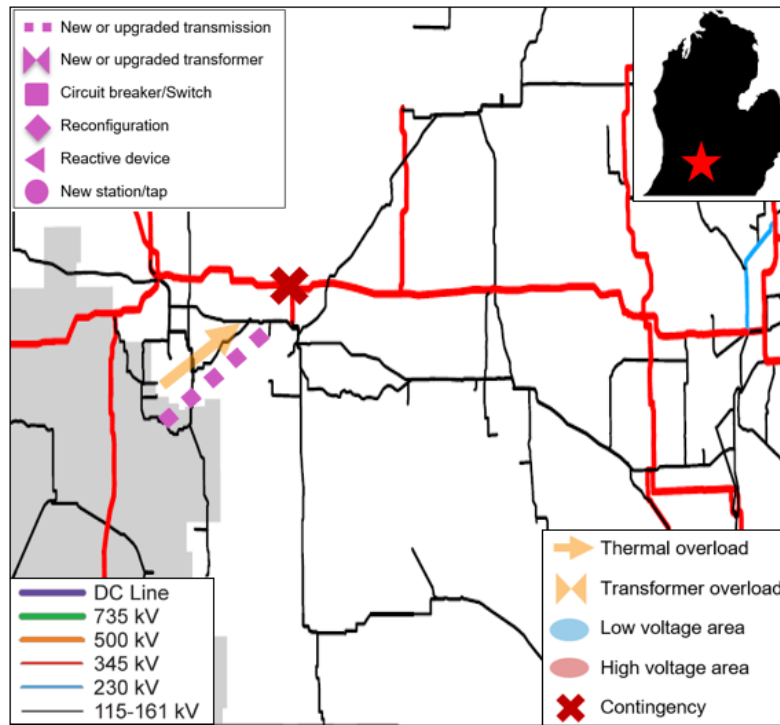
Baseline Reliability Projects

According to Attachment FF of the MISO Tariff, "Baseline Reliability Projects are Network Upgrades identified in the base case as required to ensure that the Transmission System is in compliance with applicable national Electric Reliability Organization (ERO) reliability standards and reliability standards adopted by Regional Reliability Organizations and applicable within the Transmission Provider Region."

Project 15896 - Battle Creek-Morrow #1 138 kV Rebuild

Project Description

This project will rebuild the Battle Creek – Morrow Ckt 1 138 kV line. This line rebuild is due to thermal overload caused by shutdown plus contingency events. The project's estimated cost is \$41 million and expected in-service date is June 1, 2025.



MISO, using Ventyx Velocity Suite © 2014

Figure 4.3.2-1: P15896 Geographic transmission map of project area

Project Need

The Battle Creek – Morrow #1 138 kV line is projected to overload for shutdown plus contingency scenarios. Rebuilding this line will reduce line loading as shown in Table 4.3.2-1.

Cont. Type	Limiting Element	Rating (MVA)	Pre-Project Loading (%)	Post-Project Loading (%)
P7	Battle Creek – Morrow 138 kV Ckt1	287	124	70

Table 4.3.2-1: P15896 Thermal loading drivers

Alternatives Considered

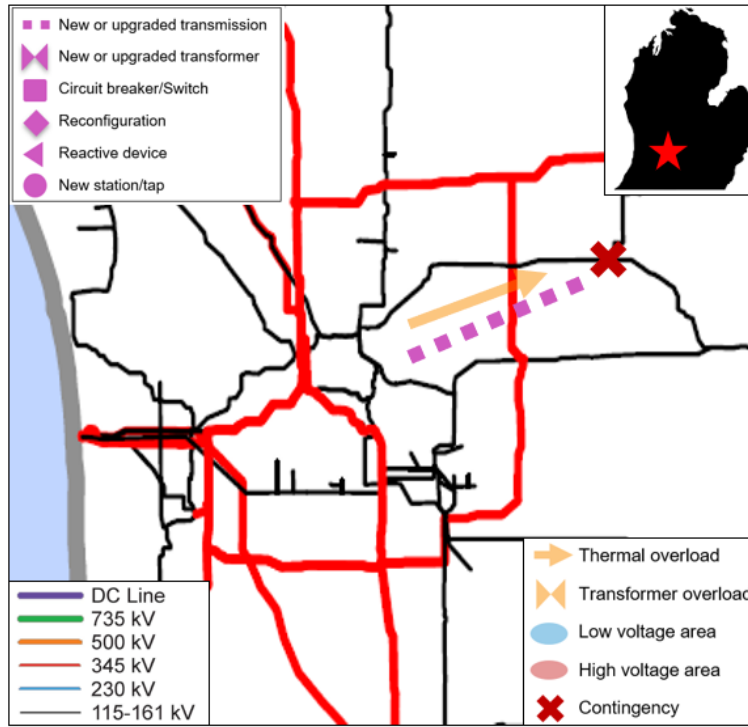
None.

Project 15809 –Four Mile-North Belding 138 kV Line Rebuild

Project Description

This project rebuilds the Alpine Jct – Rogue River Jct 138 kV and Rogue River Jct – Cannon Jct. sections of the Four Mile-North Belding 138 kV line. The project's estimated cost is \$11.7 million and expected in-service date is December 31, 2024.

Note that Project 15809 was originally proposed in MTEP19.



MISO, using Ventyx Velocity Suite © 2014

Figure 4.3.2-2: P15809 Geographic transmission map of project area

Project Need

The Alpine Jct – Rogue River Jct 138 kV and Rogue River Jct – Cannon Jct. sections of the Four Mile-North Belding 138 kV line are projected to overload for P6 contingencies. The rebuild will reduce loading as shown in Table 4.3.2-2.

Cont. Type	Limiting Element	Rating (MVA)	Pre-Project Loading (%)	Post-Project Loading (%)
P6	Alpine Jct-Rogue River Jct 138 kV Line	277	111	70
P6	Rogue River Jct-Canon Jct 138 kV Line	332	105	68

Table 4.3.2-2: P15809 Thermal loading drivers

Alternatives Considered

None.

Project 18728 –Rebuild Cobb-Sternberg 138 kV Line

Project Description

This project rebuilds the Cobb-Sternberg 138 kV line. The project’s estimated cost is \$10.45 million and expected in-service date is June 1, 2025.

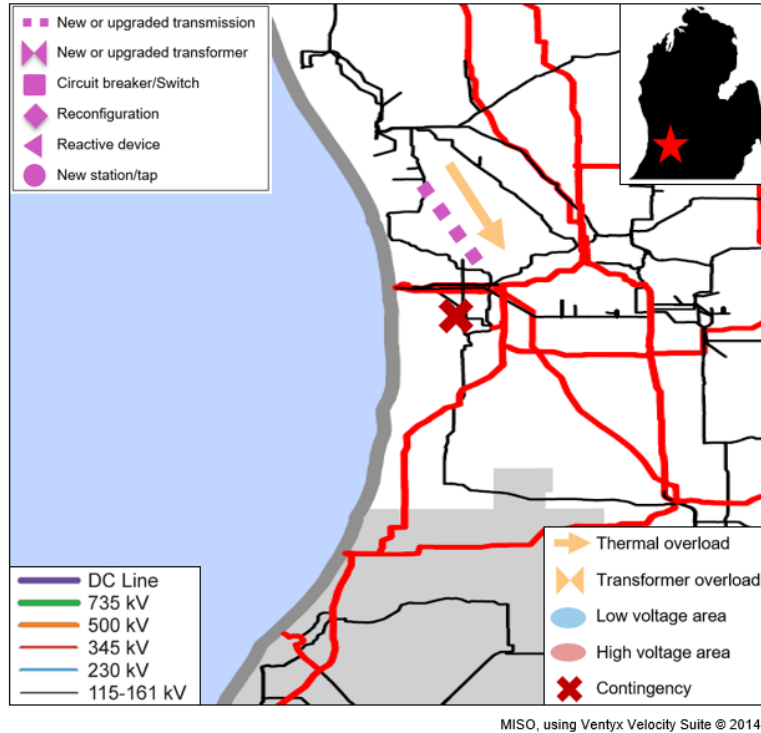


Figure 4.3.2-3: P18728 Geographic transmission map of project area

Project Need

The Cobb-Sternberg 138 kV line is projected to overload for P6 contingencies. The line rebuild will relieve the line loading as shown in Table 4.3.2-3.

Cont. Type	Limiting Element	Rating (MVA)	Pre-Project Loading (%)	Post-Project Loading (%)
P6	Cobb-Sternberg 138 kV Line	287	118	56

Table 4.3.2-3: P18728 Thermal loading drivers

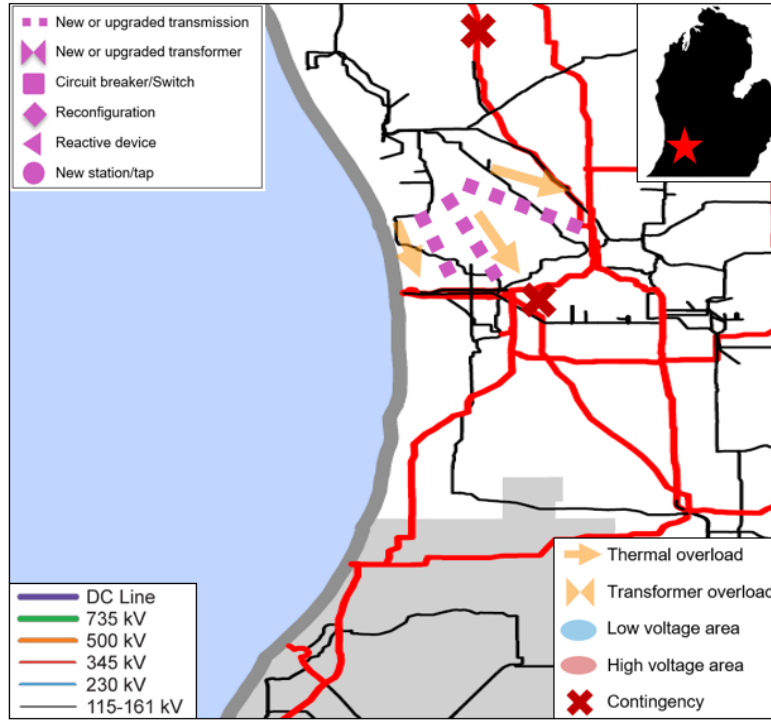
Alternatives Considered

None.

Project 19545 – Grand Rapids Area Project

Project Description

This project will construct two 138 kV circuits to cut in Chopin – Northridge 138 kV line into Meyer 138 kV Station. Three 138 kV breakers will also be installed to facilitate new lines coming into the Meyer Station. Furthermore, the project will sag remediate the limit on Chopin – Meyer 138 kV Line, Meyer – Northridge 138 kV Line and Meyer – Tallmadge #2 138 kV Line. The project's estimated cost is \$8.54 million and expected in-service date is June 1, 2025.



MISO, using Ventyx Velocity Suite © 2014

Figure 4.3.2-4: P19545 Geographic transmission map of project area

Project Need

The Chopin – Meyer 138 kV Line, Meyer – Northridge 138 kV line and Meyer – Tallmadge #2 138 kV lines are projected to overload for P6 contingencies. The sag remediation on these line sections will relieve the loading as shown in Table 4.3.2-4.

Cont. Type	Limiting Element	Rating (MVA)	Pre-Project Loading (%)	Post-Project Loading (%)
P6	Tallmadge-Meyer Ckt 2 138 kV Line	306	124	77
P6	Northridge-Meyer 138 kV Line	306	153	82
P6	Chopin-White Road 138 kV Line	218	107	84
P6	White Road-Meyer 138 kV Line	222	115	84

Table 4.3.2-4: P19545 Thermal loading drivers

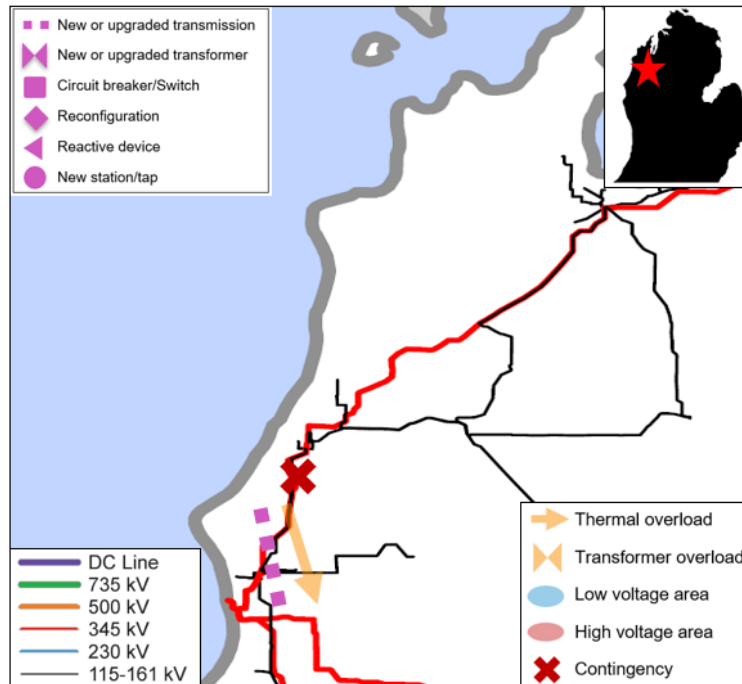
Alternatives Considered

None.

Project 18164 –Amber- Pere Marquette #2 138 kV Line

Project Description

This project will construct a new line from Amber to Pere Marquette utilizing existing right of way. The project's estimated cost is \$5.81 million and expected in-service date is June 1, 2024.



MISO, using Ventyx Velocity Suite © 2014

Figure 4.3.2-5: P18164 Geographic transmission map of project area

Project Need

The Amber – Pere Marquette Ckt 1 is projected to overload for P6 contingencies. Construction of a new line reduces the loading as shown in Table 4.3.2-5.

Cont. Type	Limiting Element	Rating (MVA)	Pre-Project Loading (%)	Post-Project Loading (%)
P6	Amber-Pere Marquette Ckt1 138 kV Line	332	109	80

Table 4.3.2-5: P18164 Thermal loading drivers

Alternatives Considered

None.

Project 15939 – Leoni-Beecher 138 kV Sixed Wire

Project Description

This project will six-wire the Mercer – Cement City Jct section of the Leoni – Beecher circuit and remediate sag limit on the sixed wired conductor section. The project's estimated cost is \$4.3 million and expected in-service date is December 31, 2023.

Note this project was previously proposed in MTEP19.

Project Need

The Mercer – Cement City Jct. section of the Leoni – Beecher 138kV line is projected to overload for shutdown plus contingency. The identified overloaded equipment on this circuit is the conductor and sag limit on the six-wired conductor section. The sag remediation on these line sections will relieve the loading as shown in Table 4.3.2-6.

Cont. Type	Limiting Element	Rating (MVA)	Pre-Project Loading (%)	Post-Project Loading (%)
P7	Mercer-Cement City Jct 138 kV Line	227	126	87

Table 4.3.2-6: P15939 Thermal loading drivers

Alternatives Considered

None.

Project 18073 –Reconductor Argenta-Riverview 138 kV Line

Project Description

This project will reconductor the Argenta – Riverview 138 kV line. The project’s estimated cost is \$2.20 million and expected in-service date is June 1, 2024.

Project Need

The Argenta – Riverview 138 kV line is projected to overload for P6 contingencies. The reconductoring will relieve the loading as shown in Table 4.3.2-7.

Cont. Type	Limiting Element	Rating (MVA)	Pre-Project Loading (%)	Post-Project Loading (%)
P6	Argenta-Riverview 138 kV Line	359	110	93

Table 4.3.2-7: P18073 Thermal loading drivers

Alternatives Considered

None.

Project 18133 –Station Equipment Upgrades Pere Marquette 138 kV 5M9 & 7W8

Project Description

This project will upgrade two breakers and other terminal station equipment at the Pere Marquette 138 kV Station. The project’s estimated cost is \$0.97 million and expected in-service date is December 31, 2022.

Project Need

The station equipment at Pere Marquette is projected to overload for P6 contingencies. The upgrades will reduce the loading as shown in Table 4.3.2-8.

Cont. Type	Limiting Element	Rating (MVA)	Pre-Project Loading (%)	Post-Project Loading (%)
P6	Pere Marquette Transformer 2	656	109	94

Table 4.3.2-8: P18133 Thermal loading drivers

Alternatives Considered

None.

Project 18043 –Sag Remediation Scott Lake-Argenta 138 kV Line

Project Description

This project will sag remediate the Scott Lake – Argenta 138 kV line. The project's estimated cost is \$0.83 million and expected in-service date is December 31, 2023.

Project Need

The Scott Lake – Argenta 138 kV line comprises of Scott Lake – Trowbridge 138 kV and Trowbridge – Argenta 138 kV line sections which are projected to overload for P6 contingencies. The sag remediation will relieve the loading as shown in Table 4.3.2-9.

Cont. Type	Limiting Element	Rating (MVA)	Pre-Project Loading (%)	Post-Project Loading (%)
P6	Argenta-Trowbridge 138 kV Line	235	111	75
P6	Trowbridge-Scott Lake 138 kV Line	219	103	74

Table 4.3.2-9: P18043 Thermal loading drivers

Alternatives Considered

None.

Project 18004 –Meyer-Briggs & Stratton Jct 138 kV Sag Remediation

Project Description

This project will sag remediate the limit on the Meyer – Briggs & Stratton Jct. 138 kV line. The project's estimated cost is \$0.81 million and expected in-service date is December 31, 2023.

Project Need

The Tanner – Briggs & Stratton Jct. 138 kV Line and Tanner – Meyer 138 kV Line are projected to overload for P6 contingencies. The sag remediation on these lines will relieve the loading as shown in Table 4.3.2-10.

Cont. Type	Limiting Element	Rating (MVA)	Pre-Project Loading (%)	Post-Project Loading (%)
P6	Tanner-Meyer 138 kV Line	306	134	87
P6	Tanner-Briggs & Stratton Jct 138 kV Line	262	116	91

Table 4.3.2-10: P18004 Thermal loading drivers

Alternatives Considered

None.

Project 18187 Pere Marquette-Stronach 138 kV Line Sag Remediation

Project Description

This project will remediate the sag on Pere Marquette – Stronach 138 kV line. The project's estimated cost is \$0.46 million and expected in-service date is December 31, 2023.

Project Need

The Pere Marquette – Stronach 138 kV line is projected to overload for P6 contingencies. The sag remediation will relieve loading on the line as shown in Table 4.3.2-11.

Cont. Type	Limiting Element	Rating (MVA)	Pre-Project Loading (%)	Post-Project Loading (%)
P6	Pere Marquette-Stronach 138 kV Line	397	106	88

Table 4.3.2-11: P18187 Thermal loading drivers

Alternatives Considered

None.

Other Projects

The “Other” projects category does not meet the criteria to be considered as Baseline Reliability Projects (BRP), New Transmission Access Projects, Market Efficiency Projects, or Multi-Value Projects. Other projects may include projects to satisfy Transmission Owner and/or state and local planning criteria other than NERC or regional reliability standards, interconnect new Loads, relocate transmission facilities, address aging transmission infrastructure, replace problematic transmission plant, improve operational performance or address other operational issues, address service reliability issues with end-use consumers, improve aesthetics including but not limited to undergrounding overhead transmission facilities, address localized economic issues, and address other miscellaneous localized needs. The tables of the project information are broken down by four general categories of project drivers, however, note that these four drivers are not defined in the MISO Tariff.

Projects Driven by Local Reliability

The following projects are driven by local planning criteria to aid operational flexibility and not for Electric Reliability Organization (ERO) standard applicability purposes.

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
18112	Kenowa 345 kV Breaker Installation	Complete row 33 of the Kenowa substation by adding 3 breakers and associated equipment. This addition creates a breaker-and-a-half scheme at Kenowa. A breaker-and-a-half scheme limits the amount of system split, apart from N-1-1 and outage + fault scenarios, reduces operational challenges, and assists relay protection efforts.	Dec. 31, 2022	\$ 3.95

Projects Driven by Age and Condition

The following projects are proposed to replace aging or degraded equipment.

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
18190	Dort 138 kV Station Rebuild	Rebuild Dort 138 kV station with a new control house at a new site approximately 0.5 miles to the south utilizing new equipment in a four-row, breaker-and-a-half scheme configuration. The new 138 kV station to be built will be called Calvary. This new 138 kV station would reduce equipment failures, unplanned outages, and repair costs. In addition, it would enable routine maintenance to be performed again in a safe and efficient manner.	June 1, 2024	\$ 21.32
18238	Morrow (Jaguar) 138 kV Station Rebuild	Rebuild Morrow 138 kV station at a new site approximately 1.55 miles to the north utilizing new equipment in a five-row, breaker-and-a-half scheme configuration. The new 138 kV station to be built will be called Jaguar. Redirect all circuits currently at the Morrow 138 kV station into the new Jaguar 138 kV station and connect CE's new replacement step-down transformers, and Kalamazoo River Generation Station via radial feeds into the new Jaguar station. Reconnector Morrow - Jaguar 795 ACSR section of the Morrow - Bradley (V22) 138 kV line using 1431 ACSR conductor and utilizing this circuit to create Cork Street - Jaguar 138 kV line.	Dec. 31, 2022	\$ 22.30
18243	Battle Creek Transformer 2 Replacement	The transformer has reached the end of its life. Replace Battle Creek 345/138kV transformer #2 with a new transformer.	Dec. 31, 2023	\$ 6.40
18244	Gallagher Transformer #2 Replacement	The transformer has reached the end of its life. Replace Gallagher 345/138kV transformer #2 with a new transformer.	Dec. 31, 2023	\$ 6.27
18245	Kenowa 345 kV Reactor 33F Replacement	The reactor has reached the end of its life. Replace Kenowa 345 kV Shunt Reactor #33F with a new reactor.	Dec. 31, 2023	\$ 2.35

Projects Driven by Load Growth

Growing load is the principal driver for the following projects.

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
17745	METC Customer Interconnections - Year 2023	Customer Requests are often in need of interconnection on an expedited basis. This blanket allows interconnection requests of smaller nature to be quickly studied and connected by METC.	Dec. 31, 2023	\$ 2.0
18055	METC EPR - Riverview Interconnection	Convert METC Riverview substation from a three-breaker straight bus configuration to a six-breaker ring. Install three 138kV breakers, bus work, and associated disconnects. A dedicated customer substation to be located approximately 1.25 miles west of Riverview Substation which will service 40MW of new load. Expansion of Riverview is needed to service the new CE substation.	July 2, 2021	\$ 6.27

Projects Driven by Other Local Needs

The following projects are proposed to due to other local needs.

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
17997	Donaldson Creek Interconnection Phase 2	Build a new breaker row 13, breakers 13B7 and 13M9 with associated switches and terminate a new WPSC line to Rodgers. WPSC MTEP17 Appendix A project 12203, 138 kV Voltage Conversion, requires two new connections to Donaldson Creek. METC will need to add a 3rd 138 kV row to its Donaldson Creek station to terminate a new 138 kV interconnection to Rodgers.	June 1, 2022	\$ 1.45

Generator Interconnection Projects

According to Attachment FF of the MISO Tariff, "Generator Interconnection Projects are New Transmission Access Projects that are associated with interconnection of new, or increase in generating capacity of existing, generation." The following Generator Interconnection Projects (GIPs) have been evaluated through the Generator Interconnection process and the associated Generator Interconnection Agreements have been approved.

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
17365	J717_J728 Generator Interconnection	Build a new 138 kV 3 Breaker ring substation cut in from the Bullock - Warren 138 kV circuit. Install OPGW from the new station to Warren 138 kV sub. J717 and J728 are a combined 385.4 MW wind farm requesting to share a point of interconnection on the Warren - Bullock 138 kV circuit in the DPP-2017-FEB cycle.	July 30 2020	\$ 8.19
17384	J796 (Goss) Network Upgrade	Network upgrades required to connect the J796 project. J796 is capable of 239 MW of solar generation. METC's Goss 138 kV station will need to be expanded by adding four 138 kV breakers to accommodate the interconnection.	Sept.1, 2021	\$ 2.81
17844	Summerton - Bullock cut into Salt River 138 kV	Extend the Summerton - Bullock 138 kV circuit by extending ~15 miles of double circuit 138 kV construction into Salt River 138 kV station. The J717 and J728 Generator Interconnections caused a stability issue and several local thermal overloads. Cutting in the Summerton - Bullock line to the J717 and J728 interconnection station to be named Salt River mitigates the stability issue and eliminates the need for other thermal mitigations.	Oct. 28, 2020	\$ 35.49
18147	Rebuild Bard Road - Warren 138 kV	Rebuild 16.4 miles of the Bard Road - Clare Junction section of the Bard Road - Warren 138 kV circuit and to rebuild 3.3 miles of the Clare Junction - Warren section of the Bard Road - Warren 138 kV circuit with 954 ACSR utilizing future 230 kV double circuit construction initially operated as 138 kV single circuit with adjacent side left vacant for future use.	May 1, 2022	\$ 23.86
18384	Sag Remediation Bullock - Saginaw River	Increase the sag limits on the Bullock to Gleaner J and Gleaner J to Hacket J sections of the Bullock - Saginaw River 138 kV circuit.	Dec. 31, 2020	\$ 1.02
18424	J758 Generator Interconnection (Network Upgrades) - Mud Lake	Build a new 138 kV 3 Breaker ring substation cut in from the Verona - Foundry 138 kV circuit. Changes will require upgraded relaying at Verona 138 kV station.	April 8, 2022	\$ 8.99
18425	Verona - Marshall cut into Mud Lake (J758 Stability Upgrade)	Cut in and extend the Verona - Marshall 138 kV line by adding 0.6 miles of 954 ACSR DCT to loop in J758. Add two 138kV 40kA breakers at positions 8M9 and 8W8 at the J758 substation 138 kV.	April 8, 2022	\$ 6.51

18426	Mud Lake - Brewer St Tap Rebuild (J758)	Rebuild the 4.4 miles of 336 ACSR from J758 Substation to Convis and Brewer St. Taps with 954 ACSR. J758 is a 200 MW solar plant looking to connect on the Verona - Foundry 138 kV line in the DPP-2017-FEB cycle. Project solves a thermal issue identified in the DPP-2017-FEB cycle.	April 8, 2022	\$ 5.70
18825	Island Rd - Replace White Bus and Position 177 Conductor	Replace the 4/0 CU and 1/0 CU bus conductor at Island Rd White bus and Position 177 to achieve a rating of at least 141 MVA (590 amps).	May 1, 2022	\$ 0.13

4.3.3 Consumers Energy (CE)

Consumers Energy proposed seven projects at an estimated cost of \$5.3 million targeting MTEP20 Appendix A. After MISO's independent reliability analysis of the Consumer Energy's area, seven of these projects were recommended by MISO to be included in MTEP20 Appendix A at an estimated cost of \$5.3 million. All these projects are classified as Other Projects. Consumers Energy did not propose any Base Line Reliability or Generation projects for METP20. All the projects' expected in-service dates and estimated costs are provided as of July 30, 2020.

Other Projects

The "Other" projects category does not meet the criteria to be considered as Baseline Reliability Projects (BRP), New Transmission Access Projects, Market Efficiency Projects, or Multi-Value Projects. Other projects may include projects to satisfy Transmission Owner and/or state and local planning criteria other than NERC or regional reliability standards, interconnect new Loads, relocate transmission facilities, address aging transmission infrastructure, replace problematic transmission plant, improve operational performance or address other operational issues, address service reliability issues with end-use consumers, improve aesthetics including but not limited to undergrounding overhead transmission facilities, address localized economic issues, and address other miscellaneous localized needs. The tables of the Other project information are broken down by four general categories of project drivers but note that these four drivers are not defined in the MISO Tariff.

Projects Driven by Age and Condition

The following projects are proposed to replace aging or degraded equipment.

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
17990	CE Substation Equipment Replacement Program - 2022	Replace damaged or defective substation equipment with new equipment.	Dec. 31, 2022	\$ 0.41
17991	CE Pole & Line Equipment Replacement Program - 2022	Replace damaged or defective poles, towers, cross arms, insulators, guying, and switches, as needed, to maintain the lines' condition.	Dec. 31, 2022	\$ 0.53
17992	Rebuild Hodenpyl Line	Rebuild the Hodenpyl 138 kV Line, approximately 0.6 miles, utilizing 138 kV single circuit construction. This line is constructed with towers and copper line conductor that date to the 1920's. Experience with this vintage of towers has shown significant ground level rusting which will lead to tower failure and line outages.	Dec. 31, 2021	\$ 0.33
17993	Muskegon #2 Line Rebuild	Rebuild a portion of the Muskegon #2 138 kV Line, approximately 2.1 miles, utilizing 138 kV single circuit construction. This portion of the line is constructed with towers installed in 1928. Experience with this vintage of towers has shown significant ground level rusting which will lead to tower failure and line outages.	Dec. 31, 2022	\$ 1.0

Projects Driven by Load Growth

Growing load is the principal driver for the following projects.

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
18886	Jaguar 138 kV Lines	Construct two new 138 kV lines from CE's new Morrow South Substation to METC's new Jaguar station. Consumers Energy plans to relocate and rebuild its distribution assets at Morrow Substation in a location south of the existing substation. This project will require two new transmission lines between the new Morrow South Substation and METC's new Jaguar substation, MTEP #18238.	Dec. 31, 2022	\$ 2.0

Projects Driven by Other Local Needs

The following projects are proposed to due to other local needs.

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
17994	Four Mile-Northridge 138 kV Lines	Construct two new 138 kV lines from Four Mile Substation to the new Northridge Substation. METC has identified a plan to install a new station named Northridge, MISO MTEP#16002. METC's plan will require Consumers Energy to reconnect its Four Mile Substation to the new station with two new transmission lines.	Dec. 31, 2022	\$ 0.5
18272	Dort-Mott Lake 138 kV Lines	Construct two new 138 kV lines from Dort Substation to the new Mott Lake Substation. METC has identified a plan to install a new station named Mott Lake, MTEP#18190. METC's plan will require Consumers Energy to reconnect its Dort Substation to the new station with two new transmission lines.	June 1, 2024	\$ 0.5

4.3.4 Wolverine Power Supply Cooperative Inc. (WPSC)

Wolverine Power Supply Cooperative Inc. proposed 12 projects at an estimated cost of \$22.37 million targeting MTEP20 Appendix A. After MISO's independent reliability analysis of the Wolverine Power Supply Cooperative Inc. area, 12 of these projects were recommended by MISO to be included in MTEP20 Appendix A at an estimated cost of \$22.37 million. Of these projects, 11 are Other Projects, and one is a Generation project with a signed Generator Interconnection Agreement. Wolverine Power Supply Cooperative Inc. did not propose any Base Line Reliability projects for MTEP20. All the projects' expected in-service dates and estimated costs are provided as of July 30, 2020.

Other Projects

The "Other" projects category does not meet the criteria to be considered as Baseline Reliability Projects (BRP), New Transmission Access Projects, Market Efficiency Projects, or Multi-Value Projects. Other projects may include projects to satisfy Transmission Owner and/or state and local planning criteria other than NERC or regional reliability standards, interconnect new Loads, relocate transmission facilities, address aging transmission infrastructure, replace problematic transmission plant, improve operational performance or address other operational issues, address service reliability issues with end-use consumers, improve aesthetics including but not limited to undergrounding overhead transmission facilities, address localized economic issues, and address other miscellaneous localized needs. The tables of the Other project information are broken down by four general categories of project drivers but note that these four drivers are not defined in the MISO Tariff.

Projects Driven by Age and Condition

The following projects are proposed to replace aging or degraded equipment.

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
18011	Rebuild Atlanta to Airport 138 kV	Rebuild and upgrade 31.5 miles of 138 kV with double circuit steel poles, 795ACSS conductor, and OPGW.	Dec. 31, 2023	\$ 15.5
18021	Eastport 69 kV BUS replacement	Rebuild the existing 69 kV bus to accommodate facility equipment replacement.	Dec. 31, 2021	\$ 0.17
18022	Westwood 69 kV BUS replacement	Rebuild the existing 69 kV bus to accommodate facility equipment replacement.	Dec. 31, 2022	\$ 0.17
18023	Eaton Rapids 69 kV BUS replacement	Rebuild the existing 69 kV bus to accommodate facility equipment replacement.	Dec. 31, 2022	\$0.17
18024	Atlanta 69 kV BUS replacement	Rebuild the existing 69 kV bus to accommodate facility equipment replacement.	Dec. 31, 2021	\$ 0.17
18026	Hawks 69 kV BUS replacement	Rebuild the existing 69 kV bus to accommodate facility equipment replacement.	Dec. 31, 2022	\$ 0.17
18027	Hammond Bay 138 kV BUS replacement	138 kV BUS replacement in order to accommodate voltage increase.	Dec. 31, 2023	\$ 0.17
18028	Hagensville 138 kV BUS replacement	Install BUS to 138 kV standard to accommodate voltage conversion.	Dec. 31, 2023	\$ 0.17

Projects Driven by Load Growth

Growing load is the principal driver for the following projects.

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
14005	Hammond Bay Interconnection Line	Construct approximately 2 miles of new transmission line to connect the Hammond Bay Distribution Station to METC's 138 kV transmission line. A transmission line to Hammond Bay station is nearing the end of its expected useful life. This project is the most efficient and cost-effective solution whereby the existing transmission line is relocated from Wolverine's transmission system to METC's transmission system, which is much closer in proximity. The result is improved reliability due to reduced line mile exposure.	Dec. 31, 2023	\$ 0.9
18015	Cadillac Substation Line Modifications	Line modifications including structure relocation and switch replacement in order to accommodate a new Cadillac distribution interconnection station.	Dec. 31, 2021	\$ 1.0

Projects Driven by Other Local Needs

The following projects are proposed to due to other local needs.

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
14004	Hagensville Interconnection Line	Construct approximately 0.25 miles of new 138 kV transmission line to connect the Grand Lake Distribution Station to METC's 138 kV Port Calcite to Rockport transmission line. A transmission line to Hagensville station is nearing the end of its expected useful life. This project is the most efficient and cost-effective solution whereby the existing transmission line is relocated from Wolverine's transmission system to METC's transmission system, which is much closer in proximity. The result is improved reliability due to reduced line mile exposure.	Dec. 31, 2023	\$ 0.26

Generator Interconnection Projects

According to Attachment FF of the MISO Tariff, "Generator Interconnection Projects are New Transmission Access Projects that are associated with interconnection of new, or increase in generating capacity of existing, generation." The following Generator Interconnection Projects (GIPs) have been evaluated through the Generator Interconnection queue process and the associated Generator Interconnection Agreements have been approved.

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
17204	Upgrade Vestaburg Transformer	Wolverine to install transformer with increased capacity at the Vestaburg Transmission station in order to support the J589 Generator Interconnection. The capacity of the new transformer will be 168MVA.	Feb. 29, 2020	\$ 3.52

4.3.5 Michigan Public Power Agency (MPPA)

Michigan Public Power Agency proposed one project at an estimated cost of \$4.0 million targeting MTEP20 Appendix A. After MISO's independent reliability analysis of the Michigan Public Power Agency area, this project was recommended by MISO to be included in MTEP20 Appendix A. This project has been categorized as an Other Project. Michigan Public Power Agency did not propose any Baseline Reliability or Generation projects for MTEP20. The project's expected in-service date and estimated cost are provided as of July 30, 2020.

Other Projects

The “Other” projects category does not meet the criteria to be considered as Baseline Reliability Projects (BRP), New Transmission Access Projects, Market Efficiency Projects, or Multi-Value Projects. Other projects may include projects to satisfy Transmission Owner and/or state and local planning criteria other than NERC or regional reliability standards, interconnect new Loads, relocate transmission facilities, address aging transmission infrastructure, replace problematic transmission plant, improve operational performance or address other operational issues, address service reliability issues with end-use consumers, improve aesthetics including but not limited to undergrounding overhead transmission facilities, address localized economic issues, and address other miscellaneous localized needs. The tables of the project information are broken down by four general categories of project drivers but note that these four drivers are not defined in the MISO Tariff.

Projects Driven by Local Reliability

The following projects are driven by local planning criteria without Electric Reliability Organization (ERO) standard applicability or by operational flexibility.

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
18003	Barlow to Parsons Line Rebuild	Rebuild 2.73 miles of 69 kV line between Barlow and Parsons. Replace 2.73 miles of #477 ACSR with .58 miles of twin #1000 MCM copper under airport flight path and 1.42 miles of #795 ACSS. Rebuild requires Switching Station upgrades at Barlow Substation.	Nov. 1, 2022	\$ 4.0

4.3.6 Northern Indiana Public Service Company (NIPSCO)

Northern Indiana Public Service Company (NIPSCO) proposed four projects at an estimated cost of \$51.0 million targeting MTEP20 Appendix A. After MISO’s independent reliability analysis of the NIPSCO area, all four projects have been recommended by MISO to be included in MTEP20 Appendix A at an estimated cost of \$51.0 million. Of these projects, two are Baseline Reliability Projects, zero are Other Projects, and two projects are Generation projects with signed Generator Interconnection Agreements. All the projects’ expected in-service dates and estimated costs are provided as of September 15, 2020.

Baseline Reliability Projects

According to Attachment FF of the MISO Tariff, “Baseline Reliability Projects are Network Upgrades identified in the base case as required to ensure that the Transmission System is in compliance with applicable national Electric Reliability Organization (ERO) reliability standards and reliability standards adopted by Regional Reliability Organizations and applicable within the Transmission Provider Region.”

Project 18276 - Rebuild Maple-LNG 138 kV Line

Project Description

The project will rebuild the Maple-LNG 138 kV Line. This replacement is due to N-1-1 (P6) potential violations. The total estimated cost of this project is \$18 million dollars. The expected in-service date for this project is 5/1/2023. The project is shown in Figure 4.3.6-1.

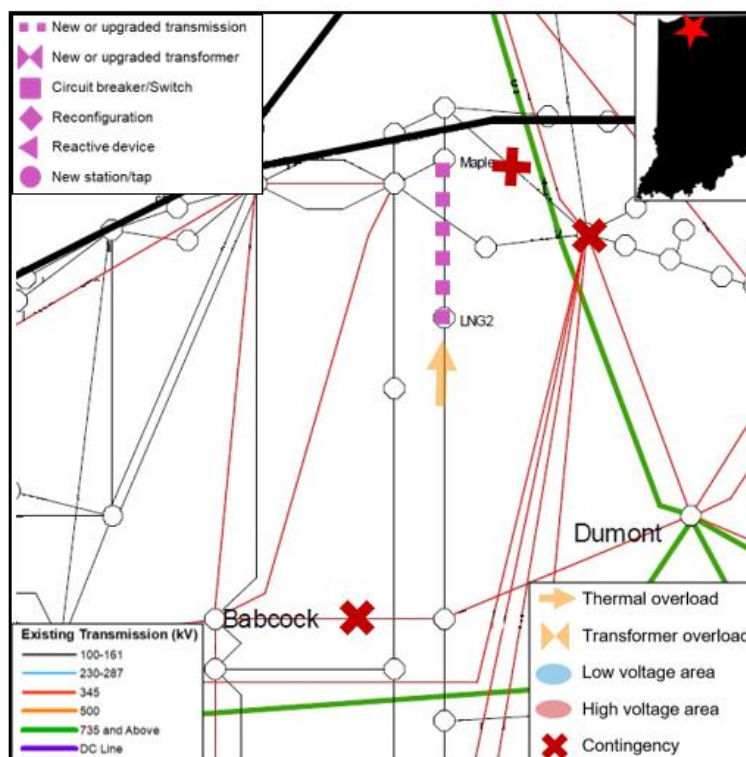


Figure 4.3.6-1: P18276 Geographic transmission map of project area

Project Need

The Maple-LNG 138 kV line becomes overloaded (117%) in year 2025 for a NERC-defined category P6-1-1 contingency event. The rebuilding of the Maple-LNG 138kV will increase the summer emergency ratings from 158 MVA to 398 MVA.

Cont. Type	Limiting Element	SE Rating (MVA)	Pre-Project Loading (%) (post-cont.)	Post-Project Loading (%) (post-cont.)
Base case	Maple - LNG 138 kV Line	158	33	33
P6-1-1	Maple - LNG 138 kV Line	398	117	26

Table 4.3.6-1: P18276 Thermal loading drivers

Project 18278 - Rebuild Stillwell-LNG 138 kV line

Project Description

The project will rebuild the Stillwell-LNG 138 kV line. This replacement is due to N-1-1 (P6) potential violations. The total estimated cost of this project is \$18 million dollars. The expected in-service date for this project is 5/1/2023. The project is shown in Figure 4.3-10.

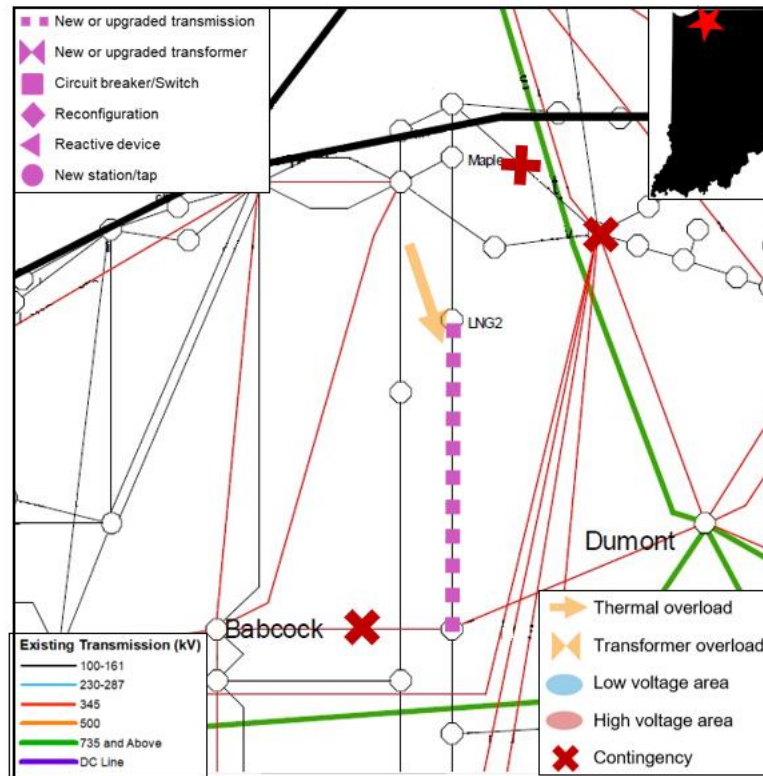


Figure 4.3.6-2: P18278 Geographic transmission map of project area

Project Need

The Stillwell-LNG 138 kV line becomes overloaded (121%) in year 2025 for a NERC-defined category P6-1-1 contingency event. The rebuilding of the Stillwell-LNG 138 kV line will increase the summer emergency ratings from 158 MVA to 478 MVA.

Cont. Type	Limiting Element	SE Rating (MVA)	Pre-Project Loading (%) (post-cont.)	Post-Project Loading (%) (post-cont.)
Base case	Stillwell-LNG 138 kV Line	158	38	38
P6-1-1	Stillwell-LNG 138 kV Line	478	121	24

Table 4.3.6-2: P18276 Thermal loading drivers

Alternatives Considered

Due to age of the line and unknown condition of the towers, line rebuild is proposed over reconductoring.

Project 18986 - Rebuild NIPSCO BES for Reinforcements

Project Description

The project will strengthen the NIPSCO transmission system ahead of the anticipated suspension of the [NIPS] Schahfer units 14, 15, 17 and 18. The total estimated cost of this project is \$48 million dollars. The expected in-service date for this project is 5/1/2023.

Project Need

The project will increase reliability with the suspension of the Schahfer units.

Cont. Type	Limiting Element	SE Rating (MVA)	Pre-Project Loading (%) (post-cont.)	Post-Project Loading (%) (post-cont.)
P7-1	[NIPS] New Carlisle—[NIPS] Bosserman 138 kV line	240	126	86
P4	[NIPS] New Carlisle—[NIPS] Bosserman 138 kV line	240	146	98
P7-1	[NIPS] New Carlisle—[NIPS] Maple 138 kV line	167	111	96

Table 4.3.6-3: P18986 Thermal loading drivers

Alternatives Considered

Other Projects

The “Other” projects category do not meet the criteria to be considered as Baseline Reliability Projects (BRP), New Transmission Access Projects, Market Efficiency Projects, or Multi-Value Projects. Other projects may include projects to satisfy Transmission Owner and/or state and local planning criteria other than NERC or regional reliability standards, interconnect new Loads, relocate transmission facilities, address aging transmission infrastructure, replace problematic transmission plant, improve operational performance or address other operational issues, address service reliability issues with end-use consumers, improve aesthetics including but not limited to undergrounding overhead transmission facilities, address localized economic issues, and address other miscellaneous localized needs. The tables of the project information are broken down by four general categories of project drivers but note that these four drivers are not defined in the MISO Tariff.

Generator Interconnection Projects

According to Attachment FF of the MISO Tariff, “Generator Interconnection Projects are New Transmission Access Projects that are associated with interconnection of new, or increase in generating capacity of existing, generation.” The following Generator Interconnection Projects (GIPs) have been evaluated through the Generator Interconnection queue and the associated Generator Interconnection Agreements have been signed.

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
18484	J837 J838 GIA Network Upgrades	Reynolds 345 kV upgrades to accommodate GI.	August 1, 2021	\$ 2.98
18685	J883 GIA Network Upgrades	Generator Interconnection. The existing East Winimac – Monticello line will be cut into at the site of the new substation.	April 15, 2023	\$ 15.99

4.4 Project Justifications – South Region

South Region Overview

The MISO South Planning Region consists of eleven Transmission-Ownning members spanning four states, Arkansas, Louisiana, Mississippi, and parts of Texas. These transmission owners are:

- Arkansas Electric Cooperative Corporative (AECC)
- City of Alexandria (AXLA), CLECO Power (CLEC)
- Cooperative Energy (SMEPA)
- East Texas Electric Cooperative (ETEC)
- Entergy Arkansas LLC (EAL)
- Entergy Louisiana LLC (ELL)
- Entergy Mississippi LLC (EML)
- Entergy New Orleans LLC (ENO)
- Entergy Texas LLC (ETL)
- Lafayette Utilities Systems (LAFA)

The region contains approximately 16,500 circuit miles of transmission lines ranging from 115 kV to 500 kV. There is also a significant 69 kV sub-transmission network interspersed across the footprint.

In the 2022 Summer Peak planning model, the region contains more than 38.1 GW of generation. The MISO South generation profile consists of mostly combine cycle, nuclear, gas, and coal fuel types, serving major load centers such as Little Rock, New Orleans, etc. Approximately 53 percent (20.2 GW) of the South region's generation capacity is made up of Combine Cycle (CC) units. Major generation centers are in central Arkansas, lower Louisiana, and western Mississippi (Figure 4.4-1).

Major load centers are typically found around larger cities in the region such as Little Rock, Jonesboro and Pine Bluff in Arkansas; Monroe, Alexandria, Lake Charles, Lafayette, New Orleans, and Baton Rouge in Louisiana; Jackson, Hattiesburg, Natchez, Vicksburg and Greenville in Mississippi. Texas major load centers in the Western load pocket include Bryan and the Woodlands area. The major load center in the WOTAB load pocket portion of Texas is in South Beaumont and the Port Arthur Area (Figure 4.4-1). According to the 2022 Summer Peak planning model, the regional load is over 36.1 GW.

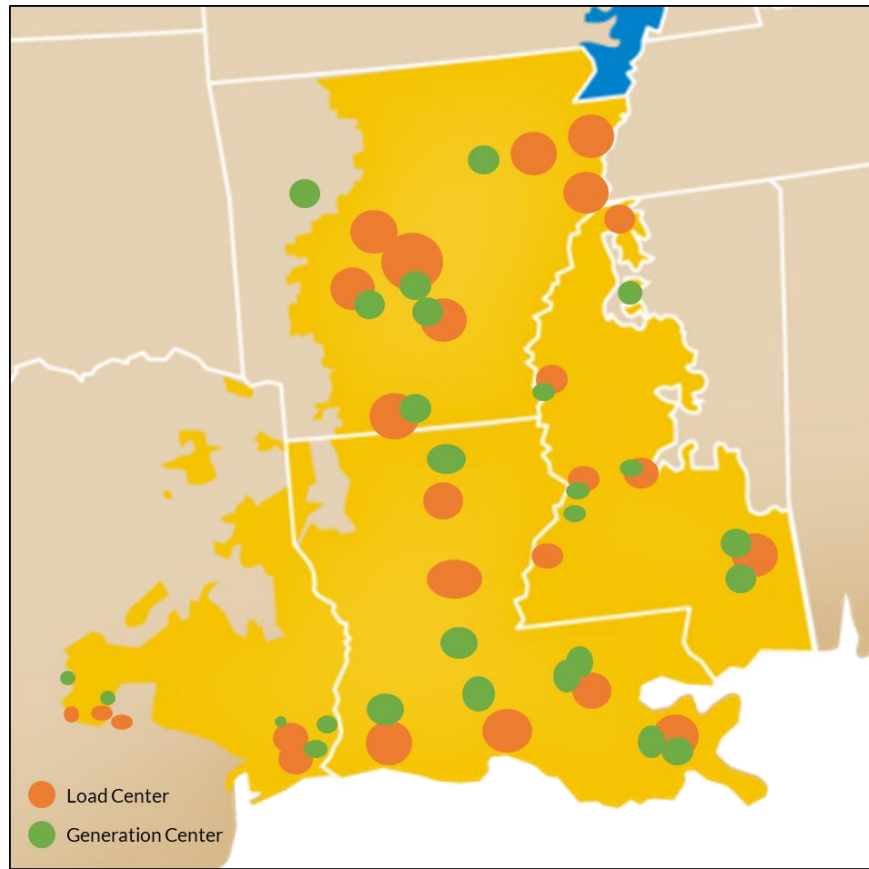


Figure 4.4-1: Generation and load centers in the South planning region

For MTEP20 MISO Transmission Planning is recommending 51 projects from the South region for inclusion in Appendix A at an estimated cost of \$578.02 million. Of these, eight are Baseline Reliability Projects, and 21 are Generator Interconnection Projects. The remaining 22 projects are classified as Other Projects because they do not meet the criteria to be considered as Baseline Reliability Projects, New Transmission Access Projects, Market Efficiency Projects, or Multi-Value Projects.

Of the 51 projects, that are being recommended to be included in MTEP20, seven have an estimated cost of less than \$1 million, 17 have an estimated cost between \$1 million and \$5 million, and the remaining 27 projects have an estimated cost greater than \$5 million (Figure 4.4-2).

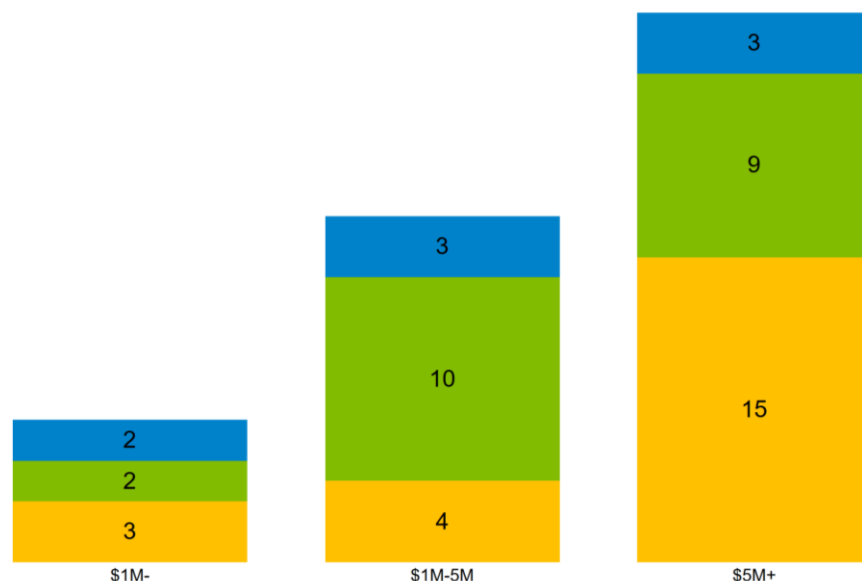


Figure 4.4-2: Cost category by Project types (Project data as of September 15, 2020)

The majority of the projects in the MISO South planning region are expected to go in service in the next three years (figure 4.4-3). There is one project that went into service in 2019. This is a GIP and the Generator Interconnection Agreement was finalized after the deadline for inclusion in the MTEP19.

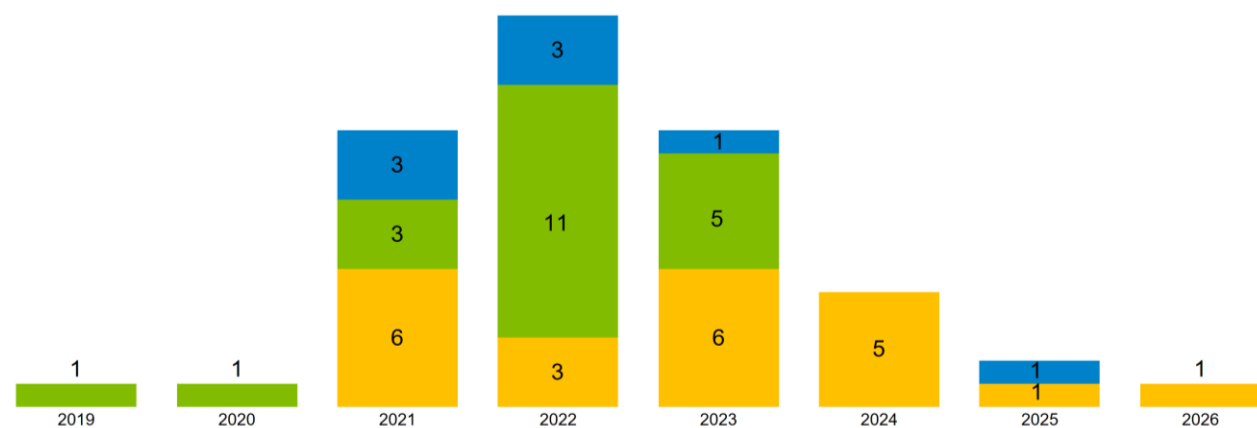


Figure 4.4- 3: Project in-service date by project type (Project data as of September 15, 2020)

In accordance with Attachment FF to the tariff, in the event a Transmission Owner determines that system conditions warrant the urgent development of system enhancements an expedited review of the impacts of the project can be requested. MISO shall use a streamlined approval process for reviewing and approving projects proposed by the Transmission Owner(s) and decisions will be provided to the Transmission Owner within 30 Days of the project's submittal to MISO unless a longer review period is mutually agreed upon. During the MTEP20 cycle, MISO did not receive any projects in the South planning region through the Expedited Project Review (EPR) process.

The ten most expensive projects account for \$375.69 million of the \$578.02 million total recommended projects for the South in MTEP20. The locations of the ten most expensive projects are shown in Figure 4.4-4 and it is seen that they are spread across the southern part of the South planning region. Projects that are blanket expenditures (relays, physical security, etc.) are excluded from this list.

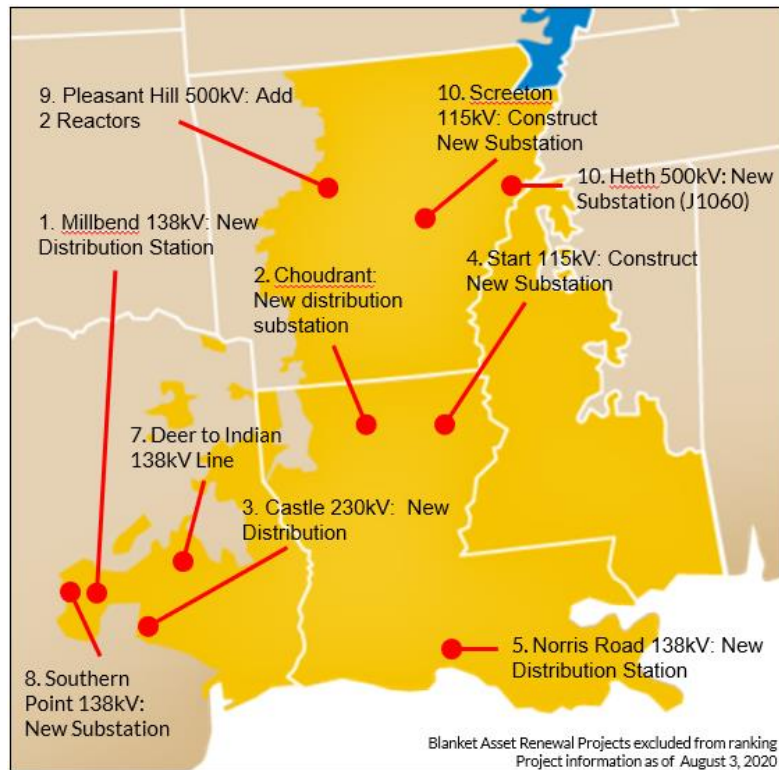


Figure 4.4-4: The ten most expensive projects excluding blanket projects. (Project data as of September 15, 2020)

4.4.1 Arkansas Electric Cooperative Corporation (AECC)

Arkansas Electric Cooperative Corporation proposed two projects at an estimated cost of \$2.1 million targeting MTEP20 Appendix A. After MISO's independent reliability analysis of the AECC area, two of these projects were recommended by MISO to be included in MTEP20 Appendix A at an estimated cost of \$2.1 million. Of these projects, one is a Baseline Reliability Project and one is an Other Type Project. All the projects expected in-service dates and estimated costs are provided as of September 15, 2020.

Baseline Reliability Projects

According to Attachment FF of the MISO Tariff, "Baseline Reliability Projects are Network Upgrades identified in the base case as required to ensure that the Transmission System is in compliance with applicable national Electric Reliability Organization (ERO) reliability standards and reliability standards adopted by Regional Reliability Organizations and applicable within the Transmission Provider Region."

Project 18805- Hickman North Cap Bank

Project Description

This project involves constructing a 60 Mvar capacitor bank that is needed to mitigate low voltages for an N-1 contingency for loss of the strongest source into Blytheville North 161 kV switching station which feeds Hickman North/NUCOR II. The project's estimated cost is \$2.0 million and expected in-service date is June 1, 2021.

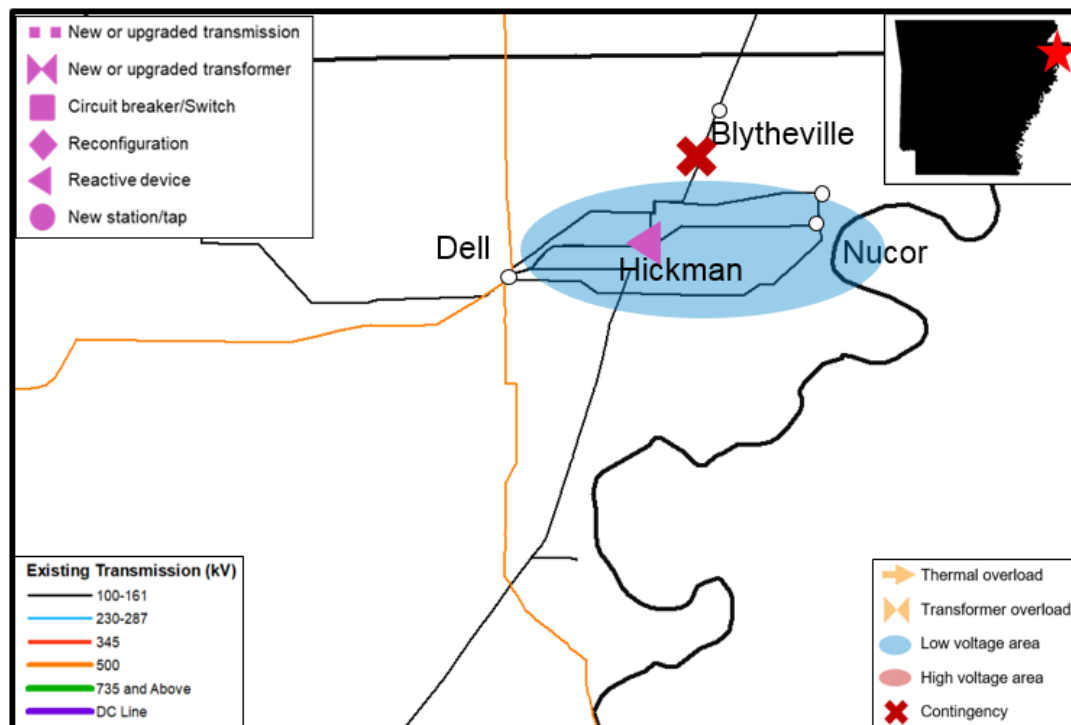


Figure 4.4.1-2: Geographic transmission map of project area

Project Need

Thermal violations caused by the single line outage from Blytheville are seen in the MTEP20 2022 Summer model (Table 4.4.1-1).

Cont. Type	Limiting Element	Voltage Limit (pu)	Pre-Project Voltage (pu)	Post-Project Voltage (pu)
P1	NUCOR Substation	0.92	0.9182	0.9905
P1	Hickman Substation	0.92	0.9191	0.9946

Table 4.4.1-1: P18805 Voltage performance drivers

Alternatives Considered

New configurations of open points and new lines were considered for construction. The cap bank is the preferred solution.

Other Projects

Other projects do not meet the criteria to be considered as Baseline Reliability Projects (BRP), New Transmission Access Projects, Market Efficiency Projects, or Multi-Value Projects. Other projects may include projects to satisfy Transmission Owner and/or state and local planning criteria other than NERC or regional reliability standards, interconnect new Loads, relocate transmission facilities, address aging transmission infrastructure, replace problematic transmission plant, improve operational performance or address other operational issues, address service reliability issues with end-use consumers, improve aesthetics including but not limited to undergrounding overhead transmission facilities, address localized economic issues, and address other miscellaneous localized needs. For the convenience of the reader, tables of project information are broken down by four general categories of project drivers but note that these four drivers are not defined in the MISO Tariff.

Projects Driven by Other Local Needs

The following projects are proposed to due to other local needs.

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
16884	Helena Industrial Breaker Removal	Remove inline breaker at the AECC Helena Industrial Switching Station. The breaker is no longer needed after reconfigurations in the area.	June 1, 2021	\$0.15

4.4.2 City of Alexandria (AXLA)

The City of Alexandria proposed one project at an estimated cost of \$1.25 million targeting MTEP20 Appendix A. After MISO's independent reliability analysis of the AXLA area, this project was recommended by MISO to be included in MTEP20 Appendix A at an estimated cost of \$1.25 million. This project is an Other Project. The project's expected in-service date and estimated costs are provided as of September 15, 2020.

Other Projects

Other projects do not meet the criteria to be considered as Baseline Reliability Projects (BRP), New Transmission Access Projects, Market Efficiency Projects, or Multi-Value Projects. Other projects may include projects to satisfy Transmission Owner and/or state and local planning criteria other than NERC or regional reliability standards, interconnect new Loads, relocate transmission facilities, address aging transmission infrastructure, replace problematic transmission plant, improve operational performance or

address other operational issues, address service reliability issues with end-use consumers, improve aesthetics including but not limited to undergrounding overhead transmission facilities, address localized economic issues, and address other miscellaneous localized needs. For the convenience of the reader, tables of project information are broken down by four general categories of project drivers but note that these four drivers are not defined in the MISO Tariff.

Projects Driven by Age and Condition

The following projects are proposed to replace aging or degraded equipment.

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
19408	Sterkx – Willow Glen 138 kV Reconductor	Line re-conductor driven by aging infrastructure. The line was originally built in 1967, which makes the conductor about 53 years old.	June 1st, 2021	\$1.25

4.4.3 CLECO Power LLC (CLEC)

Cleco Power (CLEC) did not submit any new projects for MTEP20.

4.4.4 Cooperative Energy (SMEPA)

Cooperative Energy proposed seven projects at an estimated cost of \$17.26 million targeting MTEP20 Appendix A. After MISO's independent reliability analysis of the Cooperative Energy area, seven of these projects were recommended by MISO to be included in MTEP20 Appendix A at an estimated cost of \$17.26 million. Of these projects, two are Baseline Reliability Projects, three are Other Projects, and two are Generation projects with signed Generator Interconnection Agreements. All the project's expected in-service dates and estimated costs are provided as of September 15, 2020.

Baseline Reliability Projects

According to Attachment FF of the MISO Tariff, "Baseline Reliability Projects are Network Upgrades identified in the base case as required to ensure that the Transmission System is in compliance with applicable national Electric Reliability Organization (ERO) reliability standards and reliability standards adopted by Regional Reliability Organizations and applicable within the Transmission Provider Region."

Project 18224 - Batesville 161 kV Bus Reconfigure

Project Description

This project involves reconfiguring the Batesville 161 kV substation so that the Batesville CE – Batesville TVA tie lines do not share a common breaker. The project's estimated cost is \$0.91 million and expected in-service date is June 1, 2022.

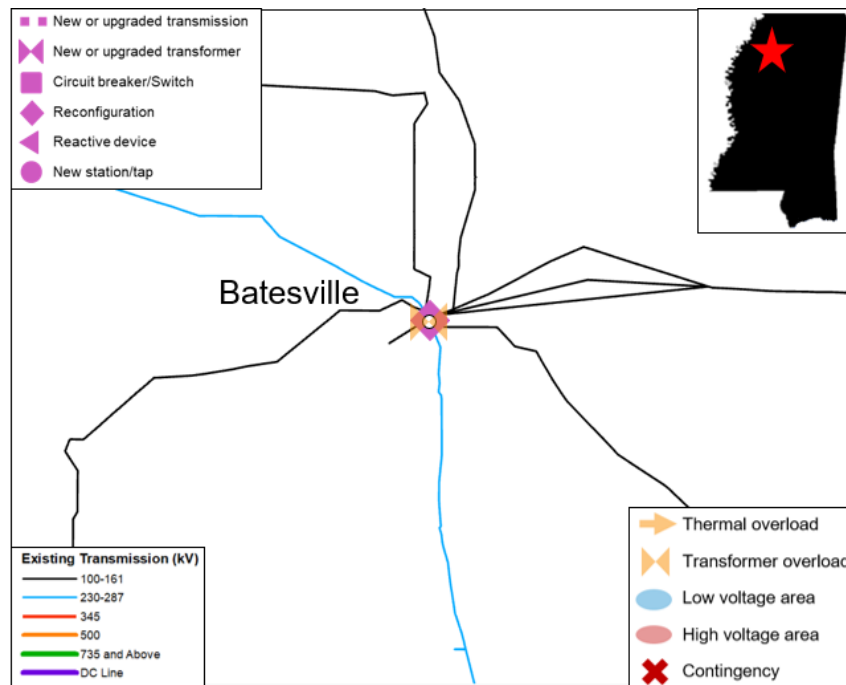


Figure 4.4.4-1: P18224 Geographic transmission map of project area

Project Need

The Entergy Batesville 230/115 kV transformer becomes overloaded to 125% in the year 2022 for the NERC defined P2-3 contingency event at Cooperative Energy's Batesville 161 kV substation. Reconfiguring the Cooperative Energy Batesville 161 kV substation so that the Batesville CE – Batesville TVA tie lines do not share a common breaker will result in this P2-3 contingency event no longer being valid (Table 4.4.4-1).

Cont. Type	Limiting Element	Rating (MVA)	Pre-Project Loading (%)	Post-Project Loading (%)
P2.3	Entergy Batesville 230/115 kV Transformer	379	125	N/A

Table 4.4.4-1: P18224 Thermal loading drivers

Alternatives Considered

Upgrading the Entergy 115 kV bus to increase the rating of the transformer to 392 MVA and establish a two-hour emergency rating was considered as an alternative. However, the loadings in the year 2025 were 130% of the new rating and exceeded the newly established emergency rating and the reconfigure of the Batesville 161 kV provided a more reliable solution.

Project 18226 – Refuge 115 kV Equipment Upgrade

Project Description

Upgrade switches at the Cooperative Energy owned Refuge 115 kV GOAB with higher amp rated switches to achieve a 319 MVA rating on the Andrus – Refuge 115 kV line. The project's estimated cost is \$25 thousand and expected in-service date is March 1, 2021.

Project Need

The Andrus – Refuge 115 kV line becomes overloaded for a NERC defined P1-2 contingency. This overload was identified using the Entergy 2019 Summer model after the line was de-rated due to

updated information regarding the switches at the Refuge 115 kV station. The loading of this line is very dependent on the output of the Gerald Andrus generation station and overloads are seen when output exceeds 725 MW. In this model, Gerald Andrus is producing 735 MW and the loading on the line reaches 103 percent (Table 4.4.4-2).

Cont. Type	Limiting Element	Rating (MVA)	Pre-Project Loading (%)	Post-Project Loading (%)
P1.2	Andrus – Refuge 115 kV Line	239	103%	77%

Table 4.4.4-2: P18226 Thermal loading drivers

Other Projects

Other projects do not meet the criteria to be considered as Baseline Reliability Projects (BRP), New Transmission Access Projects, Market Efficiency Projects, or Multi-Value Projects. Other projects may include projects to satisfy Transmission Owner and/or state and local planning criteria other than NERC or regional reliability standards, interconnect new Loads, relocate transmission facilities, address aging transmission infrastructure, replace problematic transmission plant, improve operational performance or address other operational issues, address service reliability issues with end-use consumers, improve aesthetics including but not limited to undergrounding overhead transmission facilities, address localized economic issues, and address other miscellaneous localized needs. For the convenience of the reader, tables of project information are broken down by four general categories of project drivers but note that these four drivers are not defined in the MISO Tariff.

Projects Driven by Local Reliability

The following projects are driven by local planning criteria without Electric Reliability Organization (ERO) standard applicability or by operational flexibility.

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
18180	Missionary 230 kV Reactor Bank	During periods of low load and minimal system generation, high voltage is experienced at Cooperative Energy substations. To mitigate the high voltage this project adds a new bay and breaker at Missionary 230 kV substation and a new reactor bank (single stage 30 Mvar) at Missionary 230 kV Substation.	March 1 2021	\$2.06
18182	Cole Road 69 kV Capacitor Bank	In order to provide better system voltage support to a heavily loaded area of the system this project installs a new 4.8 Mvar capacitor bank at Cole Road 69 kV substation.	March 1, 2021	\$0.66

Projects Driven by Load Growth

The following projects are proposed to replace aging or degraded equipment.

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
17086	Brandon Area Loop Project	A member cooperative has requested loop service. This project will tap the East Whitfield to South Brandon line with a 115 kV GOAB and build a new 115 kV transmission line from the South Brandon tap to the East Brandon delivery point GOAB.	December 1, 2022	\$9.95

Generator Interconnection Projects

According to Attachment FF of the MISO Tariff, “Generator Interconnection Projects are New Transmission Access Projects that are associated with interconnection of new, or increase in generating capacity of existing, generation.” The following Generator Interconnection Projects (GIPs) have been evaluated through the Generator Interconnection queue and the associated Generator Interconnection Agreements have been signed.

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
17524 ⁵	Batesville 161 kV Ties Reconductor	The Cooperative Energy Batesville to Batesville TVA 161 kV tie lines were identified as needed transmission upgrades for the following MISO DPP-2017-AUG South projects: J663, J834, J866, and J907. Cooperative Energy's Network Upgrades shall consist of reconductoring both Batesville to Batesville TVA 161 kV tie lines with Aluminum Conductor Composite Reinforced (ACCR) conductor.	June 1, 2021	\$0.65
17004	Benndale Recip Project	The Benndale combustion turbine is at the end of its useful life and is no longer competitive in Cooperative Energy's generation fleet. Therefore, the unit has been retired. Additionally, the Benndale switching station has older oil circuit breakers and relaying that needs to be replaced. The existing Benndale combustion turbine facility will have two 11MW RICE gas-fired engines installed with a new GSU. The Benndale switching station will be rebuilt and all oil circuit breakers will be replaced with gas circuit breakers and updated relaying package.	December 25, 2019	\$3.01

4.4.5 East Texas Electric Cooperative (ETEC)

East Texas Electric Cooperative proposed one project at an estimated cost of \$33.3 million targeting MTEP20 Appendix A. After MISO's independent reliability analysis of the ETEC area, their one project was recommended by MISO to be included in MTEP20 Appendix A at an estimated cost of \$33.3 million. This project is a baseline reliability project. The project's expected in-service date and estimated costs are provided as of September 15, 2020.

Baseline Reliability Projects

According to Attachment FF of the MISO Tariff, “Baseline Reliability Projects are Network Upgrades identified in the base case as required to ensure that the Transmission System is in compliance with applicable national Electric Reliability Organization (ERO) reliability standards and reliability standards adopted by Regional Reliability Organizations and applicable within the Transmission Provider Region.”

⁵ GIP project submitted after February 1, 2020 (modeling cutoff date) and not included in the MTEP20 models. This will be modeled in the following MTEP cycle.

Project 19665 - Deer to Indian 138 kV line

Project Description

This project involves constructing approximately 16-19 miles of new 138 kV transmission line from Deer substation to Indian substation. The project will consist of a 795 ACSR Drake single circuit transmission line from Deer substation to Indian Substation, or a 795 ACSR Drake single circuit transmission line from Deer Substation to a new 3 breaker ring bus switching station (cut into existing Indian to Gulf Livingston transmission line) approximately one mile to the west of Indian substation. This is contingent on the ability to acquire ROW for the transmission project. The project's estimated cost is \$33.34 million and expected in-service date is June 1, 2025.

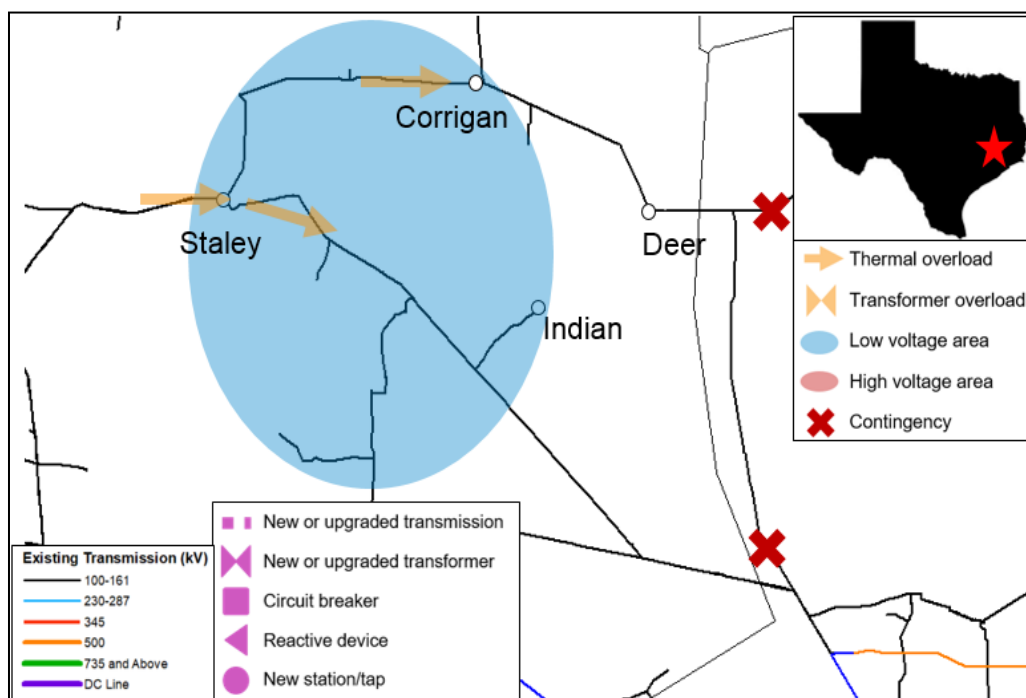


Figure 4.4.5-3: Geographic transmission map of project area

Project Need

The project is needed to mitigate violations due to the loss of two simultaneous lines in the area. (Table 4.4.5-1) or (Table 4.4.5-2).

Cont. Type	Limiting Element	Rating (MVA)	Pre-Project Loading (%)	Post-Project Loading (%)
P6	4CORIGAN!- 4MOSCOW 1	102	111.98%	16.00%
P6	4L111T411- 4MOSCOW 1	102	129.09%	31.00%
P6	4STALEY%- 4MAGROVE 1	112	131.13%	41.00%
P6	4MAGROVE- 4L111T411 1	102	139.71%	41.00%

Table 4.4.5-1: 19665 Thermal loading drivers

Cont. Type	Limiting Element	Voltage Limit (pu)	Pre-Project Voltage (pu)	Post-Project Voltage (pu)
P6	4DOUCETT!	0.92	0.815	0.9746
P6	4WARREN!	0.92	0.800	0.9631

P6	4L111T411	0.92	0.890	0.9866
P6	4CORIGAN!	0.92	0.858	0.9810
P6	4SHECORR!	0.92	0.856	0.9805
P6	4STRYKER	0.92	0.853	0.9799
P6	4CAMDEN	0.92	0.846	0.9784
P6	4MAGROVE	0.92	0.924	0.9946
P6	4MOSCOW	0.92	0.863	0.9811
P6	4FRED_SW	0.92	0.802	0.9643
P6	4FRED	0.92	0.799	0.9625
P6	4URLAND!	0.92	0.806	0.9670
P6	4WOODVILLE	0.92	0.808	0.9690
P6	4DEER 1!	0.92	0.822	0.9780
P6	4DOGWOOD	0.92	0.805	0.9664
P6	4KICKAPO	0.92	0.890	0.9864

Table 4.4.5-2: P19665 Voltage performance drivers

Alternatives Considered

Considering the longevity of the system and operational flexibility, this is the preferred solution compared to other mitigation options allowed by the standard.

4.4.6 Entergy Arkansas LLC (EAL)

Entergy Arkansas proposed 18 projects at an estimated cost of \$147.74 million targeting MTEP20 Appendix A. After MISO's independent reliability analysis of the EAL area, 18 of these projects were recommended by MISO to be included in MTEP20 Appendix A at an estimated cost of \$147.74 million. Of these projects, 3 are Baseline Reliability Projects, 3 are Other Projects, and 12 are Generation projects with signed Generator Interconnection Agreements. All the projects' expected in-service dates and estimated costs are provided as of September 15, 2020.

Baseline Reliability Projects

According to Attachment FF of the MISO Tariff, "Baseline Reliability Projects are Network Upgrades identified in the base case as required to ensure that the Transmission System is in compliance with applicable national Electric Reliability Organization (ERO) reliability standards and reliability standards adopted by Regional Reliability Organizations and applicable within the Transmission Provider Region."

The following projects are driven by Baseline reliability. However, they are not driven by steady state standards.

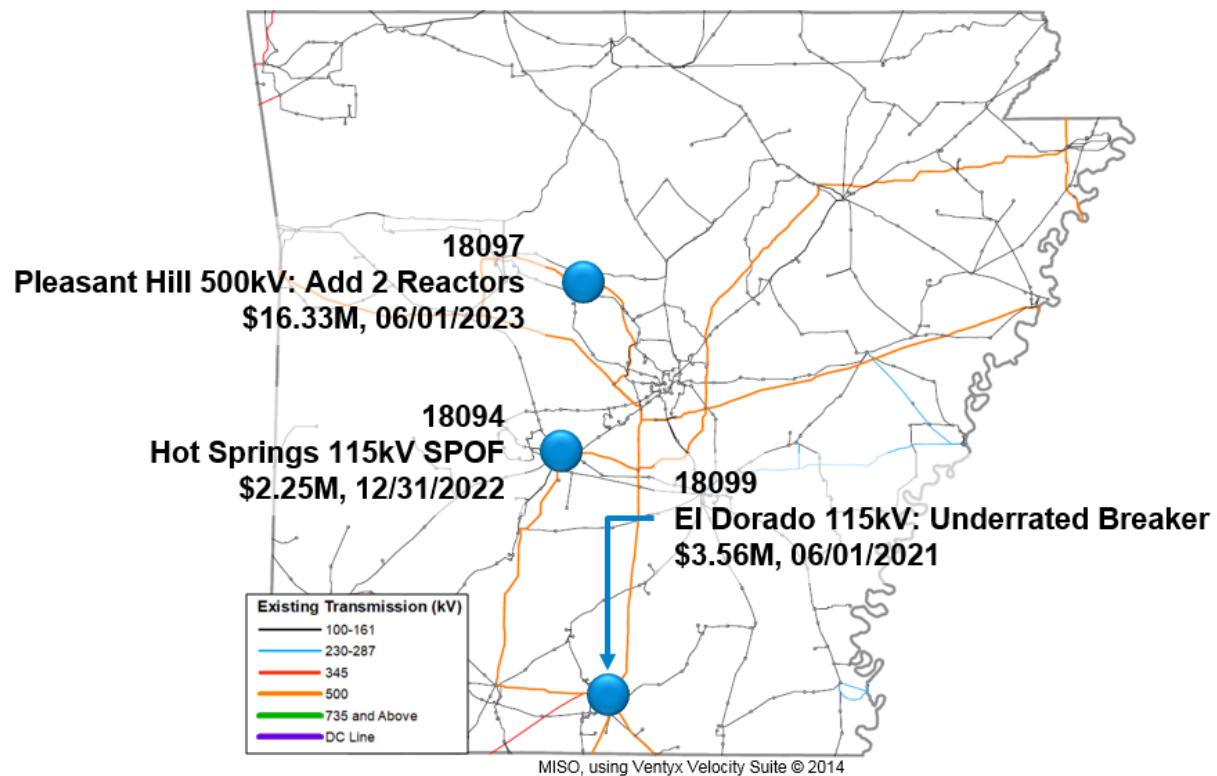


Figure 4.4.6-4: Location of BRP project submitted by EAL

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
18094	Hot Springs 115 kV SPOF	Install dual primary protection on the 115 kV bus schemes at Hot Springs. There is the potential for ~350 MW of load at risk around Hot Springs and ~150 MW of generator instability (Rommel, Blakely, Carpenter) to occur for a SLG fault on the Hot Springs 115 kV bus with non-redundant relay failure.	December 31, 2022	\$2.56
18097	Pleasant Hill 500 kV: Add 2 Reactors	Add 2 100 Mvar reactors at Pleasant Hill 500 kV bus. There is the potential for approximately 2000 MW of generation capacity to go unstable following a SLG stuck breaker.	June 1, 2023	\$16.33
18099	El Dorado 115 kV: Underrated Breakers	Replace the 115 kV breaker(s) at El Dorado with 63 kA fault interrupting capability. B5820, B6258, B6758, B7158, and B7658 will be replaced. MTEP20 short circuit analysis indicated five underrated breakers at the existing El Dorado EHV 115 kV substation.	June 1, 2021	\$3.56

Other Projects

Other projects do not meet the criteria to be considered as Baseline Reliability Projects (BRP), New Transmission Access Projects, Market Efficiency Projects, or Multi-Value Projects. Other projects may include projects to satisfy Transmission Owner and/or state and local planning criteria other than NERC or regional reliability standards, interconnect new Loads, relocate transmission facilities, address aging transmission infrastructure, replace problematic transmission plant, improve operational performance or

address other operational issues, address service reliability issues with end-use consumers, improve aesthetics including but not limited to undergrounding overhead transmission facilities, address localized economic issues, and address other miscellaneous localized needs. For the convenience of the reader, tables of project information are broken down by four general categories of project drivers but note that these four drivers are not defined in the MISO Tariff.

Projects Driven by Local Reliability

The following projects are driven by local planning criteria without Electric Reliability Organization (ERO) standard applicability or by operational flexibility.

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
18100	Danville 161 kV: Install Cap Bank	Install a 24 Mvar cap bank at Danville 161 kV. This is need due to low voltage in the City of Dardanelle and Danville areas for the loss of SPA Dardanelle Dam – City of Dardanelle 161 kV & North Magazine – Blue Mountain 161 kV line sections.	June 1 2023	\$1.76

Projects Driven by Load Growth

The following projects are proposed to replace aging or degraded equipment.

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
18101	Hot Springs Plemmons 115 kV: Construct new substation	Build a new 115 kV substation with a single 40 MVA 115/13.8 kV LTC transformer. This is needed to better serve the area around Hot Springs, AR.	June 1, 2024	\$10.90
18103	Screeton 115 kV: Construct new substation	Build a new 115 kV substation with a single 40 MVA 115/13.8kV LTC transformer between Hazen and Carlisle 115 kV substations near the town of Screeton. This will help more reliably serve the customers in the area.	June 1, 2024	\$15.13

Generator Interconnection Projects

According to Attachment FF of the MISO Tariff, “Generator Interconnection Projects are New Transmission Access Projects that are associated with interconnection of new, or increase in generating capacity of existing, generation.” The following Generator Interconnection Projects (GIPs) have been evaluated through the Generator Interconnection queue and the associated Generator Interconnection Agreements have been signed.

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
18725	Searcy Price 161 kV: Expand 161 kV Station (J893)	The Searcy Price 161 kV substation will be upgraded to incorporate a new POI for J893, a 100 MW photovoltaic generator project. Transmission Network Upgrades for this interconnection include two new 161 kV circuit breakers, 161 kV disconnect switches, expanding the existing 161 kV	August 1, 2021	\$8.05

		operating bus, substation yard, control house, and additional relay control panels.		
18726	Locust Creek 161 kV: Construct Switching Station (J919)	Construct a new 3 breaker switching station 2.62 miles north east of the Newport Air Base substation on the L962 161 kV Newport Air Base - Cash 161 kV transmission line section. This will be used as the point of interconnection for the 180 MW solar photovoltaic generator project J919.	February 1, 2023	\$13.08
18727	Marston 161 kV: Cut in Switching Station (J944)	Entergy will cut in a new 3 breaker ring bus that will be built by the Interconnection Customer on the Portageville - New Madrid 161 kV line section. This bisection will occur near the tenth structure from the New Madrid substation. Approximately 4.52 miles of ADSS fiber shall be installed between Marston and New Madrid. At the remote end Hayti South, the OCB B9150 will be replaced with a SF6 gas circuit breaker rated at 3000 A.	October 1, 2022	\$4.38
18946	Kuhn Road 161 kV: Expand Station and Line Rebuild (J934)	The Kuhn Road 161 kV bus will be expanded to provide a point of interconnection for J934 solar generating facility. Upgrades will include two 161 kV dead end structures, two 161 kV 3000 A vertical break disconnects switches, one 161 kV 3000 A gas circuit breaker, three 161 kV rated surge arresters, three metering accuracy coupling capacitor voltage transformers, revenue accuracy meter, relaying panel and revenue metering panel. Three surge arresters on the incoming line L940 will be installed. Relay settings updates are required for the 161 kV bus differential relay, and the Kuhn Road - Parkin and Kuhn Road - Ebony 161 kV transmission lines. The existing Ebony - Kuhn Road 161 kV transmission line will be rebuilt to support 666.6 kcmil ACSR Flamingo conductor at a minimum rating of 229 MVA.	June 1, 2022	\$11.16
19005	Ritchie 230 kV: Expand substation (J834)	The J834 generator interconnection project will require network upgrades at the Entergy Arkansas Ritchie 230 kV substation. This will include new interconnection facilities that will consist of two 230 kV dead end structures, one 230 kV 3000A gas circuit breaker, two 230 kV vertical break disconnect switches, 230 kV surge arrestors, CCVT's, metering CT's, and all associated relaying panels. There will be 0.4 miles of 666.6 ACSR Flamingo conductor, and 0.3 miles of 7#7 alumo-	November 15, 2021	\$5.51

		weld conductor installed for the interconnection. Relay settings will be required in all remote end locations as well as the Autotransformer, Transfer Breaker, and the East and West Bus diff.		
19065	Ritchie 230 kV: Add POI (J663)	This project will add a point of interconnection to the existing Ritchie 230 kV substation for J663. TOIF upgrades include two 230 kV dead end structures, two 230 kV 3000A disconnect switches, one 230 kV 3000A gas circuit breaker, three surge arresters, metering CT's, and relay panels. Relay settings updates will be required to all transmission lines, auto transformer, transfer breaker, and bus diffs where needed.	May 1, 2022	\$6.11
19068	Flat Fork 230 kV: Cut in Switching Station (J907)	Entergy Arkansas will construct new surge arrestors, protective relays, RTU, and other supporting equipment at Ritchie 230 kV. There will be equipment upgrades performed at Brinkley East substation such as disconnect switches, insulators, station service, protective relays, and other supporting equipment. The Brinkley East - Aubrey 230 kV transmission line cut in will consist of two new steel dead end structures, new steel tangent structures, and new monopole structures. The Interconnection Customer will construct the three-breaker ring bus substation.	June 1, 2022	\$3.09
19725	Plum Point Network Upgrades (J1146)	Network Upgrades associated with GIA J1146 for 10 MW's of incremental NRIS at Plum Point Generation Facility. This includes reviewing and updating relay settings at San Souci 500 kV Switch Station, Driver 500 kV substation, and Dell 500 kV substation.	October 10,2020	\$0.27
19726	Bob White 161 kV: Cut In Switching Station (J1007)	This project will cut in the J1007 161 kV substation on the Marion - Marked Tree 161 kV substation. J1007 is a 180 MW solar generating facility with an executed generator interconnection agreement.	June 1, 2023	\$2.28
19727	Falls 161 kV: New Cut In (J1125)	The J1125 135 MW solar project will interconnect to the Wynne South - AECC Wynne East 161 kV line section. The cut-in will comprise of four new steel dead-end structures one new steel tangent structure and the removal of two existing structures. New conductors and shield wire will also be installed. Relay panels at the remote end substations of Moses and Parkin will be replaced.	March 1, 2023	\$2.87
19728	Driver 230 kV: POI Existing Substation (J1155)	The project involves the expansion of the Driver 230 kV substation to interconnect	September 1, 2022	\$2.95

		J1155 solar generation project. Upgrades will include installing a new 230 kV breaker, CCVT's, conductor for the cut-in, surge arrestors, and CT's.		
19767	Heth 500 kV: New Substation (J1060)	The Heth 500 kV substation is to cut in the J1060 450 MW solar facility. This includes a 500 kV cut into the exiting West Memphis - Keo 500 kV line. A new 500 kV interconnection substation will be installed. Remote end upgrades at KEO and West Memphis include removing line traps, line tuners, existing relays, and replacing them with new relay panels.	May 1, 2023	\$38.04

4.4.7 Entergy Louisiana LLC (ELL)

Entergy Louisiana proposed ten projects at an estimated cost of \$170.3 million targeting MTEP20 Appendix A. After MISO's independent reliability analysis of the ELL area, ten of these projects were recommended by MISO to be included in MTEP20 Appendix A at an estimated cost of \$170.3 million. Of these projects, one is a Baseline Reliability Project, six are Other Projects, and three are Generation projects with signed Generator Interconnection Agreements. All the projects' expected in-service dates and estimated costs are provided as of September 15, 2020.

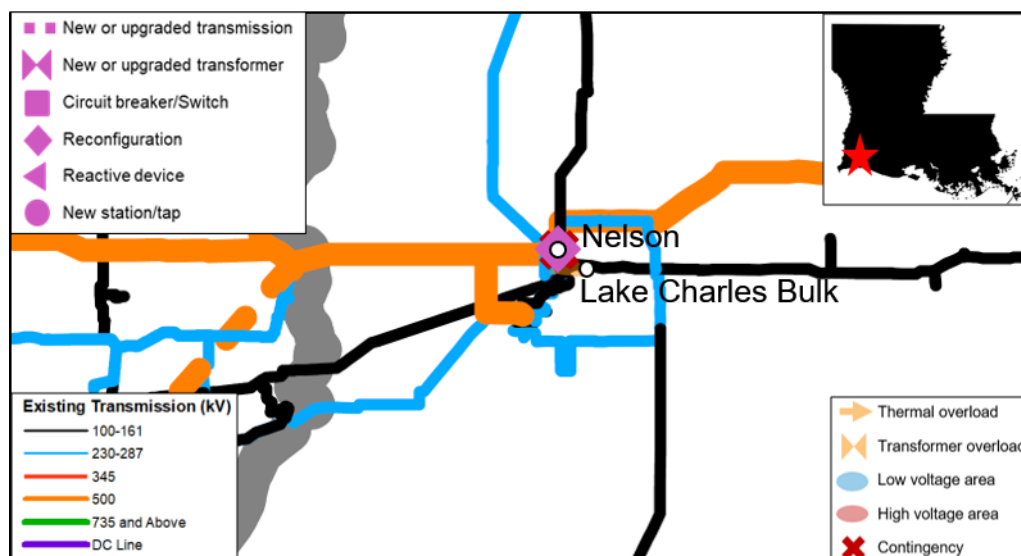
Baseline Reliability Projects

According to Attachment FF of the MISO Tariff, "Baseline Reliability Projects are Network Upgrades identified in the base case as required to ensure that the Transmission System is in compliance with applicable national Electric Reliability Organization (ERO) reliability standards and reliability standards adopted by Regional Reliability Organizations and applicable within the Transmission Provider Region."

Project 15645- Nelson 138 kV: Install transmission breakers

Project Description

This project involves re-terminating the Nelson – Lake Charles Bulk 138 kV L-654 at the open bay east of Nelson to Deridder bay and moving L-698 to the bay L-654 currently occupies. Breaker 7920 and one new IPO breaker (3000A continuous rating, 80 kA interrupting rating) would be needed to re-terminate L-654 in double-bus-double-breaker scheme. The project's estimated cost is \$10.6 million and expected in-service date is June 1, 2022.



MISO, using Ventyx Velocity Suite © 2014

Figure 4.4.7-1: P15645 Geographic transmission map of project area

Project Need

The project was originally identified in the MTEP19 analysis with an overloading up to 106% starting in the 2021 Summer model for the internal fault at breaker 18590 in the Nelson 138 kV substation.

Cont. Type	Limiting Element	Rating (MVA)	Pre-Project Loading (%)	Post-Project Loading (%)
P23	Nelson – Lake Charles Bulk North 138 kV	216	95%	59%

Table 4.4.7-1: P15645 Thermal loading drivers

Other Projects

Other projects do not meet the criteria to be considered as Baseline Reliability Projects (BRP), New Transmission Access Projects, Market Efficiency Projects, or Multi-Value Projects. Other projects may include projects to satisfy Transmission Owner and/or state and local planning criteria other than NERC or regional reliability standards, interconnect new Loads, relocate transmission facilities, address aging transmission infrastructure, replace problematic transmission plant, improve operational performance or address other operational issues, address service reliability issues with end-use consumers, improve aesthetics including but not limited to undergrounding overhead transmission facilities, address localized economic issues, and address other miscellaneous localized needs. For the convenience of the reader, tables of project information are broken down by four general categories of project drivers but note that these four drivers are not defined in the MISO Tariff.

Projects Driven by Local Reliability

The following projects are driven by local planning criteria without Electric Reliability Organization (ERO) standard applicability or by operational flexibility.

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
13938	Cheniere 115 kV: Install transmission breakers	Install transmission breakers at the existing Cheniere 115 kV substation, which is located on the Frostkraft - Sterlington 115 kV transmission circuit.	December 1, 2023	\$5.39
15568	Oak Grove 138 kV: Add Breaker	Install transmission breakers at the existing Oak Grove 138 kV substation, located on the Willow Glen to Gonzales 138 kV transmission circuit.	December 1, 2022	\$2.00
18136	Scott to Lafayette 69 kV: Upgrade Station Equipment	Upgrade all station equipment to achieve a through path rating of 588 Amps.	June 1, 2021	\$0.05

Projects Driven by Load Growth

The following projects are proposed to replace aging or degraded equipment.

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
13925	Choudrant: New distribution substation	Construct a 230 kV distribution substation named Choudrant in southern Lincoln parish near the town of Choudrant and construct a 230 kV transmission line from the new substation that terminates into the existing Mt Olive substation.	June 1, 2023	\$56.23
18227	Norris Road 138 kV: New Distribution Station	New distribution station to be cut into the CLECO Youngsville to Segura 138 kV line.	June 1, 2026	\$39.20
18233	Start 115 kV: Construct New Substation	Build new 115 kV substation named Start along with 2 miles of in-and-out 115 kV line from Alto to new substation site.	June 1, 2025	\$39.51

Generator Interconnection Projects

According to Attachment FF of the MISO Tariff, "Generator Interconnection Projects are New Transmission Access Projects that are associated with interconnection of new, or increase in generating capacity of existing, generation." The following Generator Interconnection Projects (GIPs) have been evaluated through the Generator Interconnection queue and the associated Generator Interconnection Agreements have been signed.

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
18785	J639 Station Interconnection Project	This project is to support the interconnection of solar project J639 on 230 kV line 731 (Addis to Big Cajun).	October 10, 2022	\$1.79
18786	J909 Station Interconnection Project	This project is to support the interconnection of solar facility J909 at the Amite 115 kV station.	February 15, 2022	\$4.25

18787	J697 Interconnection: Expand Oak Ridge 115 kV Substation	This project shall consist of transforming the existing 115 kV yard from single bus to dual bus, expanding the existing substation site, installing a new control house, and installing shielding and light fixtures to accommodate the connection of the Interconnection Customer's transmission line.	October 15, 2022	\$11.31
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4.4.8 Entergy Mississippi LLC (EML)

Entergy Mississippi proposed four projects at an estimated cost of \$23.29 million targeting MTEP20 Appendix A. After MISO's independent reliability analysis of the EML area, all of these projects were recommended by MISO to be included in MTEP20 Appendix A at an estimated cost of \$23.29 million. Of these projects, zero are Baseline Reliability Projects, zero are Other Projects, and four are Generation projects with signed Generator Interconnection Agreements. All the projects' expected in-service dates and estimated costs are provided as of September 15, 2020.

Baseline Reliability Projects

According to Attachment FF of the MISO Tariff, "Baseline Reliability Projects are Network Upgrades identified in the base case as required to ensure that the Transmission System is in compliance with applicable national Electric Reliability Organization (ERO) reliability standards and reliability standards adopted by Regional Reliability Organizations and applicable within the Transmission Provider Region."

Generator Interconnection Projects

According to Attachment FF of the MISO Tariff, "Generator Interconnection Projects are New Transmission Access Projects that are associated with interconnection of new, or increase in generating capacity of existing, generation." The following Generator Interconnection Projects (GIPs) have been evaluated through the Generator Interconnection queue and the associated Generator Interconnection Agreements have been signed.

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
19285	New 230 kV Switching Station Cut-in - J866	Terminal equipment upgrades required to facilitate POI for Generator Interconnection customer J866.	July 1, 2021	\$2.5
19286	Cotton Blossom 230 kV: Construct Switching Station - J830	This project will construct a new 230 kV switching station cut into the Canton South - Yandell Road line to facilitate the Generator Interconnection project J830	November 15, 2021	\$11.37
19287	Webb 115 kV: Substation Expansion - J908	This project will expand the 115 kV Webb substation to facilitate the Generator Interconnection project J908.	June 1, 2023	\$9.42
19785	Twinkletown 230 kV: Expand Substation	This project is needed to interconnect new J962 100 MW generator to the existing Twinkletown 230 kV substation.	June 30, 2022	TBD

4.4.9 Entergy New Orleans LLC (ENO)

Entergy New Orleans (ENO) proposed one project at an estimated cost of \$5.1 million targeting MTEP20 Appendix A. After MISO's independent reliability analysis of the ENO area, this project was recommended by MISO to be included in MTEP20 Appendix A at an estimated cost of \$5.1 million. This project is classified as an Other Project. The project's expected in-service date and estimated cost are provided as of September 15, 2020.

Other Projects

Other projects do not meet the criteria to be considered as Baseline Reliability Projects (BRP), New Transmission Access Projects, Market Efficiency Projects, or Multi-Value Projects. Other projects may include projects to satisfy Transmission Owner and/or state and local planning criteria other than NERC or regional reliability standards, interconnect new Loads, relocate transmission facilities, address aging transmission infrastructure, replace problematic transmission plant, improve operational performance or address other operational issues, address service reliability issues with end-use consumers, improve aesthetics including but not limited to undergrounding overhead transmission facilities, address localized economic issues, and address other miscellaneous localized needs. For the convenience of the reader, tables of project information are broken down by four general categories of project drivers but note that these four drivers are not defined in the MISO Tariff.

Projects Driven by Local Reliability

The following projects are driven by local planning criteria without Electric Reliability Organization (ERO) standard applicability or by operational flexibility.

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
18146	Sherwood Forest 115 kV: Install breakers	Install transmission line breakers at the existing Sherwood Forest 115 kV substation.	December 1st, 2023	\$5.1

4.4.10 Entergy Texas LLC (ETL)

Entergy Texas proposed six projects at an estimated cost of \$172.41 million targeting MTEP20 Appendix A. After MISO's independent reliability analysis of the ETL area, all six of these projects were recommended by MISO to be included in MTEP20 Appendix A at an estimated cost of \$172.41 million. All projects are Other Projects. All the projects' expected in-service dates and estimated costs are provided as of September 15, 2020.

Other Projects

Other projects do not meet the criteria to be considered as Baseline Reliability Projects (BRP), New Transmission Access Projects, Market Efficiency Projects, or Multi-Value Projects. Other projects may include projects to satisfy Transmission Owner and/or state and local planning criteria other than NERC or regional reliability standards, interconnect new Loads, relocate transmission facilities, address aging transmission infrastructure, replace problematic transmission plant, improve operational performance or address other operational issues, address service reliability issues with end-use consumers, improve

aesthetics including but not limited to undergrounding overhead transmission facilities, address localized economic issues, and address other miscellaneous localized needs. For the convenience of the reader, tables of project information are broken down by four general categories of project drivers but note that these four drivers are not defined in the MISO Tariff.

Projects Driven by Local Reliability

The following projects are driven by local planning criteria without Electric Reliability Organization (ERO) standard applicability or by operational flexibility.

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
18259	Lacon 138 kV: Breaker and feeders	Install a 138 kV breaker at Lacon in the bay looking toward the LNG tap. Also included in this project are distribution feeder additions. Adding a breaker will reduce the exposure to customers at Lacon to line faults and will increase the local reliability.	June 1, 2023	\$10.00

Projects Driven by Age and Condition

The following projects are proposed to replace aging or degraded equipment.

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
18251	Alligator Bayou to Fort Worth 69 kV Rebuild	Rebuild and reconductor the Alligator Bayou to Fort Worth 69 kV line. The existing Kolbs - Fort Worth - Alligator Bayou line does not meet existing NESC loading guidelines for NESC Light loading conditions. This line will need to be rebuilt to avoid catastrophic damage from both wind and icing events. Rebuilding the line will help avoid catastrophic structure failure, load loss, and potential liability issues due to failure to meet NESC design criteria. The major components being replaced are the line and a few GOABs.	March 1, 2022	\$10.40
18255	Fort Worth to Kolbs 69 kV Rebuild	Rebuild and reconductor the Fort Worth and Kolbs stations. The existing Kolbs - Fort Worth - Alligator Bayou line does not meet existing NESC loading guidelines for NESC Light loading conditions. This line will need to be rebuilt to avoid catastrophic damage from both wind and icing events. Rebuilding the line will help avoid catastrophic structure failure, load loss, and potential liability issues due to failure to meet NESC design criteria. The major components being replaced are the line and a few GOABs.	June 1, 2023	\$14.10

Projects Driven by Load Growth

The following projects are proposed to replace aging or degraded equipment.

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
13943	Southern Point 138 kV: New Substation	Distribution System Load Growth. 7.0 MVA to start with all load coming from Speedway Sub.	June 1, 2024	\$22.00
15648	Castle 230 kV: New Distribution Station	Distribution load growth. 18.5 MVA to start with all load coming from Dobbin Sub.	June 1, 2024	\$50.51
18248	Millbend 138 kV: New Distribution Station	Area load growth. 56.5 MVA to start with 35.2 MVA from New Caney Sub and 21.3 MVA from Heights Sub.	June 1, 2024	\$65.40

4.4.11 Lafayette Utilities System (LAFA)

Lafayette Utilities System proposed one project at an estimated cost of \$6.4 million targeting MTEP20 Appendix A. After MISO's independent reliability analysis of the LAFA area, this project was recommended by MISO to be included in MTEP20 Appendix A at an estimated cost of \$6.4 million. This project is classified as an Other Project. The project's expected in-service date and estimated cost are provided as of September 15, 2020.

Other Projects

Other projects do not meet the criteria to be considered as Baseline Reliability Projects (BRP), New Transmission Access Projects, Market Efficiency Projects, or Multi-Value Projects. Other projects may include projects to satisfy Transmission Owner and/or state and local planning criteria other than NERC or regional reliability standards, interconnect new Loads, relocate transmission facilities, address aging transmission infrastructure, replace problematic transmission plant, improve operational performance or address other operational issues, address service reliability issues with end-use consumers, improve aesthetics including but not limited to undergrounding overhead transmission facilities, address localized economic issues, and address other miscellaneous localized needs. For the convenience of the reader, tables of project information are broken down by four general categories of project drivers but note that these four drivers are not defined in the MISO Tariff.

Projects Driven by Local Reliability

The following projects are driven by local planning criteria without Electric Reliability Organization (ERO) standard applicability or by operational flexibility.

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
14105	Bonin 69 kV Reconfiguration	A reconfiguration of the Bonin 69 kV station	September 30, 2021	\$6.4

4.5 Project Justifications – West Region

West Region Overview

The MISO West Planning Region consists of 19 Transmission-Ownning members spanning eight states in the upper Midwest. It includes Iowa, Minnesota, Wisconsin, and parts of North Dakota, South Dakota, Montana, Michigan, and Illinois. These transmission owners are:

- American Transmission Company (ATC)
- Cedar Falls Utilities (CFU)
- Central Minnesota Municipal Power Agency (CMMPA)
- City of Ames, IA (COA)
- Great River Energy (GRE)
- ITC Midwest (ITCM)
- MidAmerican Energy Company (MEC)
- Minnesota Municipal Power Agency (MMPA)
- Minnesota Power (MP)
- Missouri River Energy Services (MRES)
- Montana-Dakota Utilities Co. (MDU)
- Muscatine Power and Water (MPW)
- Northwestern Wisconsin Electric (NWECE)
- Otter Tail Power Company (OTP)
- Rochester Public Utilities (RPU)
- Southern Minnesota Municipal Power Agency (SMMPA)
- Wilmar Municipal Utilities (WMU)
- WPPI Energy (WPPI)
- Xcel Energy (Northern States Power, XEL/NSP)

The West planning region contains approximately 27,300 miles of transmission ranging from 57 kV to 500 kV. In the 2022 Summer Peak planning model, the region contains more than 67.2 GW of generation. Installed generation capacity in the region consists mostly of coal, gas and wind. Approximately 33.4 percent (22.5 GW) of the West region's generation capacity is made up of wind units. Major generation centers are located in central North Dakota; the Twin Cities in Minnesota; and the Quad Cities in Iowa and Illinois, with wind generation located in the eastern Dakotas and western Iowa and Minnesota (Figure 4.5-1).

Major load centers are typically found around larger cities in the region: Minneapolis/Saint Paul, Milwaukee and Des Moines. According to the 2022 Summer Peak planning model, the regional load exceeds 42.2 GW. Power generally flows from generation-rich areas in the western portion of the region through Minnesota, Iowa, and Wisconsin, toward large load centers in the east. This is especially prevalent in times of high wind output.

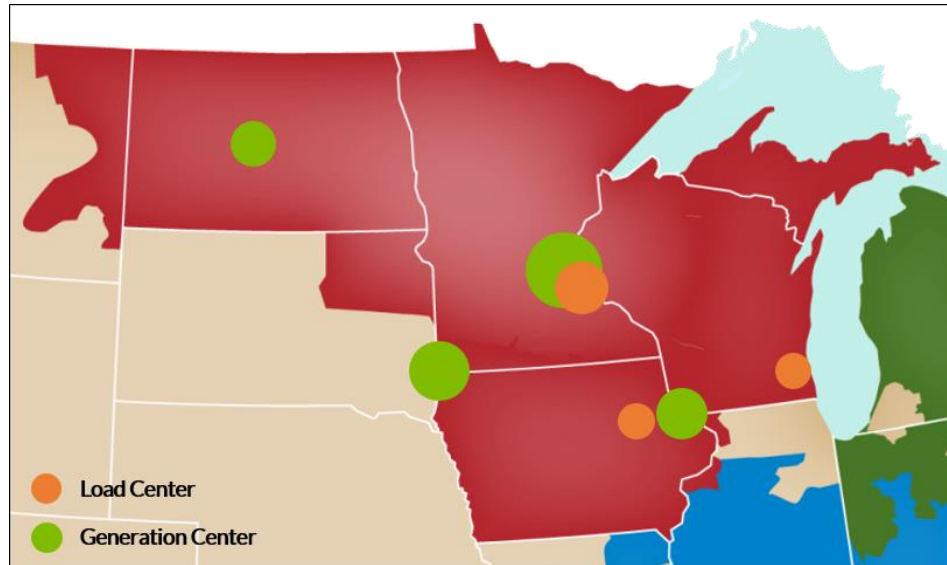


Figure 4.5-1: Generation and load centers in the West Planning Region

For MTEP20, MISO Transmission Planning is recommending 148 projects from the West region for inclusion in Appendix A at an estimated cost of \$1.06 billion. Of these, nine are Baseline Reliability Projects, 50 are Generator Interconnection Projects and the remaining 89 projects are classified as Other because they do not meet the criteria to be considered as Baseline Reliability Projects, New Transmission Access Projects, Market Efficiency Projects, or Multi-Value Projects.

Of the 148 projects, that are being recommended to be included in MTEP, 38 have an estimated cost of less than \$1 million, 53 have an estimated cost between \$1 million and \$5 million, and the remaining 57 projects are estimated to cost greater than \$5 million (Figure 4.2-2).

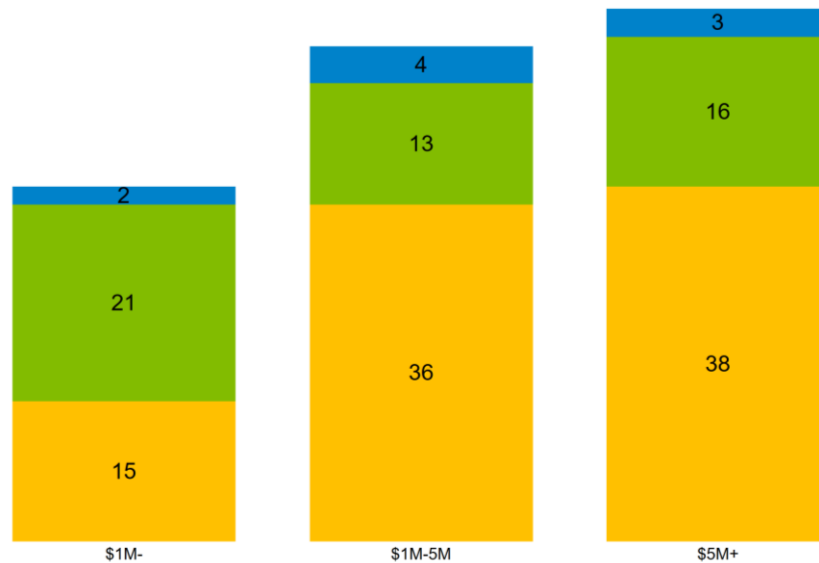


Figure 4.5-2: Project cost category by project type (Project data as of September 15, 2020)

The majority of the projects in the MISO West planning region are expected to go in service in the next three years (Figure 4.5-3). There were a handful of projects that are being approved in 2020 that either already went into service in 2019 or are expecting to go into service in 2020. Many of those were transmission network upgrades (GIP) correlated with generation projects that were justified by a separate Generator Interconnection process. Others fit into a bucket of timing or constructability considerations. Examples of these are the retirement of catastrophic failure devices, load additions which required short time frames to come online, or those that take advantage of outages already scheduled and planned the upgrade accordingly to meet the compliance.

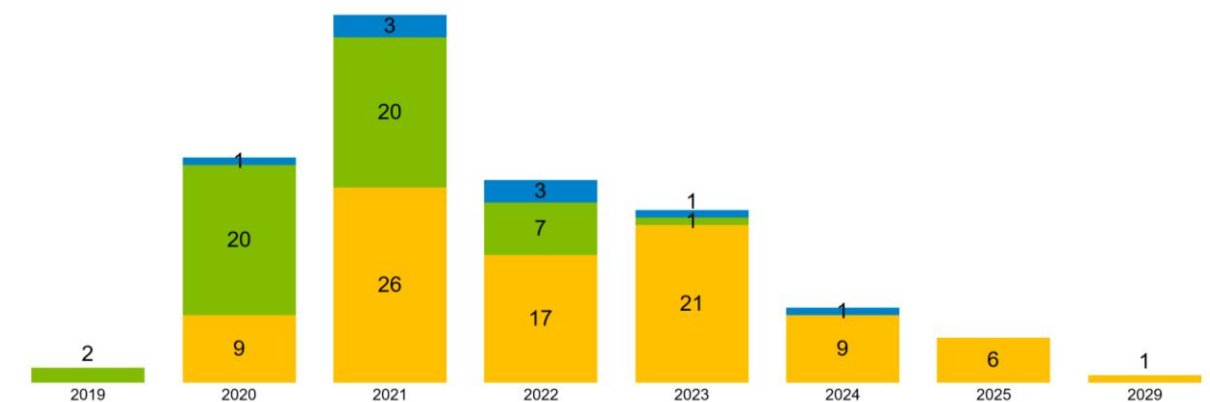


Figure 4.5-3: Project in-service date by project type (Project data as of September 15, 2020)

In accordance with Attachment FF to the tariff, in the event that a Transmission Owner determines that system conditions warrant the urgent development of system enhancements that would be jeopardized unless MISO performs an expedited review of the impacts of the project, MISO shall use a streamlined approval process for reviewing and approving projects proposed by the Transmission Owner(s) so that decisions will be provided to the Transmission Owner within 30 Days of the project's submittal to MISO unless a longer review period is mutually agreed upon. During the MTEP20 cycle, MISO received the following projects through the Expedited Project Review (EPR) process.

1. Project ID 18847 ITCM Big Cedar Load Interconnection
2. Project ID 18848 ATC Blount - East Campus, 69 kV, (6907), Replace Damaged Underground Cable
3. Project ID 19685 OTP Hoot Lake 115/41.6 kV Transformer Replacement

The ten most expensive projects represent \$384.6 million of the \$1.06 billion total recommended projects for the West in MTEP20. The locations of these projects can be seen in Figure 4.5-4 and the investment is spread across the West Planning region. Projects that are blanket expenditures (relays, physical security, etc.) are excluded from this list.

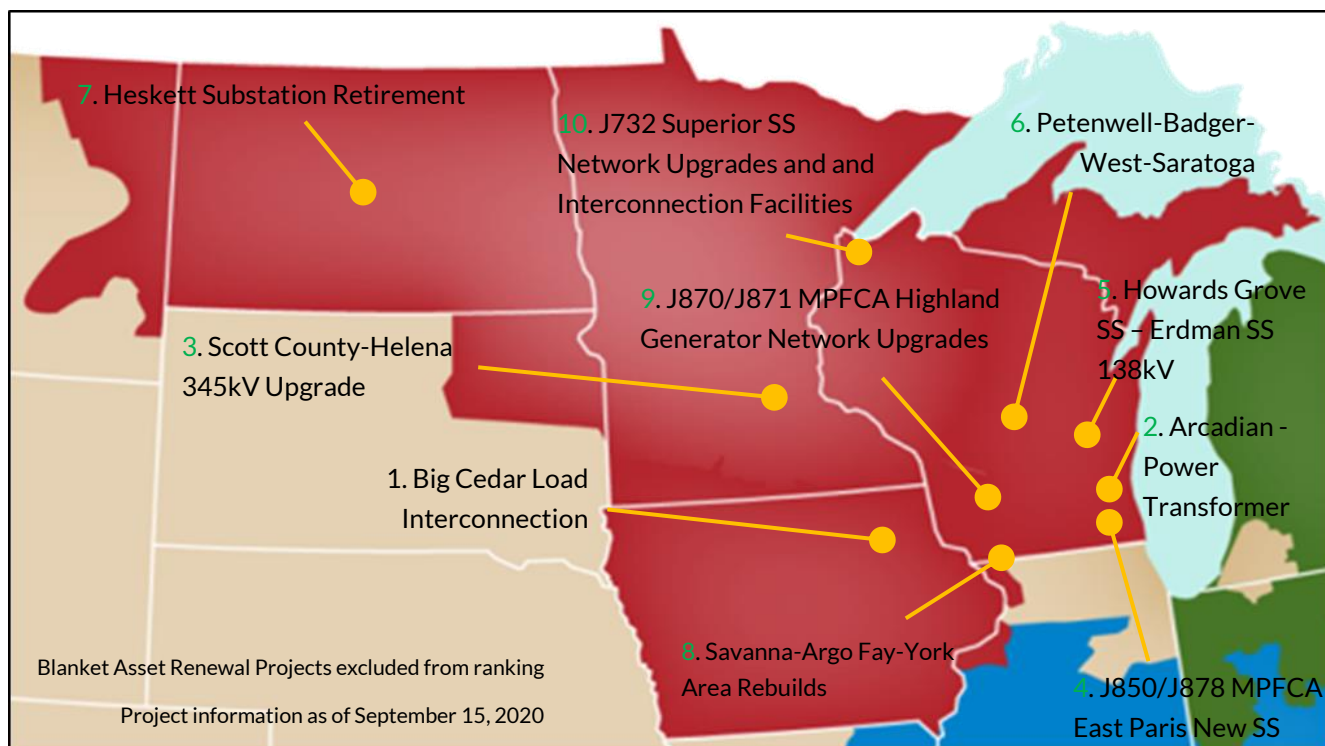


Figure 4.5-4: The ten most expensive projects in the West Planning Region for MTEP20

4.5.1 American Transmission Company (ATC)

American Transmission Company proposed 41 projects at an estimated cost of \$377 million targeting MTEP20 Appendix A. After MISO's independent reliability analysis of the ATC area, 41 of these projects were recommended by MISO to be included in MTEP20 Appendix A at an estimated cost of \$377 million. Of these projects, two are Baseline Reliability Projects, 18 are Other Projects, and 21 are Generation projects with signed Generator Interconnection Agreements. All the projects' expected in-service dates and estimated costs are provided as of September 15, 2020.

Baseline Reliability Projects

According to Attachment FF of the MISO Tariff, "Baseline Reliability Projects are Network Upgrades identified in the base case as required to ensure that the Transmission System is in compliance with applicable national Electric Reliability Organization (ERO) reliability standards and reliability standards adopted by Regional Reliability Organizations and applicable within the Transmission Provider Region."

Project 18017 – Petenwell-Badger-West-Saratoga, (X-43) 138 kV, Line Rebuild

Project Description

This project involves rebuilding 22.7 miles on the 138 kV Line X-43. The limitations of the system result in voltage collapse for forced outages during planned maintenance. X-43 will be rebuilt with a maximum rating of 1030 Amps for all seasons. The project's estimated cost is \$24.5 million, and the expected in-service date is December 31, 2022

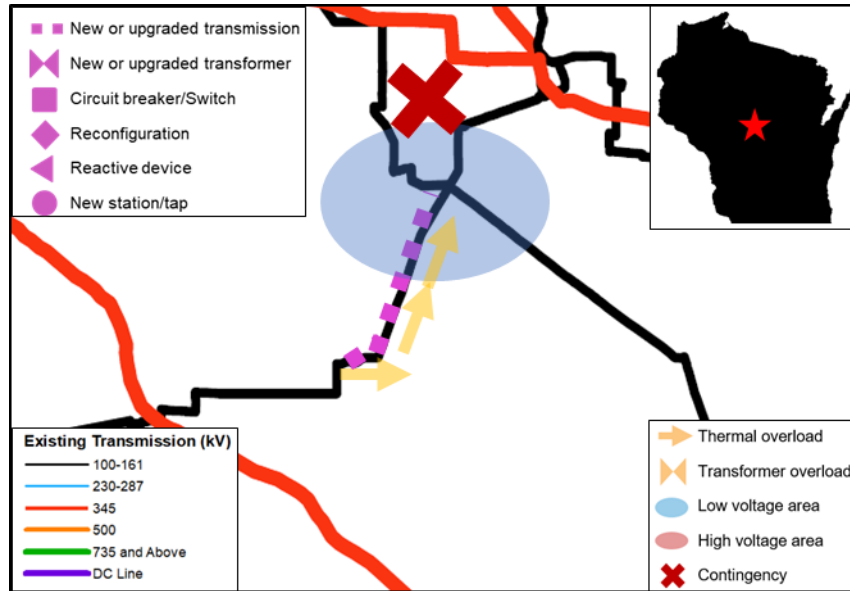


Figure 4.5.1-1: Geographic transmission map of project area

Project Need

The Council Creek SPS no longer fully mitigates market congestion due to increased system flows. Rebuilding this line will resolve system reliability, market congestion and pending asset renewal issues.

Cont. Type	Limiting Element	Rating (MVA)	Pre-Project Loading (%)	Post-Project Loading (%)
P61	698840 ACEC BADGERW - 699240 SAR 138 1	99	105.0	41.4
P61	698840 ACEC BADGERW - 699808 PETENWEL 1 ¹	99	107.4	42.3

¹This line overloads in the SUM22, SUM25, and SUM30 models, all of which are mitigated by P18017

Table 4.5.1-1: P18017 Thermal loading drivers

Cont. Type	Limiting Element	Voltage Limit (pu)	Pre-Project Voltage (pu)	Post-Project Voltage (pu)
P61	698948 HOLYWOOD	.9	0.895	.943
P61	699240 SAR 138	.9	0.895	.944
P61	699241 POE 138	.9	0.895	.942
P61	699242 SGL 138	.9	0.888	.936
P61	699243 LPV 138	.9	0.890	.938
P61	698361 VUL 138	.9	0.895	.942

Table 4.5.1-2: P18017 Voltage performance drivers

Alternatives Considered

No alternative was considered for the Petenwell-Badger-West-Saratoga, (X-43) 138 kV, Line Rebuild.

Project 18568 – Howards Grove SS – Erdman SS 138 kV, new line

Project Description

This project involves constructing approximately seven miles of 138 kV Line from Howards Grove to Erdman. The project's estimated cost is \$25.0 million, and the expected in-service date is November 30, 2023.

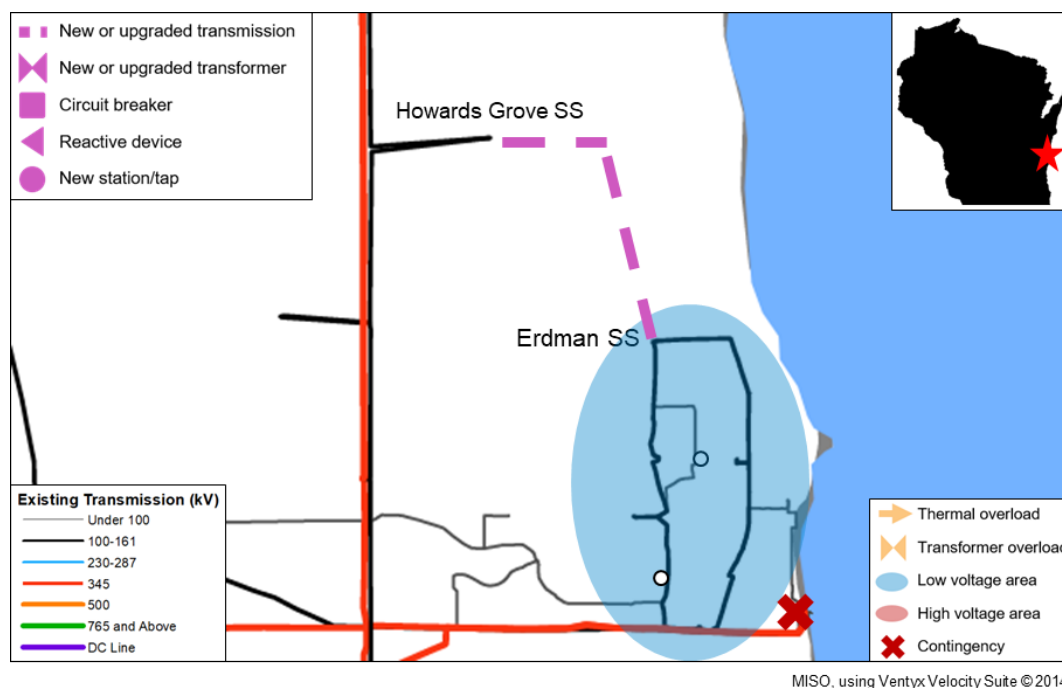


Figure 4.5.1-2: Geographic transmission map of project area

Project Need

Multiple P6 contingencies in the Sheboygan area lead to voltage instability. Previous mitigation to keep the system radial is no longer viable. Voltage instability is seen in the 2022, 2025, and 2030 Summer cases, as well as the 2025 Shoulder 40% and Winter cases. The following buses are a worst-case snapshot of the area. There are additional voltage violations at 138 kV and 69 kV buses due to multiple P61 violations. Constructing the new Howards Grove – Erdman 138 kV line provides additional support to the area and alleviates all violations.

Cont. Type	Limiting Element	Voltage Limit (pu)	Pre-Project Voltage (pu)	Post-Project Voltage (pu)
P61	699231 SKT 138	.9	0.59	.98
P61	699238 HUEBNER	.9	0.59	.98
P61	699285 EDG1 138	.9	0.6	.98
P61	699286 EDG2 138	.9	0.6	.98
P61	699225 LDR 138	.9	0.73	.97
P61	699226 KOH 138	.9	0.73	.97
P61	699228 ERD 138	.9	0.73	.97
P61	699229 NOS 138	.9	0.73	.97

Table 4.5.1-3: P18568 Voltage performance drivers

Alternatives Considered

No alternative was considered for the Howards Grove SS – Erdman SS 138 kV new line.

Other Projects

Other projects do not meet the criteria to be considered as Baseline Reliability Projects (BRP), New Transmission Access Projects, Market Efficiency Projects, or Multi-Value Projects. Other projects may include projects to satisfy Transmission Owner and/or state and local planning criteria other than NERC or regional reliability standards, interconnect new Loads, relocate transmission facilities, address aging transmission infrastructure, replace problematic transmission plant, improve operational performance or address other operational issues, address service reliability issues with end-use consumers, improve aesthetics including but not limited to undergrounding overhead transmission facilities, address localized economic issues, and address other miscellaneous localized needs. For the convenience of the reader, tables of project information are broken down by four general categories of project drivers but note that these four drivers are not defined in the MISO Tariff.

Projects Driven by Local Reliability

The following projects are driven by local planning criteria without Electric Reliability Organization (ERO) standard applicability or by operational flexibility.

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
4727	Winona – Atlantic (Winona69) 69 kV reconductor	The Winona - Atlantic (Winona69) 69 kV circuit is 22 miles long and can be a limiter for the import of energy to Houghton and Keweenaw Counties, MI. This project will re-conductor this circuit with 477 kcmil ACSR and add ATC owned and operated OPGW.	December 31, 2024	\$13.90
12175	West Middleton - Stage Coach 69 kV (6927), partial rebuild	Replace the underground cable at West Middleton with an overhead line. Replacement of jumpers, switches and other terminal equipment.	December 31, 2023	\$2.74
14913	Line Clearance Mitigation Program 2021	Line Clearance Mitigation projects have shorter project life cycles. Projects are driven by the ongoing assessment and analysis of field line clearances using Light Detection and Ranging (LiDAR) technology.	June 30, 2024	\$3.79

Projects Driven by Age and Condition

The following projects are proposed to replace aging or degraded equipment.

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
14909	Small Capital Project and Asset Renewal Program 2021	Transmission Line Projects typically (but not limited to): structure, Cross-arm, Insulator, Surge-arrester, and Pole hardware replacements. Substation Projects typically (but not limited to): relays, Circuit breakers, Switches, Instrument Transformers (CTs & PTs	June 30, 2024	\$38.91

		etc.), Batteries, RTUs, and IT/OT/Communications hardware replacements. Project drivers may be Asset Renewal or small system limit upgrades. These projects have limited scope and cost. Initial capital spend for this program is expected in 2021.		
15930	Pulliam SS, Transformer Retirement and Bus Reconfiguration	Removal of Transformer T16 coordinated with retirement of Pulliam Generation Station. Bus Reconfiguration, straight bus to ring bus. Additional asset renewal of relays and removal of unnecessary circuit breakers	December 31, 2022	\$4.00
16486	Arcadian SS - Power Transformer and Control House Replacement and Asset Renewal	Vacate and demolish existing control house buildings and procure new level-8 control house and contents (relay, RTU, etc.). Remove both Westinghouse LTC transformers (T2 & T3) and replace with one 345/138kV 500MVA transformer. T1 will be retained and relocated onto proper oil containment. Remove all three 345 kV live tan breakers and associated free standing CTs. Remove all four 138 kV Allis Chalmers BZO oil circuit breakers. Remove both 345 kV circuit switchers. Spare 230 kV transformer and towers will remain	December 31, 2023	\$40.00
16488	Tilden SS - Control House Replacement and Breaker & Relay Asset Renewal	Replace six, 138 kV oil breakers with SF6 gas breakers that utilize high creep bushings for mine dust contamination. Retire six bushing potential devices installed on two breakers and replace them with six bus voltage transformers. Install a new ATC owned control house and replace relays with ATC standard models. Replace other equipment including disconnect switches, arresters and bus voltage transformers and junction boxes	December 31, 2023	\$5.88
17585	Colley Road SS, Transformer Replacement and Asset Renewal	Replace 1975 vintage 93 MVA 138/69/12.47 kV power transformer with a 100 MVA new purchase. Replace four sets of three-phase 1960's vintage hook-stick disconnect switches	December 31, 2023	\$3.15
18019	Glenview SS, Transformer Retirement and Breaker & Relay Asset Renewal	Retire and dispose of the T1 Power Transformer including the station power. Replace five 69 kV oil circuit breakers and two 138 kV oil circuit breakers. Replace eight disconnect switches. Install line arresters on the E-57 and GLWG21 138 kV line terminals and retire bus 1 and 2 arresters	December 31, 2023	\$2.20
18020	Mackinac HVDC System Control Module (SCM) Asset Renewal	The Mackinac HVDC System Control Modules (SCMs) consist of multiple computer Servers which provide programmable control and protection functions. This project replaces the SCM servers following industry standard and OEM recommended life cycle timelines	December 31, 2022	\$4.00
18848	Blount - East Campus, 69 kV, (6907), Replace Damaged Underground Cable	Replace 1.9 miles of damaged High-Pressure Fluid Filled (HPFF) cable with new HPFF using	September 30, 2020	\$9.00

		existing pipe. Damage caused by Blount SS event in July of 2019.		
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Projects Driven by Load Growth

The following projects are proposed to replace aging or degraded equipment.

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
14907	Load Interconnection Program 2021	Load Interconnection project life cycles are customer need driven and typically have shorter project life cycle. These load Interconnections typically require limited infrastructure modifications and additions. These projects have limited scope and cost. Initial capital spend for this program is expected in 2021.	June 30, 2024	\$20.12
17628	Preble SS, DIC, add 4th transformer and substation rebuild	WPS will expand Preble Substation and install a new control house, add the 4th 138/24.9 kV, 22 MVA xfmr with a circuit breaker on the high-side, rebuild and relocate Bank 2 xfmr, upgrade the high-side to a circuit breaker, and retire and remove Henry St Substation ATC will extend 138 kV bus to the north and the south, relocate both 138 kV line terminals for line X-154 (Highway V – Tower Dr), replace existing 77/88 MODs with 138 kV line breakers, install a 138 KV bus-tie breaker, and retire the 69 kV Henry St Tap on line L-64 (Danz Ave – Finger Road)	June 1, 2022	\$3.50
18051	Sheboygan County SS, DIC, substation rebuild	Sheboygan County SS, DIC, substation rebuild	December 31, 2023	\$8.00
18052	Barneveld SS, DIC, new substation	ATC will construct an underbuilt 69 kV (138kV capable) circuit from Hooterville to Wally Road onto the ordered route of the Cardinal - Hickory Creek 345 kV project. Connect the new WPL Barneveld substation with a loop through straight bus configuration with 2-line breakers. Connect the new MHU Military Ridge substation with a radial tap with line switches. Retire the existing Y-128 section from Mt. Horeb to Wally Road	December 31, 2023	\$12.30
18053	Waupaca SS, DIC, transformer addition	ATC to expand the Waupaca Substation to accommodate a new 138 kV bus section, new bus-tie breaker, add 2- line breakers, reposition ATC's 14-Mvar capacitor banks, and replace power-line carrier with optical ground wire (OPGW) from Waupaca – Harrison North.	June 1, 2022	\$6.90

Projects Driven by Other Local Needs

The following projects are proposed due to other local needs such as communications and physical security.

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
14911	Communication Reliability Program 2021	Substation Communications Reliability Upgrade projects are typically (but not limited to): OPGW additions, replacements, relocations, or removals. Project drivers typically are communications support for SCADA, Relay protection, Security systems, Small communications network upgrades, and Telecom industry market transitions.	June 30, 2024	\$10.93
14915	Physical Security Program 2021	The Physical Security Program is guided by the NERC Standard CIP-014 and is designed to harden substation infrastructure. Specific project locations and scopes of work are not made public. Initial capital spend for this program is expected in 2021.	June 30, 2024	\$17.91

Generator Interconnection Projects

According to Attachment FF of the MISO Tariff, "Generator Interconnection Projects are New Transmission Access Projects that are associated with interconnection of new, or increase in generating capacity of existing, generation." The following Generator Interconnection Projects (GIPs) have been evaluated through the Generator Interconnection queue and the associated Generator Interconnection Agreements have been signed.

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
17525	J870/J871 MPFCA Highland Generator Network Upgrades	A 300 MW solar farm in Grant and Iowa County, Wisconsin. Customer has requested a Point of Interconnection (POI) at Highland Substation that taps the existing X-17 line between Eden and Wyoming Valley Substations.	December 31, 2021	\$22.18
17526	J886 GIC Kewaunee SS Interconnection Facilities and Network Upgrades	To modify the existing 138 kV Kewaunee Substation to provide transmission interconnection of the J866 with a total summer capacity of 150MW.	August 1, 2020	\$ 2.29
18566	J807 GIC Falcon SS Addition - Generator Interconnection Facilities and Network Upgrades	A 41.4 MW wind farm in Lafayette County, Wisconsin. Customer has requested a Point of Interconnection (POI) at a new terminal at the existing 138 kV Falcon Substation.	August 27, 2021	\$0.87
18567	J821 GIC Zenith SS - Generator Interconnection Facilities and Network Upgrades	A 99.9 MW wind farm in Marathon County, Wisconsin. Customer has requested a Point of Interconnection (POI) at a new Zenith Substation.	August 27, 2021	\$7.24

18665	J855 GIC Ebenezer SS - Generator Interconnection Facilities and Network Upgrades	J855 is a 100.8 MW, 28 turbine wind farm in Grant County, Wisconsin with a requested point of interconnection at a new 138 kV substation named Ebenezer SS tapping the existing 138 kV line, X-16.	November 6, 2020	\$5.53
18849	J870 Highland SS Interconnection Facilities	A 300 MW solar farm in Grant and Iowa County, Wisconsin. Customer has requested a Point of Interconnection (POI) at Highland Substation that taps the existing X-17 line between Eden and Wyoming Valley Substations.	August 1, 2020	\$0.23
18925	J818 Jefferson SS Generator Interconnection Facilities and Network Upgrades	Interconnection of a 149 MW solar facility located in Jefferson County, Wisconsin.	April 1, 2022	\$5.03
18985	J850/J878 MPFCA East Paris New SS	The purpose of the project is to construct a 345-138 kV substation to the east of the 138 kV Paris substation.	December 23, 2022	\$25.53
19145	J947 Tennyson SS, Generator Interconnection Facilities and Network Upgrades	Interconnection of a 188-inverter solar farm in Green County, Wisconsin with a 200 MW capacity.	September 30, 2021	\$5.61
19146	J850 North Creek SS, Generator Interconnection Facilities and Network Upgrades	Interconnection of a 250 MW solar farm located in Walworth County, Wisconsin.	September 30, 2022	\$12.57
19225	J864 Lone Rock SS Generator Interconnection Facilities and Network Upgrades	Interconnection of a 49.98 MW solar farm at the Lone Rock SS.	September 1, 2022	\$1.77
19226	J807 ¹ J819 ² J864 ⁴ J870 ⁵ J871 ⁶ J947 ⁷ MPFCA Common Use Upgrades	Each individual Customer is responsible for a proportionate percentage share of the Common Use Upgrades estimated cost based on its proportionate impact on the constrained facilities causing the need for the CUU.	March 31, 2021	\$0.44
19245	J807 ¹ J819 ² J850 ⁵ MPFCA Common Use Upgrades	Each individual Interconnection Customer is responsible for a proportionate percentage share of the Common Use Upgrades (CUU) estimated cost based on its proportionate impact on the constrained facilities causing the need for the CUU.	March 31, 2021	\$0.37
19246	J864 ⁴ J870 ⁵ J871 ⁶ J947 ⁷ MPFCA Common Use Upgrades	Each individual Customer is responsible for a proportionate percentage share of the Common Use Upgrades estimated cost based on its proportionate impact on the constrained facilities causing the need for the CUU.	March 31, 2021	\$1.41
19247	J807 ¹ J819 ² J850 ⁵ J947 ⁷ MPFCA Common Use Upgrades	Each individual Interconnection Customer is responsible for a proportionate percentage share of the Common Use Upgrades (CUU) estimated cost based on its proportionate impact on the constrained facilities causing the need for the CUU.	March 1, 2021	\$0.20
19248	J807 ¹ J819 ² J870 ⁵ J947 ⁷ MPFCA Common Use Upgrades	Each individual Interconnection Customer is responsible for a proportionate	March 1, 2021	\$0.18

		percentage share of the Common Use Upgrades (CUU) estimated cost based on its proportionate impact on the constrained facilities causing the need for the CUU.		
19265	J878 GIC Paris SS, Generator Interconnection Facilities and Network Upgrades	Interconnection Customer will install a 200 MW solar farm interconnecting at the Paris SS in Kenosha County, Wisconsin.	September 30, 2022	\$2.35
19267	J807 ¹ J864 ⁴ J870 ⁵ J871 ⁶ J947 ⁷ MPFCA Common Use Upgrades	Each individual Customer is responsible for a proportionate percentage share of the Common Use Upgrades (CUU) estimated cost based on its proportionate impact on the constrained facilities causing the need for the CUU.	March 31, 2021	\$0.54
19268	J807 ¹ J819 ² J947 ⁷ MPFCA Common Use Upgrades	Each individual Interconnection Customer is responsible for a proportionate percentage share of the Common Use Upgrades (CUU) estimated cost based on its proportionate impact on the constrained facilities causing the need for the CUU.	December 31, 2021	\$4.07
19269	J864 ⁴ J870 ⁵ J871 ⁶ MPFCA Common Use Upgrades	Each individual Customer is responsible for a proportionate percentage share of the Common Use Upgrades (CUU) estimated cost based on its proportionate impact on the constrained facilities causing the need for the CUU.	March 31, 2021	\$0.92
19729	J732 Superior SS Network Upgrades and Interconnection Facilities	Interconnection of about 560 MW the facility composed of one (1) steam turbine unit and one (1) natural gas unit in a 1x1 combined cycle facility interconnecting at the new Superior Substation.	November 30, 2023	\$21.25

¹ J807 Customer is seeking interconnection service for 41.4 MW

² J819 Customer is seeking interconnection service for 99.9 MW

³ J850 Customer is seeking interconnection service for 250 MW

⁴ J864 Customer is seeking interconnection service for 49.98 MW

⁵ J870 Customer is seeking interconnection service for 200 MW

⁶ J871 Customer is seeking interconnection service for 100 MW

⁷ J947 Customer is seeking interconnection service for 200 MW

4.5.2 Cedar Falls Utilities (CFU)

Cedar Falls Utilities did not submit any new projects for MTEP20. MISO has not identified any issues in CFU area.

4.5.3 Central Minnesota Municipal Power Agency (CMMPA)

CMMPA proposed one project at an estimated cost of \$2 million targeting MTEP20 Appendix A. After MISO's independent reliability analysis of the CMMPA area, this project is recommended by MISO to be included in MTEP20 Appendix A at an estimated cost of \$2 million. This project is classified as an Other Project. The project's expected in-service date and estimated cost are provided as of September 15, 2020.

Other Projects

Other projects do not meet the criteria to be considered as Baseline Reliability Projects (BRP), New Transmission Access Projects, Market Efficiency Projects, or Multi-Value Projects. Other projects may include projects to satisfy Transmission Owner and/or state and local planning criteria other than NERC or regional reliability standards, interconnect new Loads, relocate transmission facilities, address aging transmission infrastructure, replace problematic transmission plant, improve operational performance or address other operational issues, address service reliability issues with end-use consumers, improve aesthetics including but not limited to undergrounding overhead transmission facilities, address localized economic issues, and address other miscellaneous localized needs. For the convenience of the reader, tables of project information are broken down by four general categories of project drivers but note that these four drivers are not defined in the MISO Tariff.

Projects Driven by Age and Condition

The following projects are proposed to replace aging or degraded equipment.

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
18118	Windom Transmission Upgrade	Windom Municipal Utilities has a 1.5-mile transmission line constructed in 1971 that has been inspected and is starting to show signs of age. It would also be beneficial to relocate the line slightly while rebuilding.	December 15, 2021	\$2

4.5.4 City of Ames, IA (COA)

City of Ames, IA did not submit any new projects for MTEP20. MISO has not identified any issues in COA area.

4.5.5 Great River Energy (GRE)

Great River Energy proposed nine projects at an estimated cost of \$77.4 million targeting MTEP20 Appendix A. After MISO's independent reliability analysis of the GRE area, nine of these projects were recommended by MISO to be included in MTEP20 Appendix A at an estimated cost of \$77.4 million. Of these projects, zero are Baseline Reliability Projects, nine are Other Projects, and zero are Generation projects with signed Generator Interconnection Agreements. All the projects' expected in-service dates and estimated costs are provided as of September 15, 2020.

Other Projects

Other projects do not meet the criteria to be considered as Baseline Reliability Projects (BRP), New Transmission Access Projects, Market Efficiency Projects, or Multi-Value Projects. Other projects may include projects to satisfy Transmission Owner and/or state and local planning criteria other than NERC or regional reliability standards, interconnect new Loads, relocate transmission facilities, address aging transmission infrastructure, replace problematic transmission plant, improve operational performance or address other operational issues, address service reliability issues with end-use consumers, improve aesthetics including but not limited to undergrounding overhead transmission facilities, address localized economic issues, and address other miscellaneous localized needs. For the convenience of the reader, tables of project information are broken down by four general categories of project drivers but note that these four drivers are not defined in the MISO Tariff.

Projects Driven by Local Reliability

The following projects are driven by local planning criteria without Electric Reliability Organization (ERO) standard applicability or by operational flexibility.

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
17664	Portage Lake 115/69 kV Project	GRE will interconnect to Minnesota Power's (MP) 13 Line (Riverton – Cromwell 115 kV) with a four position, 115 kV ring bus, to be called Portage Lake, at or near the existing Mille Lacs Electric Cooperative (MLEC) Kimberly substation. The new 115 kV Portage Lake ring bus will have four positions; 115 kV line to Riverton (13 Line), 115 kV line to Cromwell (26 Line), 115/69 kV transformer with a 9.5-mile line to Palisade. (AKA Kimberly)	September 30, 2023	\$13.20
17964	Blue Earth - East Chain - Wilbert Tap (FE-WB) 69 kV Line Rebuild	Rebuild entire FE-WB line (Blue Earth to East Chain to Wilbert Tap)	December 31, 2022	\$12.27
17971	Westwood 1 115 kV Conversion	Convert the existing GRE/SEA Westwood 1 substation from 69 kV to 115 kV service. Rebuild the existing 69 kV transmission line from Le Sauk to Westwood 1 to West St. Cloud to 115 kV.	December 2, 2022	\$6.02
17988	Lake Mary 115 kV Conversion	Convert the GRE/REA Lake Mary substation from 41.6 kV to 115 kV service.	November 30, 2022	\$1.40

Projects Driven by Age and Condition

The following projects are proposed to replace aging or degraded equipment.

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
14366	Svea Tap - Litchfield Tap Line Rebuild Project	Rebuild GRE's existing 69 kV transmission line from Willmar to Litchfield Muni Tap with a larger conductor.	June 14, 2024	\$ 18.35
17944	Heron Lake - Round Lake (FE-RH) 69 kV Line Rebuild	Rebuild entire FE-RH line (from Heron Lake substation to Round Lake substation)	September 30, 2021	\$ 13.23
17970	Howard Lake - Highland - Maple Lake Line Rebuild	Rebuild the 69 kV transmission line from Howard Lake to Highland to Maple Lake.	December 29, 2023	\$ 10.40
17986	Lake Jennie Tap 69 kV Line Rebuild	Rebuild 2.77 miles of Lake Jennie Tap line.	November 1, 2021	\$ 1.55

Projects Driven by Load Growth

The following projects are proposed to replace aging or degraded equipment.

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
18025	Cannon River Park 115 kV Substation	Construct a new 115 kV transmission line from Cannon River Park to the Cannon River Park tap in the Xcel 0832 line near the Xcel Faribault Energy Park.	March 5, 2021	\$ 1.00

4.5.6 ITC Midwest (ITCM)

ITC Midwest proposed 34 projects at an estimated cost of \$308.4 million targeting MTEP20 Appendix A. After MISO's independent reliability analysis of the ITCM area, 34 of these projects were recommended by MISO to be included in MTEP20 Appendix A at an estimated cost of \$308.4 million. Of these projects, zero are Baseline Reliability Projects, 26 are Other Projects, and eight are Generation projects with signed Generator Interconnection Agreements. All the projects' expected in-service dates and estimated costs are provided as of September 15, 2020.

Other Projects

Other projects do not meet the criteria to be considered as Baseline Reliability Projects (BRP), New Transmission Access Projects, Market Efficiency Projects, or Multi-Value Projects. Other projects may include projects to satisfy Transmission Owner and/or state and local planning criteria other than NERC or regional reliability standards, interconnect new Loads, relocate transmission facilities, address aging transmission infrastructure, replace problematic transmission plant, improve operational performance or address other operational issues, address service reliability issues with end-use consumers, improve aesthetics including but not limited to undergrounding overhead transmission facilities, address localized economic issues, and address other miscellaneous localized needs. For the convenience of the reader, tables of project information are broken down by four general categories of project drivers but note that these four drivers are not defined in the MISO Tariff.

Projects Driven by Local Reliability

The following projects are driven by local planning criteria without Electric Reliability Organization (ERO) standard applicability or by operational flexibility.

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
9100	Burlington-4th St-Agency 69 kV Rebuild	Rebuild a portion of the Burlington-South Burlington 69 kV line and upgrade the Burlington 69 kV terminal so the Burlington-South Burlington line is conductor limited. Rebuild the Burlington 4th Street-Agency 69 kV line to T2-477 ACSR.	May 31, 2021	\$4.94
17732	Eddyville Industrial Complex 161 kV Substation Expansion	ITC will expand the Eddyville Industrial Complex 161 kV substation to the north. ITC will construct a five position 161 kV ring bus North of the existing Eddyville Industrial Complex substation. All new terminal equipment will be capable of 2000A minimum. ITCM will terminate the Beacon - Tri County and EIC - Tri County 161 kV circuits into these breakers.	May 31, 2023	\$7.96
18114	Adams 161 kV Breaker and 1/2 Upgrade	Add a breaker to the 161 kV bus between breakers 0681 and 0680 and relocate the Rochester and Harmony 161 kV lines to terminal positions in the newly formed breaker and 1/2 row (install switches, bus work, dead-end, CCVTs, relays, and all necessary equipment as needed). Add a breaker to the 161 kV bus between breakers 0685 and 0682 (install switches, bus work, dead-end, CCVTs, relays, and all necessary equipment as needed). Add a breaker to the 161 kV bus between breakers 0790 and 0686 (install switches, bus work, dead-end, CCVTs, relays, and all necessary equipment as needed). All equipment needs to be capable of a minimum of 2000 amp operation.	September 15, 2021	\$3.38

Projects Driven by Age and Condition

The following projects are proposed to replace aging or degraded equipment.

Project 17726 Savanna Substation Expansion

Project Description

ITC will be expanding the Savanna 69 kV Substation to a Ring Bus capable of 6-positions. . A new 161/69 kV 150MVA transformer will be installed at Savanna.

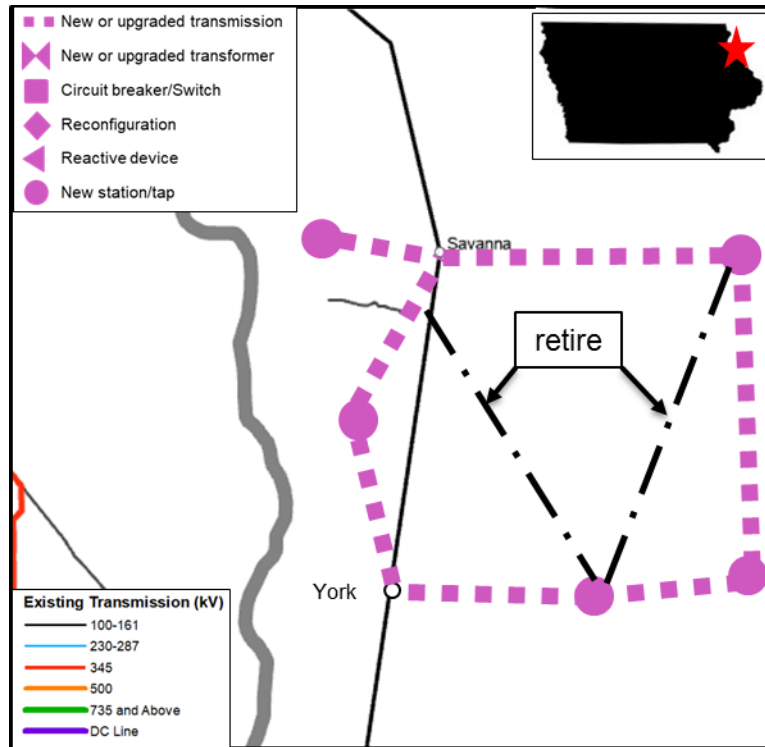


Figure 4.5.6-2: Geographic transmission map of the project area

Project Need

The 34 kV system in the Savanna area will be converted to 69 kV based on age, condition, and contractual obligations. The Savanna substation must be expanded to accommodate the conversions.

Alternatives Considered

Stakeholders submitted an alternative that would bring the existing non-ITC owned 69 kV line into the new 69 kV station. The new 69 kV station would then be built as a breaker and half configuration which would increase the cost of project 17726 by \$3.5M. Studies show that there were no changes in reliability with the alternative. MISO recommends the original project for MTEP20.

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
17724	Sixth Street 115 kV Substation Retirement	Once the remaining 34.5 kV lines at the Sixth St Substation are converted to 69 kV, the remaining substation equipment should be retired, including any control enclosures and fence, etc. The 34.5 kV lines will be connected to bypass the substation as part of the conversion work, but the 115 kV lines will need to be connected to bypass the substation as part of the substation retirement project. The communications currently routed through Sixth Street will also need to be moved/removed before the retirement can be completed.	December 31, 2023	\$1.51

17725	Savanna 161kV Substation Expansion	TC will be expanding the Savanna Substation to a Ring Bus capable of 6-positions. The new 161 kV equipment will be capable of 2000A. There will be several phases for the 161 kV.	December 31, 2021	\$8.70
17726	Savanna 69 kV Substation Expansion	ITC will be expanding the Savanna 69 kV Substation to a Ring Bus capable of 6-positions. Most new 69 kV equipment will be capable of 1200A or greater. ITC will install a 161/69 kV 150MVA transformer at Savanna. The 161/69 kV transformer position and the Metform position will be capable of 2000A or greater. There will be several phases for the 69 kV.	December 31, 2021	\$9.54
17727	York Substation 69 kV Expansion	ITC will be expanding the York Substation to include a 161/69 kV 150MVA transformer. All new 69 kV equipment will be capable of 2000A. A tie switch will be added to the 161 kV. All new 161 kV equipment will be capable of 2000A or greater. See page 2 for the electrical layout of the York Substation. ITC will be retiring the 34.5 kV out of the York Substation. No additional 161 kV equipment will be needed to accommodate this work. All new 69kV equipment will be capable of 1200A or greater.	December 31, 2025	\$9.78
17728	Savanna-Metform-York 34 kV to 69 kV Rebuild	ITC will rebuild approximately 6 miles of the Savanna – Metform – York 34.5 kV line to 69 kV standards. The overhead conductor will have an ampacity of 1200A or higher. ITC will need to investigate the Danisco to York 34.5 kV to determine if the insulators are capable of 69 kV operation. If not, ITC will reinsulate the Danisco to York 34.5 kV line with 69 kV insulators. The existing conductor is 954 ACSR and does not need to be replaced due to ampacity needs. The static is 3/8 steel.	December 31, 2021	\$10.75
17729	Savanna-Argo Fay-York Area Rebuilds	ITC will build a new 69 kV line approximately 6 miles from the York Substation to the Argo Fay Substation. The overhead conductor will have an ampacity of 600A or higher. A route study may be needed to determine the best route. ITC will build a new 69 kV line approximately 6.5 miles from the Clay Substation to the Chadwick Substation. The overhead conductor will have an ampacity of 600A or higher. A route study may be needed to determine the best route. ITC will rebuild approximately 6.5 miles of the Argo Fay Substation to the Chadwick Substation 34.5 kV line to 69 kV Standards. The	December 30, 2025	\$22.35

		overhead conductor will have an ampacity of 600A or higher. ITC will retire approximately 6 miles of 34.5 kV line between Argo Fay and the Clay substation. ITC will retire approximately 6 miles of 34.5 kV line between Argo Fay and the Savanna substation.		
17730	Savanna-Clay 34 kV to 69 kV Rebuild	ITC will rebuild approximately 7.75 miles of 34.5 kV to 69 kV standards from Savanna Substation to Clay Substation. The overhead conductor will have an ampacity of 600A or higher. A route study may be needed to determine the best route.	December 30, 2025	\$8.26
17731	Savanna City Loop 34 kV to 69 kV Rebuild	ITC will rebuild approximately 1.66 miles of the Savanna – Eaton 34.5 kV line to 69 kV standards. The overhead conductor will have an ampacity of 600A or higher. ITC will rebuild approximately 0.85 miles of the Bowen – Eaton 34.5 kV line to 69 kV standards. The overhead conductor will have an ampacity of 600A or higher. ITC will rebuild approximately 3.80 miles of the Bowen – Savanna 34.5kV line to 69 kV standards. The overhead conductor will have an ampacity of 600A or higher.	December 30, 2025	\$10.48
17740	ITCM Asset Blanket for 2023	Replace equipment that is past its expected design life, utilizes outdated functionality or whose failure rates and maintenance have increased potential for mis-operations and higher maintenance costs associated with their ongoing use.	December 31, 2023	\$37.08
17966	Metform Substation 69 kV Conversion	ITC will be converting the Metform Substation to 69 kV. All network path 69 kV equipment will be capable of 1200A or greater. All network path switches will need to be replaced to be capable of 1200A or greater. ITC will retire the bypass bus and switches.	December 31, 2021	\$0.93
17967	Bowen Substation 69 kV Conversion	ITC will be converting the Bowen Substation to 69 kV. All network path 69 kV equipment will be capable of 600A or greater.	December 30, 2029	\$2.75
17980	Hancock T3 161/69 kV Transformer Replacement	Replace the Beaver Channel T1 & T2 75MVA 161/69 kV transformers with the spare 150MVA 161/69 kV transformer (SN: WT03583) currently located at the Ames substation. Also install a high side breaker for the new transformer at Beaver channel and replace switch 1454. Any new high side equipment should be 1200A minimum and any new low side equipment should be 1600A minimum. Replace the Hancock T3 75MVA 161/69 kV transformer with the Beaver Channel T1	April 30, 2020	\$4.59

		75MVA 161/69 kV Transformer (SN: 70450A). Also replace the high side circuit switcher 1320 with a 161 kV circuit breaker. Re-use switch 1320B if feasible. Any other new required high-side equipment should be 1200A minimum.		
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Projects Driven by Load Growth

The following projects are proposed to replace aging or degraded equipment.

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
17704	Cedar Rapids Hunter Field Substation Interconnection	IPL will be building a new 69/24.9 kV substation named Cedar Rapids Hunter Field near the existing Viola's Corner substation. This new substation will replace Viola's Corner. The new substation will have one new 69/24.9 kV 50 MVA transformer installed initially, with a second transformer to be added in the future as load grows in the area. ITC will build a new 69 kV substation with two single box structures and two breakers installed, one-line breaker and one bus tie breaker, and will add an additional line breaker in the future when IPL adds a second transformer.	December 30, 2021	\$3.58
17705	Madrid North 69 kV Substation Rebuild Interconnection	IPL has requested ITC to rebuild the Madrid North Substation to a standard single box and double box with a bus tie breaker to accommodate their 25 kV plans which will include two transformers and 25 kV switchgear. ITC will also need to install a new control enclosure, as IPL will be retiring their existing control enclosure to accommodate their new switch gear. The existing 69 kV circuit switcher will be replaced with a new 69 kV breaker, which will allow for future retirement of the bus work associated with the 69/34.5 kV transformer. The existing zig-zag transformer should be removed to allow additional space for the installation of the new ITC owned control enclosure. The 69/34.5 kV transformer will need to remain until the conversion to 69 kV happens between Dawson and Madrid North (covered on EI Sketch ITCMW_16_034). The existing section of 69 kV bus that feeds the 69/34.5 kV transformer, transformer, and low-side breaker should be re-used, and then removed in the future as part of the 69 kV conversion.	December 31, 2021	\$4.14
17734	Clinton Perrin 69/24 kV Substation Interconnection	IPL requested a new Clinton Perrin 69/24 kV substation in Clinton, IA. The new substation in city of Clinton, will provide capacity for industrial load south/west of town and will support retirement of Beaver Channel sub and Mill Creek sub after 25 kV conversion. IPL will install a single transformer, 12.5 kV substation, 20/33MVA capacity. Transformer will have 25 kV option but will be replaced by a 30/50MVA unit for final 25 kV configuration.	December 31, 2023	\$3.79

		Installed XFMR will be moved elsewhere at that time. This conversion is estimated to be 10-20 years after completion of this construction. Four feeders will be used from this substation to serve load. Contingency at 12.5 kV will be up to max rating of transformer, 33MVA. Contingency at 25 kV will be reviewed and updated at a later date but may be up to future 50MVA top rating of final unit.		
17735	Clinton South 69 kV Substation Retirement	Retire all ITC owned equipment in the Clinton South 69/distribution sub and reroute the 69 kV line to bypass the Clinton South sub.	December 31, 2024	\$0.16
17736	Waukon Snakeskin 69 kV Substation Interconnection	IPL will be constructing the Waukon Snakeskin 2 transformer 69/25 kV substation the replace the nearby Waukon substation. ITC will install 2 box structures, 2-line breakers and 1 tie breaker. The cap bank and cap bank breaker at the existing Waukon substation should be moved to the Decorah side of the Waukon Snakeskin substation if possible or a new cap bank and breaker of equivalent effective rating should be installed. All substation equipment should have a minimum rating of 1200A. ITC will also install short in/out tap lines.	December 31, 2021	\$2.51
17738	Thompson MacTavish 69 kV Substation Interconnection	ITC will install an ~0.02-mile tap line to IPL's new Thompson MacTavish 69/24.9 kV Distribution Substation. This substation will be Open Air Single Bus with a 12/20MVA 69/24.9 kV transformer and high side breaker. The substation will be located on one of the two land options available to IPL. The Thompson and Buffalo Center substations will be retired on future EI sketches once IPL moves the loads from those substations to Thompson MacTavish. Coordinate with IPL and DPC for outages.	December 31, 2025	\$0.35
17739	Calmar Prairie Farmer 69 kV Substation Interconnection	ITC will install an ~0.02-mile tap line to IPL's new Calmar Prairie Farmer 69/24.9 kV Distribution Substation. The transformer will have a high-side breaker installed by IPL. This substation will be located near the existing Calmar substation. IPL has provided four proposed locations. The Calmar and Ossian substations will be retired on future EI sketches once IPL moves the loads from those substations to Calmar Prairie Farmer.	December 31, 2021	\$0.21
17741	ITCM Customer Interconnects with short lead time 2023	These projects are being done at the request of an interconnection customer in order to facilitate new load, re-distribute existing load, improve the performance of the sub-transmission and distribution systems, or to accommodate a new Transmission-to-Transmission connection request.	December 31, 2023	\$2.40
18061	Mason City Industrial 161 kV Substation Interconnection	ITCM will interconnect Alliant Energy's Mason City Industrial 161/24.9 kV distribution substation with in/out taps to the Emery-Lime Creek 161 kV line and install four (4) 161 kV breakers in a ring bus configuration at the 161 kV distribution substation. IPL will install a single distribution transformer	May 30, 2022	\$9.75

		initially to serve a load of ~18MW. IPL will install a second distribution transformer in the future to accommodate load growth and load moved from 69 kV distribution substations.		
18846	Creston Roundhouse 69 kV Substation Interconnection	Per Alliant's request, ITC will construct a 3-terminal straight bus with tie breaker at the proposed Creston Roundhouse substation. All ITC terminal equipment will be capable of 2000A minimum and Alliant will own the land and joint use assets. In addition, ITC line equipment will use T2-4/0. A new load will be added at the Big Cedar Industrial Park in Cedar Rapids, IA and ITC will construct a new Big Cedar Substation to connect this new load. The new sub will tap the Beverly – Morgan Valley 345 kV line and create a Morgan Valley – Big Cedar #1 345 kV line and a Big Cedar – Beverly 345 kV line. New line taps should utilize bundled T2-795 ACSR conductor and have OPGW. ITC will also need to add a second 345 kV circuit on the existing Beverly – Morgan Valley 345 kV line to add a 3rd 345 kV feed into the new Big Cedar substation. The new 345 kV line should utilize bundled T2-795 ACSR conductor. The new Big Cedar substation will include a new 3 position 345 kV breaker and half substation with 3 – 345 kV lines and 2 – 345/161 kV transformers (450 MVA nameplate) and all 345 kV substation equipment should be rated 3000 amps summer normal rating. The 161 kV portion of the substation will be folded breaker and ½, with each transformer connected to a dedicated position (double breaker double bus), and the new 161 kV underground feeds to the customer will be connected on breaker and a half positions. All new 161 kV substation equipment should be rated 2000 amps minimum summer normal rating. All new underground 161 kV lines from Big Cedar to the customer substations should be rated 900 amps minimum summer normal rating. ITC will install new 161 kV ring bus substations at each of the new customer substations. All substation equipment should be rated 2000 amps minimum summer normal rating. ITC will build out the Morgan Valley substation to full 4 position breaker and ½ with the addition of this load and the additional line into Morgan Valley. IPL has also requested that the existing 69 kV double circuit line be relocated as part of this project. The Beverly – Fairfax #2 and Fairfax – Oakridge 69 kV line will be moved to follow the railroad tracks up to Edgewood Rd, then follow Edgewood Rd to meet up with the existing line route.	December 15, 2022	\$7.86
18847	Big Cedar Load Interconnection		December 31, 2021	\$104.63

Generator Interconnection Projects

According to Attachment FF of the MISO Tariff, "Generator Interconnection Projects are New Transmission Access Projects that are associated with interconnection of new, or increase in generating capacity of existing, generation." The following Generator Interconnection Projects (GIPs) have been evaluated through the Generator Interconnection queue and the associated Generator Interconnection Agreements have been signed.

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
16710	J233 J514 GIA Amendment Network Upgrades	Network upgrades include raising spans on the Marshalltown-Blairstown 115 kV and the Blairstown-Prairie Creek 115 kV.	January 16, 2020	\$2.73
16844	J495 Generator Interconnection	ITCM and EDF have entered into an engineering and procurement agreement for the acquisition of materials and design of a 3-terminal, 3 breaker, 345 kV switching station to be constructed roughly 7 miles east of the Ledyard 345 kV station and connected to the Ledyard to Colby 345 kV line. The E&P agreement will cover the design and material acquisition for the 345 kV station, and construction work will occur under the GIA.	November 21, 2019	\$10.66
16864	J475 Generator Interconnection	The Poweshiek to Irvine 161 kV line is to be rebuilt on existing ROW with a conductor having a minimum summer rating of 1,595A. Upgrades at the Poweshiek 161 kV substation will also be required.	October 1, 2020	\$11.00
16865	J504 Generator Interconnection	The DAEC 161 kV substation will be expanded to accommodate the J504 terminal, including the addition of breaker position 5953. New equipment should be capable of 2000A.	October 15, 2020	\$0.27
16866	J498/J499/J500 Generator Interconnection	One span on the Poweshiek - Reasnor kV line will be raised to mitigate sag limits and increase the summer rating to 366 MVA (1313 A).	November 7, 2019	\$0.24
17733	J504 Generator Interconnection-TOIF	The GIA between ITCM, Duane Arnold Solar, LLC (NextEra Energy) and MISO has been executed to facilitate the interconnection of project J504, a 60 MVA solar generating facility rated 51 MW gross and 50MW net with a Point of Interconnection at the DAEC 161 kV substation. The Interconnection Customer is responsible for costs associated with the TOIF.	October 15, 2020	\$0.00
18111	J523 Generator Interconnection-TOIF	Install new terminal for J523 gen-tie line. This work includes installing switch, dead-end, CCVT's, switches, and relay panels (see GIA for further details).	September 15, 2021	\$0.00

18113	J523 Generator Interconnection Network Upgrades	Add two 161 kV breakers in series with the bus-tie breaker 6620 to create a breaker and 1/2 row. Relocate the line for the Adams 345/161 kV transformer from the existing terminal to a new terminal position on a breaker and 1/2 row (install dead end, coordinate relocation of metering with Xcel, and replace necessary equipment to accommodate the relocation of the transformer terminal). All new equipment must be capable of 2000-amp operation.	September 15, 2021	\$1.22
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4.5.7 MidAmerican Energy Company (MEC)

MidAmerican Energy Company proposed 20 projects at an estimated cost of \$100.5 million targeting MTEP20 Appendix A. After MISO's independent reliability analysis of the MEC area, 20 of these projects were recommended by MISO to be included in MTEP20 Appendix A at an estimated cost of \$100.5 million. Of these projects, one is a Baseline Reliability, six are Other Projects, and 13 are Generation projects with signed Generator Interconnection Agreements. All the projects' expected in-service dates and estimated costs are provided as of September 15, 2020.

Baseline Reliability Projects

According to Attachment FF of the MISO Tariff, "Baseline Reliability Projects are Network Upgrades identified in the base case as required to ensure that the Transmission System is in compliance with applicable national Electric Reliability Organization (ERO) reliability standards and reliability standards adopted by Regional Reliability Organizations and applicable within the Transmission Provider Region."

Project 19646 – Franklin to Wall Lake 161 kV Line Structure Replacement

Project Description

Line structure replacements to increase line ratings. The line shows up as overloading in recent MTEP20 results for P6 contingencies. The project's estimated cost is \$0.125 million and expected in-service date is December 31, 2021.

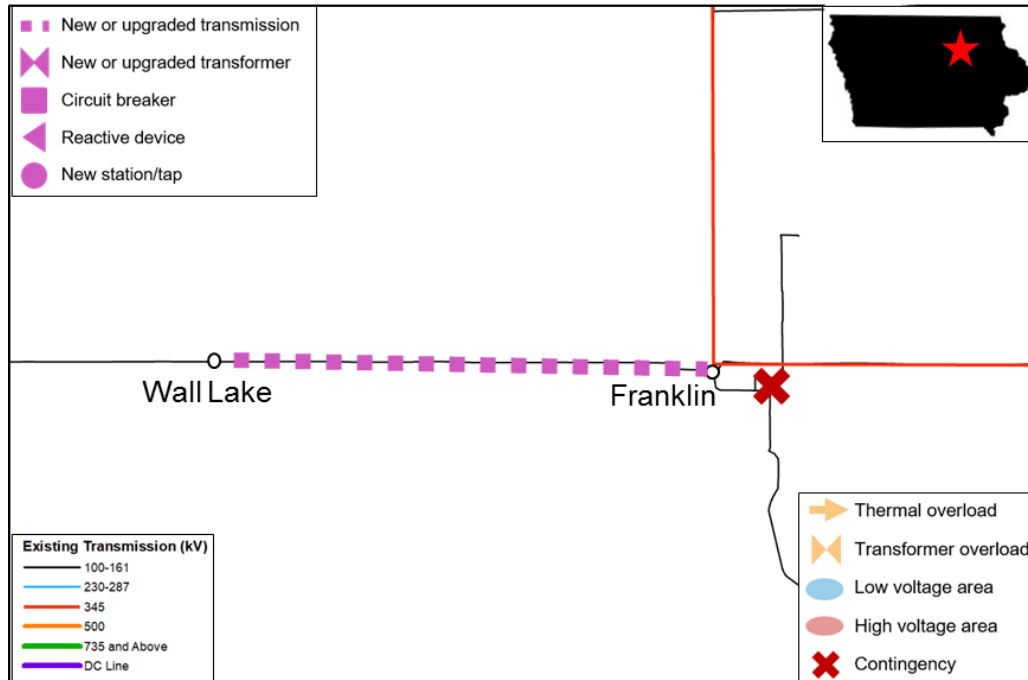


Figure 4.5.7-1: Geographic transmission map of project area

Project Need

P6 events in the area of Beaver Creek and Nuthatch cause thermal violations, shown in Table 4.5.7-1 line. These are seen in the 90% wind cases.

Cont. Type	Limiting Element	Rating (MVA)	Pre-Project Loading (%)	Post-Project Loading (%)
P6	636230 FRANKLIN5 - 636235 WALL LAKE5 1	204	1.07	0.97
P6	636230 FRANKLIN5 - 636235 WALL LAKE5 1	204	1.07	0.97
P6	636230 FRANKLIN5 - 636235 WALL LAKE5 1	204	1.06	0.96
P6	636230 FRANKLIN5 - 636235 WALL LAKE5 1	204	1.06	0.96

Table 4.5.7-1: P19646 Thermal loading drivers

Alternatives Considered

The transmission line is conductor limited and upgrading the structures will maximize the existing facilities at the lowest cost. Existing real time issues in the area limit the availability to redispatch generation following the initial outage.

Other Projects

Other projects do not meet the criteria to be considered as Baseline Reliability Projects (BRP), New Transmission Access Projects, Market Efficiency Projects, or Multi-Value Projects. Other projects may include projects to satisfy Transmission Owner and/or state and local planning criteria other than NERC or regional reliability standards, interconnect new Loads, relocate transmission facilities, address aging transmission infrastructure, replace problematic transmission plant, improve operational performance or address other operational issues, address service reliability issues with end-use consumers, improve aesthetics including but not limited to undergrounding overhead transmission facilities, address localized economic issues, and address other miscellaneous localized needs. For the convenience of the reader, tables

of project information are broken down by four general categories of project drivers but note that these four drivers are not defined in the MISO Tariff.

Projects Driven by Local Reliability

The following projects are driven by local planning criteria without Electric Reliability Organization (ERO) standard applicability or by operational flexibility.

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
18050	Sub 77: Add 161 kV Circuit Breaker	Add new 161 kV circuit breaker at Sub 77 on the 161 kV line terminal to Sub 91.	December 1, 2021	\$0.80
18128	Raun: Retire 13 kV Reactors	Retire the three 13.8 kV reactors at Raun substation and replace relaying and switching device on Raun 345 kV reactor	June 1, 2021	\$1.03
18176	Circuit Breaker Additions and Replacements 2020-2022	Blanket project to add and replace new 345 kV circuit breakers to expand to ring bus which will reduce mis-operations, increase reliability and operating flexibility.	December 1, 2022	\$31.00
18280	Floyd 161-69 kV Transformer Terminal Upgrade	Replace terminal equipment adjacent to Floyd Substation 161-69 kV Transformer AB.	November 1, 2020	\$1.20

Projects Driven by Load Growth

The following projects are proposed to replace aging or degraded equipment.

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
18129	25th Street NW 161 kV Substation	Construct a new 161 kV substation and associated 161 kV line work to source customer-owned transformation and new load.	July 1, 2022	\$12.94
18137	South Waukeee 161-13.2 kV Substation	Construct a new 161-13.2 kV distribution substation and associated 161 kV line work to accommodate area growth.	June 1, 2022	\$11.99

Generator Interconnection Projects

According to Attachment FF of the MISO Tariff, “Generator Interconnection Projects are New Transmission Access Projects that are associated with interconnection of new, or increase in generating capacity of existing, generation.” The following Generator Interconnection Projects (GIPs) have been evaluated through the Generator Interconnection queue and the associated Generator Interconnection Agreements have been signed.

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
18117	Washburn-Hazleton 161 kV Reconductor	Reconductor Washburn-Hazleton 161 kV line. Upgrade Washburn substation terminal equipment.	October 31, 2020	\$16.70
18119	Lundquist-Deere Northeast 161 kV Uprate	Replace terminal equipment at Deere Northeast substation to increase 161 kV line rating.	October 31, 2020	\$0.05
18121	Deere Northeast-Washburn 161 kV Uprate	Replace limiting substation terminal equipment at Deere Northeast and Washburn to increase the rating of the Deere Northeast-Washburn 161 kV line.	October 31, 2020	\$0.10
18124	Sub T FD-Boone Jct 161 kV Reconductor	Reconductor MidAmerican's portion of the Sub T FD-Boone Jct 161 kV line and upgrade substation terminal equipment at Sub T FD.	December 31, 2020	\$7.32
18125	Grimes-Beaver Creek 345 kV Uprate	Replace structures to uprate the Grimes-Beaver Creek 345 kV line.	December 31, 2020	\$0.75
18126	Franklin-Wall Lake 161 kV Uprate	Replace structures to uprate the Franklin-Wall Lake 161 kV line.	June 30, 2020	\$0.22
18127	Sycamore: 345 kV Equipment Upgrades	At the Sycamore substation, replace two 345 kV circuit breakers, associated disconnect switches, and jumpers to increase the rating of the Grimes-Sycamore 345 kV south line.	June 30, 2021	\$1.20
18151	Granger Tap-100th St & 54th Ave 161 kV Reconductor	Reconductor Granger Tap-100th St & 54th Ave 161 kV line with T2-795 ACSR Drake.	June 30, 2021	\$2.09
18174	Bondurant-Montezuma 345 kV Uprate	Replace structures to uprate the Bondurant-Montezuma 345 kV line.	June 30, 2020	\$0.32
18177	Webster-Sub T FD 161 kV Reconductor	Reconductor portion of Webster-Sub T Fort Dodge 161 kV line.	December 15, 2021	\$6.30
18181	Black Hawk 345 kV Capacitor	Add 345 kV, 150 Mvar capacitor bank at Black Hawk 345 kV.	December 31, 2021	\$4.50
18183	Midport 161 kV Capacitor	Add 161 kV, 50 Mvar capacitor at Midport. Includes addition of 161 kV circuit breakers on 161 kV line terminals.	June 30, 2021	\$1.55
18188	Diamond Trail-Hills 345 kV Uprate	Replace structures to uprate the Diamond Trail (J530 POI) - Hills 345 kV line.	June 30, 2020	\$0.35

4.5.8 Minnesota Municipal Power Agency (MMPA)

Minnesota Municipal Power Agency did not submit any new projects for MTEP20. MISO has not identified any issues in MMPA area.

4.5.9 Minnesota Power (MP)

Minnesota Power proposed 12 projects at an estimated cost of \$22 million targeting MTEP20 Appendix A. After MISO's independent reliability analysis of the TO area, 12 of these projects were recommended by MISO to be included in MTEP20 Appendix A at an estimated cost of \$22 million. Of these projects, three are Baseline Reliability Projects and nine are Other Projects. All the projects' expected in-service dates and estimated costs are provided as of September 15, 2020.

Baseline Reliability Projects

According to Attachment FF of the MISO Tariff, "Baseline Reliability Projects are Network Upgrades identified in the base case as required to ensure that the Transmission System is in compliance with applicable national Electric Reliability Organization (ERO) reliability standards and reliability standards adopted by Regional Reliability Organizations and applicable within the Transmission Provider Region."

Project 15592 – Hibbing 14 Line Upgrade

Project Description

This project involves upgrading the Hibbing to 14 Line Tap line. The project's estimated cost is \$1.1 million and expected in-service date is May 1, 2021. The line will be upgraded from a summer rating of 99 MVA to 144 MVA.

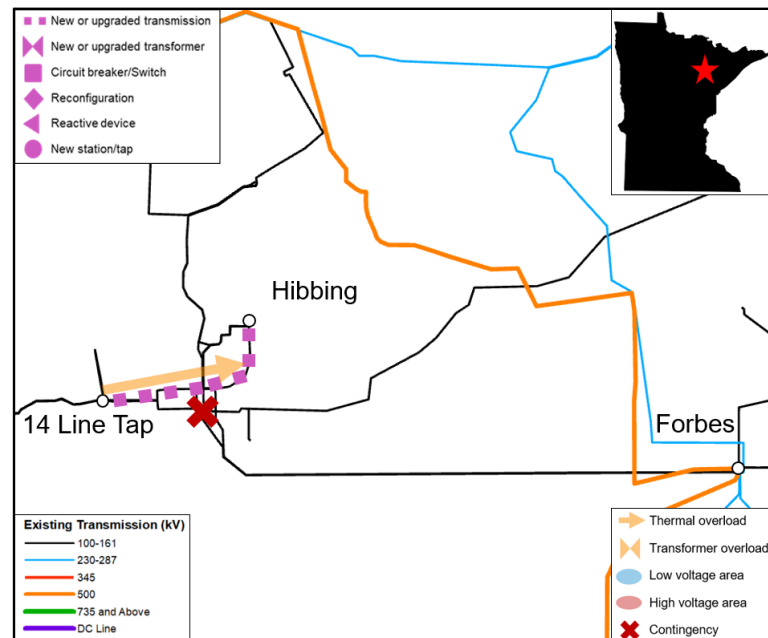


Figure 4.5.9-5: Geographic transmission map of project area

Project Need

Many events for the loss of parallel circuits around Hibbing cause the overloads, shown in Table 4.5-P15592-1, on the Hibbing line. These are seen in the 90% wind cases, and the 2022 Summer case.

Cont. Type	Limiting Element	Rating (MVA)	Pre-Project Loading (%)	Post-Project Loading (%)
P12	608724 HIBBING7 - 608731 14L TAP7 1	99	1.15	0.79
P22	608724 HIBBING7 - 608731 14L TAP7 1	99	1.23	0.84
P23	608724 HIBBING7 - 608731 14L TAP7 1	99	1.07	0.73
P24	608724 HIBBING7 - 608731 14L TAP7 1	99	1.18	0.81
P46	608724 HIBBING7 - 608731 14L TAP7 1	99	1.06	0.72
P55	608724 HIBBING7 - 608731 14L TAP7 1	99	1.15	0.79
P71	608724 HIBBING7 - 608731 14L TAP7 1	99	1.05	0.72

Table 4.5.9-1: P15592 Thermal loading drivers

Alternatives Considered

Generation redispatch is not feasible as this project is being driven by single initiating events, and for some events is over rate B. Upgrading the line is the most cost effective, and reliable option available.

Project 15593 – 25 Line Upgrade

Project Description

This project involves upgrading the Hibbing to Virginia 115 kV line. The project's estimated cost is \$0.9 million and expected in-service date is March 31, 2022. The line will be upgraded from a summer rating of 55 MVA to 99 MVA.

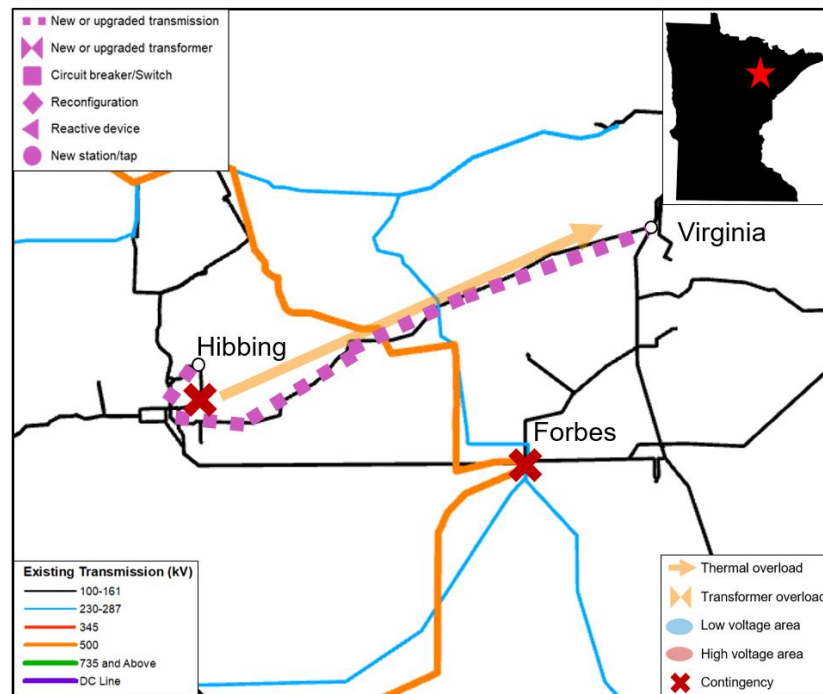


Figure 4.5.9-2: Geographic transmission map of project area

Project Need

Events which cause the loss of parallel circuits around Hibbing and bus faults at Forbes cause the overloads, shown in Table 4.5.9-2, on the Hibbing to Virginia line. These overloads are during high wind transfer scenarios.

Cont. Type	Limiting Element	Rating (MVA)	Pre-Project Loading (%)	Post-Project Loading (%)
P46	608706 KEWTNTP7 - 608724 HIBBING7 1	55	1.16	0.64
P46	608706 KEWTNTP7 - 608743 MATURI 7 1	55	1.13	0.62
P46	608707 MNTACT27 - 608743 MATURI 7 1	55	1.1	0.6

Table 4.5.9-2: P15593 Thermal loading drivers

Alternatives Considered

Generation redispatch is not feasible as this project is being driven by single initiating events and would require a large amount of redispatch. Upgrading the line is the most cost effective, and reliable option available.

Project 15598 – Badoura Transformer Replacement

Project Description

This project involves replacing the existing 230/115 kV transformer at Badoura, along with adding an additional breaker within the Substation. The transformer is non-standard and approaching end of life. The replacement transformer will include LTCs to be consistent with system spare. The project's estimated cost is \$4.5 million and expected in-service date is September 31, 2023.

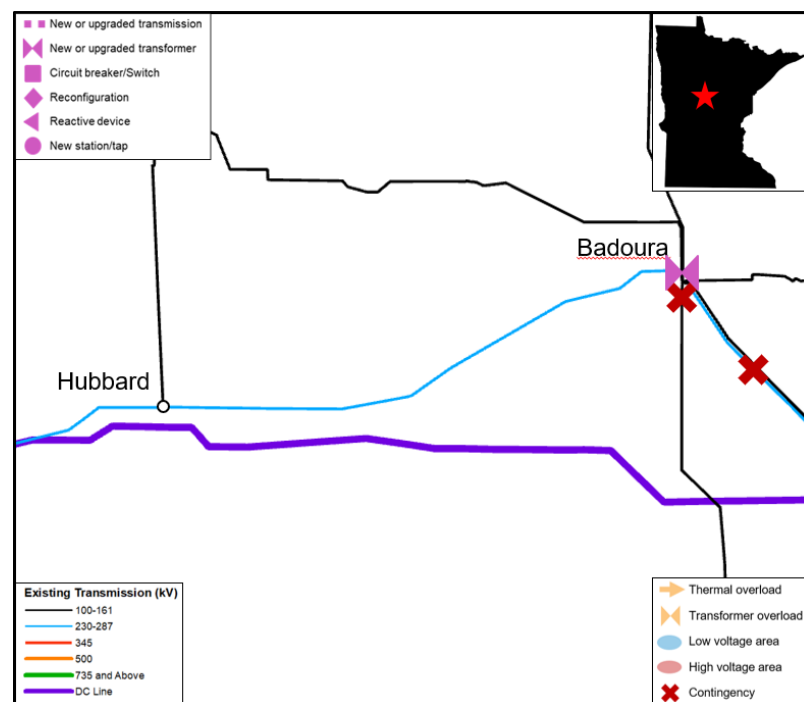


Figure 4.5.9-3: Geographic transmission map of project area

Project Need

This transformer at Badoura is old and approaching end of life. Replacing the transformer and adding an additional breaker will alleviate overloads seen in the area, along with eliminating local system exposure to P23, breaker failure, events.

Cont. Type	Limiting Element	Rating (MVA)	Pre-Project Loading (%)	Post-Project Loading (%)
P6	608610 BADOURA4 - 608640 BADOURA7 1	120	1.02	0.65
P23	608644 DOGLAKE7 - 619700 GRE-THMSTWN7 1	77	1.05	0*

*The P23 contingency is eliminated by the breaker being added with project 15598

Table 4.5.9-3: P15598 Thermal loading drivers

Alternatives Considered

Load shed, or redispatch is not allowable for the events driving this project. The breaker must be added in order to meet TPL compliance.

Other Projects

Other projects do not meet the criteria to be considered as Baseline Reliability Projects (BRP), New Transmission Access Projects, Market Efficiency Projects, or Multi-Value Projects. Other projects may include projects to satisfy Transmission Owner and/or state and local planning criteria other than NERC or regional reliability standards, interconnect new Loads, relocate transmission facilities, address aging transmission infrastructure, replace problematic transmission plant, improve operational performance or address other operational issues, address service reliability issues with end-use consumers, improve aesthetics including but not limited to undergrounding overhead transmission facilities, address localized economic issues, and address other miscellaneous localized needs. For the convenience of the reader, tables of project information are broken down by four general categories of project drivers but note that these four drivers are not defined in the MISO Tariff.

Projects Driven by Local Reliability

The following projects are driven by local planning criteria without Electric Reliability Organization (ERO) standard applicability or by operational flexibility.

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
17869	Superior Transmission Line Relocations	Relocate transmission lines around industrial site to enable development.	March 1, 2021	\$4.9
18109	15th Ave West Transformer Addition	Add 115/34.5 kV transformer to existing 15th Ave West Substation. This project includes the transmission equipment necessary to accommodate the new transformer.	December 31, 2022	\$0.25
18110	Little Falls Bus Reconfiguration	Reconfigure Little Falls 115 kV bus to eliminate line-connected transformer configuration	December 31, 2022	\$0.25

Projects Driven by Age and Condition

The following projects are proposed to replace aging or degraded equipment.

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
17870	National Taconite Breaker Replacements	Replace existing breakers and associated equipment at National Taconite Substation	December 31, 2021	\$1.8
17871	Laskin Breaker Replacements	Replace breakers and associated equipment at Laskin Substation	December 31, 2021	\$0.7
18060	8 Line Relocation	Relocate & rebuild Fond Du Lac - Thomson 115 kV Line. Add a 115 kV breaker at Fond Du Lac.	December 31, 2022	\$3.5
18064	Hibbing Breaker Replacements	Replace existing 115 kV circuit breakers and associated equipment at Hibbing Substation	December 31, 2022	\$1.0
18065	Verndale Substation Modernization	Replace existing 115 kV circuit breakers at Verndale Substation	December 31, 2022	\$0.7

Projects Driven by Load Growth

The following projects are proposed to replace aging or degraded equipment.

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
17184	Nemadji Transformer Addition	Expand 115 kV ring bus and add new 115/13.8 kV transformer at Nemadji Switching Station for customer interconnection	November 1, 2020	\$2.50

4.5.10 Missouri River Energy Services (MRES)

Missouri River Energy Services proposed two projects at an estimated cost of \$1.6 million targeting MTEP20 Appendix A. After MISO's independent reliability analysis of the MRES area, these projects were recommended by MISO to be included in MTEP20 Appendix A at an estimated cost of \$1.6 million. Of these projects, one is a Baseline Reliability Project and one is a Generation project with a signed Generator Interconnection Agreement. Both project's expected in-service dates and estimated costs are provided as of September 15, 2020.

Baseline Reliability Projects

According to Attachment FF of the MISO Tariff, "Baseline Reliability Projects are Network Upgrades identified in the base case as required to ensure that the Transmission System is in compliance with applicable national Electric Reliability Organization (ERO) reliability standards and reliability standards adopted by Regional Reliability Organizations and applicable within the Transmission Provider Region."

Project 18645 - Hutchinson Jumper Replacement

Project Description

This project involves replacing the jumpers at Hutchinson/McLeod 230/115 kV transformer to mitigate overloads on this transformer observed in the MTEP19 results. The transformer is limited by a couple sets of jumpers. To increase the transformer capacity the jumpers will be replaced. The project's estimated cost is \$20,000 and expected in-service date is June 1, 2021.

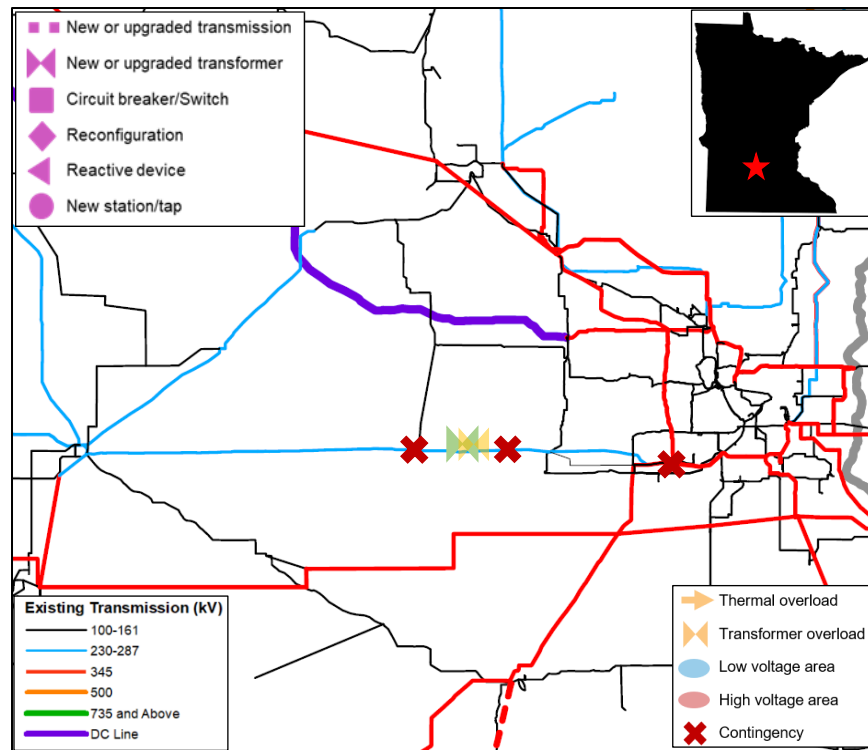


Figure 4.5.10-6: Geographic transmission map of project area

Project Need

Under high wind sensitivity cases, the Hutchinson 230/115 kV transformer can be overloaded for a variety of P1, P2 and P7 category events which result in the loss of parallel or other lines. As shown in Table 4.5-P18645-1. This project will provide Hutchinson Transformer 25% margin for emergency. This allows applicable time duration for system adjustment to bring the flow back within normal rating.

Cont. Type	Limiting Element	Rating (MVA)	Pre-Project Loading (%)	Post-Project Loading (%)
P12	658276 HUC-MCLEOD 4 - 658277 HUC-MCLEOD 7 1	177	125.5	91.5
P23	658276 HUC-MCLEOD 4 - 658277 HUC-MCLEOD 7 1	177	125.8	91.6
P71	658276 HUC-MCLEOD 4 - 658277 HUC-MCLEOD 7 1	177	109.4	80.3

Table 4.5.10-1: P18645 Thermal loading drivers

Alternatives Considered

This were no alternatives received to address Hutchinson 230/161 kV transformer overloads.

Generator Interconnection Projects

According to Attachment FF of the MISO Tariff, “Generator Interconnection Projects are New Transmission Access Projects that are associated with interconnection of new, or increase in generating capacity of existing, generation.” The following Generator Interconnection Projects (GIPs) have been evaluated through the Generator Interconnection queue and the associated Generator Interconnection Agreements have been signed.

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
18906	McLeod Cap Bank Addition	The August 2016 West DPP cycle identified a network upgrade at the McLeod 230 kV bus. The project involves adding a switchable 50 Mvar capacitor bank to mitigate low voltages.	September 1, 2021	\$ 1.60

4.5.11 Montana-Dakota Utilities Co. (MDU)

Montana-Dakota Utilities Co. proposed seven projects at an estimated cost of \$46.6 million targeting MTEP20 Appendix A. After MISO’s independent reliability analysis of the TO area, seven of these projects were recommended by MISO to be included in MTEP20 Appendix A at an estimated cost of \$46.6 million. Of these projects, zero are Baseline Reliability Projects, five are Other Projects, and two are Generation projects with signed Generator Interconnection Agreements. All the projects’ expected in-service dates and estimated costs are provided as of September 15, 2020.

Other Projects

Other projects do not meet the criteria to be considered as Baseline Reliability Projects (BRP), New Transmission Access Projects, Market Efficiency Projects, or Multi-Value Projects. Other projects may include projects to satisfy Transmission Owner and/or state and local planning criteria other than NERC or regional reliability standards, interconnect new Loads, relocate transmission facilities, address aging transmission infrastructure, replace problematic transmission plant, improve operational performance or address other operational issues, address service reliability issues with end-use consumers, improve aesthetics including but not limited to undergrounding overhead transmission facilities, address localized economic issues, and address other miscellaneous localized needs. For the convenience of the reader, tables of project information are broken down by four general categories of project drivers but note that these four drivers are not defined in the MISO Tariff.

Projects Driven by Local Reliability

The following projects are driven by local planning criteria without Electric Reliability Organization (ERO) standard applicability or by operational flexibility.

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
18257	Greenway Substation	Build three breaker 41.6 kV ring bus on at junction of three terminal Wishek-Bowdle-Glenham 41.6 kV line.	December 31, 2021	\$ 3.10

Projects Driven by Age and Condition

The following projects are proposed to replace aging or degraded equipment.

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
15784	Zahl-Corinth 60 kV line	Rebuild Zahl-Corinth 60 kV line	September 30, 2023	\$ 3.50
17466	Heskett Substation Retirement	The Heskett substation will be retired due to age and condition concerns. The existing Mandan substation will be expanded to accommodate the 230 kV and 115 kV facilities currently connected to Heskett. A new Mandan 38th St substation will be constructed to accommodate the 69 and 41.6 kV facilities.	May 31, 2023	\$ 24.00
18254	Tioga Substation Rebuild	Rebuild Tioga substation due to age and condition	December 31, 2021	\$ 2.60
18260	Greenway Tap-Herried Rebuild	Rebuild Greenway Tap-Herried 41.6 kV line	December 31, 2021	\$ 6.60

Generator Interconnection Projects

According to Attachment FF of the MISO Tariff, "Generator Interconnection Projects are New Transmission Access Projects that are associated with interconnection of new, or increase in generating capacity of existing, generation." The following Generator Interconnection Projects (GIPs) have been evaluated through the Generator Interconnection queue and the associated Generator Interconnection Agreements have been signed.

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
17304	G359, Ellendale 345 Network Upgrades	G359 Network Upgrades at Ellendale 345: four 230 kV 50 Mvar fast-switched capacitors and a new switcher for the 345 kV line reactor on the Big Stone South line.	February 28, 2020	\$ 6.43
19485*	Big Stone-Browns Valley 230 Line Structure Replacements	GI network upgrade to replace structures on Big Stone-Browns Valley 230 kV line.	June 22, 2020	\$ 0.35

* This GIP project was submitted after **February 1, 2020** (modeling cut off date) and not included in the MTEP20 models. This will be modeled in the following MTEP cycle.

4.5.12 Muscatine Power and Water (MPW)

Muscatine Power and Water did not submit any new projects for MTEP20. MISO has not identified any issues in MPW area.

4.5.13 Northwestern Wisconsin Electric (NWECE)

Northwestern Wisconsin Electric did not submit any new projects for MTEP20. MISO has not identified any issues in NWECE area.

4.5.14 Otter Tail Power Company (OTP)

Otter Tail Power proposed five projects at an estimated cost of \$19 million targeting MTEP20 Appendix A. After MISO's independent reliability analysis of the Otter Tail Power area, five of these projects were recommended by MISO to be included in MTEP20 Appendix A at an estimated cost of \$19 million. Of these projects, one is a Baseline Reliability Project, two are Other Projects, and two are Generation projects with signed Generator Interconnection Agreements. All the projects' expected in-service dates and estimated costs are provided as of September 15, 2020.

Baseline Reliability Projects

According to Attachment FF of the MISO Tariff, "Baseline Reliability Projects are Network Upgrades identified in the base case as required to ensure that the Transmission System is in compliance with applicable national Electric Reliability Organization (ERO) reliability standards and reliability standards adopted by Regional Reliability Organizations and applicable within the Transmission Provider Region."

Project 10043 – Jamestown 345 kV Breaker Addition

Project Description

This project involves constructing and additional breaker at the Jamestown 345 kV substation. The project's estimated cost is \$1.2 million and expected in-service date is July 3, 2020.

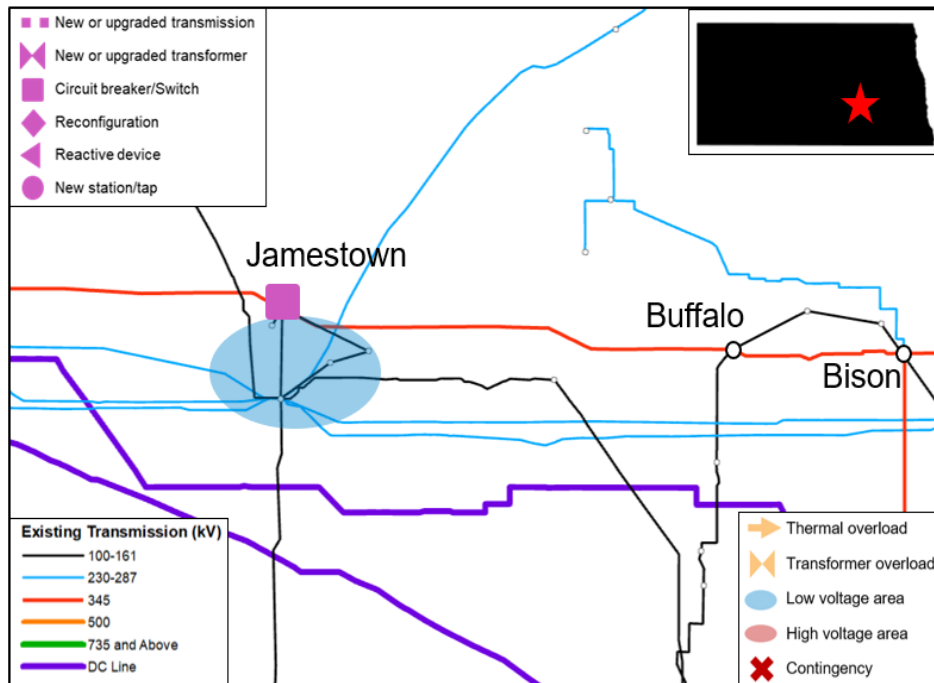


Figure 4.5.14-7: Geographic transmission map of project area

Project Need

This additional breaker is being implemented to ensure TPL-001-4 compliance by eliminating the need for load shed to mitigate low voltage on the Jamestown 115 kV system after a P23 event within the 345 kV substation. Without load shed around Jamestown after this P23, local voltage collapse would occur on the 115 kV system, as seen in the Table 4.5.14-1 below.

Cont. Type	Limiting Element	Voltage Limit (pu)	Pre-Project Voltage (pu)	Post-Project Voltage (pu)
P23	620272 JAMESPK7	0.97	Non-Converged	1.01
P23	620273 JAMETAP7	0.97	Non-Converged	1.01
P23	620270 SPIRITWD 7	0.97	Non-Converged	1.01

Table 4.5.14-1: P10043 Voltage performance drivers

Alternatives Considered

This project is being driven by breaker failure events. Adding in an additional 345 kV breaker at the Jamestown substation is the most cost-effective option in order to eliminate the voltage collapse exposure on the 115 kV system.

Other Projects

Other projects do not meet the criteria to be considered as Baseline Reliability Projects (BRP), New Transmission Access Projects, Market Efficiency Projects, or Multi-Value Projects. Other projects may include projects to satisfy Transmission Owner and/or state and local planning criteria other than NERC or regional reliability standards, interconnect new Loads, relocate transmission facilities, address aging transmission infrastructure, replace problematic transmission plant, improve operational performance or address other operational issues, address service reliability issues with end-use consumers, improve aesthetics including but not limited to undergrounding overhead transmission facilities, address localized economic issues, and address other miscellaneous localized needs. For the convenience of the reader, tables

of project information are broken down by four general categories of project drivers but note that these four drivers are not defined in the MISO Tariff.

Projects Driven by Local Reliability

The following projects are driven by local planning criteria without Electric Reliability Organization (ERO) standard applicability or by operational flexibility.

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
18364	OTP Oslo 115 kV Switch Replacement	Replace 600 Amp Switch at OTP Oslo with 1200 Amp Switch.	November 30, 2022	\$0.2

Projects Driven by Age and Condition

The following projects are proposed to replace aging or degraded equipment.

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
19685	OTP Hoot Lake 115/41.6 kV Transformer	Replace Hoot Lake 115/41.6 kV Transformer with 115/41.6/34.5 kV Transformer with higher capacity and LTC.	December 31, 2021	\$0.9

Generator Interconnection Projects

According to Attachment FF of the MISO Tariff, "Generator Interconnection Projects are New Transmission Access Projects that are associated with interconnection of new, or increase in generating capacity of existing, generation." The following Generator Interconnection Projects (GIPs) have been evaluated through the Generator Interconnection queue and the associated Generator Interconnection Agreements have been signed.

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
18504	OTP Hankinson to Wahpeton Network Upgrades-G736/J442	This project consists of upgrades to Otter Tail Power Company's substations and transmission lines to accommodate the interconnection of a wind generators, G736/J442, to the Big Stone South substation	June 2, 2020	\$0.22
18947	OTP Ellendale-Oakes-Forman-Hankinson 230 kV Upgrades J302/J503	This project consists of upgrades to Otter Tail Power Company's substations and transmission lines to accommodate the interconnection of a wind generators, J302/J503.	March 1, 2021	\$16.5

4.5.15 Rochester Public Utilities (RPU)

Rochester Public Utilities did not submit any new projects for MTEP20. MISO has not identified any issues in RPU area.

4.5.16 Southern Minnesota Municipal Power Agency (SMMPA)

Southern Minnesota Municipal Power Agency did not submit any new projects for MTEP20. MISO has not identified any issues in SMMPA area.

4.5.17 Wilmar Municipal Utilities (WMU)

Wilmar Municipal Utilities did not submit any new projects for MTEP20. MISO has not identified any issues in WMU area.

4.5.18 WPPI Energy (WPPI)

WPPI Energy did not submit any new projects for MTEP20. MISO has not identified any issues in WPPI area.

4.5.19 Xcel Energy (Northern States Power)

Xcel Energy proposed 17 projects at an estimated cost of \$111 million targeting MTEP20 Appendix A. After MISO's independent reliability analysis of the Xcel Energy area, 17 of these projects were recommended by MISO to be included in MTEP20 Appendix A at an estimated cost of \$111 million. Of these projects, one is a Baseline Reliability Project, 13 are Other Projects, and three are Generation projects with signed Generator Interconnection Agreements. All the projects' expected in-service dates and estimated costs are provided as of September 15, 2020.

Baseline Reliability Projects

According to Attachment FF of the MISO Tariff, "Baseline Reliability Projects are Network Upgrades identified in the base case as required to ensure that the Transmission System is in compliance with applicable national Electric Reliability Organization (ERO) reliability standards and reliability standards adopted by Regional Reliability Organizations and applicable within the Transmission Provider Region."

Project 15877 – East Metro Area Upgrades

Project Description

This project involves reconfiguring line connections at the Red Rock 115 kV substation as well as upgrading the 115 kV line from Baytown to Long Lake. The project's estimated cost is \$10 million and expected in-service date is June 15, 2022.

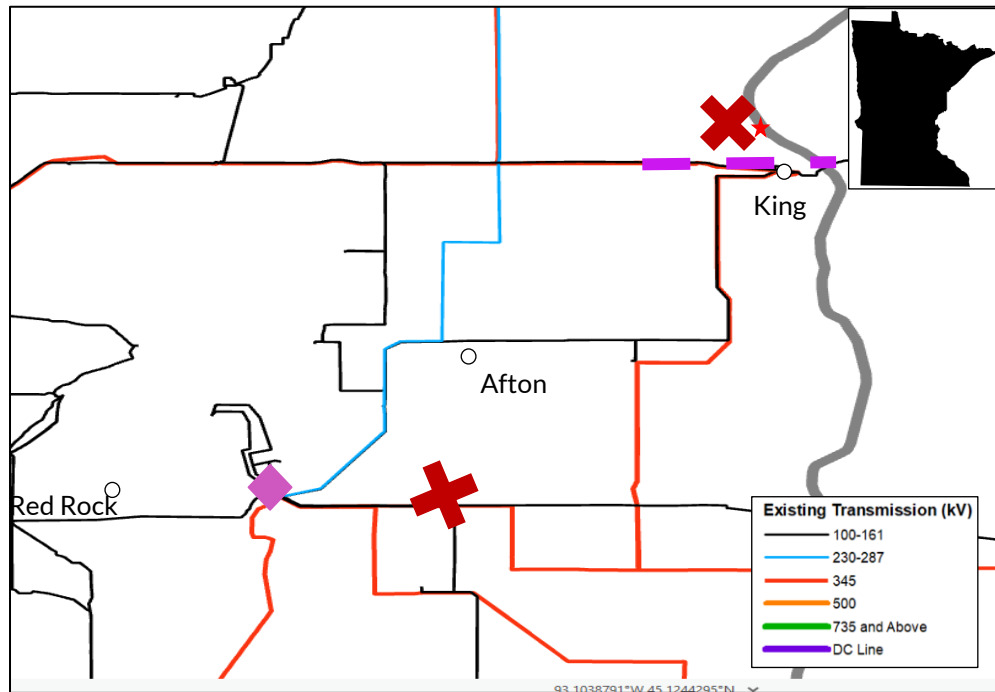


Figure 4.5.19-8: Geographic transmission map of project area

Project Need

In summer peak conditions, loss of the King transformer and either the Baytown-Long Lake 115 kV line or the Red Rock-Afton 115 kV line there are thermal overloads, with some above rate B, in the eastern Twin Cities area as well as several thermal and voltage concerns in the western Wisconsin area. This project prevents any thermal overloads above rate B so all concerns can be addressed (Table 4.5.19-1).

Cont. Type	Limiting Element	Rating (MVA)	Pre-Project Loading (%)	Post-Project Loading (%)
P6	Baytown – Long Lake 115 kV	360.7	117.4	74.4

Table 4.5.19-1: P15877 Thermal loading drivers

Alternatives Considered

Generation redispatch or load shed is not feasible for this thermal violation due the loading being above the rate B emergency rating. Upgrading this 115 kV line is the best and most reliable option to address this reliability issue.

Other Projects

Other projects do not meet the criteria to be considered as Baseline Reliability Projects (BRP), New Transmission Access Projects, Market Efficiency Projects, or Multi-Value Projects. Other projects may include projects to satisfy Transmission Owner and/or state and local planning criteria other than NERC or regional reliability standards, interconnect new Loads, relocate transmission facilities, address aging transmission infrastructure, replace problematic transmission plant, improve operational performance or address other operational issues, address service reliability issues with end-use consumers, improve aesthetics including but not limited to undergrounding overhead transmission facilities, address localized economic issues, and address other miscellaneous localized needs. For the convenience of the reader, tables of project information are broken down by four general categories of project drivers but note that these four drivers are not defined in the MISO Tariff.

Projects Driven by Local Reliability

The following projects are driven by local planning criteria without Electric Reliability Organization (ERO) standard applicability or by operational flexibility.

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
17984	New Turtle Lake Sub	A new substation will be built to replace existing distribution Turtle Lake sub allow future load. Existing Turtle Lake sub will be retired once distribution conversion is complete.	June 15, 2021	\$0.70
17985	Rosemount 2nd XFMR	Add second 115/34.5 kV transformer to Rosemount Substation.	December 15, 2020	\$1.20
18006	Osprey - Hawkins 115 kV Rebuild	Rebuild 11.36 miles of 115 kV line between Osprey to Hawkins to remediate aging conditions on-line that have resulted in reliability issues.	June 15, 2022	\$8.50
18007	Prentice - Phillips 115 kV Rebuild	Rebuild of 14.8 miles of 115 kV line from Prentice - Phillips Tap needed to remediate reliability issues caused by aging condition of the line.	January 31, 2023	\$10.00
18029	West St Cloud - Millwood 69 kV Rebuild	Rebuild 24 miles of 69 kV line from West St. Cloud - Millwood to remediate reliability issues from aging condition of line.	June 1, 2021	\$10.40
18505	Jim Falls - Holcombe Rebuild	Rebuild Jim Falls - Holcombe 115 kV line to remove overload under contingency.	March 1, 2023	\$15.00

Projects Driven by Age and Condition

The following projects are proposed to replace aging or degraded equipment.

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
18038	Kinnickinick Substation Rebuild	Existing substation equipment is nonstandard equipment and is beyond expected end of life and requires rebuild.	December 15, 2020	\$0.35
18130	Port Wing - Cornucopia 34.5 kV Rebuild	9.6 miles of 34.5 kV line from Port Wing to Cornucopia is at end of useful life, rebuild	June 15, 2023	\$5.00

		needed to remediate reliability concerns due to aging condition of line.		
18131	Orienta Tap - Iron River 34.5 kV Rebuild	Rebuild 9 miles of 34.5 kV line from former Orienta tap switch to Iron River to remediate aging condition issues.	June 15, 2023	\$5.00
19406	Chisago Capacitor Retirement	Chisago 34.5 kV capacitors on the tertiary of TR9 and TR10 have failed catastrophically. There is currently no operational or planning need for the capacitors at this time and they will be retired.	December 1, 2020	\$0.16

Projects Driven by Load Growth

The following projects are proposed to replace aging or degraded equipment.

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
17989	EREPC Wellington Interconnection	Install new 3-way switch and 4-mile 69 kV line to new EREPC Wellington substation.	September 1, 2020	\$1.32
18033	Hydro Lane Expansion	Expand Hydro Lane substation to support increased customer load.	May 15, 2020	\$0.35
18132	Cottage Grove Airgas Load Interconnection	New 115 kV in and out substation tapped on Chemolite to Cottage Grove 115 kV line to serve new Airgas load interconnection.	October 1, 2020	\$5.00

Generator Interconnection Projects

According to Attachment FF of the MISO Tariff, "Generator Interconnection Projects are New Transmission Access Projects that are associated with interconnection of new, or increase in generating capacity of existing, generation." The following Generator Interconnection Projects (GIPs) have been evaluated through the Generator Interconnection queue and the associated Generator Interconnection Agreements have been signed.

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
18344	J587 Blazing Star 2	New 200 MW wind Generator Interconnection with new 345 kV – 15 kV substation and generation facilities.	September 15, 2020	\$3.10
18345	Scott County - Helena Line Upgrade	Upgrade Helena - Scott County 345 kV line to accommodate additional flow from J587 Blazing Star 2 Generator Interconnection.	September 15, 2022	\$34.00
18444	J526 Panther - HUC-McLeod upgrade	Replace structures on HUC - McLeod 230 kV line to increase capacity to accommodate J526 wind generation.	March 31, 2020	\$0.88

